

superior conj	2000 Jan 16 01:18:55	25°☿12'16	-1°59'58		2000 Nov 07 07:27:32	30°♄	
minimum elong	2000 Jan 15 21:08:41	24°☿55'01	1°59'47	direct	2000 Nov 08 02:28:04	29°♄56'26	
	2000 Jan 18 22:19:47	0°≈			2000 Nov 08 21:42:12	0°♄	
evening rise	2000 Jan 28 13:52:18	16°≈34'14		max. declination.	2000 Nov 09 18:35:46	0°♄11'42	9S34'36"
	2000 Feb 05 08:08:33	0°✕		morning max el	2000 Nov 15 05:22:05	3°♄56'43	19°20'25
asc. node	2000 Feb 11 02:31:56	9°✕00'00		greatest brilliancy	2000 Nov 29 02:12:24	22°♄42'41	-0.7m
evening max el	2000 Feb 15 01:15:04	13°✕46'50	18°08'55		2000 Dec 03 20:25:59	0°♄	
retrograde	2000 Feb 21 12:46:24	17°✕10'42		morning set	2000 Dec 09 00:58:31	8°♄01'52	
evening set	2000 Feb 24 08:26:03	16°✕31'03		desc. node	2000 Dec 09 07:26:54	8°♄26'59	
max. declination.	2000 Feb 24 04:28:39	16°✕35'29	2S22'9"	max. Earth dist.	2000 Dec 22 13:04:32	29°♄08'43	1.44674 AU
inferior conj	2000 Mar 01 15:10:33	11°✕20'33	3°41'46		2000 Dec 23 02:02:54	0°☿	
minimum elong	2000 Mar 01 15:52:40	11°✕18'43	3°41'44				
min. Earth dist.	2000 Mar 04 11:15:57	8°✕23'59	0.62420 AU	superior conj	2000 Dec 25 19:23:03	4°☿19'15	-1°36'44
morning rise	2000 Mar 07 22:08:31	5°✕21'02		minimum elong	2000 Dec 25 10:49:31	3°☿45'11	1°35'58
direct	2000 Mar 14 20:39:18	2°✕46'44		min. declination.	2000 Dec 26 15:11:00	5°☿38'10	24S59'14"
desc. node	2000 Mar 20 09:36:40	4°✕12'34		evening rise	2001 Jan 09 04:34:18	27°☿43'31	
min. declination.	2000 Mar 20 20:37:29	4°✕26'36	9S58'10"		2001 Jan 10 13:25:44	0°≈	
morning max el	2000 Mar 28 21:13:19	10°✕42'02	27°49'30	asc. node	2001 Jan 27 23:35:19	26°≈31'20	
	2000 Apr 13 00:17:14	0°☿		evening max el	2001 Jan 28 13:38:11	27°≈08'07	18°25'53
	2000 Apr 30 03:53:17	0°♄			2001 Feb 01 07:12:51	0°✕	
morning set	2000 May 01 15:21:16	2°♄58'00		retrograde	2001 Feb 04 01:58:07	0°✕41'41	
max. Earth dist.	2000 May 07 05:42:13	14°♄40'49	1.32608 AU		2001 Feb 06 19:57:07	30°♄	
				evening set	2001 Feb 07 02:53:52	29°≈50'53	
superior conj	2000 May 09 03:49:33	18°♄49'54	0°00'54	max. declination.	2001 Feb 07 15:08:17	29°≈32'44	8S56'43"
minimum elong	2000 May 09 03:46:51	18°♄49'39	0°00'53	inferior conj	2001 Feb 13 00:17:00	24°≈21'18	3°40'48
behind sun begin	2000 May 08 22:36:52	18°♄21'38		minimum elong	2001 Feb 12 23:16:27	24°≈24'13	3°40'43
behind sun end	2000 May 09 08:56:49	19°♄17'43		min. Earth dist.	2001 Feb 15 05:09:53	21°≈48'18	0.64268 AU
asc. node	2000 May 09 01:49:07	18°♄39'00		morning rise	2001 Feb 18 19:03:50	18°≈14'04	
	2000 May 14 07:10:23	0°♄		direct	2001 Feb 25 15:41:19	15°≈24'37	
evening rise	2000 May 16 02:45:46	3°♄53'07		min. declination.	2001 Mar 04 15:03:42	17°≈46'34	15S 0'56"
max. declination.	2000 May 29 14:50:00	29°♄07'24	25N38'50"	desc. node	2001 Mar 07 06:39:15	19°≈40'35	
	2000 May 30 04:27:13	0°☿		morning max el	2001 Mar 11 06:09:48	23°≈16'39	27°27'46
evening max el	2000 Jun 09 13:31:53	13°☿01'28	24°02'57		2001 Mar 17 06:04:57	0°✕	
desc. node	2000 Jun 16 08:51:39	18°☿07'07			2001 Apr 06 07:13:33	0°☿	
retrograde	2000 Jun 23 08:31:55	19°☿57'43		morning set	2001 Apr 15 09:24:23	16°☿50'58	
evening set	2000 Jun 27 19:19:12	19°☿13'55		max. Earth dist.	2001 Apr 20 06:48:44	26°☿44'55	1.33421 AU
min. Earth dist.	2000 Jul 04 00:16:17	16°☿13'13	0.56391 AU		2001 Apr 21 20:08:08	0°♄	
inferior conj	2000 Jul 06 11:34:56	14°☿41'46	-4°39'54				
minimum elong	2000 Jul 06 06:34:56	14°☿49'31	4°39'15	superior conj	2001 Apr 23 09:24:29	3°♄17'09	-0°26'41
min. declination.	2000 Jul 08 06:24:22	13°☿36'33	17N57'17"	minimum elong	2001 Apr 23 10:44:21	3°♄24'13	0°26'23
morning rise	2000 Jul 14 20:29:59	10°☿42'12		asc. node	2001 Apr 25 22:53:18	8°♄45'39	
direct	2000 Jul 17 13:20:19	10°☿23'11		evening rise	2001 Apr 30 14:10:02	18°♄39'05	
morning max el	2000 Jul 27 09:16:30	14°☿56'50	19°47'52		2001 May 06 04:53:02	0°♄	
max. declination.	2000 Aug 02 09:00:07	22°☿06'39	21N 1'51"	max. declination.	2001 May 20 16:12:12	21°♄57'24	25N27'15"
asc. node	2000 Aug 05 01:01:40	26°☿14'25		evening max el	2001 May 22 04:33:11	23°♄30'12	22°26'44
	2000 Aug 07 05:42:02	0°♄		desc. node	2001 Jun 03 05:52:22	29°♄55'40	
morning set	2000 Aug 14 04:15:54	13°♄20'50		retrograde	2001 Jun 04 05:21:49	29°♄57'51	
				evening set	2001 Jun 07 06:05:41	29°♄37'36	
superior conj	2000 Aug 22 01:05:28	29°♄14'55	1°45'38	min. Earth dist.	2001 Jun 15 11:39:40	26°♄03'52	0.55212 AU
minimum elong	2000 Aug 22 01:25:36	29°♄16'35	1°45'38	inferior conj	2001 Jun 16 13:25:56	25°♄27'22	-3°34'14
	2000 Aug 22 10:10:48	0°♄		minimum elong	2001 Jun 16 05:30:22	25°♄38'37	3°32'06
max. Earth dist.	2000 Aug 27 19:54:32	10°♄26'24	1.37241 AU	morning rise	2001 Jun 25 07:00:02	21°♄35'42	
evening rise	2000 Aug 31 15:07:03	17°♄25'24		min. declination.	2001 Jun 26 06:09:49	21°♄24'59	18N35'19"
	2000 Sep 07 22:22:24	0°♄		direct	2001 Jun 28 05:49:04	21°♄15'55	
desc. node	2000 Sep 12 08:11:15	7°♄05'13		morning max el	2001 Jul 09 17:03:47	26°♄35'57	21°07'53
	2000 Sep 28 13:27:39	0°♄			2001 Jul 12 22:46:40	0°☿	
evening max el	2000 Oct 06 10:18:31	8°♄49'40	25°30'51	asc. node	2001 Jul 22 22:06:25	15°☿08'36	
min. declination.	2000 Oct 17 09:56:10	15°♄42'57	19S38'45"	max. declination.	2001 Jul 22 18:49:50	14°☿53'39	22N34'46"
retrograde	2000 Oct 18 13:41:01	15°♄47'57		morning set	2001 Jul 29 12:02:40	28°☿04'44	
evening set	2000 Oct 24 12:39:51	13°♄18'23			2001 Jul 30 10:17:46	0°♄	
min. Earth dist.	2000 Oct 28 22:05:50	8°♄27'42	0.67052 AU				
inferior conj	2000 Oct 30 02:09:05	6°♄57'33	-0°39'41	superior conj	2001 Aug 05 21:50:56	13°♄31'21	1°43'19
minimum elong	2000 Oct 30 03:10:18	6°♄54'16	0°39'15	minimum elong	2001 Aug 05 20:43:39	13°♄25'33	1°43'17
asc. node	2000 Nov 01 00:15:37	4°♄33'32		max. Earth dist.	2001 Aug 10 05:44:21	22°♄17'52	1.35456 AU
morning rise	2000 Nov 04 17:54:50	1°♄01'25		evening rise	2001 Aug 14 10:48:09	0°♄27'07	

	2001 Aug 14 05:03:43	0°♎		asc. node	2002 Jul 09 19:09:17	4°♊30'36	
desc. node	2001 Aug 30 05:11:47	27°♎16'51		max. declination.	2002 Jul 12 22:05:29	10°♊49'17	23°N33'52"
	2001 Sep 01 00:36:59	0°♏		morning set	2002 Jul 13 22:41:08	12°♊57'24	
evening max el	2001 Sep 18 22:24:12	22°♏28'02	26°32'00				
retrograde	2001 Oct 01 19:23:19	29°♏41'10		superior conj	2002 Jul 21 01:47:09	28°♊09'08	1°34'38
min. declination.	2001 Oct 01 17:21:28	29°♏41'08	14S46'50"	minimum elong	2002 Jul 20 23:45:11	27°♊58'20	1°34'27
evening set	2001 Oct 08 07:55:12	27°♏00'38			2002 Jul 21 22:41:20	0°♋	
min. Earth dist.	2001 Oct 12 10:00:04	22°♏46'22	0.66152 AU	max. Earth dist.	2002 Jul 24 00:16:09	4°♋20'10	1.34036 AU
inferior conj	2001 Oct 14 01:42:38	20°♏46'52	-1°37'47	evening rise	2002 Jul 28 20:55:53	14°♋11'05	
minimum elong	2001 Oct 14 04:18:59	20°♏39'01	1°36'44		2002 Aug 06 09:50:58	0°♌	
asc. node	2001 Oct 18 21:18:24	15°♏47'09		desc. node	2002 Aug 17 02:12:51	17°♌04'05	
morning rise	2001 Oct 20 01:16:11	15°♏01'32			2002 Aug 26 21:09:56	0°♍	
direct	2001 Oct 23 00:22:53	14°♏12'07		evening max el	2002 Sep 01 10:23:49	5°♍55'54	27°12'40
max. declination.	2001 Oct 25 06:08:52	14°♏39'24	4S17'45"	retrograde	2002 Sep 14 19:38:26	13°♍14'39	
morning max el	2001 Oct 29 16:30:18	17°♏52'03	18°33'55	min. declination.	2002 Sep 15 16:51:25	13°♍12'10	8S54' 4"
	2001 Nov 07 19:52:39	0°♎		evening set	2002 Sep 21 18:52:43	10°♍30'51	
morning set	2001 Nov 19 00:21:39	17°♎40'42		min. Earth dist.	2002 Sep 25 14:23:40	6°♍52'33	0.64894 AU
desc. node	2001 Nov 26 04:29:03	29°♎05'00		inferior conj	2002 Sep 27 18:31:17	4°♍28'25	-2°36'08
	2001 Nov 26 18:23:29	0°♏		minimum elong	2002 Sep 27 22:39:38	4°♍16'57	2°34'37
superior conj	2001 Dec 04 21:35:54	12°♏48'22	-0°56'08		2002 Oct 02 09:25:48	30°♎	
minimum elong	2001 Dec 04 14:34:24	12°♏20'47	0°55'16	morning rise	2002 Oct 04 03:22:02	28°♎56'52	
max. Earth dist.	2001 Dec 05 07:21:54	13°♏26'42	1.45145 AU	asc. node	2002 Oct 05 18:20:30	28°♎25'02	
	2001 Dec 15 19:54:43	0°♐		direct	2002 Oct 06 19:25:45	28°♎19'12	
min. declination.	2001 Dec 18 10:15:50	4°♐06'40	25S21'31"	max. declination.	2002 Oct 09 16:56:26	29°♎03'02	1N19'21"
evening rise	2001 Dec 20 19:54:13	7°♐55'19			2002 Oct 11 05:56:25	0°♑	
greatest brilliancy	2001 Dec 29 16:46:47	21°♐59'39	-0.8m	morning max el	2002 Oct 13 07:34:34	1°♑48'03	18°04'06
	2002 Jan 03 21:37:56	0°♒		morning set	2002 Oct 31 04:55:09	28°♑45'51	
evening max el	2002 Jan 11 23:44:16	10°♒35'00	19°00'41		2002 Oct 31 22:42:55	0°♓	
asc. node	2002 Jan 14 20:37:06	13°♒03'31		desc. node	2002 Nov 13 01:29:42	19°♓49'14	
retrograde	2002 Jan 18 20:49:56	14°♒29'05		superior conj	2002 Nov 14 04:39:36	21°♓37'50	-0°07'38
evening set	2002 Jan 22 03:54:58	13°♒26'09		minimum elong	2002 Nov 14 03:42:45	21°♓34'03	0°07'29
max. declination.	2002 Jan 23 20:48:30	12°♒06'26	14S39' 8"	behind sun begin	2002 Nov 13 18:09:30	20°♓55'53	
inferior conj	2002 Jan 27 18:55:03	7°♒39'49	3°24'31	behind sun end	2002 Nov 14 13:16:00	22°♓12'09	
minimum elong	2002 Jan 27 16:48:06	7°♒46'29	3°24'10	max. Earth dist.	2002 Nov 18 02:03:11	27°♓48'19	1.44873 AU
min. Earth dist.	2002 Jan 29 08:27:02	5°♒41'50	0.65744 AU		2002 Nov 19 11:29:25	0°♈	
morning rise	2002 Feb 02 05:21:56	1°♒28'23		evening rise	2002 Nov 30 14:10:07	17°♈18'12	
	2002 Feb 04 04:18:29	30°♒♐			2002 Dec 08 20:20:46	0°♉	
direct	2002 Feb 08 17:27:44	28°♐37'49		min. declination.	2002 Dec 10 07:23:39	2°♐13'25	25S39'57"
	2002 Feb 13 17:20:07	0°♒		greatest brilliancy	2002 Dec 14 04:59:21	8°♐05'32	-0.7m
min. declination.	2002 Feb 17 17:43:48	2°♒40'56	18S47'58"	evening max el	2002 Dec 26 05:38:12	24°♐04'26	19°51'46
morning max el	2002 Feb 21 16:07:39	6°♒15'47	26°35'20	asc. node	2003 Jan 01 17:37:30	28°♐21'53	
desc. node	2002 Feb 22 03:41:42	6°♒45'13		retrograde	2003 Jan 02 18:19:17	28°♐27'45	
	2002 Mar 11 23:33:34	0°♈		evening set	2003 Jan 06 09:02:51	27°♐11'23	
morning set	2002 Mar 29 16:07:34	0°♉06'33		inferior conj	2003 Jan 11 20:02:25	21°♐10'56	2°56'45
	2002 Mar 29 14:44:20	0°♉		minimum elong	2003 Jan 11 17:26:31	21°♐19'35	2°56'06
max. Earth dist.	2002 Apr 02 21:04:40	8°♉15'39	1.34683 AU	max. declination.	2003 Jan 11 16:08:36	21°♐23'54	18S51'37"
superior conj	2002 Apr 07 08:54:43	17°♉21'45	-0°54'38	min. Earth dist.	2003 Jan 12 19:28:49	19°♐52'58	0.66815 AU
minimum elong	2002 Apr 07 11:40:23	17°♉36'00	0°54'06	morning rise	2003 Jan 17 01:36:21	14°♐57'45	
asc. node	2002 Apr 12 19:55:54	28°♉45'31		direct	2003 Jan 23 01:08:08	12°♐17'44	
	2002 Apr 13 10:10:30	0°♊		morning max el	2003 Feb 04 01:03:57	19°♐26'59	25°20'59
evening rise	2002 Apr 14 23:16:20	3°♊13'02		min. declination.	2003 Feb 05 10:30:42	20°♐53'58	21S15'17"
	2002 Apr 30 07:15:29	0°♋		desc. node	2003 Feb 09 00:43:45	24°♐58'43	
evening max el	2002 May 04 03:41:50	4°♋19'13	20°58'15		2003 Feb 13 01:00:10	0°♌	
max. declination.	2002 May 08 23:52:12	8°♋03'35	24°N 1' 8"		2003 Mar 05 02:04:04	0°♍	
retrograde	2002 May 15 18:49:55	9°♋59'11		morning set	2003 Mar 12 06:50:27	12°♍31'10	
evening set	2002 May 17 23:06:04	9°♋47'59		max. Earth dist.	2003 Mar 15 23:36:00	19°♍20'00	1.36388 AU
desc. node	2002 May 21 02:54:26	8°♋55'25		superior conj	2003 Mar 21 23:34:26	0°♎56'03	-1°21'18
inferior conj	2002 May 27 07:09:24	5°♋49'53	-1°47'10	minimum elong	2003 Mar 22 03:29:57	1°♎15'32	1°20'43
minimum elong	2002 May 27 02:07:54	5°♋57'00	1°45'23		2003 Mar 21 12:15:44	0°♏	
min. Earth dist.	2002 May 28 00:01:43	5°♋25'57	0.54919 AU	evening rise	2003 Mar 30 03:56:18	17°♏28'34	
morning rise	2002 Jun 05 05:00:39	1°♋46'34		asc. node	2003 Mar 30 16:57:25	18°♏34'16	
direct	2002 Jun 08 15:11:25	1°♋21'16			2003 Apr 05 14:37:18	0°♐	
min. declination.	2002 Jun 10 10:49:21	1°♋28'40	16N23'59"	evening max el	2003 Apr 16 14:39:03	15°♐46'45	19°46'00
morning max el	2002 Jun 21 14:29:50	7°♋33'36	22°43'51	max. declination.	2003 Apr 24 00:45:43	20°♐15'51	20N31'31"
	2002 Jul 07 10:35:27	0°♑		retrograde	2003 Apr 26 11:58:55	20°♐33'12	

evening set	2003 Apr 28 12:35:15	20°♄22'05		max. Earth dist.	2004 Feb 25 19:00:57	0°♄26'50	1.38438 AU
inferior conj	2003 May 07 07:20:38	16°♄20'19	0°12'00				
minimum elong	2003 May 07 07:52:24	16°♄19'29	0°11'48	superior conj	2004 Mar 04 01:42:57	13°♄49'22	-1°44'21
transit middle	2003 May 07 07:52:24	16°♄19'29	0°11'48	minimum elong	2004 Mar 04 05:58:25	14°♄09'33	1°43'56
transit begin	2003 May 07 05:13:00	16°♄23'37			2004 Mar 12 09:44:18	0°♄	
transit end	2003 May 07 10:31:48	16°♄15'22		evening rise	2004 Mar 13 01:32:17	1°♄17'25	
desc. node	2003 May 07 23:55:44	15°♄54'31		asc. node	2004 Mar 16 13:58:52	8°♄06'03	
min. Earth dist.	2003 May 09 14:05:17	14°♄55'40	0.55597 AU	evening max el	2004 Mar 29 12:27:28	27°♄52'31	18°53'09
morning rise	2003 May 16 00:43:40	11°♄48'12			2004 Apr 01 02:27:26	0°♄	
direct	2003 May 20 07:32:35	11°♄07'24		retrograde	2004 Apr 06 20:27:15	1°♄55'30	
min. declination.	2003 May 24 01:25:38	11°♄39'17	12N 2'22"	max. declination.	2004 Apr 06 18:05:23	1°♄55'28	15N 9'27"
morning max el	2003 Jun 03 05:41:27	18°♄04'38	24°25'48	evening set	2004 Apr 09 01:42:17	1°♄39'26	
	2003 Jun 13 01:33:47	0°♄			2004 Apr 13 01:22:57	30°♄♄	
asc. node	2003 Jun 26 16:12:02	24°♄13'09		inferior conj	2004 Apr 17 01:04:47	27°♄21'45	1°51'46
morning set	2003 Jun 28 10:29:20	27°♄53'45		minimum elong	2004 Apr 17 04:48:43	27°♄15'02	1°50'37
	2003 Jun 29 10:16:58	0°♄		min. Earth dist.	2004 Apr 20 06:11:50	25°♄04'21	0.57091 AU
max. declination.	2003 Jul 03 05:05:19	8°♄10'32	24N15'31"	desc. node	2004 Apr 23 20:57:26	22°♄52'40	
				morning rise	2004 Apr 25 04:49:04	22°♄15'36	
superior conj	2003 Jul 05 10:20:31	13°♄00'13	1°20'44	direct	2004 Apr 30 13:05:11	21°♄06'59	
minimum elong	2003 Jul 05 07:56:35	12°♄47'12	1°20'23	min. declination.	2004 May 05 05:46:54	21°♄59'23	6N13'14"
max. Earth dist.	2003 Jul 07 03:39:27	16°♄43'37	1.33029 AU	morning max el	2004 May 14 20:35:53	28°♄33'58	25°59'31
evening rise	2003 Jul 12 17:34:03	28°♄26'03			2004 May 16 06:54:22	0°♄	
	2003 Jul 13 12:09:35	0°♄			2004 Jun 05 12:47:26	0°♄	
	2003 Jul 30 14:05:20	0°♄		morning set	2004 Jun 11 21:49:45	12°♄47'55	
desc. node	2003 Aug 03 23:13:50	6°♄16'33		asc. node	2004 Jun 12 13:14:33	14°♄09'28	
evening max el	2003 Aug 14 20:57:08	19°♄01'51	27°25'59				
retrograde	2003 Aug 28 13:41:22	26°♄19'25		superior conj	2004 Jun 18 21:24:07	27°♄56'59	1°02'31
min. declination.	2003 Aug 30 12:44:26	26°♄08'06	2S22' 6"	minimum elong	2004 Jun 18 19:08:11	27°♄44'32	1°02'06
evening set	2003 Sep 04 18:24:07	23°♄43'17		max. Earth dist.	2004 Jun 19 13:21:32	29°♄24'33	1.32416 AU
min. Earth dist.	2003 Sep 08 09:21:38	20°♄37'05	0.63291 AU		2004 Jun 19 19:49:28	0°♄	
inferior conj	2003 Sep 11 01:56:58	17°♄56'16	-3°31'39	max. declination.	2004 Jun 22 14:52:17	6°♄04'49	24N47'33"
minimum elong	2003 Sep 11 07:09:28	17°♄43'16	3°30'06	evening rise	2004 Jun 25 21:27:14	13°♄01'26	
morning rise	2003 Sep 17 21:11:54	12°♄41'42			2004 Jul 04 14:51:44	0°♄	
direct	2003 Sep 20 08:52:14	12°♄12'14		desc. node	2004 Jul 20 20:15:45	24°♄39'40	
asc. node	2003 Sep 22 15:22:31	12°♄37'21			2004 Jul 25 13:58:16	0°♄	
max. declination.	2003 Sep 24 03:54:23	13°♄21'46	6N54'47"	evening max el	2004 Jul 27 03:29:27	1°♄32'27	27°07'11
morning max el	2003 Sep 26 23:43:51	15°♄38'03	17°51'48	retrograde	2004 Aug 10 00:32:46	8°♄46'28	
	2003 Oct 07 01:28:06	0°♄		min. declination.	2004 Aug 13 10:03:33	8°♄15'11	4N23'50"
morning set	2003 Oct 13 10:28:52	11°♄03'13		evening set	2004 Aug 17 02:27:09	6°♄30'51	
	2003 Oct 24 11:19:57	0°♄		min. Earth dist.	2004 Aug 20 17:45:18	3°♄47'32	0.61394 AU
				inferior conj	2004 Aug 23 20:50:37	1°♄02'33	-4°19'27
superior conj	2003 Oct 25 09:57:48	1°♄34'14	0°36'59	minimum elong	2004 Aug 24 01:56:22	0°♄51'18	4°18'25
minimum elong	2003 Oct 25 13:32:55	1°♄49'06	0°36'29		2004 Aug 25 01:33:09	30°♄♄	
desc. node	2003 Oct 30 22:30:38	10°♄36'23		morning rise	2004 Aug 31 03:03:53	26°♄08'24	
max. Earth dist.	2003 Oct 31 18:42:41	11°♄57'37	1.43905 AU	direct	2004 Sep 02 13:09:14	25°♄44'10	
evening rise	2003 Nov 09 22:29:07	26°♄21'38		max. declination.	2004 Sep 07 17:26:50	27°♄39'31	12N 5'29"
	2003 Nov 12 07:19:09	0°♄		asc. node	2004 Sep 08 12:25:09	28°♄16'55	
min. declination.	2003 Dec 02 05:11:32	29°♄05'44	25S51'29"	morning max el	2004 Sep 09 14:00:51	29°♄15'04	17°58'07
	2003 Dec 02 21:33:56	0°♄			2004 Sep 10 07:38:15	0°♄	
evening max el	2003 Dec 09 06:11:16	7°♄34'57	20°56'29	morning set	2004 Sep 25 10:45:18	24°♄14'21	
retrograde	2003 Dec 17 16:00:58	12°♄33'45			2004 Sep 28 14:12:59	0°♄	
asc. node	2003 Dec 19 14:38:09	12°♄12'43					
evening set	2003 Dec 21 16:16:36	11°♄02'18		superior conj	2004 Oct 05 18:29:06	12°♄50'54	1°10'48
inferior conj	2003 Dec 27 01:11:05	4°♄50'45	2°20'04	minimum elong	2004 Oct 05 23:00:53	13°♄10'41	1°10'15
minimum elong	2003 Dec 26 22:39:06	4°♄59'27	2°19'15	max. Earth dist.	2004 Oct 13 07:17:34	25°♄39'22	1.42357 AU
min. Earth dist.	2003 Dec 27 12:01:11	4°♄13'33	0.67498 AU		2004 Oct 15 22:56:58	0°♄	
	2003 Dec 30 19:52:20	30°♄♄		desc. node	2004 Oct 16 19:33:16	1°♄23'21	
morning rise	2004 Jan 01 04:49:59	28°♄37'40		evening rise	2004 Oct 19 13:52:58	5°♄48'23	
max. declination.	2004 Jan 03 02:48:45	27°♄13'33	20S13'21"		2004 Nov 04 14:39:56	0°♄	
direct	2004 Jan 06 13:43:56	26°♄16'11		evening max el	2004 Nov 21 01:16:08	21°♄07'05	22°11'10
	2004 Jan 14 11:02:11	0°♄		min. declination.	2004 Nov 22 05:04:34	22°♄15'08	25S41'30"
morning max el	2004 Jan 17 09:21:14	2°♄43'08	23°54'45	retrograde	2004 Nov 30 12:17:18	26°♄44'38	
desc. node	2004 Jan 26 21:44:47	14°♄00'46		evening set	2004 Dec 04 23:56:00	24°♄56'41	
min. declination.	2004 Jan 26 22:09:15	14°♄02'07	22S42'12"	asc. node	2004 Dec 05 11:39:18	24°♄31'35	
	2004 Feb 07 04:19:56	0°♄		inferior conj	2004 Dec 10 08:21:36	18°♄37'17	1°36'18
morning set	2004 Feb 21 23:26:33	23°♄46'56		minimum elong	2004 Dec 10 06:21:05	18°♄44'15	1°35'32
	2004 Feb 25 12:58:16	0°♄		min. Earth dist.	2004 Dec 10 07:47:21	18°♄39'16	0.67810 AU

morning rise	2004 Dec 15 12:38:06	12°♂26'12		retrograde	2005 Nov 14 05:42:29	10°♂56'13	
max. declination.	2004 Dec 19 15:16:00	10°♂29'07	19S 7'17"	evening set	2005 Nov 19 06:21:45	8°♂51'10	
direct	2004 Dec 20 06:29:02	10°♂27'07		asc. node	2005 Nov 22 08:42:35	5°♂31'30	
morning max el	2004 Dec 29 20:06:12	16°♂05'58	22°26'39	min. Earth dist.	2005 Nov 24 04:28:02	3°♂06'12	0.67775 AU
	2005 Jan 10 04:08:38	0°♂		inferior conj	2005 Nov 24 15:42:52	2°♂27'43	0°46'49
desc. node	2005 Jan 12 18:46:54	3°♂37'58		minimum elong	2005 Nov 24 14:37:52	2°♂31'26	0°46'22
min. declination.	2005 Jan 17 11:44:54	10°♂25'03	23S36'33"		2005 Nov 26 11:53:19	30°♂	
	2005 Jan 30 05:36:47	0°♂		morning rise	2005 Nov 29 22:52:01	26°♂20'29	
morning set	2005 Feb 01 12:17:11	3°♂41'05		direct	2005 Dec 04 02:23:39	24°♂44'17	
max. Earth dist.	2005 Feb 06 15:41:11	12°♂13'49	1.40595 AU	max. declination.	2005 Dec 04 16:46:23	24°♂46'09	16S22'18"
				morning max el	2005 Dec 12 12:41:31	29°♂38'02	21°04'41
superior conj	2005 Feb 14 10:49:37	25°♂50'54	-2°00'19		2005 Dec 12 21:19:15	0°♂	
minimum elong	2005 Feb 14 13:42:54	26°♂03'54	2°00'12	desc. node	2005 Dec 30 15:49:30	23°♂39'57	
	2005 Feb 16 17:45:44	0°♂			2006 Jan 03 21:26:12	0°♂	
evening rise	2005 Feb 24 13:00:57	14°♂32'41		min. declination.	2006 Jan 09 05:48:57	8°♂10'59	24S14'51"
asc. node	2005 Mar 03 11:00:43	27°♂15'15		morning set	2006 Jan 11 21:21:54	12°♂18'27	
	2005 Mar 05 01:33:52	0°♂		max. Earth dist.	2006 Jan 19 18:38:08	24°♂54'58	1.42559 AU
evening max el	2005 Mar 12 18:20:15	10°♂30'13	18°20'24		2006 Jan 22 20:41:28	0°♂	
retrograde	2005 Mar 20 00:14:23	14°♂06'03		superior conj	2006 Jan 26 21:33:39	6°♂47'40	-2°04'42
max. declination.	2005 Mar 21 06:50:14	13°♂59'32	8N38' 9"	minimum elong	2006 Jan 26 20:33:39	6°♂43'24	2°04'41
evening set	2005 Mar 22 11:24:22	13°♂42'06		evening rise	2006 Feb 07 10:28:15	27°♂05'17	
inferior conj	2005 Mar 29 16:11:00	9°♂03'52	2°58'19		2006 Feb 09 01:21:39	0°♂	
minimum elong	2005 Mar 29 19:48:40	8°♂56'13	2°57'33	asc. node	2006 Feb 18 08:02:57	15°♂54'17	
min. Earth dist.	2005 Apr 02 02:36:05	6°♂12'00	0.59084 AU	evening max el	2006 Feb 24 05:03:52	23°♂31'00	18°07'32
morning rise	2005 Apr 06 01:37:24	3°♂29'21		retrograde	2006 Mar 02 20:29:06	26°♂55'28	
desc. node	2005 Apr 10 17:59:23	1°♂51'21		max. declination.	2006 Mar 05 01:04:46	26°♂33'06	1N39'39"
direct	2005 Apr 12 07:45:31	1°♂44'44		evening set	2006 Mar 05 13:02:19	26°♂22'00	
min. declination.	2005 Apr 17 12:50:50	2°♂53'15	0S13'57"	inferior conj	2006 Mar 12 02:43:42	21°♂23'03	3°33'19
morning max el	2005 Apr 26 16:23:48	9°♂29'18	27°09'43	minimum elong	2006 Mar 12 04:35:21	21°♂18'32	3°33'07
	2005 May 12 09:13:31	0°♂		min. Earth dist.	2006 Mar 15 06:10:16	18°♂21'06	0.61227 AU
morning set	2005 May 27 07:04:21	27°♂34'28		morning rise	2006 Mar 18 18:25:22	15°♂29'58	
	2005 May 28 10:44:02	0°♂		direct	2006 Mar 25 13:42:13	13°♂10'52	
asc. node	2005 May 30 10:18:03	4°♂14'38		desc. node	2006 Mar 28 15:01:11	13°♂36'53	
superior conj	2005 Jun 03 09:11:43	12°♂53'29	0°40'44	min. declination.	2006 Mar 31 05:23:48	14°♂36'43	6S34'25"
minimum elong	2005 Jun 03 07:30:19	12°♂44'11	0°40'23	morning max el	2006 Apr 08 18:38:50	21°♂05'10	27°45'49
max. Earth dist.	2005 Jun 03 01:42:05	12°♂12'14	1.32184 AU		2006 Apr 16 12:19:48	0°♂	
evening rise	2005 Jun 10 06:03:02	27°♂48'59			2006 May 05 08:28:01	0°♂	
	2005 Jun 11 07:02:41	0°♂		morning set	2006 May 11 12:21:09	12°♂07'05	
max. declination.	2005 Jun 13 06:03:47	4°♂01'14	25N13'17"	asc. node	2006 May 17 07:22:21	24°♂24'08	
	2005 Jun 28 04:01:18	0°♂		max. Earth dist.	2006 May 17 12:45:18	24°♂53'24	1.32332 AU
desc. node	2005 Jul 07 17:17:24	11°♂52'56		superior conj	2006 May 18 20:02:06	27°♂44'04	0°16'05
evening max el	2005 Jul 09 03:19:38	13°♂17'02	26°15'23	minimum elong	2006 May 18 19:17:59	27°♂40'02	0°15'56
retrograde	2005 Jul 23 03:00:04	20°♂28'11			2006 May 19 20:51:56	0°♂	
min. declination.	2005 Jul 28 19:47:58	19°♂07'53	10N50' 8"	evening rise	2006 May 25 17:14:09	12°♂41'31	
evening set	2005 Jul 29 14:18:08	18°♂45'14			2006 Jun 03 11:21:01	0°♂	
min. Earth dist.	2005 Aug 02 16:09:02	16°♂09'13	0.59336 AU	max. declination.	2006 Jun 04 06:07:00	1°♂23'49	25N32'42"
inferior conj	2005 Aug 05 23:35:50	13°♂37'26	-4°51'31	evening max el	2006 Jun 20 20:10:57	24°♂16'53	24°56'16
minimum elong	2005 Aug 06 02:27:27	13°♂31'56	4°51'16	desc. node	2006 Jun 24 14:18:36	27°♂28'19	
morning rise	2005 Aug 13 16:44:08	9°♂06'12			2006 Jun 28 19:57:08	0°♂	
direct	2005 Aug 16 03:50:33	8°♂45'10		retrograde	2006 Jul 04 19:33:23	1°♂22'26	
max. declination.	2005 Aug 23 17:10:44	12°♂16'18	16N27'54"	evening set	2006 Jul 10 02:38:59	0°♂18'23	
morning max el	2005 Aug 23 23:18:09	12°♂30'42	18°24'13		2006 Jul 10 20:18:14	30°♂	
asc. node	2005 Aug 26 09:29:28	15°♂09'11		min. declination.	2006 Jul 15 02:29:28	27°♂39'13	16N 3'21"
	2005 Sep 04 17:52:27	0°♂		min. Earth dist.	2006 Jul 15 07:37:21	27°♂31'00	0.57357 AU
morning set	2005 Sep 09 00:18:23	8°♂03'28		inferior conj	2006 Jul 18 07:07:24	25°♂32'16	-4°55'46
superior conj	2005 Sep 18 02:38:00	25°♂17'15	1°32'32	minimum elong	2006 Jul 18 05:15:36	25°♂35'23	4°55'41
minimum elong	2005 Sep 18 05:54:38	25°♂32'21	1°32'16	morning rise	2006 Jul 26 10:32:39	21°♂23'03	
	2005 Sep 20 16:39:43	0°♂		direct	2006 Jul 29 00:39:29	21°♂03'47	
max. Earth dist.	2005 Sep 25 14:19:49	8°♂38'43	1.40428 AU	morning max el	2006 Aug 07 00:32:19	25°♂16'10	19°11'11
evening rise	2005 Sep 30 01:20:03	16°♂10'40		max. declination.	2006 Aug 09 23:04:39	28°♂27'02	19N42' 6"
desc. node	2005 Oct 03 16:35:36	22°♂06'11			2006 Aug 11 04:09:26	0°♂	
	2005 Oct 08 17:14:37	0°♂		asc. node	2006 Aug 13 06:34:22	2°♂59'20	
	2005 Oct 30 09:02:03	0°♂		morning set	2006 Aug 23 23:11:12	22°♂20'17	
evening max el	2005 Nov 03 15:51:08	4°♂40'58	23°30'52		2006 Aug 27 19:30:31	0°♂	
min. declination.	2005 Nov 10 05:18:01	9°♂50'32	24S32'15"				

superior conj	2006 Sep 01 04:49:19	8° $\cap$ 37'45	1°43'29	min. Earth dist.	2007 Jun 26 19:56:24	7° $\ominus$ 50'55	0.55793 AU
minimum elong	2006 Sep 01 06:11:55	8° $\cap$ 44'25	1°43'26	inferior conj	2007 Jun 28 18:39:36	6° $\ominus$ 41'52	-4°18'07
max. Earth dist.	2006 Sep 07 17:54:35	20° $\cap$ 55'35	1.38375 AU	minimum elong	2007 Jun 28 11:48:42	6° $\ominus$ 52'02	4°16'47
evening rise	2006 Sep 11 12:59:26	27° $\cap$ 40'55		min. declination.	2007 Jul 04 09:21:19	3° $\ominus$ 43'55	18N37' 9"
	2006 Sep 12 21:07:31	0° $\underline{\cap}$		morning rise	2007 Jul 07 07:28:16	2° $\ominus$ 47'19	
desc. node	2006 Sep 20 13:37:54	12° $\underline{\cap}$ 41'11		direct	2007 Jul 10 02:15:15	2° $\ominus$ 28'24	
	2006 Oct 02 04:38:05	0° $\cap$		morning max el	2007 Jul 20 14:59:49	7° $\ominus$ 21'04	20°19'28
evening max el	2006 Oct 17 04:08:04	18° $\cap$ 18'36	24°49'14	max. declination.	2007 Jul 29 08:45:45	18° $\ominus$ 34'04	21N46'20"
min. declination.	2006 Oct 26 20:47:23	24° $\cap$ 50'04	21S53' 9"	asc. node	2007 Jul 31 03:38:22	21° $\ominus$ 33'18	
retrograde	2006 Oct 28 19:16:01	25° $\cap$ 04'41			2007 Aug 04 17:14:54	0° $\cap$	
evening set	2006 Nov 03 09:55:10	22° $\cap$ 43'23		morning set	2007 Aug 08 04:22:27	6° $\cap$ 55'57	
min. Earth dist.	2006 Nov 07 23:42:09	17° $\cap$ 32'16	0.67409 AU				
inferior conj	2006 Nov 08 21:30:35	16° $\cap$ 20'06	-0°07'07	superior conj	2007 Aug 15 19:56:15	22° $\cap$ 36'54	1°45'31
minimum elong	2006 Nov 08 21:41:04	16° $\cap$ 19'31	0°07'03	minimum elong	2007 Aug 15 19:35:33	22° $\cap$ 35'10	1°45'30
transit middle	2006 Nov 08 21:41:04	16° $\cap$ 19'31	0°07'03		2007 Aug 19 13:00:47	0° $\cap$	
transit begin	2006 Nov 08 19:12:02	16° $\cap$ 27'45		max. Earth dist.	2007 Aug 20 23:43:29	2° $\cap$ 48'16	1.36445 AU
transit end	2006 Nov 09 00:10:06	16° $\cap$ 11'18		evening rise	2007 Aug 24 22:18:02	10° $\cap$ 12'46	
asc. node	2006 Nov 09 05:46:06	15° $\cap$ 52'48			2007 Sep 05 12:02:18	0° $\underline{\cap}$	
morning rise	2006 Nov 14 09:34:46	10° $\cap$ 19'03		desc. node	2007 Sep 07 10:39:54	3° $\underline{\cap}$ 02'59	
direct	2006 Nov 18 00:25:15	9° $\cap$ 03'38			2007 Sep 27 17:17:31	0° $\cap$	
max. declination.	2006 Nov 19 08:26:06	9° $\cap$ 13'14	12S21'19"	evening max el	2007 Sep 29 16:08:46	1° $\cap$ 58'35	25°58'58
morning max el	2006 Nov 25 12:56:44	13° $\cap$ 20'49	19°54'16	min. declination.	2007 Oct 11 11:46:37	9° $\cap$ 03'10	17S45'22"
	2006 Dec 08 05:51:32	0° $\cap$		retrograde	2007 Oct 12 03:59:58	9° $\cap$ 04'49	
desc. node	2006 Dec 17 12:51:42	13° $\cap$ 59'51		evening set	2007 Oct 18 08:46:42	6° $\cap$ 30'08	
morning set	2006 Dec 21 17:19:30	20° $\cap$ 26'02		min. Earth dist.	2007 Oct 22 15:05:00	1° $\cap$ 54'32	0.66708 AU
	2006 Dec 27 20:54:37	0° $\ominus$		inferior conj	2007 Oct 23 23:54:48	0° $\cap$ 11'35	-1°04'10
min. declination.	2007 Jan 01 00:02:59	6° $\ominus$ 30'48	24S44'32"	minimum elong	2007 Oct 24 01:35:46	0° $\cap$ 06'18	1°03'28
max. Earth dist.	2007 Jan 02 04:50:06	8° $\ominus$ 25'14	1.44075 AU		2007 Oct 24 03:36:27	30° $\cap$	
				asc. node	2007 Oct 27 02:49:04	26° $\cap$ 32'06	
superior conj	2007 Jan 07 06:05:13	16° $\ominus$ 32'46	-1°52'30	morning rise	2007 Oct 29 18:46:01	24° $\cap$ 19'25	
minimum elong	2007 Jan 06 23:37:47	16° $\ominus$ 06'34	1°52'05	direct	2007 Nov 01 22:58:32	23° $\cap$ 21'31	
	2007 Jan 15 09:24:36	0° $\approx$		max. declination.	2007 Nov 03 20:49:48	23° $\cap$ 41'31	7S25'56"
evening rise	2007 Jan 20 13:33:01	8° $\approx$ 46'44		morning max el	2007 Nov 08 20:31:08	27° $\cap$ 12'16	18°58'34
	2007 Feb 02 09:19:56	0° $\cap$			2007 Nov 11 08:40:44	0° $\cap$	
asc. node	2007 Feb 05 05:05:16	3° $\cap$ 53'48		morning set	2007 Dec 01 01:32:24	29° $\cap$ 17'40	
evening max el	2007 Feb 07 17:39:42	6° $\cap$ 47'02	18°13'55		2007 Dec 01 12:20:44	0° $\cap$	
retrograde	2007 Feb 14 04:37:39	10° $\cap$ 13'45		desc. node	2007 Dec 04 09:53:57	4° $\cap$ 32'30	
evening set	2007 Feb 17 02:22:11	9° $\cap$ 29'37		max. Earth dist.	2007 Dec 15 20:55:09	22° $\cap$ 30'51	1.44961 AU
max. declination.	2007 Feb 17 04:43:05	9° $\cap$ 26'39	5S12'12"				
inferior conj	2007 Feb 23 04:45:35	4° $\cap$ 11'01	3°43'32	superior conj	2007 Dec 17 15:26:31	25° $\cap$ 18'19	-1°21'15
minimum elong	2007 Feb 23 04:40:08	4° $\cap$ 11'16	3°43'31	minimum elong	2007 Dec 17 06:39:35	24° $\cap$ 43'42	1°20'19
min. Earth dist.	2007 Feb 25 18:41:46	1° $\cap$ 22'11	0.63242 AU		2007 Dec 20 14:42:40	0° $\ominus$	
	2007 Feb 27 03:00:29	30° $\cap$		min. declination.	2007 Dec 23 18:30:36	5° $\ominus$ 01'18	25S 9' 2"
morning rise	2007 Mar 01 06:01:29	28° $\approx$ 07'29		evening rise	2008 Jan 01 18:22:15	19° $\ominus$ 31'10	
direct	2007 Mar 08 04:44:27	25° $\approx$ 24'37			2008 Jan 08 04:45:42	0° $\approx$	
min. declination.	2007 Mar 14 12:23:32	27° $\approx$ 18'16	12S13'17"	evening max el	2008 Jan 22 05:25:15	20° $\approx$ 11'57	18°38'41
desc. node	2007 Mar 15 12:03:19	27° $\approx$ 53'43		asc. node	2008 Jan 23 02:07:10	21° $\approx$ 02'11	
	2007 Mar 18 09:34:40	0° $\cap$		retrograde	2008 Jan 28 20:30:52	23° $\approx$ 52'45	
morning max el	2007 Mar 22 01:47:20	3° $\cap$ 20'37	27°44'26	evening set	2008 Jan 31 23:57:08	22° $\approx$ 56'56	
	2007 Apr 10 23:07:04	0° $\cap$		max. declination.	2008 Feb 01 21:42:07	22° $\approx$ 20'52	11S28'27"
morning set	2007 Apr 25 11:17:32	26° $\cap$ 17'18		inferior conj	2008 Feb 06 18:19:01	17° $\approx$ 20'14	3°35'28
	2007 Apr 27 07:16:02	0° $\cap$		minimum elong	2008 Feb 06 16:45:29	17° $\approx$ 24'56	3°35'18
max. Earth dist.	2007 Apr 30 18:24:02	7° $\cap$ 13'31	1.32893 AU	min. Earth dist.	2008 Feb 08 16:39:56	15° $\approx$ 01'01	0.64944 AU
				morning rise	2008 Feb 12 09:06:49	11° $\approx$ 11'05	
superior conj	2007 May 03 04:05:26	12° $\cap$ 21'56	-0°10'40	direct	2008 Feb 19 02:57:07	8° $\approx$ 19'15	
minimum elong	2007 May 03 04:36:31	12° $\cap$ 24'43	0°10'33	min. declination.	2008 Feb 26 17:55:25	11° $\approx$ 11'51	16S45'49"
behind sun begin	2007 May 03 00:38:01	12° $\cap$ 03'19		desc. node	2008 Mar 01 09:04:06	14° $\approx$ 06'51	
behind sun end	2007 May 03 08:35:02	12° $\cap$ 46'08		morning max el	2008 Mar 03 11:13:23	16° $\approx$ 06'45	27°08'45
asc. node	2007 May 04 04:24:20	14° $\cap$ 33'06			2008 Mar 14 22:45:59	0° $\cap$	
evening rise	2007 May 10 05:03:42	27° $\cap$ 31'44			2008 Apr 02 17:44:36	0° $\cap$	
	2007 May 11 09:16:45	0° $\cap$		morning set	2008 Apr 08 00:59:31	9° $\cap$ 56'06	
max. declination.	2007 May 26 14:10:33	26° $\cap$ 48'44	25N39'12"	max. Earth dist.	2008 Apr 12 14:59:58	19° $\cap$ 03'06	1.33901 AU
	2007 May 29 00:55:46	0° $\ominus$					
evening max el	2007 Jun 02 09:57:05	4° $\ominus$ 49'01	23°21'57	superior conj	2008 Apr 16 07:24:04	26° $\cap$ 41'03	-0°38'32
desc. node	2007 Jun 11 11:20:17	10° $\ominus$ 49'35		minimum elong	2008 Apr 16 09:20:42	26° $\cap$ 51'16	0°38'08
retrograde	2007 Jun 15 23:40:02	11° $\ominus$ 35'55			2008 Apr 17 21:06:58	0° $\cap$	
evening set	2007 Jun 19 18:43:40	11° $\ominus$ 04'21		asc. node	2008 Apr 20 01:26:33	4° $\cap$ 37'45	

evening rise	2008 Apr 23 15:43:27	12°♄13'44			2009 Mar 08 18:55:53	0°♄	
	2008 May 02 20:00:06	0°♄		morning set	2009 Mar 22 01:28:47	22°♄51'11	
evening max el	2008 May 14 03:51:02	15°♄23'59 21°47'34			2009 Mar 25 19:55:16	0°♄	
max. declination.	2008 May 15 20:33:58	16°♄56'25 25N 4'22"		max. Earth dist.	2009 Mar 26 00:14:04	0°♄20'54	1.35363 AU
retrograde	2008 May 26 15:47:33	21°♄32'20					
desc. node	2008 May 28 08:19:36	21°♄25'52		superior conj	2009 Mar 31 03:29:25	10°♄33'32	-1°06'12
evening set	2008 May 29 05:41:52	21°♄17'25		minimum elong	2009 Mar 31 06:48:15	10°♄50'23	1°05'38
inferior conj	2008 Jun 07 15:26:46	17°♄14'37 -2°52'35		asc. node	2009 Apr 06 22:28:06	24°♄33'22	
minimum elong	2008 Jun 07 08:03:55	17°♄24'56 2°50'17		evening rise	2009 Apr 07 23:11:35	26°♄40'34	
min. Earth dist.	2008 Jun 07 07:39:24	17°♄25'30 0.54978 AU			2009 Apr 09 14:21:23	0°♄	
morning rise	2008 Jun 16 11:35:49	13°♄20'14		evening max el	2009 Apr 26 07:48:04	26°♄27'45	20°25'12
direct	2008 Jun 19 14:31:06	12°♄58'46			2009 Apr 30 22:29:09	0°♄	
min. declination.	2008 Jun 19 12:56:48	12°♄58'47 17N58'31"		max. declination.	2009 May 02 14:11:39	0°♄48'14	22N49' 7"
morning max el	2008 Jul 01 17:54:05	18°♄41'08 21°47'03		retrograde	2009 May 07 05:00:25	1°♄44'32	
	2008 Jul 10 20:16:50	0°♄		evening set	2009 May 09 06:22:02	1°♄34'00	
asc. node	2008 Jul 17 00:40:57	10°♄40'19			2009 May 13 23:52:55	30°♄	
max. declination.	2008 Jul 18 04:59:31	12°♄56'03 23N 2'18"		desc. node	2009 May 15 05:18:56	29°♄23'29	
morning set	2008 Jul 22 13:34:57	21°♄44'09		inferior conj	2009 May 18 10:02:00	27°♄36'01	-0°55'57
	2008 Jul 26 11:48:21	0°♄		minimum elong	2009 May 18 07:23:37	27°♄39'52	0°54'59
				min. Earth dist.	2009 May 19 20:15:08	26°♄46'12	0.55098 AU
superior conj	2008 Jul 29 20:04:26	7°♄03'11 1°40'20		morning rise	2009 May 27 07:17:50	23°♄22'30	
minimum elong	2008 Jul 29 18:29:52	6°♄54'56 1°40'14		direct	2009 May 31 01:21:53	22°♄52'06	
max. Earth dist.	2008 Aug 02 13:11:51	14°♄42'54 1.34813 AU		min. declination.	2009 Jun 02 20:41:40	23°♄09'42	14N47'55"
evening rise	2008 Aug 07 00:40:51	23°♄34'01		morning max el	2009 Jun 13 11:52:20	29°♄24'14	23°27'17
	2008 Aug 10 10:50:47	0°♄			2009 Jun 14 02:47:19	0°♄	
desc. node	2008 Aug 24 07:41:13	23°♄05'12		asc. node	2009 Jul 03 21:43:30	0°♄12'02	
	2008 Aug 29 02:49:48	0°♄			2009 Jul 03 19:19:35	0°♄	
evening max el	2008 Sep 11 04:31:47	15°♄34'27 26°52'17		morning set	2009 Jul 07 00:56:45	6°♄39'25	
retrograde	2008 Sep 24 07:16:50	22°♄50'10		max. declination.	2009 Jul 08 10:26:19	9°♄36'48 23N52'39"	
min. declination.	2008 Sep 24 15:05:34	22°♄49'49 12S25' 9"					
evening set	2008 Oct 01 00:47:08	20°♄07'29		superior conj	2009 Jul 14 02:15:31	21°♄47'37	1°29'19
min. Earth dist.	2008 Oct 05 00:05:21	16°♄08'31 0.65662 AU		minimum elong	2009 Jul 14 00:00:26	21°♄35'32	1°29'03
inferior conj	2008 Oct 06 20:53:06	13°♄58'10 -2°02'41		max. Earth dist.	2009 Jul 16 11:59:42	26°♄55'29	1.33561 AU
minimum elong	2008 Oct 07 00:09:57	13°♄48'36 2°01'23			2009 Jul 17 23:07:31	0°♄	
asc. node	2008 Oct 12 23:51:54	8°♄18'30		evening rise	2009 Jul 21 15:43:24	7°♄32'33	
morning rise	2008 Oct 13 00:12:14	8°♄18'04			2009 Aug 02 23:07:01	0°♄	
direct	2008 Oct 15 20:05:49	7°♄33'57		desc. node	2009 Aug 11 04:41:25	12°♄39'25	
max. declination.	2008 Oct 18 08:02:01	8°♄07'29 1S57'49"		evening max el	2009 Aug 24 16:14:59	28°♄55'07	27°21'57
morning max el	2008 Oct 22 09:33:41	11°♄07'59 18°19'05			2009 Aug 25 20:18:01	0°♄	
	2008 Nov 04 15:59:50	0°♄		retrograde	2009 Sep 07 04:44:50	6°♄13'14	
morning set	2008 Nov 10 13:27:06	9°♄34'00		min. declination.	2009 Sep 08 12:22:34	6°♄07'54	6S13'18"
desc. node	2008 Nov 20 06:55:23	25°♄13'26		evening set	2009 Sep 14 07:19:19	3°♄31'08	
	2008 Nov 23 07:08:50	0°♄		min. Earth dist.	2009 Sep 18 00:32:55	0°♄07'28	0.64257 AU
					2009 Sep 18 03:26:07	30°♄	
superior conj	2008 Nov 25 16:51:56	3°♄47'51 -0°35'45		inferior conj	2009 Sep 20 10:04:44	27°♄35'08	-3°00'15
minimum elong	2008 Nov 25 12:14:41	3°♄29'39 0°35'07		minimum elong	2009 Sep 20 14:45:53	27°♄22'42	2°58'40
max. Earth dist.	2008 Nov 27 16:06:41	6°♄53'45 1.45122 AU		morning rise	2009 Sep 26 23:14:53	22°♄10'31	
evening rise	2008 Dec 11 23:56:15	29°♄19'48		direct	2009 Sep 29 13:13:28	21°♄36'37	
	2008 Dec 12 10:12:35	0°♄		asc. node	2009 Sep 29 20:54:04	21°♄37'09	
min. declination.	2008 Dec 14 14:25:04	3°♄24'21 25S29'46"		max. declination.	2009 Oct 02 18:52:05	22°♄29'47	3N41' 3"
greatest brilliancy	2008 Dec 23 02:33:42	16°♄38'15 -0.8m		morning max el	2009 Oct 06 01:30:45	25°♄02'49	17°56'43
	2009 Jan 01 09:51:12	0°♄			2009 Oct 10 03:45:35	0°♄	
evening max el	2009 Jan 04 13:58:13	3°♄40'12 19°20'36		morning set	2009 Oct 23 05:29:18	21°♄12'55	
asc. node	2009 Jan 08 23:08:47	7°♄04'47			2009 Oct 28 10:08:38	0°♄	
retrograde	2009 Jan 11 16:43:37	7°♄45'23					
evening set	2009 Jan 15 02:58:21	6°♄36'44		superior conj	2009 Nov 05 08:02:20	13°♄02'26	0°12'22
max. declination.	2009 Jan 17 18:29:59	4°♄15'26 16S39'23"		minimum elong	2009 Nov 05 09:27:04	13°♄08'09	0°12'10
inferior conj	2009 Jan 20 15:59:32	0°♄44'04 3°13'55		behind sun begin	2009 Nov 05 02:39:28	12°♄40'38	
minimum elong	2009 Jan 20 13:36:05	0°♄51'47 3°13'27		behind sun end	2009 Nov 05 16:14:41	13°♄35'38	
	2009 Jan 21 05:36:16	30°♄		desc. node	2009 Nov 07 03:58:43	15°♄59'40	
min. Earth dist.	2009 Jan 21 23:17:24	29°♄03'03 0.66251 AU		max. Earth dist.	2009 Nov 10 10:52:57	21°♄14'42	1.44545 AU
morning rise	2009 Jan 25 23:59:51	24°♄31'55			2009 Nov 16 00:28:02	0°♄	
direct	2009 Feb 01 07:10:20	21°♄44'47		evening rise	2009 Nov 21 12:14:55	8°♄30'26	
min. declination.	2009 Feb 11 18:42:06	27°♄11'36 19S58'41"			2009 Dec 05 17:24:15	0°♄	
morning max el	2009 Feb 13 20:42:57	29°♄11'28 26°05'52		min. declination.	2009 Dec 06 12:31:58	1°♄10'05 25S46' 0"	
	2009 Feb 14 15:39:02	0°♄		evening max el	2009 Dec 18 17:38:47	17°♄09'53	20°17'43
desc. node	2009 Feb 16 06:05:26	1°♄43'16		retrograde	2009 Dec 26 14:38:07	21°♄47'43	

asc. node	2009 Dec 26 20:09:21	21°♄47'25		evening rise	2010 Oct 31 21:27:15	17°♄38'11	
evening set	2009 Dec 30 09:14:19	20°♄24'56			2010 Nov 08 23:43:08	0°♄	
inferior conj	2010 Jan 04 19:06:16	14°♄19'07	2°42'08	min. declination.	2010 Nov 28 06:36:31	26°♄54'26	25S51'31"
minimum elong	2010 Jan 04 16:28:22	14°♄28'01	2°41'24		2010 Dec 01 00:10:28	0°♄	
min. Earth dist.	2010 Jan 05 12:54:02	13°♄18'55	0.67154 AU	evening max el	2010 Dec 01 15:41:41	0°♄40'24	21°27'14
max. declination.	2010 Jan 08 07:14:20	9°♄49'01	19S50'16"	retrograde	2010 Dec 10 12:04:27	5°♄56'09	
morning rise	2010 Jan 09 23:31:50	8°♄05'47		asc. node	2010 Dec 13 17:10:56	4°♄59'05	
direct	2010 Jan 15 16:51:48	5°♄33'16		evening set	2010 Dec 14 16:52:51	4°♄17'51	
morning max el	2010 Jan 27 05:23:21	12°♄25'43	24°45'10		2010 Dec 18 14:53:28	30°♄♂	
min. declination.	2010 Jan 31 19:02:33	17°♄28'08	21S57' 4"	inferior conj	2010 Dec 20 01:23:06	28°♄02'17	2°02'18
desc. node	2010 Feb 03 03:08:44	20°♄19'15		minimum elong	2010 Dec 19 23:01:25	28°♄10'27	2°01'28
	2010 Feb 10 09:05:49	0°♄		min. Earth dist.	2010 Dec 20 07:14:07	27°♄42'03	0.67672 AU
	2010 Mar 01 13:27:36	0°♄		morning rise	2010 Dec 25 04:59:32	21°♄49'36	
morning set	2010 Mar 04 07:32:35	4°♄48'05		max. declination.	2010 Dec 28 09:25:26	19°♄55'18	19S59'27"
max. Earth dist.	2010 Mar 07 23:12:54	11°♄22'52	1.37232 AU	direct	2010 Dec 30 07:20:38	19°♄37'29	
				morning max el	2011 Jan 09 14:25:49	25°♄44'52	23°16'55
superior conj	2010 Mar 14 13:16:21	23°♄50'59	-1°31'43		2011 Jan 13 11:24:49	0°♄	
minimum elong	2010 Mar 14 17:29:13	24°♄11'31	1°31'10	desc. node	2011 Jan 21 00:13:13	9°♄37'47	
	2010 Mar 17 16:11:40	0°♄		min. declination.	2011 Jan 22 22:11:08	12°♄16'08	23S 7'23"
evening rise	2010 Mar 23 01:11:34	10°♄45'26			2011 Feb 03 22:18:51	0°♄	
asc. node	2010 Mar 24 19:29:31	14°♄15'21		morning set	2011 Feb 13 12:59:14	15°♄30'48	
	2010 Apr 02 13:05:48	0°♄		max. Earth dist.	2011 Feb 17 17:56:03	22°♄41'57	1.39353 AU
evening max el	2010 Apr 08 23:29:45	8°♄11'10	19°21'01		2011 Feb 21 20:53:20	0°♄	
max. declination.	2010 Apr 16 20:58:45	12°♄32'46	18N27'49"				
retrograde	2010 Apr 18 04:06:02	12°♄37'54		superior conj	2011 Feb 25 08:47:38	6°♄23'38	-1°52'15
evening set	2010 Apr 20 06:00:57	12°♄25'14		minimum elong	2011 Feb 25 12:44:19	6°♄41'56	1°51'56
inferior conj	2010 Apr 28 16:43:41	8°♄17'21	0°57'59	evening rise	2011 Mar 06 18:56:54	24°♄21'05	
minimum elong	2010 Apr 28 19:03:36	8°♄13'30	0°57'11		2011 Mar 09 17:47:03	0°♄	
min. Earth dist.	2010 May 01 10:58:59	6°♄28'47	0.56142 AU	asc. node	2011 Mar 11 16:31:28	3°♄38'25	
desc. node	2010 May 02 02:20:32	6°♄04'40		evening max el	2011 Mar 23 01:09:53	20°♄31'51	18°36'50
morning rise	2010 May 07 05:12:30	3°♄30'29		retrograde	2011 Mar 30 20:47:35	24°♄21'23	
direct	2010 May 11 22:26:35	2°♄39'34		max. declination.	2011 Mar 31 10:24:57	24°♄20'13	12N31' 3"
min. declination.	2010 May 16 04:22:43	3°♄21'11	9N43'45"	evening set	2011 Apr 02 04:25:50	24°♄02'23	
morning max el	2010 May 26 02:22:32	9°♄51'24	25°07'48	inferior conj	2011 Apr 09 19:36:22	19°♄36'31	2°24'10
	2010 Jun 10 05:41:01	0°♄		minimum elong	2011 Apr 09 23:37:09	19°♄28'47	2°23'05
asc. node	2010 Jun 20 18:45:38	20°♄01'09		min. Earth dist.	2011 Apr 13 05:01:09	17°♄01'20	0.57889 AU
morning set	2010 Jun 21 12:45:27	21°♄35'42		morning rise	2011 Apr 17 15:43:13	14°♄16'51	
	2010 Jun 25 10:31:53	0°♄		desc. node	2011 Apr 18 23:22:30	13°♄43'16	
superior conj	2010 Jun 28 12:06:48	6°♄42'03	1°13'29	direct	2011 Apr 23 10:03:56	12°♄53'11	
minimum elong	2010 Jun 28 09:42:52	6°♄28'57	1°13'07	min. declination.	2011 Apr 28 08:34:04	13°♄52'42	3N32'43"
max. declination.	2010 Jun 28 18:25:07	7°♄16'28	24N29'37"	morning max el	2011 May 07 19:05:15	20°♄29'55	26°33'03
max. Earth dist.	2010 Jun 29 18:26:08	9°♄27'16	1.32714 AU		2011 May 15 23:18:19	0°♄	
evening rise	2010 Jul 05 15:45:29	21°♄57'05			2011 Jun 02 20:02:29	0°♄	
	2010 Jul 09 16:29:10	0°♄		morning set	2011 Jun 05 23:26:26	6°♄27'40	
	2010 Jul 27 21:42:43	0°♄		asc. node	2011 Jun 07 15:49:39	10°♄02'06	
desc. node	2010 Jul 29 01:42:36	1°♄33'11		superior conj	2011 Jun 12 23:44:23	21°♄39'57	0°53'41
evening max el	2010 Aug 07 01:09:58	11°♄46'46	27°22'00	minimum elong	2011 Jun 12 21:40:04	21°♄28'33	0°53'18
retrograde	2010 Aug 20 19:58:32	19°♄03'32		max. Earth dist.	2011 Jun 13 05:33:32	22°♄11'58	1.32264 AU
min. declination.	2010 Aug 23 08:03:18	18°♄45'41	0N27'49"		2011 Jun 16 19:08:43	0°♄	
evening set	2010 Aug 28 00:51:01	16°♄34'05		evening rise	2011 Jun 19 21:56:58	6°♄38'52	
min. Earth dist.	2010 Aug 31 14:54:53	13°♄39'27	0.62514 AU	max. declination.	2011 Jun 19 06:03:29	5°♄15'16	24N58'44"
inferior conj	2010 Sep 03 12:34:50	10°♄54'42	-3°53'15		2011 Jul 02 05:37:43	0°♄	
minimum elong	2010 Sep 03 17:56:36	10°♄41'59	3°51'52	desc. node	2011 Jul 15 22:42:34	19°♄29'43	
morning rise	2010 Sep 10 12:30:59	5°♄48'49		evening max el	2011 Jul 20 05:01:43	23°♄58'34	26°49'02
direct	2010 Sep 12 23:08:41	5°♄21'58			2011 Jul 28 17:58:38	0°♄	
asc. node	2010 Sep 16 17:57:00	6°♄28'10		retrograde	2011 Aug 03 03:50:02	1°♄12'04	
max. declination.	2010 Sep 17 06:25:41	6°♄47'25	9N 8'49"	min. declination.	2011 Aug 07 08:47:53	0°♄25'42	7N10'24"
morning max el	2010 Sep 19 17:20:18	8°♄48'45	17°52'13		2011 Aug 08 09:46:07	30°♄♂	
	2010 Oct 03 15:03:45	0°♄		evening set	2011 Aug 10 00:57:06	29°♄09'01	
morning set	2010 Oct 05 19:44:20	3°♄55'04		min. Earth dist.	2011 Aug 13 19:02:23	26°♄30'51	0.60526 AU
				inferior conj	2011 Aug 17 01:04:01	23°♄48'52	-4°35'36
superior conj	2010 Oct 17 01:04:33	23°♄33'46	0°52'48	minimum elong	2011 Aug 17 05:31:46	23°♄39'33	4°34'55
minimum elong	2010 Oct 17 05:26:45	23°♄52'17	0°52'13	morning rise	2011 Aug 24 12:00:29	19°♄04'18	
	2010 Oct 20 21:18:42	0°♄		direct	2011 Aug 26 22:02:55	18°♄41'48	
max. Earth dist.	2010 Oct 24 01:46:21	5°♄13'10	1.43306 AU	max. declination.	2011 Sep 01 22:38:38	21°♄08'33	14N 2' 6"
desc. node	2010 Oct 25 01:01:40	6°♄47'13		morning max el	2011 Sep 03 05:55:45	22°♄17'58	18°06'39

asc. node	2011 Sep 03 15:01:26	22°♎40'28		min. declination.	2012 Jul 22 07:25:46	10°♎30'34	13N14'31"
	2011 Sep 09 05:58:27	0°♎		min. Earth dist.	2012 Jul 25 14:16:17	8°♎27'28	0.58467 AU
morning set	2011 Sep 19 02:12:07	17°♎23'31		inferior conj	2012 Jul 28 19:57:21	6°♎07'41	-4°57'38
	2011 Sep 25 21:08:39	0°♎		minimum elong	2012 Jul 28 21:04:06	6°♎05'40	4°57'35
				morning rise	2012 Aug 05 17:30:19	1°♎45'47	
superior conj	2011 Sep 28 20:15:42	5°♎21'32	1°21'26	direct	2012 Aug 08 05:40:09	1°♎25'37	
minimum elong	2011 Sep 29 00:25:24	5°♎40'08	1°21'00	morning max el	2012 Aug 16 12:03:59	5°♎21'19	18°41'31
max. Earth dist.	2011 Oct 06 11:22:07	18°♎35'24	1.41571 AU	max. declination.	2012 Aug 17 07:32:31	6°♎09'36	17N57'48"
evening rise	2011 Oct 11 20:15:54	27°♎25'29		asc. node	2012 Aug 20 12:06:11	9°♎58'48	
desc. node	2011 Oct 11 22:03:49	27°♎32'44			2012 Sep 01 02:32:12	0°♎	
	2011 Oct 13 10:51:35	0°♎		morning set	2012 Sep 01 20:10:09	1°♎25'37	
	2011 Nov 02 16:54:14	0°♎					
evening max el	2011 Nov 14 08:40:08	14°♎12'58	22°44'52	superior conj	2012 Sep 10 12:43:48	18°♎12'27	1°38'23
min. declination.	2011 Nov 18 04:30:58	17°♎36'31	25S22'10"	minimum elong	2012 Sep 10 15:13:00	18°♎24'10	1°38'14
retrograde	2011 Nov 24 07:19:24	20°♎07'16			2012 Sep 16 23:22:04	0°♎	
evening set	2011 Nov 29 00:12:31	18°♎12'20		max. Earth dist.	2012 Sep 17 16:38:19	1°♎16'20	1.39560 AU
asc. node	2011 Nov 30 14:14:26	16°♎41'02		evening rise	2012 Sep 21 18:04:17	8°♎16'53	
inferior conj	2011 Dec 04 08:52:26	11°♎50'44	1°15'58	desc. node	2012 Sep 27 19:04:39	18°♎12'44	
minimum elong	2011 Dec 04 07:12:45	11°♎56'28	1°15'17		2012 Oct 05 10:35:14	0°♎	
min. Earth dist.	2011 Dec 04 03:47:06	12°♎08'19	0.67835 AU	evening max el	2012 Oct 26 22:11:32	27°♎49'13	24°04'52
morning rise	2011 Dec 09 14:05:03	5°♎40'46			2012 Oct 29 06:18:09	0°♎	
direct	2011 Dec 14 01:42:48	3°♎51'20		min. declination.	2012 Nov 03 23:35:34	3°♎42'27	23S36'56"
max. declination.	2011 Dec 14 01:22:25	3°♎51'20	18S 8'37"	retrograde	2012 Nov 06 23:03:48	4°♎18'03	
morning max el	2011 Dec 23 03:08:53	9°♎10'30	21°50'30	evening set	2012 Nov 12 05:36:09	2°♎06'04	
desc. node	2012 Jan 07 21:16:54	29°♎26'29			2012 Nov 14 07:42:28	30°♎	
	2012 Jan 08 06:33:41	0°♎		asc. node	2012 Nov 16 11:17:41	27°♎18'21	
min. declination.	2012 Jan 14 14:48:26	9°♎22'53	23S53'38"	min. Earth dist.	2012 Nov 17 00:08:14	26°♎35'18	0.67662 AU
morning set	2012 Jan 24 13:16:15	24°♎49'10		inferior conj	2012 Nov 17 15:46:47	25°♎42'25	0°24'26
	2012 Jan 27 18:11:38	0°♎		minimum elong	2012 Nov 17 15:11:34	25°♎44'25	0°24'10
max. Earth dist.	2012 Jan 30 16:22:47	4°♎49'19	1.41469 AU	morning rise	2012 Nov 23 00:46:24	19°♎37'29	
				direct	2012 Nov 26 22:47:32	18°♎10'14	
superior conj	2012 Feb 07 09:02:29	17°♎59'02	-2°03'55	max. declination.	2012 Nov 27 21:25:56	18°♎14'57	14S49'15"
minimum elong	2012 Feb 07 10:36:14	18°♎05'54	2°03'53	morning max el	2012 Dec 04 22:48:28	22°♎46'49	20°33'03
	2012 Feb 14 01:37:53	0°♎			2012 Dec 11 01:39:47	0°♎	
evening rise	2012 Feb 18 00:57:08	7°♎18'46		desc. node	2012 Dec 24 18:19:38	19°♎37'01	
asc. node	2012 Feb 26 13:34:11	22°♎35'36			2012 Dec 31 14:02:41	0°♎	
	2012 Mar 02 11:40:40	0°♎		morning set	2013 Jan 02 13:59:16	3°♎05'20	
evening max el	2012 Mar 05 09:34:45	3°♎20'36	18°12'35	min. declination.	2013 Jan 05 09:01:39	7°♎26'48	24S27'41"
retrograde	2012 Mar 12 07:48:59	6°♎49'21		max. Earth dist.	2013 Jan 11 22:52:47	17°♎54'24	1.43278 AU
max. declination.	2012 Mar 14 00:36:39	6°♎36'59	5N43'34"				
evening set	2012 Mar 14 21:26:35	6°♎21'28		superior conj	2013 Jan 18 08:56:11	28°♎26'02	-2°01'48
inferior conj	2012 Mar 21 19:20:39	1°♎34'38	3°16'38	minimum elong	2013 Jan 18 05:37:16	28°♎12'13	2°01'42
minimum elong	2012 Mar 21 22:19:11	1°♎27'57	3°16'08		2013 Jan 19 07:25:16	0°♎	
	2012 Mar 23 13:22:19	30°♎		evening rise	2013 Jan 30 15:08:00	19°♎30'37	
min. Earth dist.	2012 Mar 25 04:06:46	28°♎35'16	0.59999 AU		2013 Feb 05 14:55:25	0°♎	
morning rise	2012 Mar 28 20:58:12	25°♎51'17		asc. node	2013 Feb 12 10:36:31	10°♎59'02	
direct	2012 Apr 04 10:11:14	23°♎50'54		evening max el	2013 Feb 16 21:30:08	16°♎28'53	18°07'53
desc. node	2012 Apr 04 20:25:27	23°♎51'23		retrograde	2013 Feb 23 09:41:05	19°♎52'28	
min. declination.	2012 Apr 09 19:15:30	25°♎05'56	2S56'40"	evening set	2013 Feb 26 04:33:33	19°♎14'24	
	2012 Apr 16 22:42:05	0°♎		max. declination.	2013 Feb 25 22:24:09	19°♎20'59	1S15'43"
morning max el	2012 Apr 18 17:22:50	1°♎40'26	27°29'33	inferior conj	2013 Mar 04 12:58:04	14°♎06'42	3°40'13
	2012 May 09 05:14:29	0°♎		minimum elong	2013 Mar 04 13:58:09	14°♎04'08	3°40'09
morning set	2012 May 20 07:12:12	21°♎08'24		min. Earth dist.	2013 Mar 07 11:03:15	11°♎08'11	0.62123 AU
	2012 May 24 11:11:39	0°♎		morning rise	2013 Mar 10 22:05:32	8°♎08'39	
asc. node	2012 May 24 12:53:04	0°♎09'10		direct	2013 Mar 17 20:02:53	5°♎37'59	
max. Earth dist.	2012 May 26 17:41:04	4°♎57'24	1.32199 AU	desc. node	2013 Mar 22 17:28:36	6°♎44'56	
				min. declination.	2013 Mar 23 17:32:33	7°♎13'36	9S 3' 5"
superior conj	2012 May 27 11:19:20	6°♎34'16	0°30'36	morning max el	2013 Mar 31 21:50:03	13°♎32'57	27°49'43
minimum elong	2012 May 27 09:59:41	6°♎26'58	0°30'19		2013 Apr 14 02:36:56	0°♎	
evening rise	2012 Jun 03 07:52:25	21°♎29'01			2013 May 01 15:37:04	0°♎	
	2012 Jun 07 11:15:47	0°♎		morning set	2013 May 04 10:02:42	5°♎31'40	
max. declination.	2012 Jun 09 00:38:43	3°♎03'07	25N22' 3"	max. Earth dist.	2013 May 10 02:59:26	17°♎30'58	1.32526 AU
	2012 Jun 26 02:23:54	0°♎		asc. node	2013 May 11 09:55:25	20°♎18'20	
evening max el	2012 Jul 01 01:57:03	5°♎23'34	25°44'36				
desc. node	2012 Jul 01 19:42:40	6°♎05'02		superior conj	2013 May 11 21:09:55	21°♎19'26	0°04'58
retrograde	2012 Jul 15 02:15:59	12°♎32'33		minimum elong	2013 May 11 20:55:59	21°♎18'10	0°04'55
evening set	2012 Jul 21 02:55:39	11°♎06'24		behind sun begin	2013 May 11 16:00:34	20°♎51'24	

behind sun end	2013 May 12 01:51:24	21°♄44'58	morning max el	2014 Mar 14 06:29:54	26°≈03'18	27°33'10
	2013 May 15 20:41:13	0°♄		2014 Mar 17 22:23:44	0°♄	
evening rise	2013 May 18 19:32:55	6°♄20'56		2014 Apr 07 15:34:42	0°♄	
max. declination.	2013 May 31 05:04:10	29°♄51'43 25N37'36"	morning set	2014 Apr 18 05:28:48	19°♄29'45	
	2013 May 31 07:06:50	0°♄	max. Earth dist.	2014 Apr 23 05:28:40	29°♄40'12	1.33268 AU
evening max el	2013 Jun 12 16:45:12	16°♄07'44 24°17'01		2014 Apr 23 09:16:01	0°♄	
desc. node	2013 Jun 18 16:43:18	20°♄47'26				
retrograde	2013 Jun 26 13:07:34	23°♄06'51	superior conj	2014 Apr 26 03:27:18	5°♄49'52	-0°22'25
evening set	2013 Jul 01 05:24:32	22°♄18'06	minimum elong	2014 Apr 26 04:34:07	5°♄55'49	0°22'11
min. Earth dist.	2013 Jul 07 03:37:09	19°♄21'22 0.56622 AU	asc. node	2014 Apr 28 06:57:14	10°♄25'49	
inferior conj	2013 Jul 09 18:41:12	17°♄42'25 -4°45'39	evening rise	2014 May 03 07:07:24	21°♄08'26	
minimum elong	2013 Jul 09 14:28:01	17°♄49'05 4°45'11		2014 May 07 14:56:32	0°♄	
min. declination.	2013 Jul 10 02:57:35	17°♄29'23 17N32'29"	max. declination.	2014 May 22 10:02:43	23°♄32'39 25N32'25"	
morning rise	2013 Jul 18 02:12:48	13°♄40'47	evening max el	2014 May 25 07:10:04	26°♄36'29	22°40'52
direct	2013 Jul 20 18:22:05	13°♄21'42		2014 May 29 09:11:46	0°♄	
morning max el	2013 Jul 30 08:48:03	17°♄49'18 19°37'45	desc. node	2014 Jun 05 13:43:51	3°♄01'35	
max. declination.	2013 Aug 04 10:10:57	23°♄42'15 20N41'23"	retrograde	2014 Jun 07 11:56:27	3°♄10'03	
asc. node	2013 Aug 07 09:09:51	28°♄08'29	evening set	2014 Jun 10 17:01:38	2°♄47'22	
	2013 Aug 08 12:12:49	0°♄		2014 Jun 17 10:04:27	30°♄	
morning set	2013 Aug 16 22:02:27	15°♄51'10	min. Earth dist.	2014 Jun 18 15:11:04	29°♄19'17	0.55334 AU
	2013 Aug 23 22:36:30	0°♄	inferior conj	2014 Jun 19 22:50:15	28°♄34'04	-3°47'16
			minimum elong	2014 Jun 19 14:59:43	28°♄45'17	3°45'20
superior conj	2013 Aug 24 20:56:03	1°♄50'36 1°45'20	morning rise	2014 Jun 28 15:15:52	24°♄42'17	
minimum elong	2013 Aug 24 21:31:56	1°♄53'33 1°45'19	min. declination.	2014 Jun 28 17:01:21	24°♄41'20	18N41'39"
max. Earth dist.	2013 Aug 30 20:51:06	13°♄21'32 1.37525 AU	direct	2014 Jul 01 12:49:56	24°♄22'53	
evening rise	2013 Sep 03 15:22:35	20°♄14'00	morning max el	2014 Jul 12 18:22:11	29°♄35'33	20°54'48
	2013 Sep 09 07:07:05	0°♄		2014 Jul 13 04:44:35	0°♄	
desc. node	2013 Sep 14 16:04:23	8°♄42'34	max. declination.	2014 Jul 24 14:31:47	15°♄47'38	22N22' 1"
	2013 Sep 29 11:38:05	0°♄	asc. node	2014 Jul 25 06:11:59	16°♄57'36	
evening max el	2013 Oct 09 10:11:15	11°♄28'12 25°20'24		2014 Jul 31 22:45:39	0°♄	
min. declination.	2013 Oct 20 02:11:22	18°♄16'48 20S19'17"	morning set	2014 Aug 01 05:12:02	0°♄33'06	
retrograde	2013 Oct 21 10:28:46	18°♄23'39				
evening set	2013 Oct 27 07:21:48	15°♄56'01	superior conj	2014 Aug 08 16:21:04	16°♄03'01	1°44'07
min. Earth dist.	2013 Oct 31 17:51:57	11°♄00'06 0.67157 AU	minimum elong	2014 Aug 08 15:25:05	15°♄58'14	1°44'05
inferior conj	2013 Nov 01 20:19:29	9°♄34'24 -0°31'03	max. Earth dist.	2014 Aug 13 05:24:02	25°♄12'06	1.35699 AU
minimum elong	2013 Nov 01 21:06:52	9°♄31'50 0°30'42		2014 Aug 15 16:43:50	0°♄	
asc. node	2013 Nov 03 08:21:17	7°♄39'49	evening rise	2014 Aug 17 08:31:38	3°♄08'22	
morning rise	2013 Nov 07 11:05:00	3°♄37'02	desc. node	2014 Sep 01 13:05:26	28°♄57'00	
direct	2013 Nov 10 21:11:52	2°♄29'30		2014 Sep 02 05:37:43	0°♄	
max. declination.	2013 Nov 12 11:18:19	2°♄43'13 10S22'57"	evening max el	2014 Sep 21 22:09:34	25°♄06'48	26°24'00
morning max el	2013 Nov 18 02:21:45	6°♄33'43 19°28'42		2014 Sep 27 22:39:16	0°♄	
greatest brilliancy	2013 Dec 01 12:56:23	24°♄34'38 -0.7m	retrograde	2014 Oct 04 17:02:15	2°♄18'45	
	2013 Dec 05 02:42:16	0°♄	min. declination.	2014 Oct 04 11:23:13	2°♄18'33	15S38'53"
desc. node	2013 Dec 11 15:20:16	10°♄02'41		2014 Oct 10 17:26:38	30°♄	
morning set	2013 Dec 12 12:14:16	11°♄23'33	evening set	2014 Oct 11 03:36:34	29°♄39'27	
	2013 Dec 24 10:12:23	0°♄	min. Earth dist.	2014 Oct 15 06:44:05	25°♄19'39	0.66308 AU
max. Earth dist.	2013 Dec 25 12:24:06	1°♄43'31 1.44542 AU	inferior conj	2014 Oct 16 20:39:56	23°♄24'11	-1°28'57
min. declination.	2013 Dec 28 03:17:10	5°♄53'14 24S54'56"	minimum elong	2014 Oct 16 23:01:36	23°♄16'59	1°27'58
			asc. node	2014 Oct 21 05:23:19	18°♄42'47	
superior conj	2013 Dec 29 06:27:28	7°♄41'42 -1°41'29	morning rise	2014 Oct 22 18:57:49	17°♄37'01	
minimum elong	2013 Dec 28 22:16:22	7°♄08'59 1°40'48	direct	2014 Oct 25 19:16:33	16°♄45'36	
	2014 Jan 11 21:34:39	0°♄	max. declination.	2014 Oct 27 22:59:09	17°♄10'54	5S11'21"
evening rise	2014 Jan 12 09:05:14	0°♄48'06	morning max el	2014 Nov 01 12:38:50	20°♄28'08	18°39'45
asc. node	2014 Jan 30 07:38:08	28°♄38'01		2014 Nov 08 23:08:52	0°♄	
evening max el	2014 Jan 31 09:57:55	29°♄48'41 18°22'12	morning set	2014 Nov 22 07:19:36	20°♄49'01	
	2014 Jan 31 14:29:04	0°♄		2014 Nov 28 02:25:47	0°♄	
retrograde	2014 Feb 06 21:43:21	3°♄20'15	desc. node	2014 Nov 28 12:21:46	0°♄39'13	
evening set	2014 Feb 09 21:47:11	2°♄31'13				
max. declination.	2014 Feb 10 07:02:12	2°♄18'04 7S55'59"	superior conj	2014 Dec 08 09:51:12	16°♄13'03	-1°03'05
	2014 Feb 13 03:29:44	30°♄	minimum elong	2014 Dec 08 02:09:58	15°♄42'52	1°02'09
inferior conj	2014 Feb 15 20:21:44	27°♄04'21 3°41'59	max. Earth dist.	2014 Dec 08 05:58:46	15°♄57'50	1.45121 AU
minimum elong	2014 Feb 15 19:34:29	27°♄06'37 3°41'57		2014 Dec 17 03:52:56	0°♄	
min. Earth dist.	2014 Feb 18 03:35:45	24°♄26'48 0.64011 AU	min. declination.	2014 Dec 19 22:13:39	4°♄22'59 25S17'49"	
morning rise	2014 Feb 21 16:41:07	20°♄57'50	evening rise	2014 Dec 24 03:50:16	11°♄07'47	
direct	2014 Feb 28 14:00:08	18°♄09'42	greatest brilliancy	2015 Jan 01 02:54:52	23°♄50'56	-0.8m
min. declination.	2014 Mar 07 08:24:54	20°♄22'34 14S16'25"		2015 Jan 05 01:07:43	0°♄	
desc. node	2014 Mar 09 14:30:11	21°♄55'19	evening max el	2015 Jan 14 20:30:32	13°♄14'49	18°54'30

asc. node	2015 Jan 17 04:39:28	15° $\approx$ 19'57	behind sun end	2015 Nov 17 17:37:41	25° $\approx$ 06'50	
retrograde	2015 Jan 21 15:54:12	17° $\approx$ 05'16		2015 Nov 20 19:43:21	0° $\approx$	
evening set	2015 Jan 24 21:57:45	16° $\approx$ 04'17	max. Earth dist.	2015 Nov 21 00:48:52	0° $\approx$ 20'04	1.44960 AU
max. declination.	2015 Jan 26 08:35:01	14° $\approx$ 59'47 13S48'52"	evening rise	2015 Dec 04 00:49:14	20° $\approx$ 36'52	
inferior conj	2015 Jan 30 13:45:22	10° $\approx$ 20'25 3°27'45		2015 Dec 10 02:34:13	0° $\approx$	
minimum elong	2015 Jan 30 11:46:10	10° $\approx$ 26'36 3°27'28	min. declination.	2015 Dec 11 19:05:45	2° $\approx$ 35'46	25S36'59"
min. Earth dist.	2015 Feb 01 05:34:30	8° $\approx$ 16'36 0.65547 AU	greatest brilliancy	2015 Dec 17 02:01:27	10° $\approx$ 37'54	-0.7m
morning rise	2015 Feb 05 01:12:34	4° $\approx$ 09'25	evening max el	2015 Dec 29 03:11:54	26° $\approx$ 44'21	19°43'14
direct	2015 Feb 11 14:56:54	1° $\approx$ 18'01		2016 Jan 02 02:19:45	0° $\approx$	
min. declination.	2015 Feb 20 04:12:45	4° $\approx$ 57'17 18S16'21"	asc. node	2016 Jan 04 01:40:43	0° $\approx$ 50'36	
desc. node	2015 Feb 24 11:31:58	8° $\approx$ 47'07	retrograde	2016 Jan 05 13:05:36	1° $\approx$ 02'37	
morning max el	2015 Feb 24 16:22:33	8° $\approx$ 59'15 26°44'46		2016 Jan 08 19:36:26	30° $\approx$ 03	
	2015 Mar 13 03:52:00	0° $\approx$	evening set	2016 Jan 09 02:36:58	29° $\approx$ 48'19	
	2015 Mar 31 01:43:34	0° $\approx$	max. declination.	2016 Jan 13 10:03:56	25° $\approx$ 21'19	18S19'47"
morning set	2015 Apr 01 14:07:21	2° $\approx$ 51'40	inferior conj	2016 Jan 14 14:04:42	23° $\approx$ 49'55	3°01'34
max. Earth dist.	2015 Apr 05 21:19:18	11° $\approx$ 15'08 1.34462 AU	minimum elong	2016 Jan 14 11:31:09	23° $\approx$ 58'22	3°00'57
			min. Earth dist.	2016 Jan 15 15:32:34	22° $\approx$ 25'50	0.66684 AU
superior conj	2015 Apr 10 04:00:10	19° $\approx$ 58'19 -0°50'24	morning rise	2016 Jan 19 20:11:13	17° $\approx$ 36'57	
minimum elong	2015 Apr 10 06:33:17	20° $\approx$ 11'33 0°49'56	direct	2016 Jan 25 21:49:39	14° $\approx$ 54'42	
	2015 Apr 14 22:51:27	0° $\approx$	morning max el	2016 Feb 07 01:23:33	22° $\approx$ 08'49	25°33'02
asc. node	2015 Apr 15 03:58:05	0° $\approx$ 26'52	min. declination.	2016 Feb 07 09:01:27	22° $\approx$ 28'13	20S55'45"
evening rise	2015 Apr 17 16:40:21	5° $\approx$ 44'27	desc. node	2016 Feb 11 08:34:47	26° $\approx$ 51'37	
	2015 May 01 01:59:59	0° $\approx$		2016 Feb 13 22:42:39	0° $\approx$	
evening max el	2015 May 07 04:49:31	7° $\approx$ 21'14 21°10'37		2016 Mar 05 10:23:28	0° $\approx$	
max. declination.	2015 May 11 08:03:46	10° $\approx$ 14'11 24N22'42"	morning set	2016 Mar 14 07:32:12	15° $\approx$ 24'26	
retrograde	2015 May 19 01:49:07	13° $\approx$ 08'53	max. Earth dist.	2016 Mar 18 01:24:03	22° $\approx$ 21'38	1.36112 AU
evening set	2015 May 21 07:54:54	12° $\approx$ 57'06		2016 Mar 22 00:19:04	0° $\approx$	
desc. node	2015 May 23 10:45:13	12° $\approx$ 25'55				
inferior conj	2015 May 30 16:55:52	8° $\approx$ 58'21 -2°05'03	superior conj	2016 Mar 23 20:10:59	3° $\approx$ 37'29	-1°17'26
minimum elong	2015 May 30 11:09:01	9° $\approx$ 06'29 2°03'04	minimum elong	2016 Mar 23 23:58:07	3° $\approx$ 56'24	1°16'51
min. Earth dist.	2015 May 31 03:30:01	8° $\approx$ 43'27 0.54902 AU	evening rise	2016 Mar 31 22:08:18	20° $\approx$ 02'51	
morning rise	2015 Jun 08 14:33:32	4° $\approx$ 57'50	asc. node	2016 Apr 01 01:00:25	20° $\approx$ 17'24	
direct	2015 Jun 11 22:32:57	4° $\approx$ 33'47		2016 Apr 05 23:09:14	0° $\approx$	
min. declination.	2015 Jun 13 08:04:56	4° $\approx$ 38'10 16N54'34"	evening max el	2016 Apr 18 13:59:28	18° $\approx$ 42'15	19°55'33
morning max el	2015 Jun 24 17:07:34	10° $\approx$ 38'41 22°28'47	max. declination.	2016 Apr 25 18:13:47	23° $\approx$ 11'37	21N13'31"
	2015 Jul 08 18:51:35	0° $\approx$	retrograde	2016 Apr 28 17:19:43	23° $\approx$ 36'07	
asc. node	2015 Jul 12 03:13:45	6° $\approx$ 15'40	evening set	2016 Apr 30 17:51:54	23° $\approx$ 25'18	
max. declination.	2015 Jul 14 16:32:43	11° $\approx$ 21'39 23N25'11"	desc. node	2016 May 09 07:45:41	19° $\approx$ 36'20	
morning set	2015 Jul 16 15:30:52	15° $\approx$ 24'31	inferior conj	2016 May 09 15:12:09	19° $\approx$ 25'01	-0°05'25
			minimum elong	2016 May 09 14:57:25	19° $\approx$ 25'23	0°05'19
superior conj	2015 Jul 23 19:23:34	0° $\approx$ 37'55 1°36'19	transit middle	2016 May 09 14:57:25	19° $\approx$ 25'23	0°05'19
minimum elong	2015 Jul 23 17:27:48	0° $\approx$ 27'42 1°36'08	transit begin	2016 May 09 11:12:22	19° $\approx$ 31'06	
	2015 Jul 23 12:14:21	0° $\approx$	transit end	2016 May 09 18:42:28	19° $\approx$ 19'40	
max. Earth dist.	2015 Jul 26 22:28:50	7° $\approx$ 11'15 1.34226 AU	min. Earth dist.	2016 May 11 17:07:26	18° $\approx$ 09'13	0.55443 AU
evening rise	2015 Jul 31 16:47:40	16° $\approx$ 04'43	morning rise	2016 May 18 09:54:22	14° $\approx$ 58'03	
	2015 Aug 07 19:14:46	0° $\approx$	direct	2016 May 22 13:19:39	14° $\approx$ 20'16	
desc. node	2015 Aug 19 10:07:02	18° $\approx$ 48'12	min. declination.	2016 May 26 02:13:34	14° $\approx$ 48'34	12N51' 9"
	2015 Aug 27 15:44:28	0° $\approx$	morning max el	2016 Jun 05 08:45:30	21° $\approx$ 11'20	24°10'46
evening max el	2015 Sep 04 10:19:20	8° $\approx$ 36'57 27°08'12		2016 Jun 12 23:22:25	0° $\approx$	
retrograde	2015 Sep 17 18:09:29	15° $\approx$ 55'16	asc. node	2016 Jun 28 00:17:00	25° $\approx$ 15'13	
min. declination.	2015 Sep 18 11:45:41	15° $\approx$ 53'32 9S54'15"		2016 Jun 29 23:24:12	0° $\approx$	
evening set	2015 Sep 24 15:58:34	13° $\approx$ 11'33	morning set	2016 Jun 30 03:15:19	0° $\approx$ 20'18	
min. Earth dist.	2015 Sep 28 12:27:09	9° $\approx$ 27'51 0.65102 AU	max. declination.	2016 Jul 03 23:17:08	8° $\approx$ 33'51	24N 9' 0"
inferior conj	2015 Sep 30 14:38:09	7° $\approx$ 07'06 -2°27'25				
minimum elong	2015 Sep 30 18:33:41	6° $\approx$ 56'04 2°25'57	superior conj	2016 Jul 07 03:23:52	15° $\approx$ 27'03	1°23'07
morning rise	2015 Oct 06 22:00:22	1° $\approx$ 33'10	minimum elong	2016 Jul 07 01:01:18	15° $\approx$ 14'10	1°22'49
asc. node	2015 Oct 08 02:25:09	1° $\approx$ 06'27	max. Earth dist.	2016 Jul 09 00:47:42	19° $\approx$ 31'44	1.33157 AU
direct	2015 Oct 09 14:57:19	0° $\approx$ 53'58		2016 Jul 14 00:46:39	0° $\approx$	
max. declination.	2015 Oct 12 09:50:21	1° $\approx$ 34'54 0N24' 0"	evening rise	2016 Jul 14 12:05:18	0° $\approx$ 57'23	
morning max el	2015 Oct 16 03:16:14	4° $\approx$ 24'01 18°07'25		2016 Jul 30 18:18:20	0° $\approx$	
	2015 Nov 02 07:06:06	0° $\approx$	desc. node	2016 Aug 05 07:07:52	8° $\approx$ 06'24	
morning set	2015 Nov 03 07:44:36	1° $\approx$ 42'04	evening max el	2016 Aug 16 21:21:24	21° $\approx$ 47'08	27°25'58
desc. node	2015 Nov 15 09:25:14	21° $\approx$ 22'46	retrograde	2016 Aug 30 13:03:50	29° $\approx$ 04'39	
			min. declination.	2016 Sep 01 07:49:39	28° $\approx$ 55'12	3S26'49"
superior conj	2015 Nov 17 14:53:00	24° $\approx$ 55'55 -0°14'57	evening set	2016 Sep 06 17:26:02	26° $\approx$ 06'41	
minimum elong	2015 Nov 17 12:59:33	24° $\approx$ 48'24 0°14'41	min. Earth dist.	2016 Sep 10 08:55:39	23° $\approx$ 16'07	0.63549 AU
behind sun begin	2015 Nov 17 08:21:26	24° $\approx$ 29'58	inferior conj	2016 Sep 12 23:40:02	20° $\approx$ 37'19	-3°23'37

minimum elong	2016 Sep 13 04:46:16	20° $\cap$ 24'21	3°22'03	evening rise	2017 Jun 28 15:05:27	15° $\ominus$ 30'23	
morning rise	2016 Sep 19 17:18:25	15° $\cap$ 19'55			2017 Jul 06 00:19:40	0° Ω	
direct	2016 Sep 22 05:30:34	14° $\cap$ 49'21		desc. node	2017 Jul 23 04:07:36	26° Ω 38'11	
asc. node	2016 Sep 23 23:28:57	15° $\cap$ 04'30			2017 Jul 25 23:41:04	0° $\cap$	
max. declination.	2016 Sep 25 20:41:13	15° $\cap$ 54'04	6N 1'12"	evening max el	2017 Jul 30 04:38:48	4° $\cap$ 23'52	27°12'07
morning max el	2016 Sep 28 19:27:18	18° $\cap$ 15'02	17°52'31	retrograde	2017 Aug 13 01:00:30	11° $\cap$ 38'24	
	2016 Oct 07 07:55:33	0° $\underline{\cap}$		min. declination.	2017 Aug 16 04:24:45	11° $\cap$ 11'18	3N18'47"
morning set	2016 Oct 15 10:02:06	13° $\underline{\cap}$ 49'57		evening set	2017 Aug 20 04:09:09	9° $\cap$ 18'43	
	2016 Oct 24 20:46:11	0° $\cap$		min. Earth dist.	2017 Aug 23 18:51:08	6° $\cap$ 33'00	0.61692 AU
superior conj	2016 Oct 27 16:16:25	4° $\cap$ 40'28	0°30'51	inferior conj	2017 Aug 26 20:42:18	3° $\cap$ 47'36	-4°13'01
minimum elong	2016 Oct 27 19:24:46	4° $\cap$ 53'24	0°30'24	minimum elong	2017 Aug 27 01:56:11	3° $\cap$ 35'50	4°11'53
desc. node	2016 Nov 01 06:27:33	12° $\cap$ 09'31			2017 Aug 31 15:27:36	30° $\cap$ Ω	
max. Earth dist.	2016 Nov 02 18:18:06	14° $\cap$ 33'25	1.44090 AU	morning rise	2017 Sep 03 01:18:08	28° Ω 50'20	
evening rise	2016 Nov 12 09:39:43	29° $\cap$ 40'45		direct	2017 Sep 05 11:29:23	28° Ω 25'28	
	2016 Nov 12 14:39:32	0° $\propto$			2017 Sep 10 02:52:00	0° $\cap$	
min. declination.	2016 Dec 02 17:31:51	29° $\propto$ 47'03	25S50'12"	max. declination.	2017 Sep 10 09:31:25	0° $\cap$ 11'32	11N17'19"
	2016 Dec 02 21:17:52	0° $\ominus$		asc. node	2017 Sep 10 20:33:48	0° $\cap$ 32'04	
evening max el	2016 Dec 11 04:38:40	10° $\ominus$ 14'49	20°46'05	morning max el	2017 Sep 12 10:16:58	1° $\cap$ 54'51	17°55'59
retrograde	2016 Dec 19 10:55:13	15° $\ominus$ 07'52		morning set	2017 Sep 28 07:54:39	26° $\cap$ 54'12	
asc. node	2016 Dec 20 22:43:37	14° $\ominus$ 55'26			2017 Sep 30 00:42:07	0° $\underline{\cap}$	
evening set	2016 Dec 23 09:41:19	13° $\ominus$ 38'38		superior conj	2017 Oct 08 20:53:36	15° $\underline{\cap}$ 45'43	1°06'26
inferior conj	2016 Dec 28 18:47:32	7° $\ominus$ 28'30	2°26'08	minimum elong	2017 Oct 09 01:27:48	16° $\underline{\cap}$ 05'30	1°05'53
minimum elong	2016 Dec 28 16:12:58	7° $\ominus$ 37'19	2°25'19	max. Earth dist.	2017 Oct 16 07:41:16	28° $\underline{\cap}$ 20'29	1.42617 AU
min. Earth dist.	2016 Dec 29 07:22:11	6° $\ominus$ 45'28	0.67423 AU		2017 Oct 17 07:58:30	0° $\cap$	
morning rise	2017 Jan 02 22:33:56	1° $\ominus$ 15'25		desc. node	2017 Oct 19 03:29:36	2° $\cap$ 56'37	
max. declination.	2017 Jan 04 03:33:01	0° $\ominus$ 17'40	20S12'19"	evening rise	2017 Oct 22 22:59:05	9° $\cap$ 01'18	
	2017 Jan 04 14:16:57	30° $\cap$ $\propto$			2017 Nov 05 19:18:43	0° $\propto$	
direct	2017 Jan 08 09:42:43	28° $\propto$ 50'56		evening max el	2017 Nov 24 00:27:17	23° $\propto$ 46'13	21°59'32
	2017 Jan 12 14:03:17	0° $\ominus$		min. declination.	2017 Nov 24 00:41:20	23° $\propto$ 46'49	25S45'48"
morning max el	2017 Jan 19 09:43:24	5° $\ominus$ 24'29	24°07'57	retrograde	2017 Dec 03 07:34:08	29° $\propto$ 18'10	
min. declination.	2017 Jan 27 12:29:55	14° $\ominus$ 51'46	22S30'20"	evening set	2017 Dec 07 17:25:15	27° $\propto$ 32'39	
desc. node	2017 Jan 28 05:38:41	15° $\ominus$ 47'17		asc. node	2017 Dec 07 19:46:40	27° $\propto$ 27'50	
	2017 Feb 07 09:35:15	0° $\approx$		inferior conj	2017 Dec 13 01:48:44	21° $\propto$ 14'04	1°43'22
morning set	2017 Feb 24 03:51:51	26° $\approx$ 51'20		minimum elong	2017 Dec 12 23:41:55	21° $\propto$ 21'23	1°42'34
	2017 Feb 25 23:07:21	0° $\propto$		min. Earth dist.	2017 Dec 13 02:50:44	21° $\propto$ 10'29	0.67787 AU
max. Earth dist.	2017 Feb 27 21:38:12	3° $\propto$ 26'30	1.38124 AU	morning rise	2017 Dec 18 05:50:39	15° $\propto$ 02'32	
superior conj	2017 Mar 07 00:28:44	16° $\propto$ 37'27	-1°41'15	max. declination.	2017 Dec 22 03:52:42	13° $\propto$ 04'17	19S25'12"
minimum elong	2017 Mar 07 04:46:25	16° $\propto$ 57'58	1°40'45	direct	2017 Dec 23 01:50:44	13° $\propto$ 00'08	
	2017 Mar 13 21:07:11	0° $\cap$		morning max el	2018 Jan 01 19:58:28	18° $\propto$ 46'18	22°39'32
evening rise	2017 Mar 15 21:00:07	3° $\cap$ 56'08			2018 Jan 11 05:09:02	0° $\ominus$	
asc. node	2017 Mar 18 22:02:44	9° $\cap$ 51'59		desc. node	2018 Jan 15 02:43:01	5° $\ominus$ 19'45	
	2017 Mar 31 17:30:32	0° $\propto$		min. declination.	2018 Jan 19 00:13:47	10° $\ominus$ 53'11	23S28'32"
evening max el	2017 Apr 01 10:17:57	0° $\propto$ 42'12	18°59'41		2018 Jan 31 13:39:05	0° $\approx$	
retrograde	2017 Apr 09 23:14:20	4° $\propto$ 50'51		morning set	2018 Feb 04 21:07:31	6° $\approx$ 58'07	
max. declination.	2017 Apr 09 14:15:58	4° $\propto$ 50'23	16N 6'46"	max. Earth dist.	2018 Feb 09 17:41:39	15° $\approx$ 05'46	1.40279 AU
evening set	2017 Apr 12 03:35:37	4° $\propto$ 35'44		superior conj	2018 Feb 17 12:27:15	28° $\approx$ 47'13	-1°58'33
inferior conj	2017 Apr 20 05:54:08	0° $\propto$ 20'38	1°38'42	minimum elong	2018 Feb 17 15:42:10	29° $\approx$ 01'57	1°58'22
minimum elong	2017 Apr 20 09:23:04	0° $\propto$ 14'30	1°37'36		2018 Feb 18 04:28:02	0° $\propto$	
	2017 Apr 20 17:37:05	30° $\cap$ $\cap$		evening rise	2018 Feb 27 10:14:35	17° $\propto$ 16'56	
min. Earth dist.	2017 Apr 23 08:42:58	28° $\cap$ 10'18	0.56827 AU	asc. node	2018 Mar 05 19:05:09	29° $\propto$ 04'59	
desc. node	2017 Apr 26 04:47:48	26° $\cap$ 25'27			2018 Mar 06 07:34:21	0° $\cap$	
morning rise	2017 Apr 28 12:06:42	25° $\cap$ 19'27		evening max el	2018 Mar 15 15:09:34	13° $\cap$ 15'29	18°23'59
direct	2017 May 03 16:32:40	24° $\cap$ 15'49		retrograde	2018 Mar 23 00:18:52	16° $\cap$ 54'28	
min. declination.	2017 May 08 06:55:36	25° $\cap$ 05'38	7N12'54"	max. declination.	2018 Mar 24 02:52:55	16° $\cap$ 49'41	9N43'31"
	2017 May 16 04:06:52	0° $\propto$		evening set	2018 Mar 25 10:31:06	16° $\cap$ 31'56	
morning max el	2017 May 17 23:23:33	1° $\propto$ 39'02	25°46'51	inferior conj	2018 Apr 01 17:53:06	11° $\cap$ 56'51	2°50'24
	2017 Jun 06 22:15:05	0° $\cap$		minimum elong	2018 Apr 01 21:40:42	11° $\cap$ 49'02	2°49'32
morning set	2017 Jun 14 14:49:10	15° $\cap$ 15'24		min. Earth dist.	2018 Apr 05 04:27:31	9° $\cap$ 08'29	0.58765 AU
asc. node	2017 Jun 14 21:21:15	15° $\cap$ 49'55		morning rise	2018 Apr 09 06:05:26	6° $\cap$ 25'45	
	2017 Jun 21 09:57:29	0° $\ominus$		desc. node	2018 Apr 13 01:51:10	5° $\cap$ 00'46	
superior conj	2017 Jun 21 14:14:06	0° $\ominus$ 23'28	1°05'31	direct	2018 Apr 15 09:20:48	4° $\cap$ 46'39	
minimum elong	2017 Jun 21 11:55:10	0° $\ominus$ 10'46	1°05'08	min. declination.	2018 Apr 20 12:51:35	5° $\cap$ 52'52	0N48'45"
max. Earth dist.	2017 Jun 22 09:53:36	2° $\ominus$ 11'12	1.32480 AU	morning max el	2018 Apr 29 18:23:45	12° $\cap$ 29'37	27°01'20
max. declination.	2017 Jun 24 08:31:30	6° $\ominus$ 25'21	24N42'19"		2018 May 13 12:40:08	0° $\propto$	
					2018 May 29 23:48:57	0° $\cap$	

morning set	2018 May 30 00:29:19	0°♊03'31		morning rise	2019 Mar 21 19:57:44	18°♋20'24	
asc. node	2018 Jun 01 18:23:19	5°♊53'58		direct	2019 Mar 28 13:58:47	16°♋05'45	
superior conj	2018 Jun 06 02:01:47	15°♊20'27	0°44'15	desc. node	2019 Mar 30 22:53:45	16°♋21'24	
minimum elong	2018 Jun 06 00:13:34	15°♊10'31	0°43'53	min. declination.	2019 Apr 03 03:43:31	17°♋28'28	5835' 6"
max. Earth dist.	2018 Jun 05 22:02:04	14°♊58'27	1.32188 AU	morning max el	2019 Apr 11 19:41:55	23°♋59'22	27°42'46
evening rise	2018 Jun 12 23:06:57	0°♋16'28			2019 Apr 17 06:00:37	0°♌	
	2018 Jun 12 19:59:36	0°♋			2019 May 06 18:25:07	0°♌	
max. declination.	2018 Jun 14 22:38:08	4°♋23'35	25N 9' 4"	morning set	2019 May 14 06:28:00	14°♌38'38	
	2018 Jun 29 05:16:19	0°♌		asc. node	2019 May 19 15:25:23	26°♌02'59	
desc. node	2018 Jul 10 01:08:33	14°♌03'38		max. Earth dist.	2019 May 20 09:25:29	27°♌40'58	1.32284 AU
evening max el	2018 Jul 12 05:29:20	16°♌15'19	26°25'06		2019 May 21 10:51:45	0°♍	
retrograde	2018 Jul 26 05:02:23	23°♌27'22		superior conj	2019 May 21 13:06:53	0°♍12'20	0°20'00
min. declination.	2018 Jul 31 10:57:53	22°♌18'07	9N51'15"	minimum elong	2019 May 21 12:12:45	0°♍07'24	0°19'49
evening set	2018 Aug 01 19:21:33	21°♌38'47		evening rise	2019 May 28 10:01:46	15°♍08'39	
min. Earth dist.	2018 Aug 05 18:35:27	19°♌02'55	0.59644 AU		2019 Jun 04 20:04:31	0°♎	
inferior conj	2018 Aug 09 02:05:58	16°♌27'42	-4°48'11	max. declination.	2019 Jun 05 21:07:33	1°♎55'17	25N29'50"
minimum elong	2018 Aug 09 05:28:25	16°♌21'04	4°47'50	evening max el	2019 Jun 23 23:15:39	27°♎21'01	25°09'23
morning rise	2018 Aug 16 17:40:42	11°♌53'09		desc. node	2019 Jun 26 22:07:48	29°♎55'42	
direct	2018 Aug 19 04:24:44	11°♌31'48			2019 Jun 27 00:19:25	0°♏	
max. declination.	2018 Aug 26 06:34:52	14°♌42'10	15N49' 2"	retrograde	2019 Jul 07 23:14:27	4°♏27'54	
morning max el	2018 Aug 26 20:34:23	15°♌14'30	18°18'59	evening set	2019 Jul 13 11:09:42	3°♏18'14	
asc. node	2018 Aug 28 17:37:48	17°♌14'21		min. declination.	2019 Jul 17 06:50:55	1°♏18'17	15N21'25"
	2018 Sep 06 02:38:39	0°♐		min. Earth dist.	2019 Jul 18 10:58:46	0°♏33'28	0.57635 AU
morning set	2018 Sep 11 19:44:55	10°♐38'11			2019 Jul 19 07:06:14	30°♐☾	
superior conj	2018 Sep 21 01:52:01	28°♐02'36	1°29'58	inferior conj	2019 Jul 21 12:33:58	28°♐28'41	-4°57'36
minimum elong	2018 Sep 21 05:24:10	28°♐18'45	1°29'39	minimum elong	2019 Jul 21 11:31:48	28°♐30'27	4°57'34
	2018 Sep 22 03:39:22	0°♑		morning rise	2019 Jul 29 14:29:45	24°♐16'16	
max. Earth dist.	2018 Sep 28 15:14:35	11°♑25'25	1.40730 AU	direct	2019 Aug 01 03:57:47	23°♐56'51	
evening rise	2018 Oct 03 06:56:35	19°♑13'45		morning max el	2019 Aug 09 23:08:00	28°♐04'37	19°02'47
desc. node	2018 Oct 06 00:30:10	23°♑40'02			2019 Aug 11 19:45:35	0°♒	
	2018 Oct 10 00:40:09	0°♒		max. declination.	2019 Aug 12 05:33:38	0°♒28'06	19N15'11"
	2018 Oct 31 04:38:16	0°♓		asc. node	2019 Aug 15 14:40:46	4°♒56'19	
evening max el	2018 Nov 06 15:31:42	7°♓19'15	23°18'56	morning set	2019 Aug 26 17:25:10	24°♒51'01	
min. declination.	2018 Nov 12 13:35:08	12°♓07'07	24S49' 8"		2019 Aug 29 07:47:46	0°♑	
retrograde	2018 Nov 17 01:33:12	13°♓29'30		superior conj	2019 Sep 04 01:40:07	11°♑15'31	1°42'26
evening set	2018 Nov 22 00:07:50	11°♓27'02		minimum elong	2019 Sep 04 03:20:04	11°♑23'31	1°42'21
asc. node	2018 Nov 24 16:50:08	8°♓37'28		max. Earth dist.	2019 Sep 10 18:53:28	23°♑47'00	1.38681 AU
inferior conj	2018 Nov 27 09:15:03	5°♓03'49	0°54'39		2019 Sep 14 07:14:24	0°♑	
minimum elong	2018 Nov 27 08:00:13	5°♓08'06	0°54'08	evening rise	2019 Sep 14 15:02:13	0°♑33'47	
min. Earth dist.	2018 Nov 26 23:34:34	5°♓37'03	0.67799 AU	desc. node	2019 Sep 22 21:29:23	14°♑16'05	
	2018 Dec 01 11:11:59	30°♒♌			2019 Oct 03 08:14:15	0°♒	
morning rise	2018 Dec 02 15:49:22	28°♒55'47		evening max el	2019 Oct 20 04:01:45	20°♒56'13	24°38'00
direct	2018 Dec 06 21:22:05	27°♒16'14		min. declination.	2019 Oct 29 11:14:29	27°♒19'30	22S25' 7"
max. declination.	2018 Dec 07 08:20:57	27°♒17'19	16S54'32"	retrograde	2019 Oct 31 15:41:27	27°♒38'16	
	2018 Dec 12 23:42:39	0°♓		evening set	2019 Nov 06 04:10:46	25°♒19'25	
morning max el	2018 Dec 15 11:29:57	2°♓16'32	21°16'12	min. Earth dist.	2019 Nov 10 19:11:34	20°♒03'02	0.67483 AU
desc. node	2019 Jan 01 23:46:05	25°♓18'23		inferior conj	2019 Nov 11 15:21:50	18°♒55'51	0°01'16
	2019 Jan 05 03:39:49	0°♔		minimum elong	2019 Nov 11 15:19:48	18°♒55'58	0°01'16
min. declination.	2019 Jan 10 18:06:34	8°♔30'20	24S 8'50"	transit middle	2019 Nov 11 15:19:48	18°♒55'58	0°01'16
morning set	2019 Jan 15 09:50:52	15°♔45'09		transit begin	2019 Nov 11 12:35:25	19°♒05'06	
max. Earth dist.	2019 Jan 22 19:18:35	27°♔37'51	1.42284 AU	transit end	2019 Nov 11 18:04:11	18°♒46'50	
	2019 Jan 24 05:49:20	0°♕		asc. node	2019 Nov 11 13:53:26	19°♒00'45	
superior conj	2019 Jan 30 02:51:40	9°♕54'26	-2°05'00	morning rise	2019 Nov 17 02:33:38	12°♒53'36	
minimum elong	2019 Jan 30 02:36:49	9°♕53'22	2°05'01	direct	2019 Nov 20 19:11:39	11°♒35'11	
evening rise	2019 Feb 10 10:04:37	29°♕56'32		max. declination.	2019 Nov 22 00:56:19	11°♒43'27	13S 4'26"
	2019 Feb 10 10:50:35	0°♖		morning max el	2019 Nov 28 10:29:15	15°♒57'11	20°03'49
asc. node	2019 Feb 20 16:07:18	17°♖49'21			2019 Dec 09 09:41:35	0°♗	
evening max el	2019 Feb 27 01:25:09	26°♖13'48	18°08'14	desc. node	2019 Dec 19 20:46:59	15°♗35'41	
retrograde	2019 Mar 05 18:18:43	29°♖38'58		morning set	2019 Dec 25 06:06:45	23°♗51'52	
evening set	2019 Mar 08 10:05:02	29°♖07'01			2019 Dec 29 04:54:49	0°♘	
max. declination.	2019 Mar 07 19:54:17	29°♖19'29	2N47' 5"	min. declination.	2020 Jan 02 12:22:47	6°♘46'35	24S39'43"
inferior conj	2019 Mar 15 01:47:34	24°♖11'19	3°29'42	max. Earth dist.	2020 Jan 05 04:21:29	11°♘00'56	1.43884 AU
minimum elong	2019 Mar 15 03:57:31	24°♖06'09	3°29'27	superior conj	2020 Jan 10 15:19:24	19°♘49'57	-1°55'35
min. Earth dist.	2019 Mar 18 06:54:47	21°♖09'08	0.60909 AU	minimum elong	2020 Jan 10 09:38:03	19°♘26'43	1°55'16

	2020 Jan 16 18:30:50	0°♊		max. declination.	2020 Nov 05 13:41:44	26°♊12'07	8S16'34"
evening rise	2020 Jan 23 16:06:14	11°♊45'35		morning max el	2020 Nov 10 17:03:16	29°♊47'38	19°05'52
	2020 Feb 03 11:37:24	0°♋			2020 Nov 10 21:55:28	0°♋	
asc. node	2020 Feb 07 13:09:42	5°♋55'26			2020 Dec 01 19:51:07	0°♌	
evening max el	2020 Feb 10 13:55:57	9°♋27'59	18°11'48	morning set	2020 Dec 03 11:08:03	2°♌33'14	
retrograde	2020 Feb 17 00:54:01	12°♋53'23		desc. node	2020 Dec 05 17:47:45	6°♌06'38	
evening set	2020 Feb 19 21:54:29	12°♋10'51		max. Earth dist.	2020 Dec 17 20:00:05	25°♌03'19	1.44878 AU
max. declination.	2020 Feb 19 21:50:12	12°♋10'56	4S 7'49"				
inferior conj	2020 Feb 26 01:44:47	6°♋55'08	3°43'13	superior conj	2020 Dec 20 03:25:43	28°♌42'04	-1°27'07
minimum elong	2020 Feb 26 01:55:55	6°♋54'38	3°43'13	minimum elong	2020 Dec 19 18:34:19	28°♌07'03	1°26'13
min. Earth dist.	2020 Feb 28 17:55:08	4°♋03'11	0.62960 AU		2020 Dec 20 23:07:20	0°♍	
morning rise	2020 Mar 03 04:56:12	0°♋52'56		min. declination.	2020 Dec 24 06:45:08	5°♍16'29	25S 5' 3"
	2020 Mar 04 11:07:34	30°♌		evening rise	2021 Jan 04 00:21:51	22°♍38'23	
direct	2020 Mar 10 03:48:35	28°♌12'42			2021 Jan 08 11:59:40	0°♍	
	2020 Mar 16 07:42:30	0°♋		evening max el	2021 Jan 24 01:56:52	22°♌51'29	18°33'52
min. declination.	2020 Mar 16 08:09:32	0°♋00'37	11S22'23"	asc. node	2021 Jan 24 10:11:58	23°♌12'05	
desc. node	2020 Mar 16 19:56:26	0°♋17'20		retrograde	2021 Jan 30 15:51:42	26°♌29'27	
morning max el	2020 Mar 24 02:06:22	6°♋08'40	27°46'56	evening set	2021 Feb 02 18:26:51	25°♌35'20	
	2020 Apr 11 04:48:10	0°♎		max. declination.	2021 Feb 03 12:14:41	25°♌07'00	10S31'45"
morning set	2020 Apr 27 06:29:16	28°♎52'23		inferior conj	2021 Feb 08 13:47:55	20°♌01'05	3°37'37
	2020 Apr 27 19:53:02	0°♏		minimum elong	2021 Feb 08 12:25:34	20°♌05'10	3°37'30
max. Earth dist.	2020 May 02 16:04:23	10°♏04'38	1.32787 AU	min. Earth dist.	2021 Feb 10 14:25:28	17°♌36'53	0.64716 AU
				morning rise	2021 Feb 14 05:55:23	13°♌52'37	
superior conj	2020 May 04 21:41:24	14°♏52'20	-0°06'30	direct	2021 Feb 21 00:52:00	11°♌01'22	
minimum elong	2020 May 04 22:00:19	14°♏54'02	0°06'26	min. declination.	2021 Feb 28 09:10:16	13°♌40'47	16S 6'49"
behind sun begin	2020 May 04 17:11:13	14°♏28'01		desc. node	2021 Mar 03 16:59:17	16°♌14'46	
behind sun end	2020 May 05 02:49:26	15°♏20'04		morning max el	2021 Mar 06 11:22:13	18°♌50'30	27°16'00
asc. node	2020 May 05 12:26:25	16°♏12'04			2021 Mar 15 22:26:29	0°♋	
evening rise	2020 May 11 21:52:04	29°♏59'30			2021 Apr 04 03:41:13	0°♎	
	2020 May 11 21:57:46	0°♐		morning set	2021 Apr 10 21:45:30	12°♎36'05	
max. declination.	2020 May 27 04:48:35	27°♐48'57	25N39'38"	max. Earth dist.	2021 Apr 15 14:20:10	21°♎59'22	1.33724 AU
	2020 May 28 18:09:19	0°♑					
evening max el	2020 Jun 04 13:07:03	7°♑55'26	23°36'18	superior conj	2021 Apr 19 01:49:33	29°♎14'15	-0°34'17
desc. node	2020 Jun 12 19:08:33	13°♑39'24		minimum elong	2021 Apr 19 03:33:04	29°♎23'22	0°33'56
retrograde	2020 Jun 18 04:58:52	14°♑45'50			2021 Apr 19 10:29:12	0°♏	
evening set	2020 Jun 22 05:28:08	14°♑10'22		asc. node	2021 Apr 22 09:28:13	6°♏17'21	
min. Earth dist.	2020 Jun 28 23:30:10	11°♑01'36	0.55989 AU	evening rise	2021 Apr 26 08:48:52	14°♏42'54	
inferior conj	2020 Jul 01 02:52:39	9°♑44'37	-4°26'57		2021 May 04 02:49:02	0°♐	
minimum elong	2020 Jun 30 20:36:05	9°♑54'04	4°25'53	evening max el	2021 May 17 05:53:39	18°♐27'41	22°01'02
min. declination.	2020 Jul 05 09:46:00	7°♑20'35	18N26'17"	max. declination.	2021 May 17 20:07:26	19°♐01'27	25N15' 7"
morning rise	2020 Jul 09 14:19:50	5°♑48'31		retrograde	2021 May 29 22:34:02	24°♐43'01	
direct	2020 Jul 12 08:26:22	5°♑29'34		desc. node	2021 May 30 16:10:41	24°♐41'48	
morning max el	2020 Jul 22 15:12:10	10°♑15'27	20°08'00	evening set	2021 Jun 01 15:58:19	24°♐26'30	
max. declination.	2020 Jul 30 06:57:33	19°♑49'38	21N29'45"	min. Earth dist.	2021 Jun 10 10:58:46	20°♐41'20	0.55043 AU
asc. node	2020 Aug 01 11:42:10	23°♑24'07		inferior conj	2021 Jun 11 01:12:54	20°♐21'23	-3°08'12
	2020 Aug 05 03:32:00	0°♒		minimum elong	2021 Jun 10 17:30:20	20°♐32'11	3°05'55
morning set	2020 Aug 09 21:49:55	9°♒24'22		morning rise	2021 Jun 19 20:33:34	16°♐28'26	
				min. declination.	2021 Jun 22 05:32:43	16°♐08'42	18N15'43"
superior conj	2020 Aug 17 15:07:19	25°♒09'36	1°45'42	direct	2021 Jun 22 22:00:03	16°♐07'37	
minimum elong	2020 Aug 17 15:00:44	25°♒09'03	1°45'42	morning max el	2021 Jul 04 19:45:07	21°♐41'58	21°33'03
	2020 Aug 20 01:29:47	0°♓			2021 Jul 11 20:35:15	0°♑	
max. Earth dist.	2020 Aug 23 00:09:49	5°♓42'26	1.36718 AU	asc. node	2021 Jul 19 08:43:29	12°♑26'37	
evening rise	2020 Aug 26 21:24:21	12°♓57'13		max. declination.	2021 Jul 19 23:56:50	13°♑38'20	22N51'46"
	2020 Sep 05 19:46:15	0°♈		morning set	2021 Jul 25 06:34:16	24°♑11'10	
desc. node	2020 Sep 08 18:29:13	4°♈40'14			2021 Jul 28 01:11:35	0°♒	
	2020 Sep 27 07:40:40	0°♉					
evening max el	2020 Oct 01 16:05:37	4°♉36'40	25°49'28	superior conj	2021 Aug 01 14:07:32	9°♒32'31	1°41'31
min. declination.	2020 Oct 13 04:48:32	11°♉37'33	18S30'49"	minimum elong	2021 Aug 01 12:42:08	9°♒25'06	1°41'27
retrograde	2020 Oct 14 01:05:00	11°♉40'10		max. Earth dist.	2021 Aug 05 12:24:40	17°♒35'59	1.35030 AU
evening set	2020 Oct 20 03:50:24	9°♉07'17		evening rise	2021 Aug 09 21:31:27	26°♒11'47	
min. Earth dist.	2020 Oct 24 11:15:31	4°♉26'20	0.66836 AU		2021 Aug 11 21:56:45	0°♓	
inferior conj	2020 Oct 25 18:23:16	2°♉47'51	-0°55'24	desc. node	2021 Aug 26 15:30:22	24°♓45'58	
minimum elong	2020 Oct 25 19:49:43	2°♉43'17	0°54'46		2021 Aug 30 05:09:59	0°♈	
asc. node	2020 Oct 28 10:55:42	29°♈33'56		evening max el	2021 Sep 14 04:24:32	18°♈13'37	26°45'39
	2020 Oct 28 01:33:21	30°♈		retrograde	2021 Sep 27 05:10:11	25°♈28'25	
morning rise	2020 Oct 31 12:07:06	26°♈54'11		min. declination.	2021 Sep 27 09:27:07	25°♈28'19	13S20'58"
direct	2020 Nov 03 17:49:42	25°♈53'50		evening set	2021 Oct 03 21:01:35	22°♈46'21	

min. Earth dist.	2021 Oct 07 21:18:07	18° 4 42'05	0.65842 AU	minimum elong	2022 Jul 16 17:26:43	24° 5 03'38	1°31'06
inferior conj	2021 Oct 09 16:18:21	16° 4 35'27	-1°53'48	max. Earth dist.	2022 Jul 19 09:48:48	29° 5 45'27	1.33722 AU
minimum elong	2021 Oct 09 19:20:55	16° 4 26'29	1°52'35		2022 Jul 19 12:34:53	0° 0	
asc. node	2021 Oct 15 07:58:03	11° 0 08'04		evening rise	2022 Jul 24 10:58:50	10° 0 05'50	
morning rise	2021 Oct 15 18:17:09	10° 0 53'29			2022 Aug 04 06:57:50	0° 0	
direct	2021 Oct 18 15:16:50	10° 0 07'35		desc. node	2022 Aug 13 12:32:43	14° 0 25'16	
max. declination.	2021 Oct 21 01:00:15	10° 0 38'49	2S52'28"		2022 Aug 26 01:02:43	0° 0	
morning max el	2021 Oct 25 05:29:43	13° 0 43'23	18°23'57	evening max el	2022 Aug 27 16:14:21	1° 0 36'58	27°19'12
	2021 Nov 05 22:34:47	0° 0		retrograde	2022 Sep 10 03:38:03	8° 0 55'27	
morning set	2021 Nov 13 18:38:49	12° 0 36'34		min. declination.	2022 Sep 11 07:27:47	8° 0 51'16	7S15'51"
desc. node	2021 Nov 22 14:48:58	26° 0 46'34		evening set	2022 Sep 17 05:11:57	6° 0 12'27	
	2021 Nov 24 15:36:24	0° 0		min. Earth dist.	2022 Sep 20 23:09:53	2° 0 43'48	0.64491 AU
				inferior conj	2022 Sep 23 06:50:01	0° 0 14'07	-2°51'42
superior conj	2021 Nov 29 04:39:25	7° 0 10'05	-0°43'06	minimum elong	2022 Sep 23 11:20:16	0° 0 01'59	2°50'09
minimum elong	2021 Nov 28 23:06:47	6° 0 48'17	0°42'22		2022 Sep 23 12:04:26	30° 0 00	
max. Earth dist.	2021 Nov 30 15:08:23	9° 0 25'35	1.45148 AU	morning rise	2022 Sep 29 18:29:16	24° 0 47'04	
	2021 Dec 13 17:52:03	0° 0		direct	2022 Oct 02 09:07:24	24° 0 11'58	
evening rise	2021 Dec 15 09:16:29	2° 0 34'54		asc. node	2022 Oct 02 04:59:54	24° 0 12'07	
min. declination.	2021 Dec 16 02:22:56	3° 0 42'10	25S26'30"	max. declination.	2022 Oct 05 11:45:04	25° 0 01'35	2N46' 6"
greatest brilliancy	2021 Dec 25 17:46:09	18° 0 48'36	-0.8m	morning max el	2022 Oct 08 21:13:31	27° 0 38'51	17°58'55
	2022 Jan 02 07:09:40	0° 0			2022 Oct 10 23:50:47	0° 0	
evening max el	2022 Jan 07 11:03:44	6° 0 19'08	19°13'19	morning set	2022 Oct 26 06:46:40	24° 0 03'55	
asc. node	2022 Jan 11 07:14:41	9° 0 25'27			2022 Oct 29 19:22:11	0° 0	
retrograde	2022 Jan 14 11:41:30	10° 0 20'15					
evening set	2022 Jan 17 20:49:05	9° 0 13'35		superior conj	2022 Nov 08 16:42:57	16° 0 15'09	0°05'24
max. declination.	2022 Jan 20 02:22:08	7° 0 20'20	15S55'37"	minimum elong	2022 Nov 08 17:20:55	16° 0 17'42	0°05'19
inferior conj	2022 Jan 23 10:28:13	3° 0 22'59	3°17'55	behind sun begin	2022 Nov 08 07:28:54	15° 0 37'56	
minimum elong	2022 Jan 23 08:10:07	3° 0 30'22	3°17'29	behind sun end	2022 Nov 09 03:12:56	16° 0 57'24	
min. Earth dist.	2022 Jan 24 19:53:47	1° 0 36'01	0.66081 AU	desc. node	2022 Nov 09 11:52:12	17° 0 32'09	
	2022 Jan 26 03:04:43	30° 0 00		max. Earth dist.	2022 Nov 13 09:53:03	23° 0 46'51	1.44675 AU
morning rise	2022 Jan 28 19:16:02	27° 0 11'02			2022 Nov 17 08:41:50	0° 0	
direct	2022 Feb 04 04:12:57	24° 0 22'35		evening rise	2022 Nov 24 23:25:52	11° 0 49'11	
min. declination.	2022 Feb 14 00:39:10	29° 0 12'47	19S32'46"		2022 Dec 06 22:08:09	0° 0	
	2022 Feb 14 21:53:36	0° 0		min. declination.	2022 Dec 08 00:18:31	1° 0 37'28	25S43'49"
morning max el	2022 Feb 16 21:06:51	1° 0 53'30	26°16'37	greatest brilliancy	2022 Dec 09 06:58:00	3° 0 30'49	-0.6m
desc. node	2022 Feb 18 14:01:41	3° 0 40'16		evening max el	2022 Dec 21 15:31:24	19° 0 48'37	20°08'21
	2022 Mar 10 01:32:17	0° 0		retrograde	2022 Dec 29 09:31:50	24° 0 21'24	
morning set	2022 Mar 25 00:32:52	25° 0 38'37		asc. node	2022 Dec 29 04:17:26	24° 0 21'08	
	2022 Mar 27 07:44:31	0° 0		evening set	2023 Jan 02 02:44:18	23° 0 00'53	
max. Earth dist.	2022 Mar 29 01:23:54	3° 0 21'43	1.35111 AU	inferior conj	2023 Jan 07 12:57:01	16° 0 56'50	2°47'32
				minimum elong	2023 Jan 07 10:19:26	17° 0 05'40	2°46'49
superior conj	2022 Apr 02 23:10:48	13° 0 11'09	-1°02'06	min. Earth dist.	2023 Jan 08 08:41:35	15° 0 50'24	0.67044 AU
minimum elong	2022 Apr 03 02:18:19	13° 0 27'08	1°01'32	max. declination.	2023 Jan 09 14:08:12	14° 0 13'37	19S32'36"
asc. node	2022 Apr 09 06:31:49	26° 0 14'46		morning rise	2023 Jan 12 17:42:44	10° 0 43'26	
evening rise	2022 Apr 10 16:55:16	29° 0 12'32		direct	2023 Jan 18 13:11:55	8° 0 08'08	
	2022 Apr 11 02:09:28	0° 0		morning max el	2023 Jan 30 05:53:44	15° 0 06'53	24°57'52
evening max el	2022 Apr 29 08:09:00	29° 0 26'22	20°36'29	min. declination.	2023 Feb 02 13:04:49	18° 0 40'29	21S41'43"
	2022 Apr 29 22:23:05	0° 0		desc. node	2023 Feb 05 11:05:16	22° 0 08'39	
max. declination.	2022 May 05 03:38:02	3° 0 37'35	23N19'33"		2023 Feb 11 11:22:17	0° 0	
retrograde	2022 May 10 11:47:31	4° 0 51'31			2023 Mar 02 22:51:59	0° 0	
evening set	2022 May 12 13:50:57	4° 0 40'54		morning set	2023 Mar 07 09:46:13	7° 0 45'24	
desc. node	2022 May 17 13:12:33	3° 0 01'03		max. Earth dist.	2023 Mar 11 01:28:14	14° 0 23'26	1.36927 AU
inferior conj	2022 May 21 19:18:03	0° 0 43'06	-1°14'22				
minimum elong	2022 May 21 15:46:53	0° 0 48'11	1°13'04	superior conj	2023 Mar 17 10:44:37	26° 0 34'22	-1°28'07
min. Earth dist.	2022 May 22 23:30:45	0° 0 02'29	0.55013 AU	minimum elong	2023 Mar 17 14:52:43	26° 0 54'40	1°27'32
	2022 May 23 01:15:05	30° 0 00			2023 Mar 19 04:24:06	0° 0	
morning rise	2022 May 30 16:57:34	26° 0 33'34		evening rise	2023 Mar 25 19:55:07	13° 0 21'00	
direct	2022 Jun 03 08:00:17	26° 0 05'14		asc. node	2023 Mar 27 03:35:25	15° 0 59'08	
min. declination.	2022 Jun 05 19:50:52	26° 0 19'08	15N27' 7"		2023 Apr 03 16:22:12	0° 0	
	2022 Jun 13 15:26:42	0° 0		evening max el	2023 Apr 11 22:10:25	11° 0 03'45	19°29'24
morning max el	2022 Jun 16 14:55:34	2° 0 30'30	23°11'59	max. declination.	2023 Apr 19 16:10:28	15° 0 28'50	19N17'12"
	2022 Jul 05 06:25:02	0° 0		retrograde	2023 Apr 21 08:34:56	15° 0 37'17	
asc. node	2022 Jul 06 05:46:18	1° 0 55'18		evening set	2023 Apr 23 09:50:45	15° 0 25'16	
morning set	2022 Jul 09 17:45:48	9° 0 06'10		inferior conj	2023 May 01 23:27:36	11° 0 19'48	0°42'08
max. declination.	2022 Jul 10 04:45:41	10° 0 04'05	23N45'17"	minimum elong	2023 May 02 01:13:11	11° 0 16'58	0°41'30
				desc. node	2023 May 04 10:15:31	9° 0 45'18	
superior conj	2022 Jul 16 19:37:42	24° 0 15'19	1°31'19	min. Earth dist.	2023 May 04 14:02:52	9° 0 39'21	0.55930 AU

morning rise	2023 May 10 13:48:14	6°♄38'05		retrograde	2024 Apr 01 22:14:30	27°♃13'07	
direct	2023 May 15 03:16:43	5°♄50'59		max. declination.	2024 Apr 02 06:35:13	27°♃12'41	13N32'32"
min. declination.	2023 May 19 05:26:50	6°♄29'22	10N38'10"	evening set	2024 Apr 04 05:01:42	26°♃55'12	
morning max el	2023 May 29 05:33:54	12°♄58'16	24°53'24	inferior conj	2024 Apr 11 23:02:57	22°♃32'20	2°13'19
	2023 Jun 11 10:26:38	0°♄		minimum elong	2024 Apr 12 03:01:30	22°♃24'52	2°12'11
asc. node	2023 Jun 23 02:50:10	21°♄42'07		min. Earth dist.	2024 Apr 15 07:21:45	20°♃02'52	0.57601 AU
morning set	2023 Jun 24 05:37:49	24°♄02'33		morning rise	2024 Apr 19 21:53:23	17°♃17'14	
	2023 Jun 27 00:23:45	0°♄		desc. node	2024 Apr 20 07:17:46	17°♃06'52	
max. declination.	2023 Jun 30 12:25:39	7°♄37'50	24N24' 2"	direct	2024 Apr 25 12:54:08	15°♃58'52	
				min. declination.	2024 Apr 30 09:26:37	16°♃55'56	4N34'25"
superior conj	2023 Jul 01 05:05:46	9°♄08'45	1°16'10	morning max el	2024 May 09 21:29:21	23°♃32'32	26°21'56
minimum elong	2023 Jul 01 02:41:14	8°♄55'37	1°15'49		2024 May 15 17:05:10	0°♄	
max. Earth dist.	2023 Jul 02 15:12:04	12°♄14'07	1.32815 AU		2024 Jun 03 07:36:38	0°♄	
evening rise	2023 Jul 08 09:53:32	24°♄27'11		morning set	2024 Jun 07 16:33:38	8°♄55'01	
	2023 Jul 11 04:11:06	0°♄		asc. node	2024 Jun 08 23:54:26	11°♄41'18	
	2023 Jul 28 21:31:13	0°♄					
desc. node	2023 Jul 31 09:34:01	3°♄26'05		superior conj	2024 Jun 14 16:32:34	24°♄06'06	0°56'55
evening max el	2023 Aug 10 01:46:39	14°♄33'55	27°24'01	minimum elong	2024 Jun 14 14:23:32	23°♄54'16	0°56'31
retrograde	2023 Aug 23 19:59:29	21°♄51'07		max. Earth dist.	2024 Jun 15 01:49:27	24°♄57'08	1.32306 AU
min. declination.	2023 Aug 26 03:11:18	21°♄35'52	0S37'30"		2024 Jun 17 09:06:57	0°♄	
evening set	2023 Aug 31 00:57:13	19°♄19'02		max. declination.	2024 Jun 19 23:20:10	5°♄35'37	24N54'21"
min. Earth dist.	2023 Sep 03 15:11:55	16°♄20'32	0.62794 AU	evening rise	2024 Jun 21 15:17:43	9°♄06'36	
inferior conj	2023 Sep 06 11:09:09	13°♄36'50	-3°45'45		2024 Jul 02 12:50:04	0°♄	
minimum elong	2023 Sep 06 16:29:35	13°♄23'56	3°44'18	desc. node	2024 Jul 17 06:34:17	21°♄32'32	
morning rise	2023 Sep 13 09:27:30	8°♄27'52		evening max el	2024 Jul 22 06:38:46	26°♄52'37	26°56'08
direct	2023 Sep 15 20:21:11	8°♄00'11			2024 Jul 25 22:41:50	0°♄	
asc. node	2023 Sep 19 02:02:58	8°♄49'34		retrograde	2024 Aug 05 04:56:05	4°♄06'17	
max. declination.	2023 Sep 19 22:55:27	9°♄19'21	8N17'18"	min. declination.	2024 Aug 09 02:24:41	3°♄26'02	6N 7'27"
morning max el	2023 Sep 22 13:16:01	11°♄26'32	17°51'38	evening set	2024 Aug 12 03:57:25	1°♄58'38	
	2023 Oct 05 00:08:36	0°♄			2024 Aug 15 00:15:30	30°♄	
morning set	2023 Oct 08 18:07:50	6°♄37'50		min. Earth dist.	2024 Aug 15 20:51:43	29°♄18'58	0.60831 AU
				inferior conj	2024 Aug 19 01:58:06	26°♄35'36	-4°30'19
superior conj	2023 Oct 20 05:38:01	26°♄34'20	0°47'23	minimum elong	2024 Aug 19 06:42:39	26°♄25'30	4°29'29
minimum elong	2023 Oct 20 09:47:45	26°♄51'50	0°46'50	morning rise	2024 Aug 26 11:16:12	21°♄47'34	
	2023 Oct 22 06:48:49	0°♄		direct	2024 Aug 28 21:13:59	21°♄24'32	
max. Earth dist.	2023 Oct 27 01:19:10	7°♄49'05	1.43526 AU	max. declination.	2024 Sep 03 13:48:57	23°♄38'25	13N17'49"
desc. node	2023 Oct 27 08:54:38	8°♄19'43		asc. node	2024 Sep 04 23:06:53	24°♄50'34	
evening rise	2023 Nov 04 07:58:24	20°♄54'39		morning max el	2024 Sep 05 02:30:17	24°♄58'43	18°03'12
	2023 Nov 10 06:25:00	0°♄			2024 Sep 09 06:50:08	0°♄	
min. declination.	2023 Nov 29 20:36:11	27°♄52'10	25S52'11"	morning set	2024 Sep 20 22:32:06	20°♄00'40	
	2023 Dec 01 14:31:30	0°♄			2024 Sep 26 08:08:45	0°♄	
evening max el	2023 Dec 04 14:28:30	3°♄19'30	21°16'18				
retrograde	2023 Dec 13 07:09:07	8°♄29'20		superior conj	2024 Sep 30 21:09:22	8°♄11'29	1°17'52
asc. node	2023 Dec 16 01:19:07	7°♄47'10		minimum elong	2024 Oct 01 01:28:52	8°♄30'39	1°17'24
evening set	2023 Dec 17 10:17:21	6°♄53'32		max. Earth dist.	2024 Oct 08 11:57:53	21°♄18'29	1.41855 AU
inferior conj	2023 Dec 22 18:54:14	0°♄39'18	2°08'48	desc. node	2024 Oct 13 05:53:56	29°♄05'45	
minimum elong	2023 Dec 22 16:28:10	0°♄47'42	2°07'58		2024 Oct 13 19:23:24	0°♄	
min. Earth dist.	2023 Dec 23 02:29:16	0°♄13'07	0.67618 AU	evening rise	2024 Oct 14 03:56:26	0°♄34'16	
	2023 Dec 23 06:17:29	30°♄			2024 Nov 02 19:17:40	0°♄	
morning rise	2023 Dec 27 22:27:38	24°♄26'24		evening max el	2024 Nov 16 08:09:21	16°♄51'53	22°33'01
max. declination.	2023 Dec 30 17:28:06	22°♄38'51	20S 7'30"	min. declination.	2024 Nov 19 05:48:57	19°♄31'58	25S31'25"
direct	2024 Jan 02 03:07:34	22°♄10'52		retrograde	2024 Nov 26 02:42:24	22°♄40'18	
morning max el	2024 Jan 12 14:37:31	28°♄25'20	23°30'07	evening set	2024 Nov 30 17:44:12	20°♄47'51	
	2024 Jan 14 02:49:27	0°♄		asc. node	2024 Dec 01 22:21:49	19°♄41'39	
desc. node	2024 Jan 23 08:08:57	11°♄21'53		inferior conj	2024 Dec 06 02:17:54	14°♄27'02	1°23'20
min. declination.	2024 Jan 24 11:17:31	12°♄54'00	22S57'51"	minimum elong	2024 Dec 06 00:30:23	14°♄33'14	1°22'38
	2024 Feb 05 05:10:03	0°♄		min. Earth dist.	2024 Dec 05 22:47:44	14°♄39'09	0.67836 AU
morning set	2024 Feb 16 19:16:30	18°♄40'19		morning rise	2024 Dec 11 07:07:55	8°♄16'38	
max. Earth dist.	2024 Feb 20 20:12:34	25°♄37'30	1.39033 AU	direct	2024 Dec 15 20:56:19	6°♄23'46	
	2024 Feb 23 07:28:53	0°♄		max. declination.	2024 Dec 15 15:45:03	6°♄24'00	18S33'20"
				morning max el	2024 Dec 25 02:30:13	11°♄49'37	22°02'57
superior conj	2024 Feb 28 08:42:53	9°♄14'46	-1°49'37		2025 Jan 08 10:29:57	0°♄	
minimum elong	2024 Feb 28 12:48:55	9°♄33'56	1°49'16	desc. node	2025 Jan 09 05:09:47	1°♄06'34	
evening rise	2024 Mar 08 15:06:57	27°♄01'34		min. declination.	2025 Jan 15 03:06:12	9°♄46'15	23S47' 2"
	2024 Mar 10 04:02:56	0°♄		morning set	2025 Jan 26 23:51:05	28°♄10'47	
asc. node	2024 Mar 13 00:38:56	5°♄25'35			2025 Jan 28 02:52:42	0°♄	
evening max el	2024 Mar 24 22:34:05	23°♄19'22	18°42'06	max. Earth dist.	2025 Feb 01 17:53:11	7°♄37'25	1.41171 AU

superior conj	2025 Feb 09 12:08:06	20° $\approx$ 59'36	-2°02'55	max. declination.	2025 Nov 30 13:33:25	20° $\mathbb{M}$ 46'02	15S26'56"
minimum elong	2025 Feb 09 14:13:40	21° $\approx$ 08'52	2°02'52	morning max el	2025 Dec 07 21:02:55	25° $\mathbb{M}$ 24'35	20°43'46
	2025 Feb 14 12:06:24	0° $\mathbb{H}$			2025 Dec 11 22:39:43	0° $\mathbb{J}$	
evening rise	2025 Feb 19 23:04:24	10° $\mathbb{H}$ 05'30		desc. node	2025 Dec 27 02:10:49	21° $\mathbb{J}$ 14'12	
asc. node	2025 Feb 27 21:40:58	24° $\mathbb{H}$ 27'18			2026 Jan 01 21:10:42	0° $\mathbb{Z}$	
	2025 Mar 03 09:03:40	0° $\mathbb{Y}$		morning set	2026 Jan 06 03:01:19	6° $\mathbb{Z}$ 32'56	
evening max el	2025 Mar 08 06:09:18	6° $\mathbb{Y}$ 04'31	18°14'53	min. declination.	2026 Jan 06 21:20:11	7° $\mathbb{Z}$ 44'18	24S22'40"
retrograde	2025 Mar 15 06:46:06	9° $\mathbb{Y}$ 35'20		max. Earth dist.	2026 Jan 14 23:14:02	20° $\mathbb{Z}$ 34'44	1.43040 AU
max. declination.	2025 Mar 16 20:05:06	9° $\mathbb{Y}$ 25'13	6N50'27"		2026 Jan 20 16:41:11	0° $\approx$	
evening set	2025 Mar 17 19:35:09	9° $\mathbb{Y}$ 08'49		superior conj	2026 Jan 21 15:48:54	1° $\approx$ 36'53	-2°03'13
inferior conj	2025 Mar 24 19:48:27	4° $\mathbb{Y}$ 24'56	3°10'39	minimum elong	2026 Jan 21 13:20:31	1° $\approx$ 26'30	2°03'09
minimum elong	2025 Mar 24 23:02:15	4° $\mathbb{Y}$ 17'50	3°10'05	evening rise	2026 Feb 02 15:49:48	22° $\approx$ 24'40	
min. Earth dist.	2025 Mar 28 05:22:52	1° $\mathbb{Y}$ 27'42	0.59680 AU		2026 Feb 06 22:48:04	0° $\mathbb{H}$	
	2025 Mar 30 02:18:00	30° $\mathbb{R}$ $\mathbb{H}$		asc. node	2026 Feb 14 18:42:08	12° $\mathbb{H}$ 56'28	
morning rise	2025 Apr 01 00:08:55	28° $\mathbb{H}$ 44'32		evening max el	2026 Feb 19 17:41:09	19° $\mathbb{H}$ 10'15	18°07'22
direct	2025 Apr 07 11:07:37	26° $\mathbb{H}$ 49'35		retrograde	2026 Feb 26 06:48:10	22° $\mathbb{H}$ 33'55	
desc. node	2025 Apr 07 04:20:08	26° $\mathbb{H}$ 49'48		max. declination.	2026 Feb 28 16:33:42	22° $\mathbb{H}$ 05'56	0S 9' 8"
min. declination.	2025 Apr 12 18:47:34	28° $\mathbb{H}$ 02'15	1S54'48"	evening set	2026 Mar 01 00:51:57	21° $\mathbb{H}$ 57'28	
	2025 Apr 16 06:25:09	0° $\mathbb{Y}$		inferior conj	2026 Mar 07 11:01:47	16° $\mathbb{H}$ 52'43	3°38'09
morning max el	2025 Apr 21 18:49:08	4° $\mathbb{Y}$ 37'19	27°23'25	minimum elong	2026 Mar 07 12:20:08	16° $\mathbb{H}$ 49'25	3°38'04
	2025 May 10 12:15:11	0° $\mathbb{Z}$		min. Earth dist.	2026 Mar 10 11:03:35	13° $\mathbb{H}$ 52'35	0.61813 AU
morning set	2025 May 23 00:51:17	23° $\mathbb{Z}$ 37'51		morning rise	2026 Mar 13 22:22:38	10° $\mathbb{H}$ 56'13	
	2025 May 26 00:59:21	0° $\mathbb{H}$		direct	2026 Mar 20 19:32:50	8° $\mathbb{H}$ 29'27	
asc. node	2025 May 26 20:57:32	1° $\mathbb{H}$ 47'51		desc. node	2026 Mar 25 01:22:11	9° $\mathbb{H}$ 19'45	
max. Earth dist.	2025 May 29 14:05:37	7° $\mathbb{H}$ 43'39	1.32181 AU	min. declination.	2026 Mar 26 14:45:10	10° $\mathbb{H}$ 01'12	8S 6'59"
				morning max el	2026 Apr 03 22:33:35	16° $\mathbb{H}$ 24'16	27°49'10
superior conj	2025 May 30 04:12:48	9° $\mathbb{H}$ 01'17	0°34'18		2026 Apr 15 03:21:30	0° $\mathbb{Y}$	
minimum elong	2025 May 30 02:44:52	8° $\mathbb{H}$ 53'13	0°33'59		2026 May 03 02:56:56	0° $\mathbb{Z}$	
evening rise	2025 Jun 06 00:46:56	23° $\mathbb{H}$ 56'04		morning set	2026 May 07 04:33:30	8° $\mathbb{Z}$ 04'33	
	2025 Jun 08 22:58:12	0° $\mathbb{Z}$		max. Earth dist.	2026 May 13 00:08:19	20° $\mathbb{Z}$ 20'30	1.32449 AU
max. declination.	2025 Jun 10 16:33:39	3° $\mathbb{Z}$ 27'48	25N18'45"	asc. node	2026 May 13 17:58:55	21° $\mathbb{Z}$ 57'14	
	2025 Jun 26 19:09:04	0° $\mathbb{Q}$		superior conj	2026 May 14 14:24:31	23° $\mathbb{Z}$ 48'24	0°09'00
evening max el	2025 Jul 04 04:38:46	8° $\mathbb{Q}$ 24'49	25°56'01	minimum elong	2026 May 14 13:59:17	23° $\mathbb{Z}$ 46'06	0°08'54
desc. node	2025 Jul 04 03:34:23	8° $\mathbb{Q}$ 22'16		behind sun begin	2026 May 14 09:42:43	23° $\mathbb{Z}$ 22'48	
retrograde	2025 Jul 18 04:45:01	15° $\mathbb{Q}$ 34'31		behind sun end	2026 May 14 18:15:50	24° $\mathbb{Z}$ 09'25	
evening set	2025 Jul 24 09:28:31	14° $\mathbb{Q}$ 02'25			2026 May 17 10:26:24	0° $\mathbb{H}$	
min. declination.	2025 Jul 24 19:25:17	13° $\mathbb{Q}$ 50'04	12N21'28"	evening rise	2026 May 21 12:18:29	8° $\mathbb{H}$ 48'22	
min. Earth dist.	2025 Jul 28 17:06:16	11° $\mathbb{Q}$ 24'58	0.58765 AU		2026 Jun 01 11:55:44	0° $\mathbb{Z}$	
inferior conj	2025 Jul 31 23:41:18	9° $\mathbb{Q}$ 00'36	-4°56'13	max. declination.	2026 Jun 01 19:25:06	0° $\mathbb{Z}$ 31'37	25N36' 6"
minimum elong	2025 Aug 01 01:28:44	8° $\mathbb{Q}$ 57'17	4°56'07	evening max el	2026 Jun 15 19:59:51	19° $\mathbb{Z}$ 13'42	24°31'01
morning rise	2025 Aug 08 19:42:05	4° $\mathbb{Q}$ 35'27		desc. node	2026 Jun 21 00:36:11	23° $\mathbb{Z}$ 24'18	
direct	2025 Aug 11 07:29:53	4° $\mathbb{Q}$ 14'58		retrograde	2026 Jun 29 17:35:55	26° $\mathbb{Z}$ 15'26	
morning max el	2025 Aug 19 09:48:06	8° $\mathbb{Q}$ 06'51	18°35'00	evening set	2026 Jul 04 15:14:41	25° $\mathbb{Z}$ 21'28	
max. declination.	2025 Aug 19 18:34:49	8° $\mathbb{Q}$ 28'09	17N24'11"	min. Earth dist.	2026 Jul 10 06:59:30	22° $\mathbb{Z}$ 28'21	0.56866 AU
asc. node	2025 Aug 22 20:09:56	12° $\mathbb{Q}$ 00'09		min. declination.	2026 Jul 12 00:29:02	21° $\mathbb{Z}$ 22'22	17N 2'49"
	2025 Sep 02 13:23:05	0° $\mathbb{H}$		inferior conj	2026 Jul 13 01:25:48	20° $\mathbb{Z}$ 42'08	-4°50'17
morning set	2025 Sep 04 15:04:05	3° $\mathbb{H}$ 59'02		minimum elong	2026 Jul 12 22:02:05	20° $\mathbb{Z}$ 47'36	4°49'59
				morning rise	2026 Jul 21 07:32:09	16° $\mathbb{Z}$ 38'07	
superior conj	2025 Sep 13 10:51:47	20° $\mathbb{H}$ 54'40	1°36'30	direct	2026 Jul 23 22:57:52	16° $\mathbb{Z}$ 19'00	
minimum elong	2025 Sep 13 13:38:18	21° $\mathbb{H}$ 07'38	1°36'17	morning max el	2026 Aug 02 08:07:29	20° $\mathbb{Z}$ 40'55	19°28'05
	2025 Sep 18 10:05:52	0° $\mathbb{Z}$		max. declination.	2026 Aug 06 12:48:01	25° $\mathbb{Z}$ 25'32	20N19'35"
max. Earth dist.	2025 Sep 20 17:59:27	4° $\mathbb{Z}$ 07'00	1.39868 AU	asc. node	2026 Aug 09 17:11:43	0° $\mathbb{Q}$ 02'51	
evening rise	2025 Sep 24 22:08:57	11° $\mathbb{Z}$ 15'59			2026 Aug 09 16:28:23	0° $\mathbb{Q}$	
desc. node	2025 Sep 30 02:54:06	19° $\mathbb{Z}$ 47'08		morning set	2026 Aug 19 15:56:34	18° $\mathbb{Q}$ 21'38	
	2025 Oct 06 16:40:46	0° $\mathbb{M}$			2026 Aug 25 11:04:14	0° $\mathbb{H}$	
	2025 Oct 29 11:02:04	0° $\mathbb{J}$		superior conj	2026 Aug 27 17:03:49	4° $\mathbb{H}$ 26'59	1°44'50
evening max el	2025 Oct 29 22:02:14	0° $\mathbb{J}$ 27'42	23°52'58	minimum elong	2026 Aug 27 17:56:05	4° $\mathbb{H}$ 31'15	1°44'49
min. declination.	2025 Nov 06 11:04:43	6° $\mathbb{J}$ 06'56	24S 0'37"	max. Earth dist.	2026 Sep 02 21:53:04	16° $\mathbb{H}$ 15'58	1.37818 AU
retrograde	2025 Nov 09 19:01:43	6° $\mathbb{J}$ 51'43		evening rise	2026 Sep 06 16:08:12	23° $\mathbb{H}$ 03'48	
evening set	2025 Nov 14 23:30:55	4° $\mathbb{J}$ 42'07			2026 Sep 10 16:20:39	0° $\mathbb{Z}$	
asc. node	2025 Nov 18 19:23:22	0° $\mathbb{J}$ 26'19		desc. node	2026 Sep 16 23:55:18	10° $\mathbb{Z}$ 18'59	
	2025 Nov 19 03:20:18	30° $\mathbb{R}$ $\mathbb{M}$			2026 Sep 30 11:44:24	0° $\mathbb{M}$	
inferior conj	2025 Nov 20 09:23:25	28° $\mathbb{M}$ 18'28	0°32'31	evening max el	2026 Oct 12 10:03:19	14° $\mathbb{M}$ 06'17	25°09'39
minimum elong	2025 Nov 20 08:37:08	28° $\mathbb{M}$ 21'05	0°32'11	min. declination.	2026 Oct 22 18:00:41	20° $\mathbb{M}$ 49'33	20S57'44"
min. Earth dist.	2025 Nov 19 19:16:06	29° $\mathbb{M}$ 06'25	0.67716 AU				
morning rise	2025 Nov 25 17:42:21	22° $\mathbb{M}$ 12'45					
direct	2025 Nov 29 17:38:27	20° $\mathbb{M}$ 42'25					

retrograde	2026 Oct 24 07:12:43	20° $\mathbb{M}$ 58'44		superior conj	2027 Aug 11 11:02:13	18° $\mathbb{O}$ 34'45	1°44'44
evening set	2026 Oct 30 01:56:47	18° $\mathbb{M}$ 33'11		minimum elong	2027 Aug 11 10:18:18	18° $\mathbb{O}$ 31'00	1°44'44
min. Earth dist.	2026 Nov 03 13:33:38	13° $\mathbb{M}$ 31'58	0.67255 AU	max. Earth dist.	2027 Aug 16 05:10:53	28° $\mathbb{O}$ 05'42	1.35951 AU
inferior conj	2026 Nov 04 14:24:25	12° $\mathbb{M}$ 10'51	-0°22'25		2027 Aug 17 04:43:22	0° $\mathbb{M}$	
minimum elong	2026 Nov 04 14:58:24	12° $\mathbb{M}$ 09'00	0°22'10	evening rise	2027 Aug 20 06:36:47	5° $\mathbb{M}$ 50'16	
asc. node	2026 Nov 05 16:24:59	10° $\mathbb{M}$ 46'49			2027 Sep 03 11:36:55	0° $\mathbb{O}$	
morning rise	2026 Nov 10 04:11:43	6° $\mathbb{M}$ 12'11		desc. node	2027 Sep 03 20:56:33	0° $\mathbb{O}$ 35'59	
direct	2026 Nov 13 15:53:53	5° $\mathbb{M}$ 02'01		evening max el	2027 Sep 24 21:59:03	27° $\mathbb{O}$ 45'16	26°15'38
max. declination.	2026 Nov 15 03:57:31	5° $\mathbb{M}$ 14'16	11S10' 7"		2027 Sep 27 09:09:42	0° $\mathbb{M}$	
morning max el	2026 Nov 20 23:31:18	9° $\mathbb{M}$ 10'32	19°37'18	min. declination.	2027 Oct 07 05:12:41	4° $\mathbb{M}$ 55'07	16S29'25"
greatest brilliancy	2026 Dec 04 06:21:21	26° $\mathbb{M}$ 51'25	-0.7m	retrograde	2027 Oct 07 14:37:04	4° $\mathbb{M}$ 55'40	
	2026 Dec 06 08:33:20	0° $\mathbb{O}$		evening set	2027 Oct 13 23:09:37	2° $\mathbb{M}$ 17'52	
desc. node	2026 Dec 13 23:12:41	11° $\mathbb{O}$ 38'07			2027 Oct 16 07:36:26	30° $\mathbb{R}$ $\mathbb{O}$	
morning set	2026 Dec 15 23:59:50	14° $\mathbb{O}$ 46'25		min. Earth dist.	2027 Oct 18 03:22:23	27° $\mathbb{O}$ 52'29	0.66452 AU
	2026 Dec 25 18:22:30	0° $\mathbb{O}$		inferior conj	2027 Oct 19 15:30:47	26° $\mathbb{O}$ 01'16	-1°20'03
max. Earth dist.	2026 Dec 28 11:41:03	4° $\mathbb{O}$ 18'00	1.44393 AU	minimum elong	2027 Oct 19 17:37:50	25° $\mathbb{O}$ 54'45	1°19'11
min. declination.	2026 Dec 29 15:33:52	6° $\mathbb{O}$ 08'44	24S50'51"	asc. node	2027 Oct 23 13:27:27	21° $\mathbb{O}$ 40'18	
				morning rise	2027 Oct 25 12:34:19	20° $\mathbb{O}$ 12'17	
superior conj	2027 Jan 01 17:07:31	11° $\mathbb{O}$ 02'52	-1°45'49	direct	2027 Oct 28 14:10:38	19° $\mathbb{O}$ 18'43	
minimum elong	2027 Jan 01 09:27:12	10° $\mathbb{O}$ 32'03	1°45'13	max. declination.	2027 Oct 30 15:50:03	19° $\mathbb{O}$ 42'06	6S 4'21"
	2027 Jan 13 06:05:52	0° $\mathbb{O}$		morning max el	2027 Nov 04 08:54:47	23° $\mathbb{O}$ 04'02	18°46'01
evening rise	2027 Jan 15 13:05:23	3° $\mathbb{O}$ 51'05			2027 Nov 10 00:26:19	0° $\mathbb{M}$	
	2027 Feb 01 01:25:39	0° $\mathbb{H}$		morning set	2027 Nov 25 15:05:14	23° $\mathbb{M}$ 59'37	
asc. node	2027 Feb 01 15:43:13	0° $\mathbb{H}$ 43'17			2027 Nov 29 10:23:36	0° $\mathbb{O}$	
evening max el	2027 Feb 03 06:14:47	2° $\mathbb{H}$ 29'09	18°18'55	desc. node	2027 Nov 30 20:14:29	2° $\mathbb{O}$ 13'21	
retrograde	2027 Feb 09 17:36:05	5° $\mathbb{H}$ 58'53		max. Earth dist.	2027 Dec 11 04:41:11	18° $\mathbb{O}$ 29'17	1.45080 AU
evening set	2027 Feb 12 16:48:48	5° $\mathbb{H}$ 11'37					
max. declination.	2027 Feb 12 23:20:58	5° $\mathbb{H}$ 02'42	6S54'14"	superior conj	2027 Dec 11 22:16:44	19° $\mathbb{O}$ 38'24	-1°09'48
inferior conj	2027 Feb 18 16:39:11	29° $\mathbb{O}$ 47'37	3°42'51	minimum elong	2027 Dec 11 14:03:34	19° $\mathbb{O}$ 06'06	1°08'52
minimum elong	2027 Feb 18 16:06:15	29° $\mathbb{O}$ 49'10	3°42'50		2027 Dec 18 11:58:04	0° $\mathbb{O}$	
	2027 Feb 18 12:15:34	30° $\mathbb{R}$ $\mathbb{O}$		min. declination.	2027 Dec 21 10:18:48	4° $\mathbb{O}$ 39'10	25S14'24"
min. Earth dist.	2027 Feb 21 02:15:21	27° $\mathbb{O}$ 05'46	0.63747 AU	evening rise	2027 Dec 27 11:23:24	14° $\mathbb{O}$ 19'28	
morning rise	2027 Feb 24 14:37:12	23° $\mathbb{O}$ 41'59		greatest brilliancy	2028 Jan 02 14:29:44	24° $\mathbb{O}$ 11'26	-0.9m
direct	2027 Mar 03 12:32:02	20° $\mathbb{O}$ 55'23			2028 Jan 06 05:57:41	0° $\mathbb{O}$	
min. declination.	2027 Mar 10 02:18:57	23° $\mathbb{O}$ 00'04	13S30'17"	evening max el	2028 Jan 17 17:16:31	15° $\mathbb{O}$ 55'01	18°48'37
desc. node	2027 Mar 11 22:24:32	24° $\mathbb{O}$ 12'14		asc. node	2028 Jan 19 12:45:40	17° $\mathbb{O}$ 35'02	
morning max el	2027 Mar 17 06:50:00	28° $\mathbb{O}$ 50'18	27°37'48	retrograde	2028 Jan 24 11:02:26	19° $\mathbb{O}$ 41'54	
	2027 Mar 18 10:01:54	0° $\mathbb{H}$		evening set	2028 Jan 27 16:08:36	18° $\mathbb{O}$ 42'47	
	2027 Apr 08 23:20:10	0° $\mathbb{Y}$		max. declination.	2028 Jan 28 21:22:38	17° $\mathbb{O}$ 50'39	12S56'45"
morning set	2027 Apr 21 01:17:44	22° $\mathbb{Y}$ 07'29		inferior conj	2028 Feb 02 08:47:26	13° $\mathbb{O}$ 01'27	3°30'46
	2027 Apr 24 22:18:07	0° $\mathbb{O}$		minimum elong	2028 Feb 02 06:56:39	13° $\mathbb{O}$ 07'08	3°30'30
max. Earth dist.	2027 Apr 26 03:49:31	2° $\mathbb{O}$ 34'08	1.33128 AU	min. Earth dist.	2028 Feb 04 02:54:55	10° $\mathbb{O}$ 51'59	0.65346 AU
				morning rise	2028 Feb 07 21:20:36	6° $\mathbb{O}$ 51'02	
superior conj	2027 Apr 28 21:21:54	8° $\mathbb{O}$ 21'52	-0°18'12	direct	2028 Feb 14 12:37:48	3° $\mathbb{O}$ 59'07	
minimum elong	2027 Apr 28 22:15:48	8° $\mathbb{O}$ 26'41	0°18'00	min. declination.	2028 Feb 22 16:04:38	7° $\mathbb{O}$ 17'45	17S42'54"
asc. node	2027 Apr 30 15:01:16	12° $\mathbb{O}$ 05'38		desc. node	2028 Feb 26 19:27:33	10° $\mathbb{O}$ 50'52	
evening rise	2027 May 06 00:01:28	23° $\mathbb{O}$ 37'11		morning max el	2028 Feb 27 16:35:37	11° $\mathbb{O}$ 42'59	26°53'38
	2027 May 09 01:57:45	0° $\mathbb{H}$			2028 Mar 13 07:07:27	0° $\mathbb{H}$	
max. declination.	2027 May 24 02:35:56	24° $\mathbb{H}$ 57'10	25N36'15"		2028 Mar 31 12:27:40	0° $\mathbb{Y}$	
evening max el	2027 May 28 09:57:33	29° $\mathbb{H}$ 42'52	22°55'08	morning set	2028 Apr 03 11:47:55	5° $\mathbb{Y}$ 35'33	
	2027 May 28 17:06:35	0° $\mathbb{O}$		max. Earth dist.	2028 Apr 07 21:18:30	14° $\mathbb{Y}$ 13'47	1.34255 AU
desc. node	2027 Jun 07 21:37:11	6° $\mathbb{O}$ 02'59					
retrograde	2027 Jun 10 18:15:14	6° $\mathbb{O}$ 21'40		superior conj	2028 Apr 11 22:55:59	22° $\mathbb{Y}$ 34'10	-0°46'12
evening set	2027 Jun 14 04:00:48	5° $\mathbb{O}$ 56'11		minimum elong	2028 Apr 12 01:16:16	22° $\mathbb{Y}$ 46'21	0°45'43
min. Earth dist.	2027 Jun 21 18:48:39	2° $\mathbb{O}$ 33'25	0.55479 AU		2028 Apr 15 11:48:07	0° $\mathbb{O}$	
inferior conj	2027 Jun 23 07:59:39	1° $\mathbb{O}$ 39'43	-3°59'19	asc. node	2028 Apr 16 12:03:40	2° $\mathbb{O}$ 08'06	
minimum elong	2027 Jun 23 00:22:59	1° $\mathbb{O}$ 50'44	3°57'34	evening rise	2028 Apr 19 09:59:44	8° $\mathbb{O}$ 15'20	
	2027 Jun 26 07:19:13	30° $\mathbb{R}$ $\mathbb{H}$			2028 May 01 01:42:11	0° $\mathbb{H}$	
min. declination.	2027 Jul 01 01:15:08	28° $\mathbb{H}$ 01'03	18N43'56"	evening max el	2028 May 09 06:13:26	10° $\mathbb{H}$ 23'49	21°23'15
morning rise	2027 Jul 01 23:11:29	27° $\mathbb{H}$ 47'17		max. declination.	2028 May 12 14:05:44	13° $\mathbb{H}$ 11'52	24N41'20"
direct	2027 Jul 04 19:39:23	27° $\mathbb{H}$ 28'09		retrograde	2028 May 21 08:43:00	16° $\mathbb{H}$ 18'53	
	2027 Jul 12 13:47:59	0° $\mathbb{O}$		evening set	2028 May 23 17:08:35	16° $\mathbb{H}$ 06'14	
morning max el	2027 Jul 15 19:24:33	2° $\mathbb{O}$ 33'40	20°42'06	desc. node	2028 May 24 18:40:03	15° $\mathbb{H}$ 52'36	
max. declination.	2027 Jul 26 10:46:49	16° $\mathbb{O}$ 46'45	22N 8'45"	inferior conj	2028 Jun 02 02:46:15	12° $\mathbb{H}$ 06'29	-2°22'33
asc. node	2027 Jul 27 14:14:58	18° $\mathbb{O}$ 46'36		minimum elong	2028 Jun 01 20:19:03	12° $\mathbb{H}$ 15'32	2°20'22
	2027 Aug 02 10:52:23	0° $\mathbb{O}$		min. Earth dist.	2028 Jun 02 06:58:02	12° $\mathbb{H}$ 00'35	0.54910 AU
morning set	2027 Aug 03 22:26:01	3° $\mathbb{O}$ 01'12		morning rise	2028 Jun 10 23:59:56	8° $\mathbb{H}$ 08'27	

direct	2028 Jun 14 06:05:59	7°♄45'25		max. declination.	2029 Apr 28 10:52:44	26°♄06'40	21N52'46"
min. declination.	2028 Jun 15 04:15:30	7°♄47'20	17N21'57"	retrograde	2029 May 01 23:05:30	26°♄40'20	
morning max el	2028 Jun 26 19:34:14	13°♄42'38	22°13'58	evening set	2029 May 03 23:42:22	26°♄29'44	
	2028 Jul 09 01:37:31	0°♄		desc. node	2029 May 11 15:42:43	23°♄17'36	
asc. node	2028 Jul 13 11:18:41	8°♄01'01		inferior conj	2029 May 12 23:27:21	22°♄30'34	-0°23'09
max. declination.	2028 Jul 15 11:06:32	11°♄56'14	23N16'25"	minimum elong	2029 May 12 22:23:09	22°♄32'10	0°22'45
morning set	2028 Jul 18 08:24:25	17°♄51'35		min. Earth dist.	2029 May 14 20:09:39	21°♄23'49	0.55305 AU
	2028 Jul 24 01:50:31	0°♄		morning rise	2029 May 21 19:15:45	18°♄08'38	
				direct	2029 May 25 19:20:48	17°♄33'39	
superior conj	2028 Jul 25 13:07:31	3°♄06'50	1°37'51	min. declination.	2029 May 29 02:38:52	17°♄58'14	13N37'49"
minimum elong	2028 Jul 25 11:18:43	2°♄57'16	1°37'43	morning max el	2029 Jun 08 11:50:26	24°♄18'15	23°55'38
max. Earth dist.	2028 Jul 28 20:57:51	10°♄02'55	1.34422 AU		2029 Jun 13 16:46:23	0°♄	
evening rise	2028 Aug 02 12:54:54	19°♄22'58		asc. node	2029 Jun 30 08:21:51	27°♄38'00	
	2028 Aug 08 05:09:45	0°♄			2029 Jul 01 12:01:25	0°♄	
desc. node	2028 Aug 20 17:58:45	20°♄31'24		morning set	2029 Jul 02 20:02:49	2°♄47'21	
	2028 Aug 27 13:07:53	0°♄		max. declination.	2029 Jul 05 17:27:08	8°♄58'01	24N 2'35"
evening max el	2028 Sep 06 10:20:27	11°♄18'04	27°03'14				
retrograde	2028 Sep 19 16:33:49	18°♄35'39		superior conj	2029 Jul 09 20:31:41	17°♄54'26	1°25'26
min. declination.	2028 Sep 20 06:37:49	18°♄34'31	10S53'39"	minimum elong	2029 Jul 09 18:11:09	17°♄41'47	1°25'08
evening set	2028 Sep 26 12:54:05	15°♄52'11		max. Earth dist.	2029 Jul 11 22:10:13	22°♄21'05	1.33290 AU
min. Earth dist.	2028 Sep 30 10:21:54	12°♄03'07	0.65300 AU		2029 Jul 15 13:37:00	0°♄	
inferior conj	2028 Oct 02 10:37:34	9°♄45'55	-2°18'37	evening rise	2029 Jul 17 06:47:22	3°♄29'40	
minimum elong	2028 Oct 02 14:19:44	9°♄35'23	2°17'12		2029 Jul 31 23:35:08	0°♄	
morning rise	2028 Oct 08 16:32:24	4°♄09'41		desc. node	2029 Aug 07 15:00:58	9°♄55'38	
asc. node	2028 Oct 09 10:30:59	3°♄50'32		evening max el	2029 Aug 19 21:42:01	24°♄32'16	27°25'15
direct	2028 Oct 11 10:27:31	3°♄28'50			2029 Aug 27 02:20:53	0°♄	
max. declination.	2028 Oct 14 02:46:35	4°♄07'00	0S31'27"	retrograde	2029 Sep 02 12:18:03	1°♄49'51	
morning max el	2028 Oct 17 23:03:31	7°♄00'12	18°11'11	min. declination.	2029 Sep 04 03:00:45	1°♄42'01	4S31'24"
	2028 Nov 02 15:04:14	0°♄			2029 Sep 08 10:59:29	30°♄	
morning set	2028 Nov 05 11:14:15	4°♄40'27		evening set	2029 Sep 09 16:12:34	29°♄10'16	
desc. node	2028 Nov 16 17:16:34	22°♄56'10		min. Earth dist.	2029 Sep 13 08:16:36	25°♄55'13	0.63803 AU
				inferior conj	2029 Sep 15 21:11:01	23°♄18'33	-3°15'27
superior conj	2028 Nov 20 01:40:06	28°♄15'52	-0°22'22	minimum elong	2029 Sep 16 02:09:33	23°♄05'44	3°13'51
minimum elong	2028 Nov 19 22:48:09	28°♄04'31	0°21'58	morning rise	2029 Sep 22 13:14:09	17°♄58'31	
	2028 Nov 21 03:59:47	0°♄		direct	2029 Sep 25 02:01:48	17°♄26'50	
max. Earth dist.	2028 Nov 22 23:47:32	2°♄52'40	1.45032 AU	asc. node	2029 Sep 26 07:34:55	17°♄34'27	
evening rise	2028 Dec 06 11:16:36	23°♄55'24		max. declination.	2029 Sep 28 13:33:00	18°♄27'02	5N 6'55"
	2028 Dec 10 09:11:55	0°♄		morning max el	2029 Oct 01 15:11:51	20°♄52'28	17°53'37
min. declination.	2028 Dec 12 06:48:26	2°♄56'39	25S34'15"		2029 Oct 08 12:40:35	0°♄	
greatest brilliancy	2028 Dec 18 21:04:51	13°♄03'15	-0.7m	morning set	2029 Oct 18 10:01:52	16°♄38'25	
evening max el	2028 Dec 31 00:43:17	29°♄24'41	19°34'59		2029 Oct 26 06:03:48	0°♄	
	2028 Dec 31 14:49:08	0°♄					
asc. node	2029 Jan 05 09:48:31	3°♄17'45		superior conj	2029 Oct 30 23:13:46	7°♄49'03	0°24'27
retrograde	2029 Jan 07 07:56:33	3°♄38'02		minimum elong	2029 Oct 31 01:49:55	7°♄59'43	0°24'05
evening set	2029 Jan 10 20:18:24	2°♄25'43		desc. node	2029 Nov 03 14:18:00	13°♄42'37	
	2029 Jan 13 12:13:31	30°♄		max. Earth dist.	2029 Nov 05 17:51:41	17°♄09'14	1.44262 AU
max. declination.	2029 Jan 14 09:34:23	28°♄57'59	17S44'13"		2029 Nov 13 22:09:25	0°♄	
inferior conj	2029 Jan 16 08:16:42	26°♄29'20	3°06'10	evening rise	2029 Nov 15 20:58:07	3°♄00'42	
minimum elong	2029 Jan 16 05:46:02	26°♄37'34	3°05'36		2029 Dec 03 22:47:08	0°♄	
min. Earth dist.	2029 Jan 17 11:45:35	24°♄59'17	0.66546 AU	min. declination.	2029 Dec 04 05:24:22	0°♄23'24	25S48'53"
morning rise	2029 Jan 21 14:59:50	20°♄16'40		evening max el	2029 Dec 14 02:58:04	12°♄54'55	20°35'54
direct	2029 Jan 27 18:40:12	17°♄32'29		retrograde	2029 Dec 22 05:50:35	17°♄42'28	
min. declination.	2029 Feb 08 09:26:01	24°♄10'32	20S34'51"	asc. node	2029 Dec 23 06:51:23	17°♄36'24	
morning max el	2029 Feb 09 01:46:30	24°♄51'07	25°44'46	evening set	2029 Dec 26 03:08:45	16°♄15'27	
desc. node	2029 Feb 12 16:30:29	28°♄45'57		inferior conj	2029 Dec 31 12:28:07	10°♄06'45	2°32'00
	2029 Feb 13 17:52:18	0°♄		minimum elong	2029 Dec 31 09:51:59	10°♄15'37	2°31'13
	2029 Mar 06 18:15:20	0°♄		min. Earth dist.	2030 Jan 01 02:49:34	9°♄17'48	0.67339 AU
morning set	2029 Mar 17 07:48:12	18°♄16'21		morning rise	2030 Jan 05 16:25:15	3°♄53'37	
max. Earth dist.	2029 Mar 21 03:05:42	25°♄23'30	1.35845 AU	max. declination.	2030 Jan 06 00:04:20	3°♄36'54	20S 7'41"
	2029 Mar 23 12:18:35	0°♄		direct	2030 Jan 11 05:45:29	1°♄26'14	
				morning max el	2030 Jan 22 10:10:39	8°♄06'24	24°21'05
superior conj	2029 Mar 26 16:33:59	6°♄18'13	-1°13'30	min. declination.	2030 Jan 29 03:20:16	15°♄46'26	22S17'54"
minimum elong	2029 Mar 26 20:11:51	6°♄36'29	1°12'54	desc. node	2030 Jan 30 13:32:16	17°♄34'53	
asc. node	2029 Apr 03 09:06:57	22°♄00'33			2030 Feb 08 14:03:28	0°♄	
evening rise	2029 Apr 03 16:13:29	22°♄36'47		morning set	2030 Feb 27 07:42:44	29°♄54'25	
	2029 Apr 07 08:51:31	0°♄			2030 Feb 27 08:59:37	0°♄	
evening max el	2029 Apr 21 13:35:22	21°♄38'41	20°05'31	max. Earth dist.	2030 Mar 03 00:15:56	6°♄27'20	1.37809 AU

superior conj	2030 Mar 09 22:54:25	19° ✕ 24'52	-1°37'58	direct	2030 Dec 25 21:14:23	15° ✕ 33'26	
minimum elong	2030 Mar 10 03:12:19	19° ✕ 45'33	1°37'28	morning max el	2031 Jan 04 19:56:06	21° ✕ 27'06	22°52'33
	2030 Mar 15 08:41:45	0° Υ			2031 Jan 12 04:18:35	0° ☿	
evening rise	2030 Mar 18 16:17:08	6° Υ 34'43		desc. node	2031 Jan 17 10:33:29	7° ☿ 02'16	
asc. node	2030 Mar 21 06:10:29	11° Υ 38'14		min. declination.	2031 Jan 20 12:31:22	11° ☿ 22'55	23S20'19"
	2030 Apr 01 04:46:55	0° ♄			2031 Feb 01 21:14:52	0° ≈	
evening max el	2030 Apr 04 08:19:58	3° ♄ 33'11	19°06'42	morning set	2031 Feb 08 05:18:37	10° ≈ 13'56	
max. declination.	2030 Apr 12 10:16:12	7° ♄ 46'29	17N 2'31"	max. Earth dist.	2031 Feb 12 19:46:08	17° ≈ 59'17	1.39954 AU
retrograde	2030 Apr 13 02:33:25	7° ♄ 47'57			2031 Feb 19 15:01:58	0° ✕	
evening set	2030 Apr 15 05:58:43	7° ♄ 33'47					
inferior conj	2030 Apr 23 11:14:25	3° ♄ 21'15	1°24'50	superior conj	2031 Feb 20 13:36:15	1° ✕ 42'47	-1°56'31
minimum elong	2030 Apr 23 14:23:20	3° ♄ 15'50	1°23'48	minimum elong	2031 Feb 20 17:09:13	1° ✕ 59'01	1°56'18
min. Earth dist.	2030 Apr 26 11:23:36	1° ♄ 18'17	0.56573 AU	evening rise	2031 Mar 02 07:10:25	20° ✕ 00'55	
desc. node	2030 Apr 28 12:43:49	0° ♄ 01'28			2031 Mar 07 15:01:28	0° Υ	
	2030 Apr 28 13:44:26	30° ♄ 01'28		asc. node	2031 Mar 08 03:12:33	0° Υ 55'03	
morning rise	2030 May 01 19:47:05	28° Υ 25'03		evening max el	2031 Mar 18 12:06:35	16° Υ 02'08	18°28'01
direct	2030 May 06 20:14:50	27° Υ 26'11		retrograde	2031 Mar 26 00:43:38	19° Υ 44'33	
min. declination.	2030 May 11 08:08:11	28° Υ 13'23	8N11'45"	max. declination.	2031 Mar 26 22:57:18	19° Υ 41'18	10N48'24"
	2030 May 15 01:30:09	0° ♄		evening set	2031 Mar 28 09:59:24	19° Υ 23'20	
morning max el	2030 May 21 02:20:12	4° ♄ 45'35	25°33'37	inferior conj	2031 Apr 04 20:02:00	14° Υ 51'31	2°41'41
	2030 Jun 08 06:31:20	0° ♄		minimum elong	2031 Apr 04 23:56:58	14° Υ 43'38	2°40'44
morning set	2030 Jun 17 07:45:10	17° ♄ 43'27		min. Earth dist.	2031 Apr 08 06:29:37	12° Υ 07'13	0.58451 AU
asc. node	2030 Jun 17 05:25:09	17° ♄ 31'08		morning rise	2031 Apr 12 11:00:40	9° Υ 24'06	
	2030 Jun 22 23:56:23	0° ☿		desc. node	2031 Apr 15 09:45:56	8° Υ 14'56	
				direct	2031 Apr 18 11:15:42	7° Υ 50'29	
superior conj	2030 Jun 24 07:04:36	2° ☿ 50'41	1°08'28	min. declination.	2031 Apr 23 13:10:11	8° Υ 54'26	1N51'46"
minimum elong	2030 Jun 24 04:43:18	2° ☿ 37'47	1°08'04	morning max el	2031 May 02 20:31:56	15° Υ 31'40	26°52'08
max. Earth dist.	2030 Jun 25 06:28:17	4° ☿ 58'40	1.32554 AU		2031 May 14 13:59:54	0° ♄	
max. declination.	2030 Jun 26 02:08:41	6° ☿ 45'56	24N37'14"		2031 May 31 12:23:00	0° ♄	
evening rise	2030 Jul 01 08:49:28	18° ☿ 00'13		morning set	2031 Jun 01 17:47:51	2° ♄ 32'56	
	2030 Jul 07 10:22:42	0° ♄		asc. node	2031 Jun 04 02:27:32	7° ♄ 34'06	
desc. node	2030 Jul 25 12:01:51	28° ♄ 35'54					
	2030 Jul 26 14:53:44	0° ♄		superior conj	2031 Jun 08 18:49:49	17° ♄ 47'59	0°47'43
evening max el	2030 Aug 02 05:34:49	7° ♄ 14'31	27°16'13	minimum elong	2031 Jun 08 16:55:20	17° ♄ 37'28	0°47'19
retrograde	2030 Aug 16 01:20:11	14° ♄ 29'44		max. Earth dist.	2031 Jun 08 18:17:25	17° ♄ 45'00	1.32205 AU
min. declination.	2030 Aug 18 22:58:49	14° ♄ 06'19	2N13'23"		2031 Jun 14 09:09:00	0° ☿	
evening set	2030 Aug 23 05:22:41	12° ♄ 06'17		evening rise	2031 Jun 15 16:13:23	2° ☿ 44'43	
min. Earth dist.	2030 Aug 26 19:39:39	9° ♄ 17'49	0.61989 AU	max. declination.	2031 Jun 16 15:15:13	4° ☿ 45'06	25N 4'56"
inferior conj	2030 Aug 29 20:10:42	6° ♄ 32'16	-4°06'17		2031 Jun 30 08:25:17	0° ♄	
minimum elong	2030 Aug 30 01:29:55	6° ♄ 20'05	4°05'02	desc. node	2031 Jul 12 09:02:38	16° ♄ 12'53	
morning rise	2030 Sep 05 23:09:47	1° ♄ 31'59		evening max el	2031 Jul 15 07:24:45	19° ♄ 12'51	26°34'04
direct	2030 Sep 08 09:27:36	1° ♄ 06'28		retrograde	2031 Jul 29 06:47:33	26° ♄ 25'35	
max. declination.	2030 Sep 13 01:44:31	2° ♄ 44'14	10N28' 1"	min. declination.	2031 Aug 03 02:50:34	25° ♄ 25'50	8N50'56"
asc. node	2030 Sep 13 04:37:23	2° ♄ 49'16		evening set	2031 Aug 04 23:45:43	24° ♄ 31'39	
morning max el	2030 Sep 15 06:25:56	4° ♄ 34'38	17°54'14	min. Earth dist.	2031 Aug 08 20:50:59	21° ♄ 55'24	0.59954 AU
morning set	2030 Oct 01 05:18:36	29° ♄ 35'01		inferior conj	2031 Aug 12 04:04:41	19° ♄ 17'20	-4°44'16
	2030 Oct 01 10:49:48	0° ♄		minimum elong	2031 Aug 12 07:53:49	19° ♄ 09'40	4°43'47
				morning rise	2031 Aug 19 18:04:31	14° ♄ 39'20	
superior conj	2030 Oct 11 23:47:34	18° ♄ 42'18	1°01'50	direct	2031 Aug 22 04:28:20	14° ♄ 17'39	
minimum elong	2030 Oct 12 04:20:47	19° ♄ 01'53	1°01'15	max. declination.	2031 Aug 28 20:32:29	17° ♄ 10'01	15N 8'32"
	2030 Oct 18 17:03:16	0° ♄		morning max el	2031 Aug 29 17:35:18	17° ♄ 57'51	18°14'14
max. Earth dist.	2030 Oct 19 07:46:52	1° ♄ 00'22	1.42869 AU	asc. node	2031 Aug 31 01:39:30	19° ♄ 21'04	
desc. node	2030 Oct 21 11:19:23	4° ♄ 29'45			2031 Sep 07 09:56:27	0° ♄	
evening rise	2030 Oct 26 08:26:33	12° ♄ 15'27		morning set	2031 Sep 14 15:20:47	13° ♄ 13'46	
	2030 Nov 07 00:30:32	0° ♄			2031 Sep 23 14:34:55	0° ♄	
min. declination.	2030 Nov 25 18:07:59	25° ♄ 08'12	25S49' 3"				
evening max el	2030 Nov 26 23:28:36	26° ♄ 25'30	21°48'06	superior conj	2031 Sep 24 01:27:15	0° ♄ 49'25	1°27'09
	2030 Dec 01 02:27:09	0° ☿		minimum elong	2031 Sep 24 05:13:57	1° ♄ 06'33	1°26'47
retrograde	2030 Dec 06 02:47:24	1° ☿ 51'52		max. Earth dist.	2031 Oct 01 15:54:01	14° ♄ 11'02	1.41029 AU
evening set	2030 Dec 10 10:51:06	0° ☿ 08'51		evening rise	2031 Oct 06 13:02:09	22° ♄ 18'43	
asc. node	2030 Dec 10 03:53:20	0° ☿ 22'21		desc. node	2031 Oct 08 08:20:18	25° ♄ 14'03	
	2030 Dec 10 15:10:38	30° ♄ 01'28			2031 Oct 11 08:18:35	0° ♄	
inferior conj	2030 Dec 15 19:14:21	23° ♄ 51'10	1°50'14		2031 Nov 01 02:35:15	0° ♄	
minimum elong	2030 Dec 15 17:01:37	23° ♄ 58'50	1°49'26	evening max el	2031 Nov 09 15:06:38	9° ♄ 58'13	23°07'02
min. Earth dist.	2030 Dec 15 21:54:47	23° ♄ 41'54	0.67752 AU	min. declination.	2031 Nov 14 20:01:50	14° ♄ 19'36	25S 3'55"
morning rise	2030 Dec 20 23:03:44	17° ♄ 39'11		retrograde	2031 Nov 19 21:15:34	16° ♄ 03'12	
max. declination.	2030 Dec 24 15:39:24	15° ♄ 40'54	19S40'43"	evening set	2031 Nov 24 17:47:17	14° ♄ 03'23	

asc. node	2031 Nov 27 00:54:17	11°♊42'57		superior conj	2032 Sep 05 22:47:37	13°♎54'53	1°41'10
inferior conj	2031 Nov 30 02:43:02	7°♊40'35	1°02'22	minimum elong	2032 Sep 06 00:45:07	14°♎04'14	1°41'04
minimum elong	2031 Nov 30 01:18:57	7°♊45'25	1°01'48	max. Earth dist.	2032 Sep 12 19:57:17	26°♎38'53	1.38988 AU
min. Earth dist.	2031 Nov 29 18:38:33	8°♊08'24	0.67816 AU		2032 Sep 14 17:23:58	0°♏	
morning rise	2031 Dec 05 08:45:29	1°♊31'47		evening rise	2032 Sep 16 17:34:27	3°♏28'55	
	2031 Dec 08 05:36:22	30°♑♍		desc. node	2032 Sep 24 05:21:16	15°♏51'43	
direct	2031 Dec 09 16:23:25	29°♑48'48			2032 Oct 03 12:29:22	0°♑	
max. declination.	2031 Dec 09 23:41:58	29°♑49'16	17S25' 0"	evening max el	2032 Oct 22 03:56:22	23°♑34'59	24°26'34
	2031 Dec 11 04:21:35	0°♊		min. declination.	2032 Oct 31 00:55:53	29°♑48'38	22S55' 0"
morning max el	2031 Dec 18 10:25:20	4°♊55'44	21°27'59		2032 Oct 31 17:33:20	0°♊	
desc. node	2032 Jan 04 07:34:33	26°♊57'25		retrograde	2032 Nov 02 11:56:45	0°♊12'35	
	2032 Jan 06 09:13:43	0°♋			2032 Nov 04 04:38:04	30°♑♍	
min. declination.	2032 Jan 12 06:08:25	8°♋50'11	24S 2'42"	evening set	2032 Nov 07 22:19:19	27°♑56'12	
morning set	2032 Jan 18 21:53:10	19°♋11'20		min. Earth dist.	2032 Nov 12 14:34:40	22°♑34'42	0.67553 AU
	2032 Jan 25 14:40:54	0°♌		asc. node	2032 Nov 12 21:56:16	22°♑10'05	
max. Earth dist.	2032 Jan 25 20:03:34	0°♌22'09	1.42003 AU	inferior conj	2032 Nov 13 09:08:30	21°♑32'31	0°09'38
				minimum elong	2032 Nov 13 08:54:12	21°♑33'18	0°09'32
superior conj	2032 Feb 02 07:31:42	13°♌00'03	-2°04'58	transit middle	2032 Nov 13 08:54:12	21°♑33'18	0°09'32
minimum elong	2032 Feb 02 07:58:40	13°♌02'00	2°04'57	transit begin	2032 Nov 13 06:41:07	21°♑40'45	
	2032 Feb 11 20:34:17	0°♍		transit end	2032 Nov 13 11:07:18	21°♑25'52	
evening rise	2032 Feb 13 09:13:45	2°♍47'00		morning rise	2032 Nov 18 19:31:19	15°♑29'12	
asc. node	2032 Feb 23 00:14:35	19°♍44'15		direct	2032 Nov 22 14:00:49	14°♑07'43	
evening max el	2032 Feb 29 21:48:20	28°♍57'27	18°09'22	max. declination.	2032 Nov 23 17:23:13	14°♑14'41	13S46'23"
	2032 Mar 02 01:19:19	0°♎		morning max el	2032 Nov 30 08:11:11	18°♑34'34	20°13'42
retrograde	2032 Mar 07 16:21:43	2°♎23'32			2032 Dec 09 12:04:50	0°♊	
max. declination.	2032 Mar 09 14:54:28	2°♎06'45	3N54'44"	desc. node	2032 Dec 21 04:35:12	17°♊12'13	
evening set	2032 Mar 10 07:22:33	1°♎53'03		morning set	2032 Dec 27 19:05:14	27°♊19'10	
	2032 Mar 13 13:46:50	30°♑♍			2032 Dec 29 12:34:29	0°♋	
inferior conj	2032 Mar 17 01:10:53	27°♍00'31	3°25'31	min. declination.	2033 Jan 03 00:32:17	7°♋02'41	24S34'45"
minimum elong	2032 Mar 17 03:38:51	26°♍54'45	3°25'11	max. Earth dist.	2033 Jan 07 04:07:14	13°♋38'33	1.43684 AU
min. Earth dist.	2032 Mar 20 07:46:50	23°♍58'45	0.60591 AU				
morning rise	2032 Mar 23 21:54:39	21°♍12'02		superior conj	2033 Jan 12 23:56:33	23°♋06'08	-1°58'13
direct	2032 Mar 30 14:29:05	19°♍02'06		minimum elong	2033 Jan 12 19:04:31	22°♋46'07	1°57'58
desc. node	2032 Apr 01 06:46:55	19°♍09'56			2033 Jan 17 03:32:08	0°♌	
min. declination.	2032 Apr 05 02:19:58	20°♍21'50	4S34'44"	evening rise	2033 Jan 25 18:07:25	14°♌43'34	
morning max el	2032 Apr 13 20:46:26	26°♍54'34	27°38'53		2033 Feb 03 16:04:01	0°♍	
	2032 Apr 16 19:07:38	0°♎		asc. node	2033 Feb 08 21:16:49	7°♍56'29	
	2032 May 07 03:35:36	0°♏		evening max el	2033 Feb 12 10:10:24	12°♍09'30	18°10'07
morning set	2032 May 16 00:24:53	17°♏09'57		retrograde	2033 Feb 18 21:20:07	15°♍33'55	
asc. node	2032 May 20 23:30:26	27°♏42'29		evening set	2033 Feb 21 17:38:14	14°♍52'54	
max. Earth dist.	2032 May 22 05:55:16	0°♐28'13	1.32246 AU	max. declination.	2033 Feb 21 15:15:42	14°♍55'40	3S 2'31"
	2032 May 22 00:45:19	0°♐		inferior conj	2033 Feb 27 22:59:43	9°♍40'00	3°42'31
				minimum elong	2033 Feb 27 23:27:56	9°♍38'46	3°42'30
superior conj	2032 May 23 06:06:58	2°♐40'43	0°23'51	min. Earth dist.	2033 Mar 02 17:19:14	6°♍45'23	0.62672 AU
minimum elong	2032 May 23 05:03:15	2°♐34'54	0°23'37	morning rise	2033 Mar 06 04:11:39	3°♍39'13	
evening rise	2032 May 30 02:49:10	17°♐36'09		direct	2033 Mar 13 02:57:21	1°♍02'02	
	2032 Jun 05 05:43:25	0°♑		min. declination.	2033 Mar 19 04:21:35	2°♍44'47	10S29'49"
max. declination.	2032 Jun 06 12:14:24	2°♑24'12	25N26'48"	desc. node	2033 Mar 19 03:48:17	2°♍44'02	
	2032 Jun 25 15:56:59	0°♒		morning max el	2033 Mar 27 02:28:11	8°♍57'40	27°48'39
evening max el	2032 Jun 26 02:14:04	0°♒24'53	25°22'06		2033 Apr 12 09:17:48	0°♎	
desc. node	2032 Jun 28 06:03:45	2°♒20'55			2033 Apr 29 08:08:47	0°♏	
retrograde	2032 Jul 10 02:33:31	7°♒32'42		morning set	2033 Apr 30 01:29:00	1°♏27'09	
evening set	2032 Jul 15 19:07:17	6°♒17'22		max. Earth dist.	2033 May 05 13:36:10	12°♏55'42	1.32685 AU
min. declination.	2032 Jul 18 13:34:09	4°♒51'52	14N35'50"				
min. Earth dist.	2032 Jul 20 14:13:51	3°♒34'54	0.57919 AU	superior conj	2033 May 07 15:11:38	17°♏22'45	-0°02'23
inferior conj	2032 Jul 23 17:30:50	1°♒24'33	-4°58'27	minimum elong	2033 May 07 15:18:13	17°♏23'21	0°02'21
minimum elong	2032 Jul 23 17:16:36	1°♒24'58	4°58'28	behind sun begin	2033 May 07 10:08:49	16°♏55'26	
	2032 Jul 25 19:20:00	30°♑♏		behind sun end	2033 May 07 20:27:37	17°♏51'17	
morning rise	2032 Jul 31 17:55:44	27°♏08'51		asc. node	2033 May 07 20:34:12	17°♏51'53	
direct	2032 Aug 03 06:52:06	26°♏49'14			2033 May 13 10:57:22	0°♐	
	2032 Aug 10 21:46:39	0°♒		evening rise	2033 May 14 14:39:35	2°♐27'38	
morning max el	2032 Aug 11 21:26:25	0°♒52'39	18°54'53	max. declination.	2033 May 28 18:58:55	28°♐42'06	25N39'14"
max. declination.	2032 Aug 13 13:24:39	2°♒35'28	18N46'19"		2033 May 29 15:50:18	0°♑	
asc. node	2032 Aug 16 22:41:57	6°♒54'44		evening max el	2033 Jun 07 16:24:40	11°♑02'43	23°50'42
morning set	2032 Aug 28 11:47:23	27°♒23'00		desc. node	2033 Jun 15 03:04:37	16°♑26'37	
	2032 Aug 29 19:41:05	0°♓		retrograde	2033 Jun 21 10:05:17	17°♑56'22	
				evening set	2033 Jun 25 16:07:52	17°♑16'34	

min. Earth dist.	2003 Jul 02 03:00:36	14°☿12'17	0.56198 AU	evening rise	2034 Apr 29 01:52:54	17°♄12'33	
inferior conj	2003 Jul 04 10:47:52	12°☿47'26	-4°34'43		2034 May 05 11:02:51	0°♄	
minimum elong	2003 Jul 04 05:11:24	12°☿56'01	4°33'53	max. declination.	2034 May 19 17:09:52	20°♄55'40	25N23'22"
min. declination.	2003 Jul 07 07:32:45	11°☿06'12	18N10' 9"	evening max el	2034 May 20 08:12:43	21°♄33'00	22°14'50
morning rise	2003 Jul 12 20:52:27	8°☿49'35		retrograde	2034 Jun 02 05:23:35	27°♄55'09	
direct	2003 Jul 15 14:19:32	8°☿30'35		desc. node	2034 Jun 02 00:06:21	27°♄55'02	
morning max el	2003 Jul 25 15:11:48	13°☿09'51	19°56'59	evening set	2034 Jun 05 02:38:23	27°♄36'44	
max. declination.	2003 Aug 01 06:10:53	21°☿13'18	21N11'38"	min. Earth dist.	2034 Jun 13 14:24:12	23°♄57'55	0.55125 AU
asc. node	2003 Aug 03 19:45:35	25°☿16'32		inferior conj	2034 Jun 14 10:58:51	23°♄28'57	-3°23'04
	2003 Aug 06 12:31:27	0°♄		minimum elong	2034 Jun 14 03:04:55	23°♄40'05	3°20'51
morning set	2003 Aug 12 15:25:28	11°♄54'10		morning rise	2034 Jun 23 05:22:46	19°♄36'56	
				min. declination.	2034 Jun 24 20:06:59	19°♄21'06	18N28'51"
superior conj	2003 Aug 20 10:33:27	27°♄43'59	1°45'43	direct	2034 Jun 26 05:23:35	19°♄16'41	
minimum elong	2003 Aug 20 10:41:34	27°♄44'40	1°45'43	morning max el	2034 Jul 07 21:29:31	24°♄43'10	21°19'21
	2003 Aug 21 13:54:41	0°♄			2034 Jul 12 16:13:18	0°☿	
max. Earth dist.	2003 Aug 26 00:56:00	8°♄38'10	1.36997 AU	asc. node	2034 Jul 21 16:49:41	14°☿14'30	
evening rise	2003 Aug 29 20:57:01	15°♄43'49		max. declination.	2034 Jul 21 19:08:57	14°☿25'14	22N40'17"
	2003 Sep 07 03:48:48	0°♄		morning set	2034 Jul 27 23:40:52	26°☿39'27	
desc. node	2003 Sep 11 02:24:18	6°♄18'20			2034 Jul 29 14:12:49	0°♄	
	2003 Sep 28 01:43:53	0°♄					
evening max el	2003 Oct 04 16:04:15	7°♄15'30	25°39'31	superior conj	2034 Aug 04 08:23:57	12°♄03'26	1°42'35
min. declination.	2003 Oct 15 21:34:23	14°♄12'17	19S14'33"	minimum elong	2034 Aug 04 07:08:28	11°♄56'55	1°42'31
retrograde	2003 Oct 16 22:03:11	14°♄16'09		max. Earth dist.	2034 Aug 08 11:53:44	20°♄30'26	1.35258 AU
evening set	2003 Oct 22 22:46:30	11°♄45'03		evening rise	2034 Aug 12 18:44:19	28°♄51'28	
min. Earth dist.	2003 Oct 27 07:17:15	6°♄58'51	0.66960 AU		2034 Aug 13 09:14:26	0°♄	
inferior conj	2003 Oct 28 12:45:31	5°♄24'49	-0°46'36	desc. node	2034 Aug 28 23:26:59	26°♄27'18	
minimum elong	2003 Oct 28 13:57:56	5°♄20'58	0°46'05		2034 Aug 31 08:33:23	0°♄	
asc. node	2003 Oct 30 18:57:30	2°♄38'18		evening max el	2034 Sep 17 04:16:02	20°♄53'03	26°38'27
	2003 Nov 02 12:07:15	30°♄		retrograde	2034 Sep 30 02:59:57	28°♄06'59	
morning rise	2003 Nov 03 05:25:06	29°♄29'47		min. declination.	2034 Sep 30 03:47:15	28°♄06'59	14S15'35"
direct	2003 Nov 06 12:39:41	28°♄26'56		evening set	2034 Oct 06 17:06:42	25°♄25'41	
max. declination.	2003 Nov 08 06:31:00	28°♄43'34	9S 6'33"	min. Earth dist.	2034 Oct 10 18:21:55	21°♄16'03	0.66016 AU
	2003 Nov 10 18:33:52	0°♄		inferior conj	2034 Oct 12 11:35:32	19°♄13'12	-1°44'53
morning max el	2003 Nov 13 13:45:26	2°♄24'03	19°13'29	minimum elong	2034 Oct 12 14:23:44	19°♄04'50	1°43'46
	2003 Dec 03 02:51:03	0°♄		asc. node	2034 Oct 17 16:00:29	14°♄00'30	
morning set	2003 Dec 06 21:20:37	5°♄51'24		morning rise	2034 Oct 18 12:15:45	13°♄29'26	
desc. node	2003 Dec 08 01:39:11	7°♄41'32		direct	2034 Oct 21 10:22:21	12°♄41'40	
max. Earth dist.	2003 Dec 20 19:16:09	27°♄37'16	1.44780 AU	max. declination.	2034 Oct 23 17:56:21	13°♄10'43	3S46'56"
	2003 Dec 22 07:24:07	0°♄		morning max el	2034 Oct 28 01:32:07	16°♄19'32	18°29'08
					2034 Nov 07 04:05:25	0°♄	
superior conj	2003 Dec 23 15:08:40	2°♄05'46	-1°32'35	morning set	2034 Nov 17 00:29:42	15°♄41'37	
minimum elong	2003 Dec 23 06:22:43	1°♄31'00	1°31'45	desc. node	2034 Nov 24 22:43:06	28°♄20'30	
min. declination.	2003 Dec 25 18:58:14	5°♄31'56	25S 0'48"		2034 Nov 25 23:50:32	0°♄	
evening rise	2004 Jan 07 05:50:49	25°♄44'59					
	2004 Jan 09 19:35:11	0°♄		superior conj	2034 Dec 02 16:45:23	10°♄34'01	-0°50'21
asc. node	2004 Jan 26 18:18:19	25°♄21'03		minimum elong	2034 Dec 02 10:21:20	10°♄08'53	0°49'33
evening max el	2004 Jan 26 22:24:06	25°♄31'33	18°29'26	max. Earth dist.	2034 Dec 03 14:01:17	11°♄57'31	1.45155 AU
retrograde	2004 Feb 02 11:21:56	29°♄07'00			2034 Dec 15 01:35:15	0°♄	
evening set	2004 Feb 05 13:04:39	28°♄14'38		min. declination.	2034 Dec 17 14:23:54	3°♄59'55	25S22'52"
max. declination.	2004 Feb 06 03:23:00	27°♄52'46	9S33'20"	evening rise	2034 Dec 18 18:14:13	5°♄49'54	
inferior conj	2004 Feb 11 09:29:02	22°♄42'51	3°39'27	greatest brilliancy	2034 Dec 28 08:02:29	20°♄56'10	-0.8m
minimum elong	2004 Feb 11 08:18:29	22°♄46'18	3°39'21		2035 Jan 03 07:28:28	0°♄	
min. Earth dist.	2004 Feb 13 12:23:37	20°♄13'50	0.64477 AU	evening max el	2035 Jan 10 08:03:59	8°♄58'56	19°06'22
morning rise	2004 Feb 17 03:00:09	16°♄35'01		asc. node	2035 Jan 13 15:18:46	11°♄45'07	
direct	2004 Feb 23 22:54:17	13°♄44'39		retrograde	2035 Jan 17 06:44:20	12°♄56'13	
min. declination.	2004 Mar 03 01:13:38	16°♄12'37	15S25'35"	evening set	2035 Jan 20 14:45:25	11°♄51'34	
desc. node	2004 Mar 06 00:50:51	18°♄25'09		max. declination.	2035 Jan 22 12:04:50	10°♄20'13	15S 8'54"
morning max el	2004 Mar 09 11:36:02	21°♄35'28	27°22'40	inferior conj	2035 Jan 26 05:06:02	6°♄03'12	3°21'39
	2004 Mar 16 19:52:32	0°♄		minimum elong	2035 Jan 26 02:53:56	6°♄10'12	3°21'16
	2004 Apr 05 13:01:25	0°♄		min. Earth dist.	2035 Jan 27 16:43:27	4°♄10'16	0.65902 AU
morning set	2004 Apr 13 18:17:48	15°♄15'54			2035 Jan 31 10:58:18	30°♄	
max. Earth dist.	2004 Apr 18 13:28:05	24°♄55'41	1.33552 AU	morning rise	2035 Jan 31 14:45:16	29°♄51'30	
	2004 Apr 20 23:46:14	0°♄		direct	2035 Feb 07 01:24:43	27°♄01'49	
					2035 Feb 14 14:31:35	0°♄	
superior conj	2004 Apr 21 20:08:15	1°♄47'37	-0°30'03	min. declination.	2035 Feb 16 08:27:49	1°♄20'56	19S 4'28"
minimum elong	2004 Apr 21 21:38:36	1°♄55'36	0°29'43	morning max el	2035 Feb 19 21:31:39	4°♄36'45	26°26'57
asc. node	2004 Apr 24 17:37:26	7°♄57'57		desc. node	2035 Feb 20 21:54:48	5°♄39'30	

	2035 Mar 11 07:16:04	0°♄		asc. node	2035 Dec 31 12:20:30	26°♄53'24	
morning set	2035 Mar 27 23:16:41	28°♄25'44		retrograde	2036 Jan 01 04:24:31	26°♄55'53	
	2035 Mar 28 19:13:14	0°♄		evening set	2036 Jan 04 20:15:41	25°♄37'37	
max. Earth dist.	2035 Apr 01 02:18:45	6°♄22'41	1.34873 AU	inferior conj	2036 Jan 10 06:52:09	19°♄35'29	2°52'43
				minimum elong	2036 Jan 10 04:15:16	19°♄44'14	2°52'02
superior conj	2035 Apr 05 18:41:36	15°♄48'46	-0°57'57	min. Earth dist.	2036 Jan 11 04:35:44	18°♄22'48	0.66926 AU
minimum elong	2035 Apr 05 21:37:16	16°♄03'49	0°57'25	max. declination.	2036 Jan 10 22:00:19	18°♄44'48	19S 8'53"
asc. node	2035 Apr 11 14:40:32	27°♄56'44		morning rise	2036 Jan 15 12:01:41	13°♄22'08	
	2035 Apr 12 14:21:45	0°♄		direct	2036 Jan 21 09:41:56	10°♄44'13	
evening rise	2035 Apr 13 10:34:09	1°♄44'39		morning max el	2036 Feb 02 06:22:16	17°♄48'49	25°10'22
	2035 Apr 30 01:44:45	0°♄		min. declination.	2036 Feb 04 08:37:55	20°♄01'23	21S24'39"
evening max el	2035 May 02 08:44:29	2°♄26'10	20°48'05	desc. node	2036 Feb 07 18:56:51	23°♄59'35	
max. declination.	2035 May 07 15:12:06	6°♄23'20	23N46'23"		2036 Feb 12 12:02:55	0°♄	
retrograde	2035 May 13 18:40:23	7°♄59'29			2036 Mar 03 07:47:58	0°♄	
evening set	2035 May 15 21:45:55	7°♄48'38		morning set	2036 Mar 09 11:27:19	10°♄41'42	
desc. node	2035 May 19 21:07:23	6°♄36'22		max. Earth dist.	2036 Mar 13 03:35:18	17°♄24'52	1.36635 AU
inferior conj	2035 May 25 04:45:08	3°♄50'47	-1°32'39				
minimum elong	2035 May 25 00:22:44	3°♄57'01	1°31'04	superior conj	2036 Mar 19 07:54:30	29°♄17'25	-1°24'23
min. Earth dist.	2035 May 26 02:52:33	3°♄19'13	0.54952 AU	minimum elong	2036 Mar 19 11:56:22	29°♄37'19	1°23'48
	2035 Jun 02 04:33:05	30°♄			2036 Mar 19 16:31:25	0°♄	
morning rise	2035 Jun 03 02:34:15	29°♄44'46		evening rise	2036 Mar 27 14:27:08	15°♄56'21	
direct	2035 Jun 06 14:52:39	29°♄18'11		asc. node	2036 Mar 28 11:41:45	17°♄43'02	
min. declination.	2035 Jun 08 18:20:54	29°♄28'28	16N 2'56"		2036 Apr 03 22:08:55	0°♄	
	2035 Jun 10 22:32:24	0°♄		evening max el	2036 Apr 13 21:03:25	13°♄57'26	19°38'07
morning max el	2035 Jun 19 17:52:27	5°♄36'30	22°56'45	max. declination.	2036 Apr 21 10:41:20	18°♄24'54	20N 3'46"
	2035 Jul 06 16:36:34	0°♄		retrograde	2036 Apr 23 13:17:57	18°♄37'49	
asc. node	2035 Jul 08 13:55:02	3°♄39'36		evening set	2036 Apr 25 14:06:41	18°♄26'21	
morning set	2035 Jul 12 10:35:44	11°♄33'06		inferior conj	2036 May 04 06:33:43	14°♄23'01	0°25'39
max. declination.	2035 Jul 11 23:12:05	10°♄33'29	23N37'14"	minimum elong	2036 May 04 07:40:07	14°♄21'17	0°25'15
				desc. node	2036 May 05 18:09:08	13°♄26'43	
superior conj	2035 Jul 19 13:05:05	26°♄43'28	1°33'13	min. Earth dist.	2036 May 06 17:05:59	12°♄50'58	0.55742 AU
minimum elong	2035 Jul 19 10:58:58	26°♄32'16	1°33'00	morning rise	2036 May 12 22:35:53	9°♄46'29	
	2035 Jul 21 02:06:09	0°♄		direct	2036 May 17 08:29:15	9°♄02'52	
max. Earth dist.	2035 Jul 22 07:42:38	2°♄35'36	1.33889 AU	min. declination.	2036 May 21 06:32:53	9°♄37'55	11N30'24"
evening rise	2035 Jul 27 06:25:35	12°♄39'55		morning max el	2036 May 31 08:41:40	16°♄05'00	24°38'43
	2035 Aug 05 15:27:58	0°♄			2036 Jun 11 13:05:44	0°♄	
desc. node	2035 Aug 15 20:27:23	16°♄10'47		asc. node	2036 Jun 24 10:58:38	23°♄23'42	
	2035 Aug 26 12:50:54	0°♄		morning set	2036 Jun 25 22:25:41	26°♄29'04	
evening max el	2035 Aug 30 16:10:49	4°♄18'35	27°15'54		2036 Jun 27 14:02:14	0°♄	
retrograde	2035 Sep 13 02:28:16	11°♄37'18		max. declination.	2036 Jul 01 06:33:09	8°♄00'03	24N17'54"
min. declination.	2035 Sep 14 02:37:05	11°♄34'06	8S17'40"				
evening set	2035 Sep 20 02:49:25	8°♄53'44		superior conj	2036 Jul 02 22:03:45	11°♄35'19	1°18'45
min. Earth dist.	2035 Sep 23 21:35:38	5°♄19'57	0.64712 AU	minimum elong	2036 Jul 02 19:39:22	11°♄22'13	1°18'24
inferior conj	2035 Sep 26 03:22:34	2°♄53'07	-2°43'09	max. Earth dist.	2036 Jul 04 12:00:10	15°♄00'59	1.32922 AU
minimum elong	2035 Sep 26 07:41:17	2°♄41'20	2°41'36	evening rise	2036 Jul 10 04:06:22	26°♄57'32	
	2035 Sep 28 22:18:14	30°♄			2036 Jul 11 16:16:06	0°♄	
morning rise	2035 Oct 02 13:31:23	27°♄23'40			2036 Jul 28 23:16:12	0°♄	
asc. node	2035 Oct 04 13:04:43	26°♄49'31		desc. node	2036 Aug 01 17:28:01	5°♄18'06	
direct	2035 Oct 05 04:53:07	26°♄47'14		evening max el	2036 Aug 12 02:18:31	17°♄20'38	27°25'27
max. declination.	2035 Oct 08 04:38:43	27°♄33'35	1N50'52"	retrograde	2036 Aug 25 19:47:49	24°♄38'07	
	2035 Oct 11 10:48:56	0°♄		min. declination.	2036 Aug 27 22:20:44	24°♄25'11	1S42'52"
morning max el	2035 Oct 11 16:56:45	0°♄15'06	18°01'30	evening set	2036 Sep 02 00:41:31	22°♄03'43	
morning set	2035 Oct 29 08:34:05	26°♄56'49		min. Earth dist.	2036 Sep 05 15:16:18	19°♄01'10	0.63062 AU
	2035 Oct 31 04:18:20	0°♄		inferior conj	2036 Sep 08 09:25:56	16°♄18'52	-3°38'06
desc. node	2035 Nov 11 19:43:39	19°♄05'05		minimum elong	2036 Sep 08 14:42:59	16°♄05'52	3°36'35
				morning rise	2036 Sep 15 06:05:45	11°♄06'51	
superior conj	2035 Nov 12 01:57:09	19°♄30'03	-0°01'46	direct	2036 Sep 17 17:20:29	10°♄38'16	
minimum elong	2035 Nov 12 01:43:41	19°♄29'09	0°01'44	asc. node	2036 Sep 20 10:08:52	11°♄13'15	
behind sun begin	2035 Nov 11 15:14:18	18°♄47'05		max. declination.	2036 Sep 21 15:35:26	11°♄51'49	7N24'44"
behind sun end	2035 Nov 12 12:13:04	20°♄11'09		morning max el	2036 Sep 24 09:06:27	14°♄04'18	17°51'31
max. Earth dist.	2035 Nov 16 08:44:45	26°♄18'54	1.44787 AU		2036 Oct 05 08:20:41	0°♄	
	2035 Nov 18 16:53:00	0°♄		morning set	2036 Oct 10 16:54:12	9°♄22'14	
evening rise	2035 Nov 28 10:27:56	15°♄08'16					
	2035 Dec 08 03:26:49	0°♄		superior conj	2036 Oct 22 10:46:26	29°♄37'05	0°41'43
min. declination.	2035 Dec 09 12:03:24	2°♄03'05	25S41' 5"	minimum elong	2036 Oct 22 14:38:48	29°♄53'15	0°41'11
greatest brilliancy	2035 Dec 12 11:15:29	6°♄28'55	-0.6m		2036 Oct 22 16:15:44	0°♄	
evening max el	2035 Dec 24 13:18:09	22°♄28'14	19°59'17	desc. node	2036 Oct 28 16:45:13	9°♄52'31	

max. Earth dist.	2036 Oct 29 00:51:56	10° ℓ 25'11	1.43736 AU	direct	2037 Aug 31 20:07:12	24° Ω 06'40	
evening rise	2036 Nov 06 18:45:15	24° ℓ 12'23		max. declination.	2037 Sep 06 05:25:12	26° Ω 09'19	12N31'50"
	2036 Nov 10 13:20:04	0° ♊		asc. node	2037 Sep 07 07:12:13	27° Ω 02'21	
min. declination.	2036 Nov 30 09:43:49	28° ♊ 42'53	25S51'47"	morning max el	2037 Sep 07 22:55:33	27° Ω 39'01	18°00'16
	2036 Dec 01 09:40:25	0° ♋			2037 Sep 10 02:17:33	0° ♌	
evening max el	2036 Dec 06 13:07:59	5° ♋ 58'59	21°05'32	morning set	2037 Sep 23 19:09:10	22° ♌ 38'55	
retrograde	2036 Dec 15 02:06:10	11° ♋ 02'51			2037 Sep 27 18:59:04	0° ♌	
asc. node	2036 Dec 17 09:22:38	10° ♋ 33'20					
evening set	2036 Dec 19 03:39:58	9° ♋ 29'25		superior conj	2037 Oct 03 22:34:21	11° ♌ 03'13	1°14'01
inferior conj	2036 Dec 24 12:25:03	3° ♋ 16'37	2°15'07	minimum elong	2037 Oct 04 03:01:30	11° ♌ 22'47	1°13'31
minimum elong	2036 Dec 24 09:55:39	3° ♋ 25'11	2°14'18	max. Earth dist.	2037 Oct 11 12:33:22	24° ♌ 01'13	1.42132 AU
min. Earth dist.	2036 Dec 24 21:44:48	2° ♋ 44'30	0.67560 AU		2037 Oct 15 04:07:25	0° ♌	
	2036 Dec 26 23:08:58	30° ♌		desc. node	2037 Oct 15 13:46:11	0° ♌ 38'57	
morning rise	2036 Dec 29 15:59:39	27° ♌ 03'35		evening rise	2037 Oct 17 12:09:41	3° ♌ 44'45	
max. declination.	2036 Dec 31 23:25:06	25° ♌ 27'40	20S12' 5"		2037 Nov 03 22:40:18	0° ♌	
direct	2037 Jan 03 22:57:25	24° ♌ 44'46		evening max el	2037 Nov 19 07:33:07	19° ♌ 30'59	22°21'12
	2037 Jan 13 11:52:32	0° ♍		min. declination.	2037 Nov 21 04:50:28	21° ♌ 19'01	25S38'24"
morning max el	2037 Jan 14 14:49:14	1° ♍ 06'00	23°43'19	retrograde	2037 Nov 28 22:00:48	25° ♌ 13'29	
desc. node	2037 Jan 24 15:57:55	13° ♍ 06'39		evening set	2037 Dec 03 11:13:35	23° ♌ 23'27	
min. declination.	2037 Jan 25 00:40:28	13° ♍ 35'43	22S47'18"	asc. node	2037 Dec 04 06:25:01	22° ♌ 40'51	
	2037 Feb 05 11:25:42	0° ♍		inferior conj	2037 Dec 08 19:42:36	17° ♌ 03'22	1°30'36
morning set	2037 Feb 19 00:48:58	21° ♍ 47'52		minimum elong	2037 Dec 08 17:47:35	17° ♌ 10'01	1°29'52
max. Earth dist.	2037 Feb 22 22:31:30	28° ♍ 34'18	1.38716 AU	min. Earth dist.	2037 Dec 08 17:46:42	17° ♌ 10'04	0.67831 AU
	2037 Feb 23 17:52:10	0° ♎		morning rise	2037 Dec 14 00:13:02	10° ♌ 52'36	
				max. declination.	2037 Dec 18 05:35:06	8° ♌ 57'20	18S55'20"
superior conj	2037 Mar 02 08:09:13	12° ♎ 04'45	-1°46'47	direct	2037 Dec 18 16:12:17	8° ♌ 56'22	
minimum elong	2037 Mar 02 12:21:44	12° ♎ 24'35	1°46'24	morning max el	2037 Dec 28 02:00:55	14° ♌ 29'02	22°15'33
evening rise	2037 Mar 11 10:57:55	29° ♎ 41'18			2038 Jan 09 13:31:38	0° ♍	
	2037 Mar 11 14:48:15	0° ♏		desc. node	2038 Jan 11 12:59:04	2° ♍ 46'54	
asc. node	2037 Mar 15 08:42:38	7° ♏ 12'15		min. declination.	2038 Jan 16 15:19:56	10° ♍ 10'46	23S39'42"
evening max el	2037 Mar 27 20:04:58	26° ♏ 07'38	18°47'48		2038 Jan 29 11:21:51	0° ♍	
	2037 Apr 03 16:23:40	0° ♏		morning set	2038 Jan 30 09:48:31	1° ♍ 30'40	
retrograde	2037 Apr 05 00:05:32	0° ♏ 06'06		max. Earth dist.	2038 Feb 04 19:35:47	10° ♍ 26'52	1.40864 AU
max. declination.	2037 Apr 05 02:43:26	0° ♏ 06'03	14N32'33"				
	2037 Apr 06 08:16:21	30° ♏		superior conj	2038 Feb 12 14:39:58	23° ♍ 58'27	-2°01'38
evening set	2037 Apr 07 06:01:49	29° ♏ 49'13		minimum elong	2038 Feb 12 17:13:36	24° ♍ 09'54	2°01'32
inferior conj	2037 Apr 15 02:55:50	25° ♏ 29'09	2°01'39		2038 Feb 15 22:41:06	0° ♎	
minimum elong	2037 Apr 15 06:47:54	25° ♏ 22'03	2°00'31	evening rise	2038 Feb 22 20:49:54	12° ♎ 51'13	
min. Earth dist.	2037 Apr 18 09:42:41	23° ♏ 05'58	0.57323 AU	asc. node	2038 Mar 02 05:44:18	26° ♎ 18'07	
desc. node	2037 Apr 22 15:10:05	20° ♏ 33'57			2038 Mar 04 10:56:00	0° ♏	
morning rise	2037 Apr 23 04:26:13	20° ♏ 18'50		evening max el	2038 Mar 11 02:47:14	8° ♏ 48'52	18°17'40
direct	2037 Apr 28 15:55:53	19° ♏ 05'43		retrograde	2038 Mar 18 06:07:21	12° ♏ 22'16	
min. declination.	2037 May 03 10:31:44	20° ♏ 00'23	5N35'26"	max. declination.	2038 Mar 19 15:49:39	12° ♏ 14'12	7N56'57"
morning max el	2037 May 12 23:56:52	26° ♏ 35'51	26°10'13	evening set	2038 Mar 20 18:03:22	11° ♏ 57'08	
	2037 May 16 05:45:52	0° ♐		inferior conj	2038 Mar 27 20:41:15	7° ♏ 16'16	3°04'03
	2037 Jun 04 18:27:01	0° ♑		minimum elong	2038 Mar 28 00:08:53	7° ♏ 08'50	3°03'21
morning set	2037 Jun 10 09:34:25	11° ♑ 22'11		min. Earth dist.	2038 Mar 31 06:49:58	4° ♏ 21'41	0.59355 AU
asc. node	2037 Jun 11 08:00:23	13° ♑ 21'02		morning rise	2038 Apr 04 03:46:04	1° ♏ 38'58	
					2038 Apr 08 13:07:52	30° ♑	
superior conj	2037 Jun 17 09:18:21	26° ♑ 32'11	1°00'05	desc. node	2038 Apr 09 12:10:22	29° ♑ 52'17	
minimum elong	2037 Jun 17 07:05:14	26° ♑ 19'59	0°59'41	direct	2038 Apr 10 12:11:53	29° ♑ 49'35	
max. Earth dist.	2037 Jun 17 22:07:16	27° ♑ 42'36	1.32357 AU		2038 Apr 12 11:56:17	0° ♒	
	2037 Jun 18 23:10:47	0° ♒		min. declination.	2038 Apr 15 18:27:50	0° ♒ 59'57	0S52'24"
max. declination.	2037 Jun 21 16:47:13	5° ♒ 56'07	24N49'26"	morning max el	2038 Apr 24 20:27:13	7° ♒ 35'39	27°16'28
evening rise	2037 Jun 24 08:42:18	11° ♒ 34'40			2038 May 11 18:00:45	0° ♒	
	2037 Jul 03 21:00:47	0° ♓		morning set	2038 May 25 18:24:06	26° ♒ 07'18	
desc. node	2037 Jul 19 14:29:04	23° ♓ 33'53			2038 May 27 14:31:19	0° ♓	
evening max el	2037 Jul 25 08:06:15	29° ♓ 45'47	27°02'29	asc. node	2038 May 29 05:02:21	3° ♓ 26'55	
	2037 Jul 25 14:08:47	0° ♓					
retrograde	2037 Aug 08 05:45:45	6° ♓ 59'35		superior conj	2038 Jun 01 21:04:05	11° ♓ 28'19	0°37'55
min. declination.	2037 Aug 11 20:16:55	6° ♓ 24'43	5N 3'26"	minimum elong	2038 Jun 01 19:28:23	11° ♓ 19'32	0°37'36
evening set	2037 Aug 15 06:27:24	4° ♓ 47'34		max. Earth dist.	2038 Jun 01 10:31:44	10° ♓ 30'20	1.32177 AU
min. Earth dist.	2037 Aug 18 22:23:48	2° ♓ 06'05	0.61133 AU	evening rise	2038 Jun 08 17:45:08	26° ♓ 23'25	
	2037 Aug 21 08:56:09	30° ♓			2038 Jun 10 11:12:39	0° ♓	
inferior conj	2037 Aug 22 02:27:48	29° ♓ 21'42	-4°24'34	max. declination.	2038 Jun 12 08:43:49	3° ♓ 51'27	25N14'52"
minimum elong	2037 Aug 22 07:25:47	29° ♓ 10'55	4°23'37		2038 Jun 27 15:58:37	0° ♓	
morning rise	2037 Aug 29 10:07:09	24° ♓ 30'19		desc. node	2038 Jul 06 11:29:52	10° ♓ 37'00	

evening max el	2038 Jul 07 07:07:51	11°♎24'48	26°06'44	superior conj	2039 May 17 07:36:07	26°♎17'02	0°12'59
retrograde	2038 Jul 21 06:58:36	18°♎35'11		minimum elong	2039 May 17 07:00:11	26°♎13'46	0°12'51
evening set	2038 Jul 27 15:25:18	16°♎57'08		behind sun begin	2039 May 17 03:59:20	25°♎57'18	
min. declination.	2038 Jul 27 08:37:06	17°♎05'30	11N25'54"	behind sun end	2039 May 17 10:01:03	26°♎30'14	
min. Earth dist.	2038 Jul 31 19:44:23	14°♎20'42	0.59072 AU		2039 May 19 00:21:49	0°♐	
inferior conj	2038 Aug 04 02:55:16	11°♎52'10	-4°54'00	evening rise	2039 May 24 05:05:31	11°♐15'34	
minimum elong	2038 Aug 04 05:19:58	11°♎47'37	4°53'50		2039 Jun 02 18:31:15	0°♑	
morning rise	2038 Aug 11 21:24:12	7°♎23'44		max. declination.	2039 Jun 03 09:57:33	1°♑07'35	25N33'51"
direct	2038 Aug 14 08:49:11	7°♎02'57		evening max el	2039 Jun 18 23:10:16	22°♑18'46	24°44'42
morning max el	2038 Aug 22 07:19:43	10°♎51'17	18°28'59	desc. node	2039 Jun 23 08:31:28	25°♑57'24	
max. declination.	2038 Aug 22 06:43:34	10°♎49'51	16N48'26"	retrograde	2039 Jul 02 21:47:33	29°♑22'43	
asc. node	2038 Aug 25 04:14:51	14°♎02'23		evening set	2039 Jul 08 00:38:42	28°♑23'20	
	2038 Sep 03 23:41:02	0°♐		min. Earth dist.	2039 Jul 13 10:24:07	25°♑33'25	0.57127 AU
morning set	2038 Sep 07 10:08:20	6°♐32'29		min. declination.	2039 Jul 14 00:01:18	25°♑11'40	16N28' 0"
				inferior conj	2039 Jul 16 07:43:32	23°♑40'20	-4°53'52
superior conj	2038 Sep 16 09:22:05	23°♐37'33	1°34'21	minimum elong	2039 Jul 16 05:10:24	23°♑44'32	4°53'42
minimum elong	2038 Sep 16 12:25:26	23°♐51'43	1°34'08	morning rise	2039 Jul 24 12:24:02	19°♑33'37	
	2038 Sep 19 21:03:32	0°♑		direct	2039 Jul 27 03:06:36	19°♑14'26	
max. Earth dist.	2038 Sep 23 19:12:33	6°♑55'55	1.40174 AU	morning max el	2039 Aug 05 07:12:34	23°♑31'04	19°18'49
evening rise	2038 Sep 28 02:44:47	14°♑15'59		max. declination.	2039 Aug 08 17:02:14	27°♑16'08	19N55'45"
desc. node	2038 Oct 02 10:48:53	21°♑21'09			2039 Aug 10 17:38:05	0°♒	
	2038 Oct 07 23:25:37	0°♒		asc. node	2039 Aug 12 01:18:00	1°♒57'48	
	2038 Oct 29 23:57:39	0°♒		morning set	2039 Aug 22 09:56:59	20°♒51'37	
evening max el	2038 Nov 01 21:47:59	3°♒05'41	23°41'03		2039 Aug 26 23:37:33	0°♓	
min. declination.	2038 Nov 08 21:32:59	8°♒28'45	24S21'30"	superior conj	2039 Aug 30 13:26:13	7°♓03'18	1°44'08
retrograde	2038 Nov 12 14:56:14	9°♒24'56		minimum elong	2039 Aug 30 14:35:11	7°♓08'54	1°44'06
evening set	2038 Nov 17 17:22:04	7°♒17'45		max. Earth dist.	2039 Sep 05 22:48:07	19°♓08'22	1.38115 AU
asc. node	2038 Nov 21 03:27:12	3°♒33'29		evening rise	2039 Sep 09 17:19:30	25°♓54'04	
min. Earth dist.	2038 Nov 22 14:21:42	1°♒37'06	0.67755 AU		2039 Sep 12 02:05:08	0°♈	
inferior conj	2038 Nov 23 02:57:35	0°♒54'08	0°40'32	desc. node	2039 Sep 19 07:52:20	11°♈54'40	
minimum elong	2038 Nov 23 02:00:42	0°♒57'22	0°40'08		2039 Oct 01 13:30:13	0°♉	
	2038 Nov 23 18:53:46	30°♉		evening max el	2039 Oct 15 09:55:12	16°♉43'42	24°58'46
morning rise	2038 Nov 28 10:38:03	24°♉47'39		min. declination.	2039 Oct 25 09:29:03	23°♉20'55	21S33'33"
direct	2038 Dec 02 12:30:01	23°♉14'10		retrograde	2039 Oct 27 03:54:19	23°♉32'59	
max. declination.	2038 Dec 03 05:27:26	23°♉16'46	16S 2'22"	evening set	2039 Nov 01 20:26:54	21°♉09'41	
morning max el	2038 Dec 10 19:31:22	28°♉02'23	20°54'50	min. Earth dist.	2039 Nov 06 09:12:49	16°♉03'08	0.67339 AU
	2038 Dec 12 15:41:17	0°♊		inferior conj	2039 Nov 07 08:26:11	14°♉46'45	-0°13'51
desc. node	2038 Dec 29 10:01:28	22°♊51'19		minimum elong	2039 Nov 07 08:46:52	14°♉45'37	0°13'42
	2039 Jan 03 04:04:56	0°♋		transit middle	2039 Nov 07 08:46:52	14°♉45'37	0°13'42
min. declination.	2039 Jan 08 09:33:55	8°♋01'53	24S17' 5"	transit begin	2039 Nov 07 07:18:05	14°♉50'29	
morning set	2039 Jan 09 15:59:23	10°♋00'21		transit end	2039 Nov 07 10:15:39	14°♉40'44	
max. Earth dist.	2039 Jan 17 23:45:14	23°♋15'51	1.42786 AU	asc. node	2039 Nov 08 00:29:34	13°♉54'10	
	2039 Jan 22 01:59:49	0°♌		morning rise	2039 Nov 12 21:16:40	8°♉46'47	
superior conj	2039 Jan 24 22:09:28	4°♌46'05	-2°04'14	direct	2039 Nov 16 10:38:02	7°♉33'51	
minimum elong	2039 Jan 24 20:30:39	4°♌39'07	2°04'12	max. declination.	2039 Nov 17 20:32:59	7°♉44'40	11S55'35"
evening rise	2039 Feb 05 16:09:06	25°♌17'34		morning max el	2039 Nov 23 20:51:08	11°♉46'57	19°46'17
	2039 Feb 08 07:26:49	0°♍		greatest brilliancy	2039 Dec 07 05:57:01	29°♉30'32	-0.7m
asc. node	2039 Feb 17 02:46:10	14°♍52'44			2039 Dec 07 13:52:33	0°♎	
evening max el	2039 Feb 22 13:55:41	21°♍51'55	18°07'17	desc. node	2039 Dec 16 07:05:29	13°♎13'17	
retrograde	2039 Mar 01 04:11:53	25°♍15'56		morning set	2039 Dec 19 12:15:11	18°♎10'26	
evening set	2039 Mar 03 21:26:39	24°♍41'06			2039 Dec 27 02:34:12	0°♏	
max. declination.	2039 Mar 03 11:04:34	24°♍51'09	0N57'53"	min. declination.	2039 Dec 31 03:50:51	6°♏23'57	24S46'19"
inferior conj	2039 Mar 10 09:27:33	19°♍39'27	3°35'37	max. Earth dist.	2039 Dec 31 11:01:40	6°♏52'29	1.44228 AU
minimum elong	2039 Mar 10 11:04:16	19°♍35'28	3°35'28				
min. Earth dist.	2039 Mar 13 11:21:33	16°♍38'08	0.61496 AU	superior conj	2040 Jan 05 03:22:46	14°♏22'36	-1°49'42
morning rise	2039 Mar 16 23:06:51	13°♍44'40		minimum elong	2040 Jan 04 20:19:59	13°♏54'09	1°49'13
direct	2039 Mar 23 19:19:39	11°♍21'59			2040 Jan 14 14:58:17	0°♐	
desc. node	2039 Mar 27 09:12:50	11°♍57'46		evening rise	2040 Jan 18 16:37:36	6°♐52'25	
min. declination.	2039 Mar 29 12:19:12	12°♍50'08	7S 9'35"		2040 Feb 01 21:16:09	0°♑	
morning max el	2039 Apr 06 23:27:38	19°♍16'41	27°47'46	asc. node	2040 Feb 03 23:48:41	2°♑46'50	
	2039 Apr 16 02:00:17	0°♒		evening max el	2040 Feb 06 02:33:15	5°♑09'26	18°16'03
	2039 May 04 13:48:46	0°♓		retrograde	2040 Feb 12 13:38:33	8°♑37'27	
morning set	2039 May 09 22:56:54	10°♓36'58		evening set	2040 Feb 15 12:02:34	7°♑51'54	
max. Earth dist.	2039 May 15 21:08:31	23°♓09'23	1.32382 AU	max. declination.	2040 Feb 15 16:01:26	7°♑46'42	5S51'31"
asc. node	2039 May 16 02:04:13	23°♓36'09		inferior conj	2040 Feb 21 13:13:13	2°♑30'48	3°43'17
				minimum elong	2040 Feb 21 12:55:27	2°♑31'38	3°43'17

	2040 Feb 23 19:37:12	30° 			2040 Dec 18 20:17:07	0° 		
min. Earth dist.	2040 Feb 24 01:11:03	29°  45'00	0.63479 AU		min. declination.	2040 Dec 21 22:32:31	4°  54'52	25S10'38"
morning rise	2040 Feb 27 12:56:08	26°  26'12			evening rise	2040 Dec 29 18:26:46	17°  29'20	
direct	2040 Mar 05 11:19:40	23°  41'27				2041 Jan 06 11:59:23	0° 	
min. declination.	2040 Mar 11 21:04:05	25°  39'05	12S42'23"		evening max el	2041 Jan 19 14:00:17	18°  34'45	18°43'05
desc. node	2040 Mar 13 06:14:48	26°  30'47			asc. node	2041 Jan 20 20:51:20	19°  47'52	
	2040 Mar 17 14:43:24	0° 			retrograde	2041 Jan 26 06:14:20	22°  18'10	
morning max el	2040 Mar 19 07:10:46	1°  37'08	27°41'40		evening set	2041 Jan 29 10:26:54	21°  20'50	
	2040 Apr 09 06:33:21	0° 			max. declination.	2041 Jan 30 10:55:13	20°  39'00	12S 2'52"
morning set	2040 Apr 22 20:54:01	24°  43'44			inferior conj	2041 Feb 04 03:59:37	15°  42'02	3°33'24
	2040 Apr 25 11:18:56	0° 			minimum elong	2041 Feb 04 02:18:16	15°  47'10	3°33'13
max. Earth dist.	2040 Apr 28 01:53:55	5°  26'17	1.32998 AU		min. Earth dist.	2041 Feb 06 00:24:30	13°  27'10	0.65138 AU
					morning rise	2041 Feb 09 17:43:46	9°  32'13	
superior conj	2040 Apr 30 15:10:24	10°  52'43	-0°14'00		direct	2041 Feb 16 10:27:07	6°  40'13	
minimum elong	2040 Apr 30 15:51:30	10°  56'24	0°13'50		min. declination.	2041 Feb 24 05:32:36	9°  41'41	17S 7'22"
behind sun begin	2040 Apr 30 13:13:00	10°  42'13			desc. node	2041 Feb 28 03:17:35	12°  55'18	
behind sun end	2040 Apr 30 18:29:59	11°  10'36			morning max el	2041 Mar 01 16:46:18	14°  26'09	27°01'57
asc. node	2040 May 01 23:07:45	13°  44'47				2041 Mar 14 09:20:59	0° 	
evening rise	2040 May 07 16:53:20	26°  04'58				2041 Apr 01 23:03:43	0° 	
	2040 May 09 13:51:19	0° 			morning set	2041 Apr 06 09:09:43	8°  17'27	
max. declination.	2040 May 24 18:06:23	26°  11'00	25N38'28"		max. Earth dist.	2041 Apr 10 21:01:07	17°  10'46	1.34059 AU
	2040 May 27 20:41:50	0° 						
evening max el	2040 May 30 12:55:14	2°  48'52	23°09'31		superior conj	2041 Apr 14 17:42:28	25°  08'36	-0°41'57
desc. node	2040 Jun 09 05:31:46	8°  59'40			minimum elong	2041 Apr 14 19:49:43	25°  19'43	0°41'31
retrograde	2040 Jun 13 00:19:13	9°  32'16				2041 Apr 17 01:03:41	0° 	
evening set	2040 Jun 16 15:00:15	9°  03'40			asc. node	2041 Apr 18 20:10:56	3°  48'23	
min. Earth dist.	2040 Jun 23 22:28:09	5°  46'00	0.55641 AU		evening rise	2041 Apr 22 03:14:49	10°  54'03	
inferior conj	2040 Jun 25 16:53:15	4°  43'59	-4°10'15			2041 May 02 04:57:28	0° 	
minimum elong	2040 Jun 25 09:38:16	4°  54'38	4°08'43		evening max el	2041 May 12 07:54:10	13°  12'6'31	21°36'14
min. declination.	2040 Jul 02 06:30:05	1°  25'38	18N41'15"		max. declination.	2041 May 14 17:37:59	15°  12'33'32	24N56'40"
morning rise	2040 Jul 04 06:47:54	0°  50'34			retrograde	2041 May 24 15:37:49	19°  12'28'51	
direct	2040 Jul 07 02:21:00	0°  31'33			evening set	2041 May 27 02:48:23	19°  12'15'07	
morning max el	2040 Jul 17 20:11:41	5°  30'08	20°29'46		desc. node	2041 May 27 02:30:50	19°  12'15'15	
max. declination.	2040 Jul 27 07:47:03	17°  51'50	21N54'12"		inferior conj	2041 Jun 05 12:39:23	15°  12'13'57	-2°39'28
asc. node	2040 Jul 28 22:21:24	20°  35'59			minimum elong	2041 Jun 05 05:38:12	15°  12'23'46	2°37'12
	2040 Aug 02 22:31:28	0° 			min. Earth dist.	2041 Jun 05 10:23:25	15°  12'17'07	0.54933 AU
morning set	2040 Aug 05 15:44:24	5°  02'28'53			morning rise	2041 Jun 14 09:19:32	11°  12'18'07	
					direct	2041 Jun 17 13:41:10	10°  12'55'57	
superior conj	2040 Aug 13 05:53:16	21°  06'18	1°45'14		min. declination.	2041 Jun 17 23:18:17	10°  12'56'19	17N45'24"
minimum elong	2040 Aug 13 05:22:03	21°  03'40	1°45'13		morning max el	2041 Jun 29 21:48:47	16°  12'45'16	21°59'24
	2040 Aug 17 17:01:55	0° 				2041 Jul 10 06:29:24	0° 	
max. Earth dist.	2040 Aug 18 05:10:23	0°  58'58	1.36211 AU		asc. node	2041 Jul 15 19:25:41	9°  12'46'45	
evening rise	2040 Aug 22 05:02:41	8°  32'35			max. declination.	2041 Jul 17 05:52:26	12°  12'33'43	23N 6'58"
	2040 Sep 03 18:28:53	0° 			morning set	2041 Jul 21 01:20:19	20°  12'18'30	
desc. node	2040 Sep 05 04:54:38	2°  14'19				2041 Jul 25 15:27:27	0° 	
	2040 Sep 26 12:23:15	0° 						
evening max el	2040 Sep 26 21:54:31	0°  12'23'18	26°06'52		superior conj	2041 Jul 28 06:58:03	5°  12'35'42	1°39'17
min. declination.	2040 Oct 08 22:51:52	7°  13'30'34	17S17'54"		minimum elong	2041 Jul 28 05:16:59	5°  12'26'51	1°39'10
retrograde	2040 Oct 09 12:05:08	7°  13'31'39			max. Earth dist.	2041 Jul 31 19:47:49	12°  12'55'25	1.34627 AU
evening set	2040 Oct 15 18:34:33	4°  15'55'32			evening rise	2041 Aug 05 09:17:54	21°  12'59'38	
min. Earth dist.	2040 Oct 19 23:54:06	0°  12'24'37	0.66590 AU			2041 Aug 09 15:36:57	0° 	
	2040 Oct 20 07:52:34	30°  12'00			desc. node	2041 Aug 23 01:54:36	22°  12'13'44	
inferior conj	2040 Oct 21 10:15:55	28°  12'37'46	-1°11'13			2041 Aug 28 12:52:05	0° 	
minimum elong	2040 Oct 21 12:08:24	28°  12'31'56	1°10'25		evening max el	2041 Sep 09 10:20:04	13°  12'58'13	26°57'35
asc. node	2040 Oct 24 21:33:21	24°  12'39'14			retrograde	2041 Sep 22 14:43:46	21°  12'14'49	
morning rise	2040 Oct 27 06:06:57	22°  12'47'03			min. declination.	2041 Sep 23 01:17:41	21°  12'14'11	11S51'34"
direct	2040 Oct 30 09:05:55	21°  12'51'12			evening set	2041 Sep 29 09:33:51	18°  12'31'45	
max. declination.	2040 Nov 01 08:42:19	22°  12'43'12	6S56'19"		min. Earth dist.	2041 Oct 03 08:02:14	14°  12'37'19	0.65499 AU
morning max el	2040 Nov 06 05:17:08	25°  12'39'22	18°52'37		inferior conj	2041 Oct 05 06:24:35	12°  12'23'48	-2°09'47
	2040 Nov 09 22:57:04	0°			minimum elong	2041 Oct 05 09:52:52	12° 12'13'47	2°08'26
morning set	2040 Nov 27 23:31:05	27° 12'11'38			morning rise	2041 Oct 11 10:54:30	6° 12'45'24	
	2040 Nov 29 18:20:28	0°			asc. node	2041 Oct 11 18:36:00	6° 12'36'15	
desc. node	2040 Dec 02 04:08:58	3° 12'47'08			direct	2041 Oct 14 05:51:36	6° 12'02'49	
max. Earth dist.	2040 Dec 13 03:33:21	21° 12'00'48	1.45025 AU		max. declination.	2041 Oct 16 19:45:21	6° 12'38'24	1S26'28"
					morning max el	2041 Oct 20 18:51:38	9° 12'35'32	18°15'16
superior conj	2040 Dec 14 10:39:27	23° 12'03'08	-1°16'17			2041 Nov 03 22:43:29	0°	
minimum elong	2040 Dec 14 02:03:25	22° 12'29'17	1°15'19		morning set	2041 Nov 08 15:15:56	7° 12'39'36	

desc. node	2041 Nov 19 01:12:33	24°♄29'10		min. Earth dist.	2042 Sep 16 07:17:07	28°♄33'00	0.64054 AU
	2041 Nov 22 12:31:23	0°♄		inferior conj	2042 Sep 18 18:23:06	25°♄58'35	-3°07'06
				minimum elong	2042 Sep 18 23:12:39	25°♄45'56	3°05'30
superior conj	2041 Nov 23 12:48:46	1°♄36'00	-0°29'47	morning rise	2042 Sep 25 08:53:25	20°♄36'03	
minimum elong	2041 Nov 23 08:58:08	1°♄20'49	0°29'15	direct	2042 Sep 27 22:18:19	20°♄03'12	
max. Earth dist.	2041 Nov 25 22:45:40	5°♄24'18	1.45091 AU	asc. node	2042 Sep 28 15:39:23	20°♄05'52	
evening rise	2041 Dec 09 21:21:20	27°♄12'10		max. declination.	2042 Oct 01 06:27:54	20°♄59'20	4N12'27"
	2041 Dec 11 16:23:50	0°♄		morning max el	2042 Oct 04 10:53:21	23°♄28'59	17°55'04
min. declination.	2041 Dec 13 18:48:04	3°♄16'25	25S31'15"		2042 Oct 09 15:18:21	0°♄	
greatest brilliancy	2041 Dec 21 14:43:05	15°♄22'32	-0.7m	morning set	2042 Oct 21 10:24:34	19°♄27'08	
	2042 Jan 01 01:19:46	0°♄			2042 Oct 27 15:27:00	0°♄	
evening max el	2042 Jan 02 22:04:30	2°♄03'52	19°27'03				
asc. node	2042 Jan 07 17:52:28	5°♄41'56		superior conj	2042 Nov 03 06:44:44	10°♄58'30	0°17'51
retrograde	2042 Jan 10 02:47:46	6°♄12'39		minimum elong	2042 Nov 03 08:43:22	11°♄06'33	0°17'34
evening set	2042 Jan 13 14:01:24	5°♄02'17		desc. node	2042 Nov 05 22:13:28	15°♄15'20	
max. declination.	2042 Jan 16 13:00:03	2°♄18'36	17S 5'18"	max. Earth dist.	2042 Nov 08 17:10:12	19°♄43'05	1.44422 AU
	2042 Jan 18 10:25:22	30°♄			2042 Nov 15 06:05:16	0°♄	
inferior conj	2042 Jan 19 02:31:54	29°♄07'53	3°10'32	evening rise	2042 Nov 19 08:13:49	6°♄19'41	
minimum elong	2042 Jan 19 00:05:11	29°♄15'50	3°10'01		2042 Dec 05 01:51:20	0°♄	
min. Earth dist.	2042 Jan 20 08:03:17	27°♄31'55	0.66396 AU	min. declination.	2042 Dec 05 17:21:43	0°♄56'04	25S47'13"
morning rise	2042 Jan 24 09:55:24	22°♄55'30		evening max el	2042 Dec 17 01:04:57	15°♄33'56	20°26'01
direct	2042 Jan 30 15:31:39	20°♄09'37		retrograde	2042 Dec 25 00:43:57	20°♄16'14	
min. declination.	2042 Feb 10 12:18:03	26°♄00'46	20S12'11"	asc. node	2042 Dec 25 14:53:32	20°♄14'17	
morning max el	2042 Feb 12 02:07:02	27°♄32'36	25°56'14	evening set	2042 Dec 28 20:34:57	18°♄51'28	
	2042 Feb 14 09:52:17	0°♄		inferior conj	2043 Jan 03 06:10:07	12°♄44'16	2°37'44
desc. node	2042 Feb 15 00:21:20	0°♄40'22		minimum elong	2043 Jan 03 03:32:45	12°♄53'10	2°36'57
	2042 Mar 08 01:46:50	0°♄		min. Earth dist.	2043 Jan 03 22:21:30	11°♄49'17	0.67241 AU
morning set	2042 Mar 20 07:38:32	21°♄06'13		max. declination.	2043 Jan 07 15:03:02	7°♄17'28	19S58'40"
max. Earth dist.	2042 Mar 24 04:37:58	28°♄24'41	1.35581 AU	morning rise	2043 Jan 08 10:20:29	6°♄31'02	
	2042 Mar 25 00:20:50	0°♄		direct	2043 Jan 14 01:49:37	4°♄00'52	
				morning max el	2043 Jan 25 10:40:05	10°♄47'45	24°34'09
superior conj	2042 Mar 29 12:42:24	8°♄57'27	-1°09'28	min. declination.	2043 Jan 30 19:30:05	16°♄47'53	22S 4'32"
minimum elong	2042 Mar 29 16:10:11	9°♄14'58	1°08'54	desc. node	2043 Feb 01 21:24:41	19°♄22'37	
asc. node	2042 Apr 05 17:12:35	23°♄42'41			2043 Feb 09 17:53:29	0°♄	
evening rise	2042 Apr 06 10:10:56	25°♄09'39			2043 Feb 28 18:44:25	0°♄	
	2042 Apr 08 19:35:29	0°♄		morning set	2043 Mar 02 10:58:55	2°♄55'16	
evening max el	2042 Apr 24 13:25:18	24°♄35'24	20°15'55	max. Earth dist.	2043 Mar 06 02:54:22	9°♄28'30	1.37496 AU
max. declination.	2042 May 01 02:29:12	28°♄59'46	22N28'41"				
retrograde	2042 May 05 05:13:00	29°♄45'11		superior conj	2043 Mar 12 20:58:32	22°♄10'36	-1°34'33
evening set	2042 May 07 06:06:18	29°♄34'40		minimum elong	2043 Mar 13 01:14:36	22°♄31'17	1°34'01
desc. node	2042 May 13 23:31:46	26°♄57'38			2043 Mar 16 20:35:17	0°♄	
inferior conj	2042 May 16 08:03:59	25°♄36'18	-0°41'15	evening rise	2043 Mar 21 11:21:22	9°♄12'02	
minimum elong	2042 May 16 06:08:00	25°♄39'09	0°40'32	asc. node	2043 Mar 23 14:13:44	13°♄23'08	
min. Earth dist.	2042 May 17 23:14:06	24°♄38'40	0.55181 AU		2043 Apr 02 00:54:42	0°♄	
morning rise	2042 May 25 04:43:39	21°♄19'07		evening max el	2043 Apr 07 06:33:24	6°♄24'21	19°14'10
direct	2042 May 29 01:31:14	20°♄46'42		max. declination.	2043 Apr 15 06:03:22	10°♄42'19	17N55'57"
min. declination.	2042 Jun 01 02:47:47	21°♄07'36	14N21'40"	retrograde	2043 Apr 16 06:17:31	10°♄45'30	
morning max el	2042 Jun 11 14:54:46	27°♄24'38	23°40'27	evening set	2043 Apr 18 08:50:43	10°♄32'11	
	2042 Jun 14 03:57:04	0°♄		inferior conj	2043 Apr 26 17:04:34	6°♄22'10	1°10'11
asc. node	2042 Jul 02 16:28:30	29°♄20'57		minimum elong	2043 Apr 26 19:48:12	6°♄17'35	1°09'16
	2042 Jul 03 00:11:04	0°♄		min. Earth dist.	2043 Apr 29 14:11:43	4°♄26'52	0.56335 AU
morning set	2042 Jul 05 12:50:49	5°♄14'07		desc. node	2043 Apr 30 20:33:13	3°♄39'14	
max. declination.	2042 Jul 07 11:45:48	9°♄23'29	23N55'49"	morning rise	2043 May 05 03:47:51	1°♄30'59	
				direct	2043 May 10 00:19:39	0°♄36'33	
superior conj	2042 Jul 12 13:43:34	20°♄21'39	1°27'38	min. declination.	2043 May 14 09:18:51	1°♄20'54	9N 8'58"
minimum elong	2042 Jul 12 11:25:54	20°♄09'18	1°27'22	morning max el	2043 May 24 05:24:47	7°♄51'57	25°19'54
max. Earth dist.	2042 Jul 14 19:43:32	25°♄10'37	1.33431 AU		2043 Jun 09 13:46:45	0°♄	
	2042 Jul 17 02:46:47	0°♄		asc. node	2043 Jun 19 13:31:25	19°♄11'57	
evening rise	2042 Jul 20 01:39:35	6°♄02'00		morning set	2043 Jun 20 00:37:37	20°♄10'23	
	2042 Aug 02 05:59:00	0°♄			2043 Jun 24 14:00:17	0°♄	
desc. node	2042 Aug 09 22:53:57	11°♄43'28					
evening max el	2042 Aug 22 21:52:27	27°♄15'50	27°23'45	superior conj	2043 Jun 26 23:55:38	5°♄17'00	1°11'18
	2042 Aug 26 00:53:43	0°♄		minimum elong	2043 Jun 26 21:32:41	5°♄03'58	1°10'55
retrograde	2042 Sep 05 11:20:50	4°♄33'39		max. Earth dist.	2043 Jun 28 03:03:07	7°♄45'05	1.32636 AU
min. declination.	2042 Sep 06 22:03:51	4°♄27'17	5S35'10"	max. declination.	2043 Jun 27 20:02:35	7°♄06'53	24N32' 4"
evening set	2042 Sep 12 14:36:41	1°♄52'38		evening rise	2043 Jul 04 02:38:48	20°♄29'20	
	2042 Sep 14 19:45:10	30°♄			2043 Jul 08 21:14:56	0°♄	

	2043 Jul 27 10:17:58	0°♍		superior conj	2044 Jun 10 11:37:43	20°♊14'34	0°51'04
desc. node	2043 Jul 27 19:52:54	0°♍31'12		minimum elong	2044 Jun 10 09:37:33	20°♊03'32	0°50'41
evening max el	2043 Aug 05 06:18:59	10°♍03'08	27°19'37	max. Earth dist.	2044 Jun 10 14:28:09	20°♊30'12	1.32233 AU
retrograde	2043 Aug 19 01:33:04	17°♍19'09			2044 Jun 14 22:45:07	0°♋	
min. declination.	2043 Aug 21 17:41:07	16°♍59'02	1N 8' 2"	evening rise	2044 Jun 17 09:24:49	5°♋12'17	
evening set	2043 Aug 26 06:09:23	14°♍52'17		max. declination.	2044 Jun 17 08:16:41	5°♋06'19	25N 0'51"
min. Earth dist.	2043 Aug 29 20:14:07	12°♍00'40	0.62276 AU		2044 Jun 30 13:28:44	0°♌	
inferior conj	2043 Sep 01 19:16:42	9°♍15'23	-3°59'15	desc. node	2044 Jul 13 16:53:50	18°♌19'17	
minimum elong	2043 Sep 02 00:38:37	9°♍02'52	3°57'55	evening max el	2044 Jul 17 09:14:18	22°♌08'44	26°42'27
morning rise	2043 Sep 08 20:38:44	4°♍12'05		retrograde	2044 Jul 31 08:21:44	29°♌22'01	
direct	2043 Sep 11 07:04:00	3°♍45'53		min. declination.	2044 Aug 04 19:34:18	28°♌30'19	7N49'29"
asc. node	2043 Sep 15 12:43:09	5°♍07'32		evening set	2044 Aug 07 03:41:50	27°♌23'05	
max. declination.	2043 Sep 15 18:05:47	5°♍16'19	9N37'57"	min. Earth dist.	2044 Aug 10 23:01:38	24°♌45'58	0.60258 AU
morning max el	2043 Sep 18 02:28:05	7°♍13'12	17°52'55	inferior conj	2044 Aug 14 05:40:56	22°♌05'37	-4°39'49
	2043 Oct 02 20:44:41	0°♎		minimum elong	2044 Aug 14 09:53:19	21°♌56'59	4°39'13
morning set	2043 Oct 04 03:00:06	2°♎15'56		morning rise	2044 Aug 21 18:03:27	17°♌24'08	
				direct	2044 Aug 24 04:12:11	17°♌02'04	
superior conj	2043 Oct 15 03:13:20	21°♎39'49	0°56'55	max. declination.	2044 Aug 30 11:00:18	19°♌38'36	14N26'38"
minimum elong	2043 Oct 15 07:41:54	21°♎58'54	0°56'21	morning max el	2044 Aug 31 14:27:17	20°♌40'03	18°09'59
	2043 Oct 20 02:23:11	0°♏		asc. node	2044 Sep 01 09:46:38	21°♌28'55	
max. Earth dist.	2043 Oct 22 07:38:12	3°♏38'20	1.43108 AU		2044 Sep 07 15:30:09	0°♐	
desc. node	2043 Oct 23 19:13:22	6°♏02'34		morning set	2044 Sep 16 11:14:00	15°♐49'50	
evening rise	2043 Oct 29 18:15:22	15°♏29'59			2044 Sep 24 01:37:46	0°♑	
	2043 Nov 08 06:25:01	0°♒					
min. declination.	2043 Nov 27 10:06:23	26°♒19'57	25S51'14"	superior conj	2044 Sep 26 01:31:55	3°♑37'19	1°24'03
evening max el	2043 Nov 29 22:21:39	29°♒04'14	21°36'51	minimum elong	2044 Sep 26 05:31:47	3°♑55'18	1°23'40
	2043 Nov 30 20:54:18	0°♓		max. Earth dist.	2044 Oct 03 16:33:30	16°♑55'42	1.41322 AU
retrograde	2043 Dec 08 21:57:32	4°♓25'00		evening rise	2044 Oct 08 19:45:52	25°♑25'11	
asc. node	2043 Dec 12 11:56:13	3°♓14'22		desc. node	2044 Oct 09 16:14:54	26°♑47'53	
evening set	2043 Dec 13 04:14:57	2°♓44'31			2044 Oct 11 16:21:35	0°♒	
	2043 Dec 15 20:44:27	30°♒♑			2044 Nov 01 02:34:58	0°♒	
inferior conj	2043 Dec 18 12:40:18	26°♒27'54	1°57'01	evening max el	2044 Nov 11 14:42:08	12°♒37'08	22°55'05
minimum elong	2043 Dec 18 10:22:30	26°♒35'51	1°56'12	min. declination.	2044 Nov 16 00:45:33	16°♒26'37	25S16'37"
min. Earth dist.	2043 Dec 18 17:01:52	26°♒12'48	0.67710 AU	retrograde	2044 Nov 21 16:51:31	18°♒36'30	
morning rise	2043 Dec 23 16:20:46	20°♒15'31		evening set	2044 Nov 26 11:24:08	16°♒39'20	
max. declination.	2043 Dec 27 02:14:14	18°♒18'56	19S53'34"	asc. node	2044 Nov 28 08:59:05	14°♒46'49	
direct	2043 Dec 28 16:45:27	18°♒06'20		inferior conj	2044 Dec 01 20:10:57	10°♒17'07	1°10'00
morning max el	2044 Jan 07 19:59:58	24°♒07'31	23°05'39	minimum elong	2044 Dec 01 18:37:55	10°♒22'28	1°09'22
	2044 Jan 13 01:21:57	0°♔		min. Earth dist.	2044 Dec 01 13:43:00	10°♒39'26	0.67831 AU
desc. node	2044 Jan 19 18:26:13	8°♔45'04		morning rise	2044 Dec 07 01:44:47	4°♒07'38	
min. declination.	2044 Jan 22 01:14:47	11°♔55'43	23S11'49"	direct	2044 Dec 11 11:31:09	2°♒21'10	
	2044 Feb 03 04:39:04	0°♕		max. declination.	2044 Dec 11 14:42:10	2°♒21'16	17S53'25"
morning set	2044 Feb 11 12:47:59	13°♕27'13		morning max el	2044 Dec 20 09:29:32	7°♒34'38	21°39'58
max. Earth dist.	2044 Feb 15 21:52:31	20°♕53'08	1.39630 AU	desc. node	2045 Jan 05 15:28:58	28°♒36'49	
	2044 Feb 21 01:40:08	0°♖			2045 Jan 06 14:20:24	0°♔	
				min. declination.	2045 Jan 12 18:20:19	9°♔11'09	23S56'38"
superior conj	2044 Feb 23 14:16:11	4°♖36'23	-1°54'17	morning set	2045 Jan 21 09:25:05	22°♔35'31	
minimum elong	2044 Feb 23 18:03:54	4°♖53'52	1°54'00		2045 Jan 25 23:34:17	0°♕	
evening rise	2044 Mar 04 03:47:57	22°♖43'18		max. Earth dist.	2045 Jan 27 20:58:25	3°♕06'57	1.41720 AU
	2044 Mar 07 23:48:51	0°♗					
asc. node	2044 Mar 09 11:15:24	2°♗43'31		superior conj	2045 Feb 04 11:36:24	16°♕03'22	-2°04'33
evening max el	2044 Mar 20 09:13:27	18°♗48'53	18°32'33	minimum elong	2045 Feb 04 12:41:57	16°♕08'08	2°04'32
retrograde	2044 Mar 28 01:30:52	22°♗34'59			2045 Feb 12 06:44:07	0°♘	
max. declination.	2044 Mar 28 19:08:48	22°♗32'58	11N52' 6"	evening rise	2045 Feb 15 07:59:13	5°♘35'33	
evening set	2044 Mar 30 09:52:25	22°♗15'00		asc. node	2045 Feb 24 08:17:45	21°♘37'14	
inferior conj	2044 Apr 06 22:40:14	17°♗46'25	2°32'13		2045 Mar 02 04:45:18	0°♙	
minimum elong	2044 Apr 07 02:39:37	17°♗38'35	2°31'11	evening max el	2045 Mar 03 18:15:44	1°♙40'46	18°10'56
min. Earth dist.	2044 Apr 10 08:41:40	15°♙06'48	0.58148 AU	retrograde	2045 Mar 10 14:41:55	5°♙08'07	
morning rise	2044 Apr 14 16:25:42	12°♙23'01		max. declination.	2045 Mar 12 10:05:13	4°♙53'51	5N 2' 0"
desc. node	2044 Apr 16 17:35:39	11°♙32'31		evening set	2045 Mar 13 04:57:49	4°♙39'03	
direct	2044 Apr 20 13:34:10	10°♙54'46			2045 Mar 19 20:22:09	30°♒♘	
min. declination.	2044 Apr 25 13:37:26	11°♙56'18	2N54'20"	inferior conj	2045 Mar 20 00:56:23	29°♘49'35	3°20'45
morning max el	2044 May 04 22:46:37	18°♙33'43	26°42'09	minimum elong	2045 Mar 20 03:41:34	29°♘43'18	3°20'20
	2044 May 14 13:08:37	0°♚		min. Earth dist.	2045 Mar 23 08:48:47	26°♘48'52	0.60273 AU
	2044 Jun 01 00:41:41	0°♛		morning rise	2045 Mar 27 00:17:37	24°♘03'48	
morning set	2044 Jun 03 11:01:05	5°♛01'06		direct	2045 Apr 02 15:09:39	21°♘58'54	
asc. node	2044 Jun 05 10:33:36	9°♛13'38		desc. node	2045 Apr 03 14:38:29	22°♘01'32	

min. declination.	2045 Apr 08 01:17:55	23° Υ 15'52	3S33'56"	evening max el	2046 Feb 15 06:24:41	14° Υ 50'49	18°08'47
morning max el	2045 Apr 16 21:56:12	29° Υ 49'51	27°34'13	retrograde	2046 Feb 21 18:00:26	18° Υ 14'38	
	2045 Apr 17 02:01:57	0° Υ		evening set	2046 Feb 24 13:33:58	17° Υ 35'11	
	2045 May 08 12:06:44	0° Υ		max. declination.	2046 Feb 24 08:56:10	17° Υ 40'20	1S56'45"
morning set	2045 May 18 18:16:36	19° Υ 40'22		inferior conj	2046 Mar 02 20:31:27	12° Υ 25'06	3°41'20
asc. node	2045 May 23 07:36:03	29° Υ 21'27		minimum elong	2046 Mar 02 21:17:07	12° Υ 23'07	3°41'19
	2045 May 23 14:41:56	0° Π		min. Earth dist.	2046 Mar 05 16:55:10	9° Υ 28'11	0.62375 AU
max. Earth dist.	2045 May 25 02:24:27	3° Π 14'56	1.32216 AU	morning rise	2046 Mar 09 03:48:21	6° Υ 25'45	
				direct	2046 Mar 16 02:11:59	3° Υ 52'01	
superior conj	2045 May 25 23:06:24	5° Π 08'28	0°27'39	desc. node	2046 Mar 21 11:41:27	5° Υ 13'30	
minimum elong	2045 May 25 21:53:35	5° Π 01'48	0°27'24	min. declination.	2046 Mar 22 01:04:34	5° Υ 30'18	9S35'58"
evening rise	2045 Jun 01 19:41:12	20° Π 03'22		morning max el	2046 Mar 30 02:59:40	11° Υ 47'25	27°49'41
	2045 Jun 06 16:19:38	0° Ξ			2046 Apr 13 12:34:32	0° Υ	
max. declination.	2045 Jun 08 03:47:13	2° Ξ 51'32	25N23'52"		2046 Apr 30 20:06:28	0° Υ	
	2045 Jun 25 21:26:54	0° Ω		morning set	2046 May 02 20:21:35	4° Υ 01'26	
evening max el	2045 Jun 29 05:12:58	3° Ω 27'59	25°34'26	max. Earth dist.	2046 May 08 11:04:52	15° Υ 46'38	1.32592 AU
desc. node	2045 Jun 30 13:55:29	4° Ω 43'00					
retrograde	2045 Jul 13 05:37:56	10° Ω 36'26		superior conj	2046 May 10 08:38:28	19° Υ 52'49	0°01'46
evening set	2045 Jul 19 02:38:32	9° Ω 15'19		minimum elong	2046 May 10 08:33:31	19° Υ 52'22	0°01'45
min. declination.	2045 Jul 20 22:41:14	8° Ω 19'30	13N47'13"	behind sun begin	2046 May 10 03:24:40	19° Υ 24'26	
min. Earth dist.	2045 Jul 23 17:23:41	6° Ω 34'54	0.58209 AU	behind sun end	2046 May 10 13:42:22	20° Υ 20'20	
inferior conj	2045 Jul 26 22:04:49	4° Ω 19'20	-4°58'30	asc. node	2046 May 10 04:39:00	19° Υ 31'09	
minimum elong	2045 Jul 26 22:36:34	4° Ω 18'23	4°58'29		2046 May 15 00:17:40	0° Π	
morning rise	2045 Aug 03 20:57:33	0° Ω 00'18		evening rise	2046 May 17 07:28:51	4° Π 55'46	
	2045 Aug 03 21:25:07	30° $\Re$ Ξ		max. declination.	2046 May 30 09:10:34	29° Π 29'46	25N38'34"
direct	2045 Aug 06 09:27:39	29° Ξ 40'23			2046 May 30 16:51:35	0° Ξ	
	2045 Aug 08 19:51:37	0° Ω		evening max el	2046 Jun 10 19:45:55	14° Ξ 09'53	24°04'59
morning max el	2045 Aug 14 19:34:39	3° Ω 39'40	18°47'31	desc. node	2046 Jun 17 10:56:31	19° Ξ 10'11	
max. declination.	2045 Aug 15 22:32:00	4° Ω 47'24	18N15'48"	retrograde	2046 Jun 24 14:57:39	21° Ξ 06'31	
asc. node	2045 Aug 19 06:50:08	8° Ω 54'18		evening set	2046 Jun 29 02:36:25	20° Ξ 21'59	
morning set	2045 Aug 31 06:24:42	29° Ω 55'32		min. Earth dist.	2046 Jul 05 06:29:27	17° Ξ 21'56	0.56423 AU
	2045 Aug 31 07:19:35	0° η		inferior conj	2046 Jul 07 18:22:52	15° Ξ 49'22	-4°41'21
				minimum elong	2046 Jul 07 13:30:44	15° Ξ 56'57	4°40'44
superior conj	2045 Sep 08 20:19:38	16° η 35'15	1°39'41	min. declination.	2046 Jul 09 04:12:22	14° Ξ 57'28	17N49' 1"
minimum elong	2045 Sep 08 22:34:41	16° η 45'55	1°39'33	morning rise	2046 Jul 16 03:04:44	11° Ξ 49'33	
max. Earth dist.	2045 Sep 15 21:15:01	29° η 30'40	1.39295 AU	direct	2046 Jul 18 19:50:53	11° Ξ 30'30	
	2045 Sep 16 03:52:34	0° $\underline{\Delta}$		morning max el	2046 Jul 28 14:59:20	16° Ξ 03'17	19°46'23
evening rise	2045 Sep 19 20:43:31	6° $\underline{\Delta}$ 25'32		max. declination.	2046 Aug 03 06:30:11	22° Ξ 43'59	20N52'23"
desc. node	2045 Sep 26 13:16:47	17° $\underline{\Delta}$ 27'02		asc. node	2046 Aug 06 03:53:36	27° Ξ 09'51	
	2045 Oct 04 17:34:39	0° $\mathbb{M}$			2046 Aug 07 20:07:47	0° Ω	
evening max el	2045 Oct 25 03:56:00	26° $\mathbb{M}$ 13'53	24°14'55	morning set	2046 Aug 15 09:07:58	14° Ω 24'05	
	2045 Oct 29 11:42:17	0° $\mathcal{A}$					
min. declination.	2045 Nov 02 14:00:11	2° $\mathcal{A}$ 16'31	23S22'43"	superior conj	2046 Aug 23 06:13:52	0° η 18'50	1°45'34
retrograde	2045 Nov 05 08:07:26	2° $\mathcal{A}$ 46'46		minimum elong	2046 Aug 23 06:37:18	0° η 20'47	1°45'33
evening set	2045 Nov 10 16:25:08	0° $\mathcal{A}$ 32'47			2046 Aug 23 02:26:17	0° η	
	2045 Nov 11 07:00:37	30° $\Re$ $\mathbb{M}$		max. Earth dist.	2046 Aug 29 01:50:25	11° η 33'24	1.37283 AU
asc. node	2045 Nov 15 06:02:20	25° $\mathbb{M}$ 19'14		evening rise	2046 Sep 01 20:53:59	18° η 31'13	
min. Earth dist.	2045 Nov 15 09:54:01	25° $\mathbb{M}$ 06'18	0.67619 AU		2046 Sep 08 12:25:17	0° $\underline{\Delta}$	
inferior conj	2045 Nov 16 02:53:41	24° $\mathbb{M}$ 09'05	0°17'55	desc. node	2046 Sep 13 10:18:35	7° $\underline{\Delta}$ 55'38	
minimum elong	2045 Nov 16 02:27:36	24° $\mathbb{M}$ 10'33	0°17'44		2046 Sep 28 22:43:31	0° $\mathbb{M}$	
morning rise	2045 Nov 21 12:30:36	18° $\mathbb{M}$ 04'52		evening max el	2046 Oct 07 16:00:18	9° $\mathbb{M}$ 53'46	25°29'19
direct	2045 Nov 25 08:54:07	16° $\mathbb{M}$ 40'16		min. declination.	2046 Oct 18 14:03:21	16° $\mathbb{M}$ 46'12	19S56'25"
max. declination.	2045 Nov 26 09:45:13	16° $\mathbb{M}$ 45'59	14S26'53"	retrograde	2046 Oct 19 18:56:03	16° $\mathbb{M}$ 51'37	
morning max el	2045 Dec 03 06:06:46	21° $\mathbb{M}$ 12'10	20°23'55	evening set	2046 Oct 25 17:36:17	14° $\mathbb{M}$ 22'21	
	2045 Dec 10 12:57:24	0° $\mathcal{A}$		min. Earth dist.	2046 Oct 30 03:11:45	9° $\mathbb{M}$ 30'56	0.67072 AU
desc. node	2045 Dec 23 12:32:16	18° $\mathcal{A}$ 49'21		inferior conj	2046 Oct 31 07:02:40	8° $\mathbb{M}$ 01'21	-0°37'55
	2045 Dec 30 20:06:40	0° Ξ		minimum elong	2046 Oct 31 08:00:59	7° $\mathbb{M}$ 58'13	0°37'30
morning set	2045 Dec 31 08:14:44	0° Ξ 46'57		asc. node	2046 Nov 02 03:05:14	5° $\mathbb{M}$ 43'20	
min. declination.	2046 Jan 04 12:42:55	7° Ξ 18'51	24S30' 0"	morning rise	2046 Nov 05 22:39:41	2° $\mathbb{M}$ 05'04	
max. Earth dist.	2046 Jan 10 04:12:14	16° Ξ 17'20	1.43474 AU	direct	2046 Nov 09 07:26:20	0° $\mathbb{M}$ 59'43	
				max. declination.	2046 Nov 10 23:18:31	1° $\mathbb{M}$ 14'46	9S55'33"
superior conj	2046 Jan 16 08:01:49	26° Ξ 20'32	-2°00'24	morning max el	2046 Nov 16 10:37:29	5° $\mathbb{M}$ 00'27	19°21'29
minimum elong	2046 Jan 16 04:00:33	26° Ξ 03'53	2°00'14	greatest brilliancy	2046 Nov 30 05:48:08	23° $\mathbb{M}$ 39'05	-0.7m
	2046 Jan 18 12:42:29	0° $\approx$			2046 Dec 04 09:29:55	0° $\mathcal{A}$	
evening rise	2046 Jan 28 19:43:09	17° $\approx$ 40'09		morning set	2046 Dec 10 08:09:42	9° $\mathcal{A}$ 11'22	
	2046 Feb 04 22:12:28	0° Υ		desc. node	2046 Dec 10 09:35:09	9° $\mathcal{A}$ 16'54	
asc. node	2046 Feb 11 05:19:58	9° Υ 55'45			2046 Dec 23 15:40:00	0° Ξ	

max. Earth dist.	2046 Dec 23 18:36:45	0°☾11'38	1.44661 AU	inferior conj	2047 Oct 15 06:42:18	21°♊50'30	-1°36'02
				minimum elong	2047 Oct 15 09:15:48	21°♊42'47	1°34'59
superior conj	2046 Dec 27 02:34:39	5°☾28'51	-1°37'41	asc. node	2047 Oct 20 00:08:02	16°♊54'28	
minimum elong	2046 Dec 26 18:03:45	4°☾54'57	1°36'55	morning rise	2047 Oct 21 06:05:06	16°♊04'56	
min. declination.	2046 Dec 27 07:05:15	5°☾46'49	24S56'52"	direct	2047 Oct 24 05:20:55	15°♊15'16	
evening rise	2047 Jan 10 10:51:01	28°☾50'34		max. declination.	2047 Oct 26 10:50:14	15°♊42'17	4S40'55"
	2047 Jan 11 03:33:23	0°☾		morning max el	2047 Oct 30 21:37:34	18°♊55'31	18°34'40
asc. node	2047 Jan 29 02:21:55	27°☾28'37			2047 Nov 08 08:21:59	0°♊	
evening max el	2047 Jan 29 18:46:53	28°☾11'49	18°25'22	morning set	2047 Nov 20 06:56:18	18°♊48'26	
	2047 Jan 31 19:33:49	0°☾		desc. node	2047 Nov 27 06:37:30	29°♊54'45	
retrograde	2047 Feb 05 07:01:07	1°☾45'04			2047 Nov 27 07:57:15	0°☾	
evening set	2047 Feb 08 07:50:15	0°☾54'29					
max. declination.	2047 Feb 08 18:56:47	0°☾38'11	8S33'35"	superior conj	2047 Dec 06 04:58:54	13°♊58'36	-0°57'28
	2047 Feb 09 16:50:52	30°☾		minimum elong	2047 Dec 05 21:49:17	13°♊30'29	0°56'35
inferior conj	2047 Feb 14 05:22:01	25°☾25'18	3°40'56	max. Earth dist.	2047 Dec 06 12:47:46	14°♊29'16	1.45142 AU
minimum elong	2047 Feb 14 04:24:13	25°☾28'05	3°40'53		2047 Dec 16 09:26:24	0°☾	
min. Earth dist.	2047 Feb 16 10:35:50	22°☾51'35	0.64229 AU	min. declination.	2047 Dec 19 02:16:37	4°☾16'29	25S19'32"
morning rise	2047 Feb 20 00:21:52	19°☾18'08		evening rise	2047 Dec 22 02:41:14	9°☾03'50	
direct	2047 Feb 26 21:04:36	16°☾28'53		greatest brilliancy	2047 Dec 30 20:54:17	22°☾58'26	-0.8m
min. declination.	2047 Mar 05 18:05:37	18°☾46'58	14S42'32"		2048 Jan 04 09:54:16	0°☾	
desc. node	2047 Mar 08 08:44:53	20°☾38'01		evening max el	2048 Jan 13 04:57:10	11°☾38'56	18°59'47
morning max el	2047 Mar 12 11:53:15	24°☾21'19	27°28'40	asc. node	2048 Jan 15 23:24:00	14°☾03'11	
	2047 Mar 17 14:28:43	0°☾		retrograde	2048 Jan 20 01:47:25	15°☾32'30	
	2047 Apr 06 21:46:25	0°☾		evening set	2048 Jan 23 08:44:39	14°☾29'49	
morning set	2047 Apr 16 14:34:22	17°☾55'04		max. declination.	2048 Jan 24 23:06:59	13°☾15'59	14S20' 5"
max. Earth dist.	2047 Apr 21 12:21:03	27°☾51'24	1.33392 AU	inferior conj	2048 Jan 28 23:49:52	8°☾43'51	3°25'06
	2047 Apr 22 12:59:10	0°☾		minimum elong	2048 Jan 28 21:44:54	8°☾50'24	3°24'46
				min. Earth dist.	2048 Jan 30 13:42:42	6°☾44'57	0.65716 AU
superior conj	2047 Apr 24 14:17:26	4°☾20'28	-0°25'47	morning rise	2048 Feb 03 10:25:24	2°☾32'26	
minimum elong	2047 Apr 24 15:34:41	4°☾27'19	0°25'31		2048 Feb 07 18:20:47	30°☾	
asc. node	2047 Apr 27 01:40:28	9°☾37'55		direct	2048 Feb 09 22:44:33	29°☾41'43	
evening rise	2047 May 01 18:52:40	19°☾41'55			2048 Feb 12 05:33:56	0°☾	
	2047 May 06 20:31:01	0°☾		min. declination.	2048 Feb 18 17:51:58	3°☾34'17	18S34'21"
max. declination.	2047 May 21 12:12:00	22°☾38'22	25N29'57"	morning max el	2048 Feb 22 21:50:01	7°☾20'15	26°36'44
evening max el	2047 May 23 10:39:34	24°☾38'36	22°28'48	desc. node	2048 Feb 23 05:47:01	7°☾40'25	
	2047 May 31 02:38:05	0°☾			2048 Mar 11 12:08:00	0°☾	
desc. node	2047 Jun 04 07:57:49	1°☾04'03			2048 Mar 29 06:23:23	0°☾	
retrograde	2047 Jun 05 12:04:40	1°☾07'11		morning set	2048 Mar 29 21:32:26	1°☾11'27	
evening set	2047 Jun 08 13:27:51	0°☾46'34		max. Earth dist.	2048 Apr 03 02:48:37	9°☾22'34	1.34649 AU
	2047 Jun 11 03:04:58	30°☾					
min. Earth dist.	2047 Jun 16 17:53:33	27°☾13'44	0.55230 AU	superior conj	2048 Apr 07 13:55:10	18°☾25'29	-0°53'47
inferior conj	2047 Jun 17 20:33:59	26°☾35'55	-3°36'56	minimum elong	2048 Apr 07 16:38:33	18°☾39'33	0°53'16
minimum elong	2047 Jun 17 12:37:53	26°☾47'11	3°34'51	asc. node	2048 Apr 12 22:42:25	29°☾37'58	
morning rise	2047 Jun 26 13:55:21	22°☾44'16			2048 Apr 13 02:54:46	0°☾	
min. declination.	2047 Jun 27 08:44:03	22°☾35'16	18N38'12"	evening rise	2048 Apr 15 04:01:55	4°☾16'01	
direct	2047 Jun 29 12:35:20	22°☾24'31			2048 Apr 29 16:21:05	0°☾	
morning max el	2047 Jul 10 23:00:34	27°☾43'24	21°05'56	evening max el	2048 May 04 09:31:29	5°☾26'34	21°00'00
	2047 Jul 13 04:57:26	0°☾		max. declination.	2048 May 09 00:51:19	9°☾04'09	24N10'13"
max. declination.	2047 Jul 23 14:33:37	15°☾15'52	22N28'22"	retrograde	2048 May 16 01:31:46	11°☾07'43	
asc. node	2047 Jul 24 00:56:29	16°☾02'54		evening set	2048 May 18 06:01:59	10°☾56'28	
morning set	2047 Jul 30 16:47:45	29°☾07'31		desc. node	2048 May 21 04:57:44	10°☾08'25	
	2047 Jul 31 02:56:58	0°☾		inferior conj	2048 May 27 14:15:35	6°☾58'17	-1°50'42
				minimum elong	2048 May 27 09:04:52	7°☾05'37	1°48'53
superior conj	2047 Aug 07 02:46:12	14°☾34'28	1°43'29	min. Earth dist.	2048 May 28 06:12:27	6°☾35'42	0.54915 AU
minimum elong	2047 Aug 07 01:41:25	14°☾28'54	1°43'26	morning rise	2048 Jun 05 12:01:34	2°☾55'26	
max. Earth dist.	2047 Aug 11 11:24:33	23°☾24'17	1.35494 AU	direct	2048 Jun 08 21:56:45	2°☾30'17	
	2047 Aug 14 20:50:48	0°☾		min. declination.	2048 Jun 10 16:08:46	2°☾37'12	16N35'47"
evening rise	2047 Aug 15 16:11:18	1°☾31'37		morning max el	2048 Jun 21 20:34:50	8°☾41'32	22°41'42
desc. node	2047 Aug 31 07:19:24	28°☾07'37			2048 Jul 07 01:42:30	0°☾	
	2047 Sep 01 13:03:43	0°☾		asc. node	2048 Jul 09 21:58:51	5°☾24'00	
evening max el	2047 Sep 20 04:03:13	23°☾31'51	26°30'52	max. declination.	2048 Jul 12 17:34:41	11°☾04'14	23N29' 8"
	2047 Sep 29 07:04:08	0°☾		morning set	2048 Jul 14 03:23:02	13°☾59'54	
retrograde	2047 Oct 03 00:45:30	0°☾44'51					
min. declination.	2047 Oct 02 21:56:31	0°☾44'48	15S 8'45"	superior conj	2048 Jul 21 06:34:42	29°☾11'47	1°34'58
	2047 Oct 06 12:31:06	30°☾		minimum elong	2048 Jul 21 04:34:09	29°☾01'07	1°34'48
evening set	2047 Oct 09 12:59:45	28°☾04'32			2048 Jul 21 15:39:34	0°☾	
min. Earth dist.	2047 Oct 13 15:14:13	23°☾49'29	0.66176 AU	max. Earth dist.	2048 Jul 24 05:42:54	5°☾26'00	1.34064 AU

evening rise	2048 Jul 29 02:02:24	15°♎14'40		morning rise	2049 May 16 07:33:10	12°♊55'58	
	2048 Aug 06 00:32:42	0°♎		direct	2049 May 20 13:57:03	12°♊15'34	
desc. node	2048 Aug 17 04:18:52	17°♎55'30		min. declination.	2049 May 24 07:22:50	12°♊47'06	12N20'50"
	2048 Aug 26 05:12:32	0°♎		morning max el	2049 Jun 03 11:44:06	19°♊12'00	24°23'52
evening max el	2048 Sep 01 16:05:20	7°♎00'05	27°12'02		2049 Jun 12 12:53:16	0°♊	
retrograde	2048 Sep 15 01:08:58	14°♎18'43		asc. node	2049 Jun 26 19:01:24	25°♊05'47	
min. declination.	2048 Sep 15 21:36:58	14°♎16'24	9S18'44"	morning set	2049 Jun 28 15:11:45	28°♊56'07	
evening set	2048 Sep 22 00:09:29	11°♎34'58			2049 Jun 29 03:16:07	0°♊	
min. Earth dist.	2048 Sep 25 19:49:27	7°♎55'52	0.64925 AU	max. declination.	2049 Jul 03 00:35:14	8°♊22'46	24N11'45"
inferior conj	2048 Sep 27 23:40:50	5°♎32'12	-2°34'28				
minimum elong	2048 Sep 28 03:47:12	5°♎20'49	2°32'58	superior conj	2049 Jul 05 15:03:58	14°♊02'35	1°21'14
morning rise	2048 Oct 04 08:19:57	0°♎00'18		minimum elong	2049 Jul 05 12:40:23	13°♊49'36	1°20'55
	2048 Oct 04 08:35:34	30°♎		max. Earth dist.	2049 Jul 07 08:58:23	17°♊49'07	1.33040 AU
asc. node	2048 Oct 05 21:11:42	29°♎29'22		evening rise	2049 Jul 12 22:28:46	29°♊29'03	
direct	2048 Oct 07 00:29:40	29°♎22'27			2049 Jul 13 04:35:03	0°♊	
	2048 Oct 09 16:42:32	0°♎			2049 Jul 30 02:28:47	0°♎	
max. declination.	2048 Oct 09 21:31:59	0°♎05'45	0N55'25"	desc. node	2049 Aug 04 01:18:10	7°♎09'19	
morning max el	2048 Oct 13 12:38:47	2°♎51'27	18°04'29	evening max el	2049 Aug 15 02:45:27	20°♎07'13	27°26'05
morning set	2048 Oct 31 10:53:40	29°♎51'41		retrograde	2049 Aug 28 19:21:16	27°♎24'47	
	2048 Oct 31 12:53:26	0°♎		min. declination.	2049 Aug 30 17:29:17	27°♎13'54	2S48' 9"
desc. node	2048 Nov 13 03:37:55	20°♎38'41		evening set	2049 Sep 05 00:01:52	24°♎48'24	
				min. Earth dist.	2049 Sep 08 15:04:42	21°♎41'34	0.63327 AU
superior conj	2048 Nov 14 11:42:02	22°♎46'50	-0°09'00	inferior conj	2049 Sep 11 07:23:54	19°♎01'03	-3°30'11
minimum elong	2048 Nov 14 10:34:54	22°♎42'22	0°08'51	minimum elong	2049 Sep 11 12:35:51	18°♎48'02	3°28'38
behind sun begin	2048 Nov 14 01:32:01	22°♎06'16		morning rise	2049 Sep 18 02:26:19	13°♎46'04	
behind sun end	2048 Nov 14 19:37:47	23°♎18'27		direct	2049 Sep 20 14:09:23	13°♎16'28	
max. Earth dist.	2048 Nov 18 07:31:27	28°♎50'57	1.44887 AU	asc. node	2049 Sep 22 18:14:37	13°♎39'28	
	2048 Nov 19 01:02:50	0°♎		max. declination.	2049 Sep 24 08:18:32	14°♎24'49	6N31'20"
evening rise	2048 Nov 30 21:17:37	18°♎27'19		morning max el	2049 Sep 27 04:52:36	16°♎42'15	17°51'50
	2048 Dec 08 09:17:16	0°♎			2049 Oct 06 15:25:03	0°♎	
min. declination.	2048 Dec 09 23:38:48	2°♎26'24	25S38'29"	morning set	2049 Oct 13 16:06:04	12°♎08'28	
greatest brilliancy	2048 Dec 14 10:35:25	9°♎09'36	-0.7m		2049 Oct 24 01:37:19	0°♎	
evening max el	2048 Dec 26 10:57:44	25°♎08'14	19°50'31				
asc. node	2049 Jan 01 20:26:58	29°♎23'54		superior conj	2049 Oct 25 16:33:11	2°♎42'13	0°35'46
retrograde	2049 Jan 02 23:14:16	29°♎30'48		minimum elong	2049 Oct 25 20:03:05	2°♎56'44	0°35'18
evening set	2049 Jan 06 13:48:21	28°♎14'42		desc. node	2049 Oct 31 00:39:12	11°♎25'57	
max. declination.	2049 Jan 11 11:34:19	22°♎58'30	18S40' 9"	max. Earth dist.	2049 Nov 01 00:22:56	13°♎01'22	1.43937 AU
inferior conj	2049 Jan 12 00:50:31	22°♎14'33	2°57'41	evening rise	2049 Nov 10 05:46:18	27°♎31'22	
minimum elong	2049 Jan 11 22:15:26	22°♎23'09	2°57'03		2049 Nov 11 20:24:54	0°♎	
min. Earth dist.	2049 Jan 13 00:34:15	20°♎55'42	0.66801 AU	min. declination.	2049 Dec 01 22:10:43	29°♎27'28	25S51' 1"
morning rise	2049 Jan 17 06:28:44	16°♎01'22			2049 Dec 02 07:51:45	0°♎	
direct	2049 Jan 23 06:19:03	13°♎21'00		evening max el	2049 Dec 09 11:40:17	8°♎38'57	20°54'58
morning max el	2049 Feb 04 06:45:23	20°♎30'58	25°22'38	retrograde	2049 Dec 17 20:59:06	13°♎36'53	
min. declination.	2049 Feb 05 05:47:12	21°♎30'19	21S 6'20"	asc. node	2049 Dec 19 17:29:17	13°♎17'43	
desc. node	2049 Feb 09 02:49:10	25°♎51'56		evening set	2049 Dec 21 21:02:54	12°♎05'43	
	2049 Feb 12 10:55:06	0°♎		inferior conj	2049 Dec 27 05:58:38	5°♎54'21	2°21'19
	2049 Mar 04 16:16:49	0°♎		minimum elong	2049 Dec 27 03:26:05	6°♎03'05	2°20'29
morning set	2049 Mar 12 12:34:51	13°♎36'36		min. Earth dist.	2049 Dec 27 17:03:12	5°♎16'20	0.67494 AU
max. Earth dist.	2049 Mar 16 05:27:15	20°♎26'23	1.36352 AU	morning rise	2050 Jan 01 09:37:49	29°♎41'17	
	2049 Mar 21 04:32:04	0°♎			2050 Jan 01 01:29:52	30°♎	
superior conj	2049 Mar 22 04:45:11	1°♎59'50	-1°20'35	max. declination.	2050 Jan 03 02:26:20	28°♎24'39	20S13'30"
minimum elong	2049 Mar 22 08:39:32	2°♎19'15	1°19'59	direct	2050 Jan 06 18:51:33	27°♎19'20	
evening rise	2049 Mar 30 08:46:56	18°♎31'21			2050 Jan 13 08:19:20	0°♎	
asc. node	2049 Mar 30 19:44:27	19°♎26'41		morning max el	2050 Jan 17 15:04:45	3°♎47'10	23°56'32
	2049 Apr 05 05:43:17	0°♎		min. declination.	2050 Jan 26 14:29:07	14°♎22'01	22S36' 9"
evening max el	2049 Apr 16 20:09:14	16°♎52'20	19°47'17	desc. node	2050 Jan 26 23:52:25	14°♎52'48	
max. declination.	2049 Apr 24 04:38:31	21°♎21'04	20N47'52"		2050 Feb 06 16:59:59	0°♎	
retrograde	2049 Apr 26 18:19:51	21°♎39'51		morning set	2050 Feb 22 05:44:45	24°♎54'05	
evening set	2049 Apr 28 18:52:05	21°♎28'48			2050 Feb 25 04:00:28	0°♎	
inferior conj	2049 May 07 14:01:24	17°♎27'17	0°08'40	max. Earth dist.	2050 Feb 26 00:59:32	1°♎32'59	1.38396 AU
minimum elong	2049 May 07 14:24:29	17°♎26'41	0°08'32				
transit middle	2049 May 07 14:24:29	17°♎26'41	0°08'32	superior conj	2050 Mar 05 07:13:24	14°♎54'11	-1°43'49
transit begin	2049 May 07 11:04:00	17°♎31'51		minimum elong	2050 Mar 05 11:30:02	15°♎14'29	1°43'22
transit end	2049 May 07 17:44:58	17°♎21'30			2050 Mar 13 01:51:47	0°♎	
desc. node	2049 May 08 01:58:09	17°♎08'45		evening rise	2050 Mar 14 06:35:45	2°♎20'58	
min. Earth dist.	2049 May 09 20:06:46	16°♎03'52	0.55570 AU	asc. node	2050 Mar 17 16:46:52	8°♎59'05	
				evening max el	2050 Mar 30 17:47:02	28°♎57'22	18°54'01

	2050 Mar 31 21:29:59	0°♄		morning set	2051 Feb 02 19:13:50	4°♄49'59	
retrograde	2050 Apr 08 02:27:58	3°♄01'09		max. Earth dist.	2051 Feb 07 21:36:40	13°♄18'58	1.40548 AU
max. declination.	2050 Apr 07 22:54:21	3°♄01'04	15N31'20"				
evening set	2050 Apr 10 07:32:34	2°♄45'15		superior conj	2051 Feb 15 16:44:45	26°♄56'52	-2°00'04
	2050 Apr 16 02:13:12	30°♄		minimum elong	2051 Feb 15 19:43:00	27°♄10'15	1°59'56
inferior conj	2050 Apr 18 07:20:57	28°♄27'53	1°49'14		2051 Feb 17 09:10:14	0°♄	
minimum elong	2050 Apr 18 11:01:57	28°♄21'17	1°48'05	evening rise	2051 Feb 25 18:20:20	15°♄37'06	
min. Earth dist.	2050 Apr 21 12:10:27	26°♄11'27	0.57049 AU	asc. node	2051 Mar 04 13:49:53	28°♄09'13	
desc. node	2050 Apr 24 23:01:21	24°♄05'08			2051 Mar 05 15:28:33	0°♄	
morning rise	2050 Apr 26 11:25:18	23°♄22'28		evening max el	2051 Mar 13 23:34:49	11°♄34'46	18°20'54
direct	2050 May 01 19:09:55	22°♄14'32		retrograde	2051 Mar 21 05:56:02	15°♄11'04	
min. declination.	2050 May 06 11:40:12	23°♄06'42	6N36' 0"	max. declination.	2051 Mar 22 11:46:36	15°♄04'54	9N 3'13"
morning max el	2050 May 16 02:35:57	29°♄41'00	25°57'56	evening set	2051 Mar 23 16:56:36	14°♄47'21	
	2050 May 16 10:24:03	0°♄		inferior conj	2051 Mar 30 22:05:17	10°♄09'34	2°56'40
	2050 Jun 06 04:17:58	0°♄		minimum elong	2051 Mar 31 01:44:42	10°♄01'53	2°55'53
morning set	2050 Jun 13 02:33:59	13°♄50'18		min. Earth dist.	2051 Apr 03 08:32:36	7°♄18'09	0.59033 AU
asc. node	2050 Jun 13 16:02:56	15°♄01'38		morning rise	2051 Apr 07 07:55:39	4°♄35'29	
				desc. node	2051 Apr 11 20:04:27	2°♄59'42	
superior conj	2050 Jun 20 02:05:53	28°♄59'14	1°03'09	direct	2051 Apr 13 13:37:22	2°♄51'43	
minimum elong	2050 Jun 19 23:49:18	28°♄46'44	1°02'46	min. declination.	2051 Apr 18 18:26:29	3°♄59'49	0N10'34"
	2050 Jun 20 13:10:02	0°♄		morning max el	2051 Apr 27 22:21:58	10°♄36'08	27°08'40
max. Earth dist.	2050 Jun 20 18:34:25	0°♄29'39	1.32417 AU		2051 May 12 22:14:11	0°♄	
max. declination.	2050 Jun 23 10:13:34	6°♄16'36	24N44'28"	morning set	2051 May 28 11:52:52	28°♄37'11	
evening rise	2050 Jun 27 02:14:13	14°♄04'00			2051 May 29 03:39:53	0°♄	
	2050 Jul 05 05:52:28	0°♄		asc. node	2051 May 31 13:06:18	5°♄06'38	
desc. node	2050 Jul 21 22:19:55	25°♄34'09					
	2050 Jul 25 18:35:00	0°♄		superior conj	2051 Jun 04 13:54:50	13°♄55'55	0°41'30
evening max el	2050 Jul 28 09:24:38	2°♄38'41	27°08'00	minimum elong	2051 Jun 04 12:11:55	13°♄46'28	0°41'09
retrograde	2050 Aug 11 06:22:20	9°♄52'45		max. Earth dist.	2051 Jun 04 06:55:49	13°♄17'28	1.32181 AU
min. declination.	2050 Aug 14 14:28:34	9°♄22'29	3N58'34"	evening rise	2051 Jun 11 10:46:49	28°♄51'25	
evening set	2050 Aug 18 08:29:24	7°♄36'31			2051 Jun 11 23:49:19	0°♄	
min. Earth dist.	2050 Aug 21 23:40:24	4°♄52'55	0.61437 AU	max. declination.	2051 Jun 14 01:03:55	4°♄14'19	25N10'49"
inferior conj	2050 Aug 25 02:35:55	2°♄07'51	-4°18'24		2051 Jun 28 15:38:04	0°♄	
minimum elong	2050 Aug 25 07:44:24	1°♄56'28	4°17'21	desc. node	2051 Jul 08 19:20:42	12°♄49'41	
	2050 Aug 27 14:50:33	30°♄		evening max el	2051 Jul 10 09:24:18	14°♄24'07	26°16'50
morning rise	2050 Sep 01 08:37:36	27°♄13'14		retrograde	2051 Jul 24 09:02:07	21°♄35'20	
direct	2050 Sep 03 18:42:00	26°♄48'57		min. declination.	2051 Jul 29 23:03:22	20°♄18'00	10N28'12"
max. declination.	2050 Sep 08 21:17:02	28°♄41'29	11N44'23"	evening set	2051 Jul 30 20:50:01	19°♄51'28	
asc. node	2050 Sep 09 15:17:27	29°♄16'31		min. Earth dist.	2051 Aug 03 22:15:08	17°♄15'32	0.59381 AU
	2050 Sep 10 11:01:42	0°♄		inferior conj	2051 Aug 07 05:42:50	14°♄43'15	-4°51'11
morning max el	2050 Sep 10 19:15:28	0°♄19'38	17°57'45	minimum elong	2051 Aug 07 08:41:11	14°♄37'31	4°50'55
morning set	2050 Sep 26 16:03:28	25°♄18'36		morning rise	2051 Aug 14 22:39:22	10°♄11'32	
	2050 Sep 29 05:31:01	0°♄		direct	2051 Aug 17 09:41:01	9°♄50'28	
				morning max el	2051 Aug 25 04:40:41	13°♄35'36	18°23'25
superior conj	2050 Oct 07 00:31:32	13°♄57'10	1°09'54	max. declination.	2051 Aug 24 19:38:27	13°♄14'30	16N10'44"
minimum elong	2050 Oct 07 05:03:28	14°♄16'56	1°09'23	asc. node	2051 Aug 27 12:20:16	16°♄06'30	
max. Earth dist.	2050 Oct 14 13:05:02	26°♄43'46	1.42402 AU		2051 Sep 05 09:01:36	0°♄	
	2050 Oct 16 12:56:25	0°♄		morning set	2051 Sep 10 05:22:40	9°♄06'53	
desc. node	2050 Oct 17 21:40:44	2°♄12'41					
evening rise	2050 Oct 20 20:52:12	6°♄57'06		superior conj	2051 Sep 19 08:14:09	26°♄22'07	1°32'00
	2050 Nov 05 02:43:30	0°♄		minimum elong	2051 Sep 19 11:33:38	26°♄37'25	1°31'43
evening max el	2050 Nov 22 06:50:56	22°♄10'42	22°09'30		2051 Sep 21 07:58:02	0°♄	
min. declination.	2050 Nov 23 01:37:53	22°♄57'14	25S43'47"	max. Earth dist.	2051 Sep 26 20:13:21	9°♄43'57	1.40476 AU
retrograde	2050 Dec 01 17:19:10	27°♄47'27		evening rise	2051 Oct 01 07:51:21	17°♄18'02	
evening set	2050 Dec 06 04:43:35	25°♄59'49		desc. node	2051 Oct 04 18:42:28	22°♄55'30	
asc. node	2050 Dec 06 14:31:49	25°♄39'00			2051 Oct 09 06:29:17	0°♄	
inferior conj	2050 Dec 11 13:08:57	19°♄40'31	1°37'46		2051 Oct 30 17:19:13	0°♄	
minimum elong	2050 Dec 11 11:07:15	19°♄47'32	1°36'59	evening max el	2051 Nov 04 21:30:31	5°♄44'29	23°29'10
min. Earth dist.	2050 Dec 11 12:47:44	19°♄41'44	0.67813 AU	min. declination.	2051 Nov 11 06:44:05	10°♄48'24	24S40' 0"
morning rise	2050 Dec 16 17:22:47	13°♄29'22		retrograde	2051 Nov 15 10:50:24	11°♄59'03	
direct	2050 Dec 21 11:32:13	11°♄29'48		evening set	2051 Nov 20 11:12:02	9°♄54'21	
max. declination.	2050 Dec 20 18:44:19	11°♄32'14	19S14'58"	asc. node	2051 Nov 23 11:34:54	6°♄40'52	
morning max el	2050 Dec 31 01:45:45	17°♄09'37	22°28'22	inferior conj	2051 Nov 25 20:31:55	3°♄30'53	0°48'26
	2051 Jan 10 15:13:57	0°♄		minimum elong	2051 Nov 25 19:24:48	3°♄34'43	0°47'59
desc. node	2051 Jan 13 20:56:01	4°♄28'53		min. Earth dist.	2051 Nov 25 09:29:15	4°♄08'42	0.67783 AU
min. declination.	2051 Jan 18 03:34:14	10°♄37'34	23S32' 2"		2051 Nov 28 13:03:55	30°♄	
	2051 Jan 30 19:24:20	0°♄		morning rise	2051 Dec 01 03:35:25	27°♄23'34	

direct	2051 Dec 05 07:24:43	25° $\mathbb{M}$ 46'51		min. declination.	2052 Oct 27 00:19:27	25° $\mathbb{M}$ 52'04	22S 7' 9"
max. declination.	2051 Dec 05 21:12:10	25° $\mathbb{M}$ 48'34	16S36' 1"	retrograde	2052 Oct 29 00:29:43	26° $\mathbb{M}$ 07'47	
	2051 Dec 13 01:18:34	0° $\mathbb{A}$		evening set	2052 Nov 03 14:49:36	23° $\mathbb{M}$ 46'52	
morning max el	2051 Dec 13 18:13:05	0° $\mathbb{A}$ 41'28	21°06'12	min. Earth dist.	2052 Nov 08 04:46:56	18° $\mathbb{M}$ 35'01	0.67420 AU
desc. node	2051 Dec 31 17:58:36	24° $\mathbb{A}$ 30'06		inferior conj	2052 Nov 09 02:22:21	17° $\mathbb{M}$ 23'30	-0°05'23
	2052 Jan 04 10:23:11	0° $\mathbb{B}$		minimum elong	2052 Nov 09 02:30:16	17° $\mathbb{M}$ 23'03	0°05'19
min. declination.	2052 Jan 09 21:41:42	8° $\mathbb{B}$ 20'46	24S11'12"	transit middle	2052 Nov 09 02:30:16	17° $\mathbb{M}$ 23'03	0°05'19
morning set	2052 Jan 13 04:47:56	13° $\mathbb{B}$ 28'31		transit begin	2052 Nov 08 23:54:00	17° $\mathbb{M}$ 31'42	
max. Earth dist.	2052 Jan 21 00:22:57	25° $\mathbb{B}$ 58'46	1.42521 AU	transit end	2052 Nov 09 05:06:31	17° $\mathbb{M}$ 14'26	
	2052 Jan 23 11:01:18	0° $\mathbb{A}$		asc. node	2052 Nov 09 08:36:52	17° $\mathbb{M}$ 02'50	
				morning rise	2052 Nov 14 14:18:24	11° $\mathbb{M}$ 22'17	
superior conj	2052 Jan 28 03:59:09	7° $\mathbb{A}$ 54'54	-2°04'51	direct	2052 Nov 18 05:24:37	10° $\mathbb{M}$ 06'26	
minimum elong	2052 Jan 28 03:07:42	7° $\mathbb{A}$ 51'15	2°04'51	max. declination.	2052 Nov 19 13:08:34	10° $\mathbb{M}$ 15'52	12S39'49"
evening rise	2052 Feb 08 16:07:52	28° $\mathbb{A}$ 10'29		morning max el	2052 Nov 25 18:18:25	14° $\mathbb{M}$ 24'14	19°55'35
	2052 Feb 09 16:28:43	0° $\mathbb{H}$			2052 Dec 07 18:07:47	0° $\mathbb{A}$	
asc. node	2052 Feb 19 10:53:02	16° $\mathbb{H}$ 49'10		desc. node	2052 Dec 17 15:01:07	14° $\mathbb{A}$ 49'39	
evening max el	2052 Feb 25 10:15:03	24° $\mathbb{H}$ 34'56	18°07'37	morning set	2052 Dec 22 00:51:35	21° $\mathbb{A}$ 36'21	
retrograde	2052 Mar 03 01:51:45	27° $\mathbb{H}$ 59'28			2052 Dec 27 10:27:56	0° $\mathbb{B}$	
max. declination.	2052 Mar 05 05:46:36	27° $\mathbb{H}$ 37'44	2N 5'28"	min. declination.	2052 Dec 31 16:01:52	6° $\mathbb{B}$ 39'43	24S41'29"
evening set	2052 Mar 05 18:17:39	27° $\mathbb{H}$ 26'14		max. Earth dist.	2053 Jan 02 10:27:49	9° $\mathbb{B}$ 28'17	1.44053 AU
inferior conj	2052 Mar 12 08:15:28	22° $\mathbb{H}$ 27'45	3°32'27				
minimum elong	2052 Mar 12 10:10:35	22° $\mathbb{H}$ 23'06	3°32'15	superior conj	2053 Jan 07 13:08:01	17° $\mathbb{B}$ 41'48	-1°53'10
min. Earth dist.	2052 Mar 15 11:56:45	19° $\mathbb{H}$ 25'44	0.61180 AU	minimum elong	2053 Jan 07 06:48:11	17° $\mathbb{B}$ 16'06	1°52'46
morning rise	2052 Mar 19 00:18:59	16° $\mathbb{H}$ 34'55			2053 Jan 14 23:50:40	0° $\mathbb{A}$	
direct	2052 Mar 25 19:24:18	14° $\mathbb{H}$ 16'28		evening rise	2053 Jan 20 19:39:32	9° $\mathbb{A}$ 53'07	
desc. node	2052 Mar 28 17:07:59	14° $\mathbb{H}$ 40'02			2053 Feb 01 21:31:26	0° $\mathbb{H}$	
min. declination.	2052 Mar 31 10:24:13	15° $\mathbb{H}$ 41'23	6S10'44"	asc. node	2053 Feb 05 07:55:03	4° $\mathbb{H}$ 49'55	
morning max el	2052 Apr 09 00:27:57	22° $\mathbb{H}$ 10'45	27°45'28	evening max el	2053 Feb 07 22:50:07	7° $\mathbb{H}$ 50'34	18°13'32
	2052 Apr 15 21:42:39	0° $\mathbb{Y}$		retrograde	2053 Feb 14 09:45:44	11° $\mathbb{H}$ 17'01	
	2052 May 05 00:00:58	0° $\mathbb{B}$		evening set	2053 Feb 17 07:24:36	10° $\mathbb{H}$ 33'06	
morning set	2052 May 11 17:12:49	13° $\mathbb{B}$ 09'31		max. declination.	2053 Feb 17 08:57:19	10° $\mathbb{H}$ 31'10	4S47'32"
asc. node	2052 May 17 10:09:17	25° $\mathbb{B}$ 15'33		inferior conj	2053 Feb 23 09:59:31	5° $\mathbb{H}$ 14'55	3°43'19
max. Earth dist.	2052 May 17 17:57:32	25° $\mathbb{B}$ 57'59	1.32325 AU	minimum elong	2053 Feb 23 09:57:34	5° $\mathbb{H}$ 15'00	3°43'19
				min. Earth dist.	2053 Feb 26 00:16:33	2° $\mathbb{H}$ 25'33	0.63206 AU
superior conj	2052 May 19 00:45:26	28° $\mathbb{B}$ 46'04	0°16'56		2053 Feb 28 12:07:06	30° $\mathbb{R}$ $\mathbb{A}$	
minimum elong	2052 May 18 23:59:10	28° $\mathbb{B}$ 41'50	0°16'45	morning rise	2053 Mar 01 11:33:07	29° $\mathbb{A}$ 11'31	
	2052 May 19 14:15:56	0° $\mathbb{I}$		direct	2053 Mar 08 10:16:26	26° $\mathbb{A}$ 29'01	
evening rise	2052 May 25 21:53:54	13° $\mathbb{I}$ 43'15		min. declination.	2053 Mar 14 16:26:28	28° $\mathbb{A}$ 20'25	11S52'26"
	2052 Jun 03 02:24:39	0° $\mathbb{B}$		desc. node	2053 Mar 15 14:11:34	28° $\mathbb{A}$ 52'35	
max. declination.	2052 Jun 04 00:43:10	1° $\mathbb{B}$ 40'52	25N31' 7"		2053 Mar 17 05:56:51	0° $\mathbb{H}$	
evening max el	2052 Jun 21 02:17:46	25° $\mathbb{B}$ 23'52	24°58'07	morning max el	2053 Mar 22 07:30:04	4° $\mathbb{H}$ 25'00	27°44'48
desc. node	2052 Jun 24 16:21:47	28° $\mathbb{B}$ 27'45			2053 Apr 10 12:46:47	0° $\mathbb{Y}$	
	2052 Jun 26 23:08:48	0° $\mathbb{Q}$		morning set	2053 Apr 25 16:17:37	27° $\mathbb{Y}$ 19'50	
retrograde	2052 Jul 05 01:44:48	2° $\mathbb{Q}$ 29'35			2053 Apr 26 23:59:56	0° $\mathbb{B}$	
evening set	2052 Jul 10 09:35:47	1° $\mathbb{Q}$ 24'42		max. Earth dist.	2053 Apr 30 23:43:29	8° $\mathbb{B}$ 18'10	1.32877 AU
	2052 Jul 13 07:37:26	30° $\mathbb{R}$ $\mathbb{B}$					
min. declination.	2052 Jul 15 02:19:10	28° $\mathbb{B}$ 56'01	15N48'34"	superior conj	2053 May 03 08:52:33	13° $\mathbb{B}$ 23'51	-0°09'48
min. Earth dist.	2052 Jul 15 13:48:08	28° $\mathbb{B}$ 37'41	0.57395 AU	minimum elong	2053 May 03 09:21:12	13° $\mathbb{B}$ 26'25	0°09'42
inferior conj	2052 Jul 18 13:34:08	26° $\mathbb{B}$ 38'09	-4°56'29	behind sun begin	2053 May 03 05:08:56	13° $\mathbb{B}$ 03'47	
minimum elong	2052 Jul 18 11:51:24	26° $\mathbb{B}$ 41'02	4°56'24	behind sun end	2053 May 03 13:33:29	13° $\mathbb{B}$ 49'05	
morning rise	2052 Jul 26 16:46:37	22° $\mathbb{B}$ 28'29		asc. node	2053 May 04 07:11:38	15° $\mathbb{B}$ 24'17	
direct	2052 Jul 29 06:48:18	22° $\mathbb{B}$ 09'12		evening rise	2053 May 10 09:42:56	28° $\mathbb{B}$ 33'12	
morning max el	2052 Aug 07 06:02:22	26° $\mathbb{B}$ 20'59	19°10'01		2053 May 11 02:11:03	0° $\mathbb{I}$	
max. declination.	2052 Aug 09 22:38:06	29° $\mathbb{B}$ 13'58	19N29'52"	max. declination.	2053 May 26 08:56:37	27° $\mathbb{I}$ 16'46	25N39'22"
	2052 Aug 10 13:35:54	0° $\mathbb{Q}$			2053 May 28 09:40:12	0° $\mathbb{B}$	
asc. node	2052 Aug 13 09:23:03	3° $\mathbb{Q}$ 54'27		evening max el	2053 Jun 02 16:02:47	5° $\mathbb{B}$ 55'58	23°23'58
morning set	2052 Aug 24 04:04:24	23° $\mathbb{Q}$ 22'44		desc. node	2053 Jun 11 13:23:16	11° $\mathbb{B}$ 53'15	
	2052 Aug 27 11:54:00	0° $\mathbb{H}$		retrograde	2053 Jun 16 06:04:42	12° $\mathbb{B}$ 43'21	
				evening set	2053 Jun 20 01:56:30	12° $\mathbb{B}$ 11'14	
superior conj	2052 Sep 01 10:03:32	9° $\mathbb{H}$ 41'09	1°43'16	min. Earth dist.	2053 Jun 27 02:05:40	8° $\mathbb{B}$ 58'29	0.55818 AU
minimum elong	2052 Sep 01 11:29:38	9° $\mathbb{H}$ 48'06	1°43'12	inferior conj	2053 Jun 29 01:28:25	7° $\mathbb{B}$ 48'20	-4°20'04
max. Earth dist.	2052 Sep 07 23:44:58	22° $\mathbb{H}$ 01'02	1.38416 AU	minimum elong	2053 Jun 28 18:42:39	7° $\mathbb{B}$ 58'23	4°18'48
evening rise	2052 Sep 11 18:58:34	28° $\mathbb{H}$ 46'33		min. declination.	2053 Jul 04 08:50:40	4° $\mathbb{B}$ 57'50	18N33'32"
	2052 Sep 12 11:54:23	0° $\mathbb{A}$		morning rise	2053 Jul 07 14:02:58	3° $\mathbb{B}$ 53'35	
desc. node	2052 Sep 20 15:43:03	13° $\mathbb{A}$ 30'31		direct	2053 Jul 10 08:48:25	3° $\mathbb{B}$ 34'37	
	2052 Oct 01 16:14:04	0° $\mathbb{M}$		morning max el	2053 Jul 20 20:41:43	8° $\mathbb{B}$ 26'23	20°17'53
evening max el	2052 Oct 17 09:49:36	19° $\mathbb{M}$ 22'10	24°47'40	max. declination.	2053 Jul 29 05:23:02	19° $\mathbb{B}$ 03'40	21N38'11"

asc. node	2053 Jul 31 06:25:43	22°♄26'55		minimum elong	2054 Jun 08 15:06:44	18°♄32'35	2°53'28
	2053 Aug 04 09:13:51	0°♄		min. Earth dist.	2054 Jun 08 13:48:46	18°♄34'24	0.54981 AU
morning set	2053 Aug 08 09:07:43	7°♄57'45		morning rise	2054 Jun 17 18:33:06	14°♄28'05	
				direct	2054 Jun 20 21:18:22	14°♄06'41	
superior conj	2053 Aug 16 00:54:25	23°♄39'15	1°45'32	min. declination.	2054 Jun 20 17:02:33	14°♄06'45	18N 5' 0"
minimum elong	2053 Aug 16 00:36:44	23°♄37'46	1°45'32	morning max el	2054 Jul 02 23:53:01	19°♄47'52	21°45'03
	2053 Aug 19 05:18:57	0°♄			2054 Jul 11 08:35:28	0°♄	
max. Earth dist.	2053 Aug 21 05:28:01	3°♄53'50	1.36479 AU	asc. node	2054 Jul 18 03:28:16	11°♄33'20	
evening rise	2053 Aug 25 03:49:35	11°♄16'50		max. declination.	2054 Jul 19 00:38:35	13°♄14'17	22N56'34"
	2053 Sep 05 01:44:20	0°♄		morning set	2054 Jul 23 18:18:31	22°♄46'11	
desc. node	2053 Sep 07 12:42:30	3°♄52'27			2054 Jul 27 04:50:29	0°♄	
	2053 Sep 26 23:17:51	0°♄					
evening max el	2053 Sep 29 21:50:16	3°♄02'00	25°57'41	superior conj	2054 Jul 31 00:55:15	8°♄05'25	1°40'34
min. declination.	2053 Oct 11 16:05:43	10°♄06'00	18S 4'37"	minimum elong	2054 Jul 30 23:22:45	7°♄57'21	1°40'29
retrograde	2053 Oct 12 09:18:24	10°♄07'52		max. Earth dist.	2054 Aug 03 18:51:06	15°♄48'55	1.34843 AU
evening set	2053 Oct 18 13:46:16	7°♄33'29		evening rise	2054 Aug 08 05:55:45	24°♄37'30	
min. Earth dist.	2053 Oct 22 20:13:53	2°♄57'09	0.66726 AU		2054 Aug 11 02:22:39	0°♄	
inferior conj	2053 Oct 24 04:50:33	1°♄14'46	-1°02'23	desc. node	2054 Aug 25 09:43:02	23°♄55'23	
minimum elong	2053 Oct 24 06:28:30	1°♄09'38	1°01'41		2054 Aug 29 14:12:51	0°♄	
	2053 Oct 25 04:53:04	30°♄		evening max el	2054 Sep 12 10:13:42	16°♄38'01	26°51'21
asc. node	2053 Oct 27 05:39:02	27°♄40'25		retrograde	2054 Sep 25 12:41:05	23°♄53'35	
morning rise	2053 Oct 29 23:31:39	25°♄22'23		min. declination.	2054 Sep 25 19:44:48	23°♄53'18	12S48'13"
direct	2053 Nov 02 03:57:14	24°♄24'09		evening set	2054 Oct 02 05:57:15	21°♄11'01	
max. declination.	2053 Nov 04 01:34:18	24°♄43'55	7S47'38"	min. Earth dist.	2054 Oct 06 05:24:55	17°♄11'16	0.65691 AU
morning max el	2053 Nov 09 01:42:42	28°♄15'16	18°59'34	inferior conj	2054 Oct 08 01:57:25	15°♄01'26	-2°00'55
	2053 Nov 10 16:31:46	0°♄		minimum elong	2054 Oct 08 05:11:42	14°♄51'59	1°59'38
morning set	2053 Dec 01 08:33:09	0°♄26'06		morning rise	2054 Oct 14 05:05:23	9°♄21'04	
	2053 Dec 01 01:53:03	0°♄		asc. node	2054 Oct 14 02:41:17	9°♄24'13	
desc. node	2053 Dec 04 12:01:21	5°♄21'37		direct	2054 Oct 17 01:07:42	8°♄36'43	
max. Earth dist.	2053 Dec 16 02:30:17	23°♄33'19	1.44956 AU	max. declination.	2054 Oct 19 12:43:24	9°♄09'53	2S21'17"
				morning max el	2054 Oct 23 14:40:06	12°♄10'54	18°19'42
superior conj	2053 Dec 17 22:50:12	26°♄27'57	-1°22'25		2054 Nov 05 05:42:36	0°♄	
minimum elong	2053 Dec 17 14:01:18	25°♄53'11	1°21'29	morning set	2054 Nov 11 19:48:13	10°♄40'24	
	2053 Dec 20 04:30:02	0°♄		desc. node	2054 Nov 21 09:01:17	26°♄02'08	
min. declination.	2053 Dec 23 10:38:46	5°♄10'22	25S 6'31"		2054 Nov 23 20:57:08	0°♄	
evening rise	2054 Jan 02 00:54:48	20°♄38'19					
	2054 Jan 07 18:39:05	0°♄		superior conj	2054 Nov 27 00:12:17	4°♄57'08	-0°37'09
evening max el	2054 Jan 22 10:36:38	21°♄14'58	18°37'57	minimum elong	2054 Nov 26 19:24:15	4°♄38'13	0°36'31
asc. node	2054 Jan 23 04:56:44	21°♄59'42		max. Earth dist.	2054 Nov 28 21:38:23	7°♄55'52	1.45130 AU
retrograde	2054 Jan 29 01:29:57	24°♄55'16			2054 Dec 12 23:47:22	0°♄	
evening set	2054 Feb 01 04:50:12	23°♄59'39		evening rise	2054 Dec 13 06:56:30	0°♄28'01	
max. declination.	2054 Feb 02 01:06:43	23°♄26'23	11S 6'52"	min. declination.	2054 Dec 15 06:40:07	3°♄34'56	25S27'49"
inferior conj	2054 Feb 06 23:19:19	18°♄23'19	3°35'49	greatest brilliancy	2054 Dec 24 07:06:14	17°♄37'41	-0.8m
minimum elong	2054 Feb 06 21:48:23	18°♄27'52	3°35'39		2055 Jan 01 19:36:24	0°♄	
min. Earth dist.	2054 Feb 08 22:00:48	16°♄03'18	0.64916 AU	evening max el	2055 Jan 05 19:14:55	4°♄43'15	19°19'32
morning rise	2054 Feb 12 14:19:10	12°♄14'12		asc. node	2055 Jan 10 01:58:50	8°♄04'36	
direct	2054 Feb 19 08:18:53	9°♄22'28		retrograde	2055 Jan 12 21:41:11	8°♄47'49	
min. declination.	2054 Feb 26 20:04:30	12°♄09'16	16S29'22"	evening set	2055 Jan 16 07:47:29	7°♄39'23	
desc. node	2054 Mar 02 11:13:48	15°♄02'23		max. declination.	2055 Jan 18 19:25:00	5°♄28'49	16S22'56"
morning max el	2054 Mar 04 16:55:26	17°♄10'15	27°09'48	inferior conj	2055 Jan 21 20:52:49	1°♄47'01	3°14'42
	2054 Mar 15 09:53:24	0°♄		minimum elong	2055 Jan 21 18:30:38	1°♄54'39	3°14'14
	2054 Apr 03 09:05:50	0°♄		min. Earth dist.	2055 Jan 23 04:29:11	0°♄05'07	0.66230 AU
morning set	2054 Apr 09 06:13:50	10°♄59'14			2055 Jan 23 06:05:25	30°♄	
max. Earth dist.	2054 Apr 13 20:33:07	20°♄08'18	1.33871 AU	morning rise	2055 Jan 27 04:59:51	25°♄34'52	
				direct	2055 Feb 02 12:25:29	22°♄47'31	
superior conj	2054 Apr 17 12:18:56	27°♄43'20	-0°37'41	min. declination.	2055 Feb 12 17:01:09	27°♄58'19	19S47'12"
minimum elong	2054 Apr 17 14:13:03	27°♄53'20	0°37'17		2055 Feb 14 20:32:10	0°♄	
	2054 Apr 18 14:14:17	0°♄		morning max el	2055 Feb 15 02:27:20	0°♄14'50	26°07'24
asc. node	2054 Apr 21 04:14:11	5°♄29'02		desc. node	2055 Feb 17 08:15:33	2°♄36'46	
evening rise	2054 Apr 24 20:25:38	13°♄15'24			2055 Mar 09 08:34:30	0°♄	
	2054 May 03 10:25:04	0°♄		morning set	2055 Mar 23 07:03:30	23°♄55'25	
evening max el	2054 May 15 09:50:04	16°♄30'52	21°49'32		2055 Mar 26 12:06:46	0°♄	
max. declination.	2054 May 16 18:45:45	17°♄46'39	25N 8'57"	max. Earth dist.	2055 Mar 27 06:04:52	1°♄26'52	1.35325 AU
retrograde	2054 May 27 22:30:58	22°♄40'21					
desc. node	2054 May 29 10:24:44	22°♄35'17		superior conj	2055 Apr 01 08:36:01	11°♄36'26	-1°05'24
evening set	2054 May 30 12:53:51	22°♄25'14		minimum elong	2055 Apr 01 11:52:55	11°♄53'07	1°04'50
inferior conj	2054 Jun 08 22:34:37	18°♄22'08	-2°55'47	asc. node	2055 Apr 08 01:16:31	25°♄24'59	

evening rise	2055 Apr 09 04:01:18	27° Υ 42'41		morning max el	2056 Jan 28 11:10:11	13° Z 29'11	24°47'01
	2055 Apr 10 06:54:24	0° B		min. declination.	2056 Feb 01 12:42:22	17° Z 55'58	21S49'40"
evening max el	2055 Apr 27 13:31:17	27° B 33'35	20°26'46	desc. node	2056 Feb 04 05:17:06	21° Z 11'20	
	2055 Apr 30 10:53:11	0° II			2056 Feb 10 20:45:34	0° $\approx$	
max. declination.	2055 May 03 16:54:26	1° II 51'25	23N 1' 6"		2056 Mar 01 04:12:33	0° K	
retrograde	2055 May 08 11:40:59	2° II 51'41		morning set	2056 Mar 04 13:37:51	5° K 54'10	
evening set	2055 May 10 13:05:26	2° II 41'10		max. Earth dist.	2056 Mar 08 05:20:11	12° K 29'32	1.37189 AU
desc. node	2055 May 16 07:27:12	0° II 36'49					
	2055 May 17 11:09:32	30° R B		superior conj	2056 Mar 14 18:40:43	24° K 55'03	-1°31'02
inferior conj	2055 May 19 17:03:02	28° B 43'14	-0°59'33	minimum elong	2056 Mar 14 22:53:17	25° K 15'35	1°30'28
minimum elong	2055 May 19 14:14:16	28° B 47'20	0°58'31		2056 Mar 17 08:38:26	0° Y	
min. Earth dist.	2055 May 21 02:23:58	27° B 54'47	0.55082 AU	evening rise	2056 Mar 23 06:13:07	11° Y 48'25	
morning rise	2055 May 28 14:19:19	24° B 30'22		asc. node	2056 Mar 24 22:20:14	15° Y 07'42	
direct	2055 Jun 01 07:57:50	24° B 00'16			2056 Apr 02 02:00:58	0° B	
min. declination.	2055 Jun 04 02:22:10	24° B 17'22	15N 2'36"	evening max el	2056 Apr 09 05:00:20	9° B 16'14	19°22'07
	2055 Jun 14 04:23:35	0° II		max. declination.	2056 Apr 17 01:27:29	13° B 38'12	18N46'41"
morning max el	2055 Jun 14 18:00:18	0° II 31'23	23°25'09	retrograde	2056 Apr 18 10:25:05	13° B 43'56	
	2055 Jul 04 11:40:07	0° S		evening set	2056 Apr 20 12:11:51	13° B 31'23	
asc. node	2055 Jul 05 00:30:50	1° S 04'06		inferior conj	2056 Apr 28 23:22:13	9° B 23'51	0°54'54
morning set	2055 Jul 08 05:39:18	7° S 41'04		minimum elong	2056 Apr 29 01:35:36	9° B 20'11	0°54'07
max. declination.	2055 Jul 09 06:02:51	9° S 50'01	23N48'22"	min. Earth dist.	2056 May 01 17:06:09	7° B 36'26	0.56113 AU
				desc. node	2056 May 02 04:28:55	7° B 18'32	
superior conj	2055 Jul 15 07:01:12	22° S 49'19	1°29'44	morning rise	2056 May 07 12:05:58	4° B 37'42	
minimum elong	2055 Jul 15 04:47:03	22° S 37'19	1°29'29	direct	2056 May 12 04:50:23	3° B 47'19	
max. Earth dist.	2055 Jul 17 17:24:40	28° S 00'29	1.33584 AU	min. declination.	2056 May 16 10:24:39	4° B 28'36	10N 4'16"
	2055 Jul 18 16:07:47	0° Q		morning max el	2056 May 26 08:32:29	10° B 58'29	25°05'48
evening rise	2055 Jul 22 20:44:28	8° Q 34'59			2056 Jun 09 19:37:19	0° II	
	2055 Aug 03 13:17:14	0° P		asc. node	2056 Jun 20 21:34:37	20° II 52'34	
desc. node	2055 Aug 12 06:44:49	13° P 30'16		morning set	2056 Jun 21 17:29:51	22° II 37'10	
	2055 Aug 25 22:32:40	0° A			2056 Jun 25 03:58:56	0° S	
evening max el	2055 Aug 25 21:56:50	29° P 58'35	27°21'34				
retrograde	2055 Sep 08 10:17:13	7° A 16'42		superior conj	2056 Jun 28 16:51:14	7° S 43'26	1°14'04
min. declination.	2055 Sep 09 17:10:11	7° A 11'35	6S38'19"	minimum elong	2056 Jun 28 14:27:11	7° S 30'19	1°13'41
evening set	2055 Sep 15 12:42:51	4° A 34'28		max. declination.	2056 Jun 28 14:01:31	7° S 27'59	24N26'18"
min. Earth dist.	2055 Sep 19 06:03:20	1° A 10'05	0.64293 AU	max. Earth dist.	2056 Jun 29 23:41:55	10° S 31'26	1.32728 AU
	2055 Sep 20 08:41:18	30° R P		evening rise	2056 Jul 05 20:38:29	22° S 58'53	
inferior conj	2055 Sep 21 15:19:47	28° P 38'06	-2°58'38		2056 Jul 09 08:42:08	0° Q	
minimum elong	2055 Sep 21 19:59:23	28° P 25'43	2°57'04		2056 Jul 27 08:37:20	0° P	
morning rise	2055 Sep 28 04:18:18	23° P 13'08		desc. node	2056 Jul 29 03:46:52	2° P 25'18	
direct	2055 Sep 30 18:20:58	22° P 39'06		evening max el	2056 Aug 07 06:58:20	12° P 50'58	27°22'19
asc. node	2055 Sep 30 23:43:52	22° P 39'22		retrograde	2056 Aug 21 01:41:36	20° P 07'44	
max. declination.	2055 Oct 03 23:22:03	23° P 31'30	3N17'34"	min. declination.	2056 Aug 23 12:45:00	19° P 50'28	0N 2'28"
morning max el	2055 Oct 07 06:35:16	26° P 05'21	17°56'57	evening set	2056 Aug 28 06:35:20	17° P 37'53	
	2055 Oct 10 14:27:09	0° A		min. Earth dist.	2056 Aug 31 20:41:04	14° P 42'43	0.62554 AU
morning set	2055 Oct 24 11:15:12	22° A 17'10		inferior conj	2056 Sep 03 18:06:13	11° P 58'07	-3°51'58
	2055 Oct 29 00:42:36	0° M		minimum elong	2056 Sep 03 23:28:35	11° P 45'21	3°50'34
				morning rise	2056 Sep 10 17:50:29	6° P 51'47	
superior conj	2055 Nov 06 14:51:59	14° M 09'50	0°11'02	direct	2056 Sep 13 04:27:46	6° P 24'51	
minimum elong	2055 Nov 06 16:08:00	14° M 14'57	0°10'52	asc. node	2056 Sep 16 20:47:22	7° P 27'40	
behind sun begin	2055 Nov 06 08:28:58	13° M 43'59		max. declination.	2056 Sep 17 10:35:15	7° P 48'37	8N46'45"
behind sun end	2055 Nov 06 23:47:02	14° M 45'53		morning max el	2056 Sep 19 22:28:41	9° P 51'35	17°52'04
desc. node	2055 Nov 08 06:02:58	16° M 47'49			2056 Oct 03 06:11:13	0° A	
max. Earth dist.	2055 Nov 11 16:21:35	22° M 16'24	1.44563 AU	morning set	2056 Oct 06 01:08:12	4° A 58'13	
	2055 Nov 16 14:09:38	0° X					
evening rise	2055 Nov 22 19:26:27	9° X 38'43		superior conj	2056 Oct 17 07:20:01	24° A 39'24	0°51'44
	2055 Dec 06 05:56:11	0° Z		minimum elong	2056 Oct 17 11:39:38	24° A 57'43	0°51'10
min. declination.	2055 Dec 07 05:04:21	1° Z 25'13	25S44'51"		2056 Oct 20 11:48:43	0° M	
evening max el	2055 Dec 19 23:02:42	18° Z 12'56	20°16'25	max. Earth dist.	2056 Oct 24 07:21:21	6° M 15'25	1.43338 AU
retrograde	2055 Dec 27 19:36:47	22° Z 50'05		desc. node	2056 Oct 25 03:04:33	7° M 35'10	
asc. node	2055 Dec 27 23:01:06	22° Z 49'59		evening rise	2056 Nov 01 04:32:23	18° M 45'58	
evening set	2055 Dec 31 14:01:58	21° Z 27'36			2056 Nov 08 12:48:52	0° X	
inferior conj	2056 Jan 05 23:55:13	15° Z 22'03	2°43'14	min. declination.	2056 Nov 28 00:39:05	27° X 22'33	25S52' 2"
minimum elong	2056 Jan 05 21:17:40	15° Z 30'56	2°42'30		2056 Nov 30 06:37:10	0° Z	
min. Earth dist.	2056 Jan 06 17:59:53	14° Z 20'56	0.67137 AU	evening max el	2056 Dec 01 21:11:30	1° Z 43'09	21°25'43
max. declination.	2056 Jan 09 00:19:50	11° Z 26'12	19S44'11"	retrograde	2056 Dec 10 17:04:28	6° Z 58'07	
morning rise	2056 Jan 11 04:22:37	9° Z 08'41		asc. node	2056 Dec 13 20:04:12	6° Z 04'07	
direct	2056 Jan 16 22:00:57	6° Z 35'45		evening set	2056 Dec 14 21:38:58	5° Z 20'09	

	2056 Dec 19 14:07:45	30° 		evening rise	2057 Oct 12 03:05:42	28°  33'10"	
inferior conj	2056 Dec 20 06:09:34	29°  04'45"	2°03'37"	desc. node	2057 Oct 12 00:07:24	28°  21'12"	
minimum elong	2056 Dec 20 03:46:55	29°  12'58"	2°02'48"		2057 Oct 13 00:44:58	0° 	
min. Earth dist.	2056 Dec 20 12:14:39	28°  43'42"	0.67665 AU		2057 Nov 02 04:11:35	0° 	
morning rise	2056 Dec 25 09:44:17	22°  52'03"		evening max el	2057 Nov 14 14:17:52	15°  16'02"	22°43'08"
max. declination.	2056 Dec 28 11:27:58	20°  59'41"	20S 3' 7"	min. declination.	2057 Nov 18 03:21:26	18°  26'42"	25S26'44"
direct	2056 Dec 30 12:26:00	20°  39'24"		retrograde	2057 Nov 24 12:21:26	21°  09'28"	
morning max el	2057 Jan 09 20:08:04	26°  47'50"	23°18'45"	evening set	2057 Nov 29 04:59:27	19°  14'52"	
	2057 Jan 12 19:22:55	0° 		asc. node	2057 Nov 30 17:05:32	17°  48'50"	
desc. node	2057 Jan 21 02:18:59	10°  32'28"20		inferior conj	2057 Dec 04 13:39:03	12°  53'21"	1°17'28"
min. declination.	2057 Jan 22 14:13:54	12°  31'29"	23S 2'22"	minimum elong	2057 Dec 04 11:57:49	12°  59'11"	1°16'49"
	2057 Feb 03 11:46:29	0° 		min. Earth dist.	2057 Dec 04 08:46:51	13°  10'12"	0.67843 AU
morning set	2057 Feb 13 19:35:25	16°  38'04"		morning rise	2057 Dec 09 18:47:59	6°  43'19"	
max. Earth dist.	2057 Feb 17 23:56:12	23°  47'06"	1.39308 AU	direct	2057 Dec 14 06:45:06	4°  53'23"	
	2057 Feb 21 12:18:29	0° 		max. declination.	2057 Dec 14 05:25:59	4°  53'24"	18S19'11"
superior conj	2057 Feb 25 14:29:02	7°  28'13"	-1°51'48"	morning max el	2057 Dec 23 08:43:31	10°  13'21"	21°52'09"
minimum elong	2057 Feb 25 18:28:24	7°  46'45"	1°51'29"	desc. node	2058 Jan 07 23:22:12	0° 	
evening rise	2057 Mar 07 00:07:48	25°  12'41'15"			2058 Jan 07 18:52:22	0° 	
	2057 Mar 09 09:36:44	0° 		min. declination.	2058 Jan 14 06:42:46	9°  33'33"	23S49'55"
asc. node	2057 Mar 11 19:23:09	4°  31'14"		morning set	2058 Jan 24 20:25:31	25°  57'48"	
evening max el	2057 Mar 23 06:28:34	21°  35'48"	18°37'32"		2058 Jan 27 08:26:26	0° 	
retrograde	2057 Mar 31 02:39:38	25°  25'54"		max. Earth dist.	2058 Jan 30 22:10:12	5°  52'53"	1.41429 AU
max. declination.	2057 Mar 31 15:17:43	25°  24'54"	12N54'18"	superior conj	2058 Feb 07 15:07:45	19°  04'38"	-2°03'48"
evening set	2057 Apr 02 10:08:23	25°  07'04"		minimum elong	2058 Feb 07 16:48:00	19°  11'59"	2°03'46"
inferior conj	2057 Apr 10 01:44:20	20°  41'37"	2°21'58"		2058 Feb 13 17:11:43	0° 	
minimum elong	2057 Apr 10 05:44:31	20°  33'56"	2°20'53"	evening rise	2058 Feb 18 06:21:52	8°  12'30"	
min. Earth dist.	2057 Apr 13 11:00:32	18°  07'14"	0.57848 AU	asc. node	2058 Feb 26 16:25:12	23°  39'12"	
morning rise	2057 Apr 17 22:14:46	15°  22'35"			2058 Mar 02 21:57:42	0° 	
desc. node	2057 Apr 19 01:31:03	14°  53'21"		evening max el	2058 Mar 06 14:46:00	4°  23'56"	18°12'56"
direct	2057 Apr 23 16:08:06	13°  59'40"		retrograde	2058 Mar 13 13:19:21	7°  53'00"	
min. declination.	2057 Apr 28 14:25:24	14°  58'54"	3N56'14"	max. declination.	2058 Mar 15 05:24:33	7°  41'05"	6N 8'53"
morning max el	2057 May 08 01:05:34	21°  35'58"	26°31'36"	evening set	2058 Mar 16 02:49:50	7°  25'18"	
	2057 May 15 09:18:28	0° 		inferior conj	2058 Mar 23 01:03:05	2°  38'53"	3°15'20"
	2057 Jun 02 12:32:58	0° 		minimum elong	2058 Mar 23 04:04:34	2°  32'07"	3°14'50"
morning set	2057 Jun 06 04:12:56	7°  29'03"			2058 Mar 26 00:28:09	30° 	
asc. node	2057 Jun 07 18:39:00	10°  25'30'4"		min. Earth dist.	2058 Mar 26 09:57:42	29°  13'46"	0.59949 AU
superior conj	2057 Jun 13 04:27:46	22°  22'41'12"	0°54'23"	morning rise	2058 Mar 30 03:04:32	26°  15'56"	
minimum elong	2057 Jun 13 02:22:25	22°  22'49'42"	0°53'59"	direct	2058 Apr 05 15:56:45	24°  15'21"	
max. Earth dist.	2057 Jun 13 10:43:31	23°  15'39"	1.32265 AU	desc. node	2058 Apr 05 22:33:35	24°  15'33"	
	2057 Jun 16 12:34:49	0° 		min. declination.	2058 Apr 11 00:40:06	26°  10'52"	2S32'36"
max. declination.	2057 Jun 19 01:25:43	5°  26'55"	24N56'12"		2058 Apr 16 22:16:55	0° 	
evening rise	2057 Jun 20 02:43:17	7°  24'01'15"		morning max el	2058 Apr 19 23:14:03	2°  45'41"	27°28'50"
	2057 Jul 01 19:54:47	0° 			2058 May 09 19:46:01	0° 	
desc. node	2057 Jul 16 00:48:27	20°  24'04"		morning set	2058 May 21 12:01:57	22°  08'10'14"	
evening max el	2057 Jul 20 11:01:09	25°  04'04"	26°50'10"		2058 May 25 04:33:45	0° 	
	2057 Jul 26 20:14:22	0° 		asc. node	2058 May 25 15:41:12	1°  00'13"	
retrograde	2057 Aug 03 09:45:58	2°  17'37"		max. Earth dist.	2058 May 27 22:54:06	6°  01'40"	1.32191 AU
min. declination.	2057 Aug 07 12:56:10	1°  17'32'48"	6N46'43"	superior conj	2058 May 28 16:02:40	7°  25'49"	0°31'24"
evening set	2057 Aug 10 07:11:22	0° 		minimum elong	2058 May 28 14:41:09	7°  28'21"	0°31'07"
	2057 Aug 10 16:27:37	30° 		evening rise	2058 Jun 04 12:34:34	22°  23'32"	
min. Earth dist.	2057 Aug 14 01:04:44	27°  35'34"	0.60565 AU		2058 Jun 08 03:39:03	0° 	
inferior conj	2057 Aug 17 06:58:00	24°  35'24"	-4°34'52"	max. declination.	2058 Jun 09 19:31:33	3°  17'01"	25N20'20"
minimum elong	2057 Aug 17 11:30:00	24°  35'55"	4°34'08"		2058 Jun 26 10:44:18	0° 	
morning rise	2057 Aug 24 17:42:12	20°  08'22"		evening max el	2058 Jul 02 08:07:18	6°  30'22"	25°46'19"
direct	2057 Aug 27 03:41:45	19°  45'49"		desc. node	2058 Jul 02 21:50:35	7°  02'34"	
max. declination.	2057 Sep 02 02:01:34	22°  08'29"	13N42'57"	retrograde	2058 Jul 16 08:23:30	13°  39'24"	
morning max el	2057 Sep 03 11:13:05	23°  21'45"	18°06'09"	evening set	2058 Jul 22 09:42:25	12°  12'19"	
asc. node	2057 Sep 03 17:50:34	23°  38'03"		min. declination.	2058 Jul 23 09:34:55	11°  42'29"	12N55'29"
	2057 Sep 08 18:27:52	0° 		min. Earth dist.	2058 Jul 26 20:26:27	9°  33'41"	0.58510 AU
morning set	2057 Sep 19 07:24:59	18°  26'43"		inferior conj	2058 Jul 30 02:16:11	7°  13'13"	-4°57'40"
	2057 Sep 25 12:42:04	0° 		minimum elong	2058 Jul 30 03:31:12	7°  10'57"	4°57'37"
superior conj	2057 Sep 29 02:06:45	6°  26'30"	1°20'42"	morning rise	2058 Aug 06 23:36:52	2°  50'51"	
minimum elong	2057 Sep 29 06:18:06	6°  45'12"	1°20'15"	direct	2058 Aug 09 11:42:36	2°  30'38"	
max. Earth dist.	2057 Oct 06 17:13:51	19° 39'36"	1.41614 AU	morning max el	2058 Aug 17 17:31:20	6° 25'50"	18°40'31"
				max. declination.	2058 Aug 18 09:01:06	7° 03'55"	17N43' 8"

asc. node	2058 Aug 21 14:53:25	10°♎54'32		evening set	2059 Jul 02 12:41:22	23°♏25'26	
	2058 Sep 01 18:34:15	0°♎		min. Earth dist.	2059 Jul 08 09:51:51	20°♏29'17	0.56660 AU
morning set	2058 Sep 03 01:10:18	2°♎28'15		inferior conj	2059 Jul 11 01:27:11	18°♏49'14	-4°46'52
				minimum elong	2059 Jul 10 21:22:57	18°♏55'41	4°46'26
superior conj	2058 Sep 11 18:10:45	19°♎16'16	1°37'59	min. declination.	2059 Jul 11 00:47:44	18°♏50'17	17N22'31"
minimum elong	2058 Sep 11 20:43:20	19°♎28'14	1°37'48	morning rise	2059 Jul 19 08:46:04	14°♏47'15	
	2058 Sep 17 14:37:45	0°♏		direct	2059 Jul 22 00:49:51	14°♏28'10	
max. Earth dist.	2058 Sep 18 22:37:10	2°♏21'39	1.39608 AU	morning max el	2059 Jul 31 14:28:35	18°♏54'52	19°36'16
evening rise	2058 Sep 23 00:23:14	9°♏23'09		max. declination.	2059 Aug 05 08:20:01	24°♏22'16	20N31'29"
desc. node	2058 Sep 28 21:09:14	19°♏01'28		asc. node	2059 Aug 08 11:56:47	29°♏02'58	
	2058 Oct 05 23:26:24	0°♎			2059 Aug 09 01:59:37	0°♎	
evening max el	2058 Oct 28 03:52:45	28°♎52'13	24°03'07	morning set	2059 Aug 18 02:53:37	16°♎53'27	
	2058 Oct 29 07:59:47	0°♏			2059 Aug 24 15:04:33	0°♎	
min. declination.	2058 Nov 05 02:07:36	4°♏42'07	23S47'36"	superior conj	2059 Aug 26 02:05:37	2°♎53'38	1°45'12
retrograde	2058 Nov 08 04:11:44	5°♏20'19		minimum elong	2059 Aug 26 02:44:56	2°♎56'52	1°45'12
evening set	2058 Nov 13 10:26:03	3°♏08'41		max. Earth dist.	2059 Sep 01 02:44:22	14°♎27'14	1.37572 AU
	2058 Nov 16 09:26:59	30°♎		evening rise	2059 Sep 04 21:13:09	21°♎18'59	
asc. node	2058 Nov 17 14:06:38	28°♎27'35			2059 Sep 09 21:36:15	0°♏	
min. Earth dist.	2058 Nov 18 05:08:12	27°♎37'15	0.67677 AU	desc. node	2059 Sep 15 18:09:22	9°♏31'41	
inferior conj	2058 Nov 18 20:35:09	26°♎45'00	0°26'07		2059 Sep 29 22:00:28	0°♎	
minimum elong	2058 Nov 18 19:57:34	26°♎47'07	0°25'51	evening max el	2059 Oct 10 15:52:00	12°♎31'08	25°18'51
morning rise	2058 Nov 24 05:28:28	20°♎39'59		min. declination.	2059 Oct 21 06:08:03	19°♎18'47	20S35'53"
direct	2058 Nov 28 03:46:41	19°♎12'16		retrograde	2059 Oct 22 15:43:17	19°♎26'12	
max. declination.	2058 Nov 29 02:01:09	19°♎16'49	15S 5'19"	evening set	2059 Oct 28 12:17:30	16°♎58'54	
morning max el	2058 Dec 06 04:13:32	23°♎49'32	20°34'27	min. Earth dist.	2059 Nov 01 22:57:28	12°♎02'14	0.67174 AU
	2058 Dec 11 11:48:37	0°♏		inferior conj	2059 Nov 03 01:11:55	10°♎37'08	-0°29'15
desc. node	2058 Dec 25 20:25:15	20°♏26'10		minimum elong	2059 Nov 03 01:56:40	10°♎34'42	0°28'56
	2059 Jan 01 03:28:47	0°♏		asc. node	2059 Nov 04 11:08:59	8°♎48'50	
morning set	2059 Jan 03 21:27:44	4°♏14'48		morning rise	2059 Nov 08 15:48:43	4°♎39'35	
min. declination.	2059 Jan 06 01:03:29	7°♏35'52	24S24'47"	direct	2059 Nov 12 02:08:26	3°♎31'41	
max. Earth dist.	2059 Jan 13 04:32:16	18°♏57'04	1.43246 AU	max. declination.	2059 Nov 13 16:00:08	3°♎45'14	10S43' 6"
superior conj	2059 Jan 19 15:33:06	29°♏33'07	-2°02'10	morning max el	2059 Nov 19 07:37:51	7°♎36'27	19°29'50
minimum elong	2059 Jan 19 12:23:10	29°♏19'55	2°02'04	greatest brilliancy	2059 Dec 02 17:05:08	25°♎32'13	-0.7m
	2059 Jan 19 21:59:20	0°♏			2059 Dec 05 15:50:11	0°♏	
evening rise	2059 Jan 31 20:53:44	20°♏35'25		desc. node	2059 Dec 12 17:26:06	10°♏51'38	
	2059 Feb 06 05:33:57	0°♏		morning set	2059 Dec 13 19:31:54	12°♏32'34	
asc. node	2059 Feb 13 13:26:11	11°♏53'50			2059 Dec 24 23:59:32	0°♏	
evening max el	2059 Feb 18 02:37:59	17°♏31'58	18°07'50	max. Earth dist.	2059 Dec 26 17:56:53	2°♏45'42	1.44523 AU
retrograde	2059 Feb 24 14:55:59	20°♏55'33		min. declination.	2059 Dec 28 19:25:34	6°♏02'05	24S52'38"
evening set	2059 Feb 27 09:41:30	20°♏17'43		superior conj	2059 Dec 30 13:39:03	8°♏50'36	-1°42'22
max. declination.	2059 Feb 27 02:55:36	20°♏24'53	0S50'25"	minimum elong	2059 Dec 30 05:32:25	8°♏18'10	1°41'42
inferior conj	2059 Mar 05 18:19:45	15°♏10'27	3°39'42		2060 Jan 12 11:58:19	0°♏	
minimum elong	2059 Mar 05 19:23:27	15°♏07'44	3°39'38	evening rise	2060 Jan 13 15:21:17	1°♏54'28	
min. Earth dist.	2059 Mar 08 16:42:44	12°♏11'38	0.62072 AU	asc. node	2060 Jan 31 10:27:42	29°♏34'28	
morning rise	2059 Mar 12 03:46:35	9°♏12'35			2060 Jan 31 19:33:28	0°♏	
direct	2059 Mar 19 01:35:43	6°♏42'32		evening max el	2060 Feb 01 15:06:34	0°♏51'43	18°21'40
desc. node	2059 Mar 23 19:35:43	7°♏45'41		retrograde	2060 Feb 08 02:47:11	4°♏22'59	
min. declination.	2059 Mar 24 22:04:40	8°♏16'43	8S40'43"	evening set	2060 Feb 11 02:43:50	3°♏34'11	
morning max el	2059 Apr 02 03:36:47	14°♏37'39	27°49'50	max. declination.	2060 Feb 11 11:00:43	3°♏22'31	7S32'39"
	2059 Apr 14 14:29:21	0°♏			2060 Feb 15 09:07:56	30°♎	
	2059 May 02 07:46:19	0°♏		inferior conj	2060 Feb 17 01:27:39	28°♏07'45	3°42'05
morning set	2059 May 05 14:59:53	6°♏34'19		minimum elong	2060 Feb 17 00:43:33	28°♏09'51	3°42'02
max. Earth dist.	2059 May 11 08:20:07	18°♏36'08	1.32509 AU	min. Earth dist.	2060 Feb 19 09:03:19	25°♏29'29	0.63975 AU
asc. node	2059 May 12 12:43:50	21°♏09'49		morning rise	2060 Feb 22 22:01:42	22°♏01'17	
superior conj	2059 May 13 01:56:24	22°♏21'38	0°05'50	direct	2060 Feb 29 19:24:35	19°♏13'22	
minimum elong	2059 May 13 01:39:47	22°♏20'07	0°05'46	min. declination.	2060 Mar 07 11:36:57	21°♏22'43	13S57'38"
behind sun begin	2059 May 12 20:50:22	21°♏53'53		desc. node	2060 Mar 09 16:38:09	22°♏52'33	
behind sun end	2059 May 13 06:29:13	22°♏46'23		morning max el	2060 Mar 14 12:11:51	27°♏07'15	27°33'54
	2059 May 16 13:59:35	0°♏			2060 Mar 17 05:28:30	0°♏	
evening rise	2059 May 20 00:13:38	7°♏22'50			2060 Apr 07 06:04:39	0°♏	
	2059 May 31 20:32:52	0°♏		morning set	2060 Apr 18 10:34:09	20°♏32'35	
max. declination.	2059 May 31 23:28:14	0°♏12'02	25N37' 3"		2060 Apr 23 02:14:14	0°♏	
evening max el	2059 Jun 13 22:59:42	17°♏15'32	24°19'06	max. Earth dist.	2060 Apr 23 10:54:46	0°♏45'16	1.33245 AU
desc. node	2059 Jun 19 18:49:49	21°♏49'13		superior conj	2060 Apr 26 08:16:44	6°♏51'56	-0°21'34
retrograde	2059 Jun 27 19:31:38	24°♏15'01					

minimum elong	2060 Apr 26 09:20:56	6°♄57'39 0°21'20	direct	2061 Feb 11 20:17:55	2°♁21'28
asc. node	2060 Apr 28 09:45:55	11°♄17'11	min. declination.	2061 Feb 20 04:54:40	5°♁51'33 18S 2'19"
evening rise	2060 May 03 11:47:05	22°♄10'00	desc. node	2061 Feb 24 13:40:29	9°♁42'05
	2060 May 07 07:10:09	0°♄	morning max el	2061 Feb 24 22:06:02	10°♁03'07 26°46'00
max. declination.	2060 May 22 05:39:55	24°♄09'12 25N34'29"		2061 Mar 12 16:16:08	0°♁
evening max el	2060 May 25 13:12:29	27°♄43'27 22°42'54		2061 Mar 30 17:26:49	0°♁
	2060 May 28 03:05:14	0°♄	morning set	2061 Apr 01 19:28:29	3°♁55'21
desc. node	2060 Jun 05 15:50:31	4°♄07'40	max. Earth dist.	2061 Apr 06 02:57:49	12°♁20'41 1.34434 AU
retrograde	2060 Jun 07 18:31:08	4°♄17'52			
evening set	2060 Jun 11 00:17:44	3°♄54'46	superior conj	2061 Apr 10 08:58:47	21°♁00'54 -0°49'35
min. Earth dist.	2060 Jun 18 21:22:09	0°♄27'30 0.55353 AU	minimum elong	2061 Apr 10 11:29:32	21°♁13'56 0°49'05
	2060 Jun 19 16:36:53	30°♄		2061 Apr 14 15:52:43	0°♄
inferior conj	2060 Jun 20 05:49:35	29°♄41'03 -3°49'51	asc. node	2061 Apr 15 06:48:22	1°♄18'34
minimum elong	2060 Jun 19 22:00:10	29°♄52'16 3°47'55	evening rise	2061 Apr 17 21:24:35	6°♄46'16
morning rise	2060 Jun 28 22:01:54	25°♄49'15		2061 Apr 30 13:30:36	0°♄
min. declination.	2060 Jun 28 18:45:01	25°♄51'04 18N43' 5"	evening max el	2061 May 07 10:38:26	8°♄27'24 21°12'20
direct	2060 Jul 01 19:29:34	25°♄29'52	max. declination.	2061 May 11 08:24:56	11°♄38'39 24N30'39"
	2060 Jul 12 05:39:41	0°♄	retrograde	2061 May 19 08:27:18	14°♄16'09
morning max el	2060 Jul 13 00:13:09	0°♄41'31 20°53'00	evening set	2061 May 21 14:48:57	14°♄04'16
max. declination.	2060 Jul 24 10:32:50	16°♄11'20 22N15'35"	desc. node	2061 May 23 12:53:05	13°♄36'45
asc. node	2060 Jul 25 08:59:55	17°♄50'52	inferior conj	2061 May 30 23:56:44	10°♄05'26 -2°08'29
	2060 Jul 31 15:26:13	0°♄	minimum elong	2061 May 30 18:01:29	10°♄13'46 2°06'27
morning set	2060 Aug 01 09:56:56	1°♄34'55	min. Earth dist.	2061 May 31 09:37:05	9°♄51'49 0.54899 AU
			morning rise	2061 Jun 08 21:27:10	6°♄05'22
superior conj	2060 Aug 08 21:16:57	17°♄05'16 1°44'13	direct	2061 Jun 12 05:15:25	5°♄41'24
minimum elong	2060 Aug 08 20:23:34	17°♄00'42 1°44'13	min. declination.	2061 Jun 13 12:57:05	5°♄45'20 17N 5' 7"
max. Earth dist.	2060 Aug 13 11:03:40	26°♄17'31 1.35734 AU	morning max el	2061 Jun 24 23:07:28	11°♄45'17 22°26'47
	2060 Aug 15 08:48:43	0°♄		2061 Jul 08 09:35:14	0°♄
evening rise	2060 Aug 17 13:56:30	4°♄12'05	asc. node	2061 Jul 12 06:02:45	7°♄08'19
desc. node	2060 Sep 01 15:09:12	29°♄46'40	max. declination.	2061 Jul 14 12:08:11	11°♄37'05 23N20'39"
	2060 Sep 01 18:38:03	0°♄	morning set	2061 Jul 16 20:13:15	16°♄26'17
evening max el	2060 Sep 22 03:50:48	26°♄10'05 26°22'51		2061 Jul 23 05:21:25	0°♄
	2060 Sep 26 15:12:19	0°♄			
retrograde	2060 Oct 04 22:26:50	3°♄21'52	superior conj	2061 Jul 24 00:11:13	1°♄39'54 1°36'37
min. declination.	2060 Oct 04 15:57:44	3°♄21'37 16S 0'12"	minimum elong	2061 Jul 23 22:17:01	1°♄29'49 1°36'28
evening set	2060 Oct 11 08:41:48	0°♄42'49	max. Earth dist.	2061 Jul 27 03:58:39	8°♄16'39 1.34248 AU
	2060 Oct 12 04:02:59	30°♄	evening rise	2061 Jul 31 21:54:49	17°♄49'45
min. Earth dist.	2060 Oct 15 11:58:33	26°♄22'15 0.66328 AU		2061 Aug 07 10:16:53	0°♄
inferior conj	2060 Oct 17 01:40:13	24°♄27'20 -1°27'09	desc. node	2061 Aug 19 12:10:49	19°♄38'57
minimum elong	2060 Oct 17 03:59:09	24°♄20'16 1°26'11		2061 Aug 27 01:04:30	0°♄
asc. node	2060 Oct 21 08:12:08	19°♄50'06	evening max el	2061 Sep 04 16:02:39	9°♄40'55 27°07'35
morning rise	2060 Oct 22 23:47:27	18°♄39'56	retrograde	2061 Sep 17 23:40:32	16°♄59'10
direct	2060 Oct 26 00:16:45	17°♄48'13	min. declination.	2061 Sep 18 16:35:12	16°♄57'34 10S18'41"
max. declination.	2060 Oct 28 03:41:00	18°♄13'14 5S34' 8"	evening set	2061 Sep 24 21:16:24	14°♄15'31
morning max el	2060 Nov 01 17:49:29	21°♄31'07 18°40'35	min. Earth dist.	2061 Sep 28 17:54:11	10°♄31'03 0.65130 AU
	2060 Nov 08 11:06:28	0°♄	inferior conj	2061 Sep 30 19:49:20	8°♄10'46 -2°25'43
morning set	2060 Nov 22 14:05:02	21°♄56'42	minimum elong	2061 Sep 30 23:42:33	7°♄59'50 2°24'15
	2060 Nov 27 16:07:49	0°♄	morning rise	2061 Oct 07 02:59:47	2°♄36'30
desc. node	2060 Nov 28 14:26:34	1°♄28'05	asc. node	2061 Oct 08 05:15:33	2°♄11'12
			direct	2061 Oct 09 20:03:30	1°♄57'06
superior conj	2060 Dec 08 17:20:15	17°♄23'03 -1°04'23	max. declination.	2061 Oct 12 14:27:22	2°♄37'30 0N 0' 7"
minimum elong	2060 Dec 08 09:31:58	16°♄52'24 1°03'27	morning max el	2061 Oct 16 08:23:58	5°♄27'18 18°07'53
max. Earth dist.	2060 Dec 08 11:31:15	17°♄00'13 1.45117 AU		2061 Nov 01 21:15:53	0°♄
	2060 Dec 16 17:36:03	0°♄	morning set	2061 Nov 03 13:51:51	2°♄48'01
min. declination.	2060 Dec 19 14:25:14	4°♄32'41 25S16' 3"	desc. node	2061 Nov 15 11:27:23	22°♄11'23
evening rise	2060 Dec 24 10:38:41	12°♄15'51			
greatest brilliancy	2061 Jan 01 06:34:43	24°♄47'11 -0.8m	superior conj	2061 Nov 17 22:03:14	26°♄04'55 -0°16'22
	2061 Jan 04 14:06:27	0°♄	minimum elong	2061 Nov 17 19:58:55	25°♄56'41 0°16'04
evening max el	2061 Jan 15 01:45:59	14°♄18'17 18°53'35	behind sun begin	2061 Nov 17 18:29:47	25°♄50'47
asc. node	2061 Jan 17 07:29:48	16°♄18'47	behind sun end	2061 Nov 17 21:28:04	26°♄02'35
retrograde	2061 Jan 21 20:53:37	18°♄08'09		2061 Nov 20 09:25:54	0°♄
evening set	2061 Jan 25 02:49:47	17°♄07'25	max. Earth dist.	2061 Nov 21 06:21:32	1°♄22'31 1.44976 AU
max. declination.	2061 Jan 26 11:20:46	16°♄07'35 13S29'21"	evening rise	2061 Dec 04 07:56:59	21°♄45'34
inferior conj	2061 Jan 30 18:43:42	11°♄23'56 3°28'18		2061 Dec 09 15:48:30	0°♄
minimum elong	2061 Jan 30 16:46:20	11°♄30'01 3°28'00	min. declination.	2061 Dec 11 11:25:50	2°♄47'58 25S35'48"
min. Earth dist.	2061 Feb 01 10:53:25	9°♄19'15 0.65524 AU	greatest brilliancy	2061 Dec 17 07:08:00	11°♄39'36 -0.7m
morning rise	2061 Feb 05 06:20:34	5°♄12'58	evening max el	2061 Dec 29 08:32:48	27°♄47'41 19°42'03

	2061 Dec 31 18:37:16	0°♊			2062 Oct 07 21:22:27	0°♎		
asc. node	2062 Jan 04 04:32:58	1°♊51'46		morning set	2062 Oct 16 15:42:33	14°♎55'04		
retrograde	2062 Jan 05 18:03:31	2°♊05'13			2062 Oct 25 11:09:20	0°♎		
evening set	2062 Jan 09 07:26:01	0°♊51'10						
	2062 Jan 10 09:17:12	30°♊♌		superior conj	2062 Oct 28 22:56:52	5°♎48'22	0°29'35	
max. declination.	2062 Jan 13 07:43:22	26°♊47'12 18S 7' 8"		minimum elong	2062 Oct 29 01:58:59	6°♎00'52	0°29'10	
inferior conj	2062 Jan 14 18:57:05	24°♊53'03 3°02'26		desc. node	2062 Nov 02 08:30:26	12°♎58'22		
minimum elong	2062 Jan 14 16:24:13	25°♊01'28 3°01'51		max. Earth dist.	2062 Nov 03 23:56:00	15°♎36'37	1.44123 AU	
min. Earth dist.	2062 Jan 15 20:41:57	23°♊28'07 0.66670 AU			2062 Nov 13 03:57:14	0°♊		
morning rise	2062 Jan 20 01:08:19	18°♊40'08		evening rise	2062 Nov 13 16:56:57	0°♊50'06		
direct	2062 Jan 26 03:04:23	15°♊57'35			2062 Dec 03 08:29:18	0°♊		
min. declination.	2062 Feb 07 05:03:56	23°♊07'10 20S46'37"		min. declination.	2062 Dec 03 10:20:02	0°♊06'24 25S49'59"		
morning max el	2062 Feb 07 07:07:55	23°♊12'22 25°34'38		evening max el	2062 Dec 12 10:06:36	11°♊18'20 20°44'37		
desc. node	2062 Feb 11 10:41:46	27°♊44'33		retrograde	2062 Dec 20 15:53:39	16°♊10'34		
	2062 Feb 13 07:55:13	0°♊		asc. node	2062 Dec 22 01:35:39	15°♊59'34		
	2062 Mar 06 00:32:33	0°♊		evening set	2062 Dec 24 14:28:49	14°♊41'37		
morning set	2062 Mar 15 13:16:08	16°♊29'25		inferior conj	2062 Dec 29 23:36:14	8°♊31'41 2°27'17		
max. Earth dist.	2062 Mar 19 07:15:27	23°♊27'38 1.36073 AU		minimum elong	2062 Dec 29 21:01:36	8°♊40'30 2°26'30		
	2062 Mar 22 16:37:53	0°♊		min. Earth dist.	2062 Dec 30 12:26:04	7°♊47'48 0.67417 AU		
				morning rise	2063 Jan 04 03:23:40	2°♊18'35		
superior conj	2062 Mar 25 01:22:48	4°♊40'58 -1°16'41		max. declination.	2063 Jan 05 01:57:56	1°♊32'19 20S11'28"		
minimum elong	2062 Mar 25 05:08:38	4°♊59'47 1°16'05			2063 Jan 08 10:28:40	30°♊♌		
evening rise	2062 Apr 02 03:00:37	21°♊05'24		direct	2063 Jan 09 14:50:57	29°♊53'40		
asc. node	2062 Apr 02 03:52:04	21°♊09'45			2063 Jan 10 20:03:00	0°♊		
	2062 Apr 06 14:49:40	0°♊		morning max el	2063 Jan 20 15:27:55	6°♊28'08 24°09'43		
evening max el	2062 Apr 19 19:35:47	19°♊47'59 19°56'56		min. declination.	2063 Jan 28 04:59:03	15°♊12'55 22S24'28"		
max. declination.	2062 Apr 26 21:52:40	24°♊16'35 21N29' 3"		desc. node	2063 Jan 29 07:43:50	16°♊38'57		
retrograde	2062 Apr 29 23:50:11	24°♊43'02			2063 Feb 07 22:04:52	0°♊		
evening set	2062 May 02 00:18:56	24°♊32'16		morning set	2063 Feb 25 10:06:20	27°♊58'04		
desc. node	2062 May 10 09:55:08	20°♊50'31			2063 Feb 26 14:06:54	0°♊		
inferior conj	2062 May 10 22:01:43	20°♊32'11 -0°08'49		max. Earth dist.	2063 Mar 01 03:39:59	4°♊32'44 1.38079 AU		
minimum elong	2062 May 10 21:37:35	20°♊32'47 0°08'41						
transit middle	2062 May 10 21:37:35	20°♊32'47 0°08'41		superior conj	2063 Mar 08 05:58:02	17°♊42'03 -1°40'38		
transit begin	2062 May 10 18:17:09	20°♊37'52		minimum elong	2063 Mar 08 10:16:32	18°♊02'39 1°40'10		
transit end	2062 May 11 00:58:00	20°♊27'43			2063 Mar 14 13:22:14	0°♊		
min. Earth dist.	2062 May 12 23:13:34	19°♊17'41 0.55416 AU		evening rise	2063 Mar 17 02:03:47	4°♊59'35		
morning rise	2062 May 19 16:50:30	16°♊06'02		asc. node	2063 Mar 20 00:55:19	10°♊45'10		
direct	2062 May 23 19:49:01	15°♊28'38			2063 Mar 31 23:06:11	0°♊		
min. declination.	2062 May 27 08:07:25	15°♊56'31 13N 8'58"		evening max el	2063 Apr 02 15:42:02	1°♊47'20 19°00'39		
morning max el	2062 Jun 06 14:51:38	22°♊18'48 24°08'46		retrograde	2063 Apr 11 05:22:51	5°♊56'54		
	2062 Jun 13 09:16:06	0°♊		max. declination.	2063 Apr 10 18:59:38	5°♊56'17 16N28'16"		
asc. node	2062 Jun 29 03:06:19	26°♊47'48		evening set	2063 Apr 13 09:33:49	5°♊41'57		
	2062 Jun 30 16:14:59	0°♊		inferior conj	2063 Apr 21 12:18:37	1°♊27'13 1°35'58		
morning set	2062 Jul 01 07:59:35	1°♊22'42		minimum elong	2063 Apr 21 15:43:41	1°♊21'14 1°34'52		
max. declination.	2062 Jul 04 18:47:05	8°♊46'09 24N 5'33"			2063 Apr 23 14:04:40	30°♊♌		
				min. Earth dist.	2063 Apr 24 14:45:16	29°♊17'56 0.56787 AU		
superior conj	2062 Jul 08 08:09:28	16°♊29'28 1°23'38		desc. node	2063 Apr 27 06:57:17	27°♊38'41		
minimum elong	2062 Jul 08 05:47:23	16°♊16'38 1°23'19		morning rise	2063 Apr 29 18:50:10	26°♊26'45		
max. Earth dist.	2062 Jul 10 06:10:52	20°♊37'26 1.33169 AU		direct	2063 May 04 22:41:33	25°♊23'49		
	2062 Jul 14 17:22:19	0°♊		min. declination.	2063 May 09 12:58:36	26°♊13'28 7N35'27"		
evening rise	2062 Jul 15 17:03:23	2°♊00'31			2063 May 16 01:30:00	0°♊		
	2062 Jul 31 07:14:15	0°♊		morning max el	2063 May 19 05:28:08	2°♊46'30 25°45'05		
desc. node	2062 Aug 06 09:14:01	8°♊58'57			2063 Jun 07 13:21:42	0°♊		
evening max el	2062 Aug 18 03:10:21	22°♊52'20 27°25'55		morning set	2063 Jun 15 19:33:00	16°♊17'45		
	2062 Aug 29 22:08:52	0°♊		asc. node	2063 Jun 16 00:09:44	16°♊42'06		
retrograde	2062 Aug 31 18:43:36	0°♊09'49			2063 Jun 22 03:19:33	0°♊		
min. declination.	2062 Sep 02 12:40:04	0°♊00'43 3S52'44"						
	2062 Sep 02 14:16:07	30°♊♌		superior conj	2063 Jun 22 18:56:23	1°♊25'40 1°06'10		
evening set	2062 Sep 07 23:01:55	27°♊31'39		minimum elong	2063 Jun 22 16:36:57	1°♊12'55 1°05'45		
min. Earth dist.	2062 Sep 11 14:36:42	24°♊20'26 0.63589 AU		max. Earth dist.	2063 Jun 23 15:08:07	3°♊16'20 1.32488 AU		
inferior conj	2062 Sep 14 05:05:46	21°♊41'54 -3°22'08		max. declination.	2063 Jun 25 03:53:46	6°♊36'51 24N39'38"		
minimum elong	2062 Sep 14 10:11:10	21°♊28'58 3°20'32		evening rise	2063 Jun 29 19:54:17	16°♊32'56		
morning rise	2062 Sep 20 22:31:59	16°♊24'07			2063 Jul 06 15:41:55	0°♊		
direct	2062 Sep 23 10:48:02	15°♊53'25		desc. node	2063 Jul 24 06:16:28	27°♊32'16		
asc. node	2062 Sep 25 02:17:59	16°♊06'52			2063 Jul 26 06:51:12	0°♊		
max. declination.	2062 Sep 27 01:06:44	16°♊56'59 5N37'43"		evening max el	2063 Jul 31 10:30:34	5°♊29'28 27°12'44		
morning max el	2062 Sep 30 00:36:22	19°♊19'03 17°52'35		retrograde	2063 Aug 14 06:45:32	12°♊44'01		

min. declination.	2063 Aug 17 08:50:04	12°  17'48	2N53'40"	minimum elong	2064 Jun 06 04:53:48	16°  12'37	0°44'37
evening set	2063 Aug 21 10:03:03	10°  23'46		max. Earth dist.	2064 Jun 06 03:11:57	16°  03'16	1.32190 AU
min. Earth dist.	2063 Aug 25 00:41:17	7°  37'41	0.61735 AU		2064 Jun 12 12:56:58	0° 	
inferior conj	2063 Aug 28 02:20:45	4°  52'17	-4°11'55	evening rise	2064 Jun 13 03:49:52	1°  518'42	
minimum elong	2063 Aug 28 07:36:40	4°  40'24	4°10'45	max. declination.	2064 Jun 14 17:40:29	4°  536'00	25N 7' 0"
morning rise	2063 Sep 04 06:44:58	29°  54'35			2064 Jun 28 17:46:37	0° 	
	2063 Sep 04 00:36:27	30°  R 		desc. node	2064 Jul 10 03:17:18	14°  59'54	
direct	2063 Sep 06 16:54:59	29°  529'39		evening max el	2064 Jul 12 11:27:03	17°  521'42	26°26'23
	2063 Sep 09 07:52:45	0° 		retrograde	2064 Jul 26 10:57:18	24°  533'48	
max. declination.	2063 Sep 11 13:25:52	1°  13'17	10N56' 8"	min. declination.	2064 Jul 31 14:20:12	23°  527'10	9N29' 5"
asc. node	2063 Sep 11 23:20:32	1°  31'27		evening set	2064 Aug 02 01:42:08	22°  544'24	
morning max el	2063 Sep 13 15:27:33	2°  58'52	17°55'42	min. Earth dist.	2064 Aug 06 00:36:51	20°  508'30	0.59686 AU
morning set	2063 Sep 29 13:12:47	27°  58'10		inferior conj	2064 Aug 09 08:03:27	17°  532'56	-4°47'43
	2063 Sep 30 15:58:48	0° 		minimum elong	2064 Aug 09 11:31:41	17°  526'05	4°47'21
				morning rise	2064 Aug 16 23:26:14	12°  557'53	
superior conj	2063 Oct 10 02:58:22	16°  51'53	1°05'32	direct	2064 Aug 19 10:05:43	12°  536'31	
minimum elong	2063 Oct 10 07:32:06	17°  511'38	1°04'58	max. declination.	2064 Aug 26 09:17:40	15°  540'55	15N31'35"
max. Earth dist.	2063 Oct 17 13:25:20	29°  524'28	1.42656 AU	morning max el	2064 Aug 27 01:50:40	16°  518'54	18°18'18
	2063 Oct 17 22:04:56	0° 		asc. node	2064 Aug 28 20:23:45	18°  511'34	
desc. node	2063 Oct 20 05:33:58	3°  M45'38			2064 Sep 05 17:19:05	0° 	
evening rise	2063 Oct 24 05:59:22	10°  M09'54		morning set	2064 Sep 12 00:47:24	11°  M41'28	
	2063 Nov 06 07:37:39	0° 					
evening max el	2063 Nov 25 06:00:07	24°  549'54	21°57'56	superior conj	2064 Sep 21 07:27:32	29°  M07'27	1°29'23
min. declination.	2063 Nov 24 20:16:49	24°  524'54	25S47'57"	minimum elong	2064 Sep 21 11:02:23	29°  M23'48	1°29'04
	2063 Dec 02 10:17:48	0° 			2064 Sep 21 18:59:08	0° 	
retrograde	2063 Dec 04 12:35:38	0°  521'04		max. Earth dist.	2064 Sep 28 21:00:49	12°  530'14	1.40771 AU
	2063 Dec 06 12:32:34	30°  R 		evening rise	2064 Oct 03 13:27:35	20°  521'13	
evening set	2063 Dec 08 22:11:47	28°  535'53		desc. node	2064 Oct 06 02:35:17	24°  529'27	
asc. node	2063 Dec 08 22:36:12	28°  535'04			2064 Oct 09 14:00:55	0° 	
inferior conj	2063 Dec 14 06:35:14	22°  517'23	1°44'46		2064 Oct 30 13:53:37	0° 	
minimum elong	2063 Dec 14 04:27:06	22°  524'47	1°43'59	evening max el	2064 Nov 06 21:07:38	8°  523'05	23°17'15
min. Earth dist.	2063 Dec 14 07:50:07	22°  513'04	0.67784 AU	min. declination.	2064 Nov 12 14:19:05	13°  503'49	24S56'24"
morning rise	2063 Dec 19 10:34:30	16°  505'50		retrograde	2064 Nov 17 06:37:45	14°  532'40	
max. declination.	2063 Dec 23 07:08:21	14°  507'47	19S32'16"	evening set	2064 Nov 22 04:55:03	12°  530'35	
direct	2063 Dec 24 06:53:53	14°  502'57		asc. node	2064 Nov 24 19:37:30	9°  547'10	
morning max el	2064 Jan 03 01:39:50	19°  550'10	22°41'20	min. Earth dist.	2064 Nov 27 04:33:27	6°  539'55	0.67804 AU
	2064 Jan 11 15:35:18	0° 		inferior conj	2064 Nov 27 14:01:07	6°  507'24	0°56'16
desc. node	2064 Jan 16 04:46:24	6°  510'37		minimum elong	2064 Nov 27 12:44:24	6°  511'48	0°55'44
min. declination.	2064 Jan 19 15:51:14	11°  505'48	23S24'18"	morning rise	2064 Dec 02 20:30:12	29°  M59'15	
	2064 Feb 01 03:16:08	0° 			2064 Dec 02 20:07:32	30°  R 	
morning set	2064 Feb 06 04:00:45	8°  507'05		direct	2064 Dec 07 02:20:49	28°  M19'12	
max. Earth dist.	2064 Feb 10 23:41:22	16°  511'29	1.40232 AU	max. declination.	2064 Dec 07 12:41:15	28°  M20'10	17S 7'52"
					2064 Dec 11 18:46:04	0° 	
superior conj	2064 Feb 18 18:19:49	29°  553'16	-1°58'14	morning max el	2064 Dec 15 17:00:53	3°  520'23	21°17'48
minimum elong	2064 Feb 18 21:38:55	0°  508'21	1°58'03	desc. node	2065 Jan 02 01:47:38	26°  508'37	
	2064 Feb 18 19:48:43	0° 			2065 Jan 04 16:19:15	0° 	
evening rise	2064 Feb 28 15:32:52	18°  521'29		min. declination.	2065 Jan 10 09:43:49	8°  539'53	24S 5'29"
asc. node	2064 Mar 05 21:57:48	29°  559'14		morning set	2065 Jan 15 17:13:01	16°  555'27	
	2064 Mar 05 22:08:15	0° 		max. Earth dist.	2065 Jan 23 01:02:47	28°  542'04	1.42249 AU
evening max el	2064 Mar 15 20:25:41	14°  520'16	18°24'31		2065 Jan 23 19:59:32	0° 	
retrograde	2064 Mar 23 06:03:50	17°  559'45					
max. declination.	2064 Mar 24 07:48:45	17°  555'16	10N 8'27"	superior conj	2065 Jan 30 09:10:11	11°  501'40	-2°05'06
evening set	2064 Mar 25 16:06:03	17°  537'25		minimum elong	2065 Jan 30 09:03:24	11°  501'11	2°05'05
inferior conj	2064 Apr 01 23:51:25	13°  502'49	2°48'34		2065 Feb 10 02:01:06	0° 	
minimum elong	2064 Apr 02 03:40:23	12°  554'59	2°47'42	evening rise	2065 Feb 10 15:38:23	1°  501'42	
min. Earth dist.	2064 Apr 05 10:24:40	10°  515'02	0.58719 AU	asc. node	2065 Feb 20 18:58:25	18°  544'19	
morning rise	2064 Apr 09 12:28:12	7°  532'12		evening max el	2065 Feb 27 06:33:37	27°  517'45	18°08'21
desc. node	2064 Apr 13 03:58:20	6°  510'02			2065 Mar 02 22:45:54	0° 	
direct	2064 Apr 15 15:17:29	5°  553'55		retrograde	2065 Mar 05 23:40:02	0°  Y43'00	
min. declination.	2064 Apr 20 18:31:54	6°  559'46	1N13'19"	max. declination.	2065 Mar 08 00:37:32	0°  Y24'04	3N12'53"
morning max el	2064 Apr 30 00:23:06	13°  536'40	27°00'06	evening set	2065 Mar 08 15:19:20	0°  Y11'16	
	2064 May 13 01:00:01	0° 			2065 Mar 09 02:01:20	30°  R 	
	2064 May 29 16:36:09	0° 		inferior conj	2065 Mar 15 07:19:32	25°  516'00	3°28'46
morning set	2064 May 30 05:15:30	1°  506'01		minimum elong	2065 Mar 15 09:32:54	25°  510'43	3°28'30
asc. node	2064 Jun 01 21:13:23	6°  546'06		min. Earth dist.	2065 Mar 18 12:41:05	22°  513'48	0.60866 AU
				morning rise	2065 Mar 22 01:52:32	19°  525'22	
superior conj	2064 Jun 06 06:43:24	16°  522'40	0°45'00	direct	2065 Mar 28 19:39:57	17°  511'24	

desc. node	2065 Mar 31 00:59:26	17° ✕ 25'09			2066 Feb 03 00:40:15	0° ✕	
min. declination.	2065 Apr 03 08:42:28	18° ✕ 33'11	5S11'15"		asc. node	2066 Feb 07 15:58:44	6° ✕ 51'42
morning max el	2065 Apr 12 01:29:07	25° ✕ 04'55	27°42'19		evening max el	2066 Feb 10 19:03:48	10° ✕ 31'50 18°11'30
	2065 Apr 16 13:55:43	0° Υ			retrograde	2066 Feb 17 06:00:41	13° ✕ 56'59
	2065 May 06 09:40:28	0° ♄			evening set	2066 Feb 20 02:56:05	13° ✕ 14'39
morning set	2065 May 14 11:17:52	15° ♄ 41'07			max. declination.	2066 Feb 20 02:09:27	13° ✕ 15'35 3S42'52"
asc. node	2065 May 19 18:16:31	26° ♄ 55'00			inferior conj	2066 Feb 26 06:58:47	7° ✕ 59'20 3°42'58
max. Earth dist.	2065 May 20 14:34:39	28° ♄ 45'32	1.32274 AU		minimum elong	2066 Feb 26 07:13:17	7° ✕ 58'41 3°42'58
	2065 May 21 04:12:38	0° ♁			min. Earth dist.	2066 Feb 28 23:29:06	5° ✕ 06'55 0.62922 AU
					morning rise	2066 Mar 04 10:28:15	1° ✕ 57'16
superior conj	2065 May 21 17:48:49	1° ♁ 14'28	0°20'49			2066 Mar 07 18:37:35	30° ♁
minimum elong	2065 May 21 16:52:35	1° ♁ 09'20	0°20'37		direct	2066 Mar 11 09:19:16	29° ♁ 17'29
evening rise	2065 May 28 14:40:14	16° ♁ 10'34				2066 Mar 15 04:35:02	0° ✕
	2065 Jun 04 11:29:26	0° ♄			min. declination.	2066 Mar 17 12:14:04	1° ✕ 03'15 11S 1' 5"
max. declination.	2065 Jun 05 15:40:05	2° ♄ 11'07	25N28'30"		desc. node	2066 Mar 17 22:00:53	1° ✕ 16'56
evening max el	2065 Jun 24 05:23:20	28° ♄ 28'31	25°11'17		morning max el	2066 Mar 25 07:48:19	7° ✕ 13'27 27°47'14
	2065 Jun 25 21:35:33	0° ♁				2066 Apr 11 17:56:22	0° Υ
desc. node	2065 Jun 27 00:16:49	0° ♁ 55'20			morning set	2066 Apr 28 11:27:50	29° Υ 55'24
retrograde	2065 Jul 08 05:27:03	5° ♁ 35'34				2066 Apr 28 12:22:36	0° ♄
evening set	2065 Jul 13 18:05:26	4° ♁ 25'03			max. Earth dist.	2066 May 03 21:22:16	11° ♄ 09'47 1.32767 AU
min. declination.	2065 Jul 17 07:11:38	2° ♁ 34'35	15N 5'23"				
min. Earth dist.	2065 Jul 18 17:09:13	1° ♁ 40'41	0.57671 AU		superior conj	2066 May 06 02:28:06	15° ♄ 54'52 -0°05'39
	2065 Jul 21 04:30:10	30° ♁			minimum elong	2066 May 06 02:44:15	15° ♄ 56'19 0°05'35
inferior conj	2065 Jul 21 18:59:18	29° ♄ 35'07	-4°58'06		behind sun begin	2066 May 05 21:48:38	15° ♄ 29'42
minimum elong	2065 Jul 21 18:06:17	29° ♄ 36'38	4°58'05		behind sun end	2066 May 06 07:39:53	16° ♄ 22'57
morning rise	2065 Jul 29 20:42:00	25° ♄ 22'16			asc. node	2066 May 06 15:18:15	17° ♄ 04'16
direct	2065 Aug 01 10:05:11	25° ♄ 02'50				2066 May 12 14:54:28	0° ♁
morning max el	2065 Aug 10 04:37:12	29° ♄ 10'04	19°01'39		evening rise	2066 May 13 02:31:04	1° ♁ 01'38
	2065 Aug 11 00:55:26	0° ♁			max. declination.	2066 May 27 23:11:57	28° ♁ 14'27 25N39'43"
max. declination.	2065 Aug 12 05:39:19	1° ♁ 18'04	19N 2'22"			2066 May 29 04:33:11	0° ♄
asc. node	2065 Aug 15 17:27:19	5° ♁ 52'13			evening max el	2066 Jun 05 19:18:35	9° ♄ 03'38 23°38'26
morning set	2065 Aug 26 22:19:10	25° ♁ 54'15			desc. node	2066 Jun 13 21:17:42	14° ♄ 43'37
	2065 Aug 28 23:58:42	0° ♁			retrograde	2066 Jun 19 11:27:57	15° ♄ 54'33
					evening set	2066 Jun 23 12:47:36	15° ♄ 18'26
superior conj	2065 Sep 04 06:56:17	12° ♁ 19'45	1°42'09		min. Earth dist.	2066 Jun 30 05:42:20	12° ♄ 10'25 0.56015 AU
minimum elong	2065 Sep 04 08:39:44	12° ♁ 28'02	1°42'05		inferior conj	2066 Jul 02 09:45:18	10° ♄ 52'17 -4°28'48
max. Earth dist.	2065 Sep 11 00:46:57	24° ♁ 53'21	1.38723 AU		minimum elong	2066 Jul 02 03:35:07	11° ♄ 01'35 4°27'45
	2065 Sep 13 21:58:33	0° ♁			min. declination.	2066 Jul 06 08:06:43	8° ♄ 38'35 18N21' 4"
evening rise	2065 Sep 14 21:05:40	1° ♁ 40'23			morning rise	2066 Jul 10 20:58:34	6° ♄ 55'58
desc. node	2065 Sep 22 23:36:43	15° ♁ 06'17			direct	2066 Jul 13 15:00:29	6° ♄ 37'00
	2065 Oct 02 20:01:02	0° ♁			morning max el	2066 Jul 23 20:56:13	11° ♄ 21'57 20°06'24
evening max el	2065 Oct 20 09:43:33	22° ♁ 00'32	24°36'22		max. declination.	2066 Jul 31 03:50:37	20° ♄ 22'38 21N21'10"
min. declination.	2065 Oct 29 14:31:21	28° ♁ 21'48	22S38'38"		asc. node	2066 Aug 02 14:31:01	24° ♄ 19'02
retrograde	2065 Oct 31 20:53:20	28° ♁ 41'59				2066 Aug 05 18:53:50	0° ♁
evening set	2065 Nov 06 09:02:53	26° ♁ 23'31			morning set	2066 Aug 11 02:37:58	10° ♁ 27'21
min. Earth dist.	2065 Nov 11 00:13:52	21° ♁ 06'26	0.67497 AU				
inferior conj	2065 Nov 11 20:11:43	19° ♁ 59'52	0°03'04		superior conj	2066 Aug 18 20:09:32	26° ♁ 13'09 1°45'41
minimum elong	2065 Nov 11 20:07:01	20° ♁ 00'08	0°03'01		minimum elong	2066 Aug 18 20:05:59	26° ♁ 12'51 1°45'41
transit middle	2065 Nov 11 20:07:01	20° ♁ 00'08	0°03'01			2066 Aug 20 17:41:27	0° ♁
transit begin	2065 Nov 11 17:25:04	20° ♁ 09'08			max. Earth dist.	2066 Aug 24 06:02:18	6° ♁ 49'24 1.36758 AU
transit end	2065 Nov 11 22:48:58	19° ♁ 51'08			evening rise	2066 Aug 28 03:02:19	14° ♁ 02'34
asc. node	2065 Nov 11 16:38:34	20° ♁ 11'44				2066 Sep 06 09:30:07	0° ♁
morning rise	2065 Nov 17 07:15:44	13° ♁ 57'29			desc. node	2066 Sep 09 20:37:51	5° ♁ 30'50
direct	2065 Nov 21 00:10:39	12° ♁ 38'37				2066 Sep 27 15:09:37	0° ♁
max. declination.	2065 Nov 22 05:35:06	12° ♁ 46'41	13S22'44"		evening max el	2066 Oct 02 21:47:47	5° ♁ 40'45 25°48'03
morning max el	2065 Nov 28 15:50:31	17° ♁ 01'13	20°05'09		min. declination.	2066 Oct 14 09:05:17	12° ♁ 40'58 18S49'42"
	2065 Dec 08 21:21:34	0° ♁			retrograde	2066 Oct 15 06:21:34	12° ♁ 43'51
desc. node	2065 Dec 19 22:50:07	16° ♁ 25'47			evening set	2066 Oct 21 08:49:01	10° ♁ 11'15
morning set	2065 Dec 25 13:37:05	25° ♁ 02'39			min. Earth dist.	2066 Oct 25 16:24:12	5° ♁ 29'33 0.66856 AU
	2065 Dec 28 18:15:21	0° ♄			inferior conj	2066 Oct 26 23:18:08	3° ♁ 51'41 -0°53'35
min. declination.	2066 Jan 02 04:08:29	6° ♄ 55'20	24S36'54"		minimum elong	2066 Oct 27 00:41:52	3° ♁ 47'15 0°52'59
max. Earth dist.	2066 Jan 05 09:59:51	12° ♄ 04'38	1.43863 AU		asc. node	2066 Oct 29 13:41:27	0° ♁ 43'25
						2066 Oct 30 06:02:37	30° ♁
superior conj	2066 Jan 10 22:13:27	20° ♄ 59'05	-1°56'09		morning rise	2066 Nov 01 16:52:21	27° ♄ 57'49
minimum elong	2066 Jan 10 16:40:40	20° ♄ 36'24	1°55'51		direct	2066 Nov 04 22:47:49	26° ♄ 57'06
	2066 Jan 16 08:49:20	0° ♁			max. declination.	2066 Nov 06 18:23:36	27° ♄ 15'10 8S38'13"
evening rise	2066 Jan 23 22:04:53	12° ♁ 52'03				2066 Nov 11 00:55:16	0° ♁

morning max el	2066 Nov 11 22:15:26	0°♄51'18	19°06'52	minimum elong	2067 Oct 11 00:22:03	17°♄30'25	1°50'51
	2066 Dec 02 09:06:06	0°♄		asc. node	2067 Oct 16 10:46:08	12°♄14'53	
morning set	2066 Dec 04 18:12:36	3°♄42'36		morning rise	2067 Oct 16 23:10:07	11°♄57'05	
desc. node	2066 Dec 06 19:52:55	6°♄56'25		direct	2067 Oct 19 20:18:47	11°♄10'55	
max. Earth dist.	2066 Dec 19 01:33:10	26°♄06'28	1.44865 AU	max. declination.	2067 Oct 22 05:41:27	11°♄41'48	3S16' 0"
				morning max el	2067 Oct 26 10:37:15	14°♄46'56	18°24'35
superior conj	2066 Dec 21 10:47:07	29°♄52'24	-1°28'12		2067 Nov 06 11:50:03	0°♄	
minimum elong	2066 Dec 21 01:55:10	29°♄17'21	1°27'19	morning set	2067 Nov 15 01:04:01	13°♄43'48	
	2066 Dec 21 12:42:16	0°♄		desc. node	2067 Nov 23 16:54:44	27°♄36'01	
min. declination.	2066 Dec 24 22:42:35	5°♄25'38	25S 2'38"		2067 Nov 25 05:13:10	0°♄	
evening rise	2067 Jan 05 06:50:51	23°♄46'20					
	2067 Jan 09 01:49:57	0°♄		superior conj	2067 Nov 30 12:00:42	8°♄20'09	-0°44'29
evening max el	2067 Jan 25 07:08:36	23°♄55'39	18°33'12	minimum elong	2067 Nov 30 06:18:01	7°♄57'42	0°43'44
asc. node	2067 Jan 25 12:59:46	24°♄10'20		max. Earth dist.	2067 Dec 01 20:36:56	10°♄28'16	1.45149 AU
retrograde	2067 Jan 31 20:53:49	27°♄33'12			2067 Dec 14 07:18:26	0°♄	
evening set	2067 Feb 03 23:22:00	26°♄39'19		evening rise	2067 Dec 16 16:12:31	3°♄43'47	
max. declination.	2067 Feb 04 15:50:12	26°♄13'21	10S 9'18"	min. declination.	2067 Dec 16 18:31:43	3°♄52'54	25S24'31"
inferior conj	2067 Feb 09 18:50:45	21°♄05'25	3°37'55	greatest brilliancy	2067 Dec 26 22:27:53	19°♄49'35	-0.8m
minimum elong	2067 Feb 09 17:30:52	21°♄09'22	3°37'47		2068 Jan 02 18:03:28	0°♄	
min. Earth dist.	2067 Feb 11 19:48:23	18°♄40'29	0.64682 AU	evening max el	2068 Jan 08 16:20:59	7°♄23'17	19°12'20
morning rise	2067 Feb 15 11:10:24	14°♄57'00		asc. node	2068 Jan 12 10:02:47	10°♄25'57	
direct	2067 Feb 22 06:15:05	12°♄05'53		retrograde	2068 Jan 15 16:41:10	11°♄23'51	
min. declination.	2067 Mar 01 11:38:32	14°♄40'16	15S49'26"	evening set	2068 Jan 19 01:40:16	10°♄17'26	
desc. node	2067 Mar 04 19:03:20	17°♄11'43		max. declination.	2068 Jan 21 03:55:00	8°♄32'43	15S37'56"
morning max el	2067 Mar 07 17:05:35	19°♄55'20	27°17'03	inferior conj	2068 Jan 24 15:23:39	4°♄27'10	3°18'36
	2067 Mar 16 08:27:32	0°♄		minimum elong	2068 Jan 24 13:06:58	4°♄34'28	3°18'12
	2067 Apr 04 18:34:45	0°♄		min. Earth dist.	2068 Jan 26 01:08:30	2°♄39'18	0.66055 AU
morning set	2067 Apr 12 03:00:22	13°♄40'26			2068 Jan 28 06:27:49	30°♄	
max. Earth dist.	2067 Apr 16 19:55:58	23°♄05'59	1.33697 AU	morning rise	2068 Jan 30 00:18:12	28°♄15'14	
				direct	2068 Feb 05 09:29:02	25°♄26'35	
superior conj	2067 Apr 20 06:46:03	0°♄17'47	-0°33'25		2068 Feb 14 22:15:39	0°♄	
minimum elong	2067 Apr 20 08:26:56	0°♄26'40	0°33'04	min. declination.	2068 Feb 14 23:42:35	0°♄03'04	19S20'20"
	2067 Apr 20 03:23:47	0°♄		morning max el	2068 Feb 18 02:52:29	2°♄58'09	26°18'08
asc. node	2067 Apr 23 12:20:55	7°♄10'07		desc. node	2068 Feb 19 16:05:59	4°♄35'00	
evening rise	2067 Apr 27 13:33:43	15°♄45'51			2068 Mar 09 14:36:52	0°♄	
	2067 May 04 17:40:01	0°♄		morning set	2068 Mar 25 06:05:29	26°♄43'52	
evening max el	2067 May 18 11:56:35	19°♄35'48	22°03'00		2068 Mar 26 23:36:16	0°♄	
max. declination.	2067 May 18 17:16:45	19°♄48'39	25N18'58"	max. Earth dist.	2068 Mar 29 07:17:32	4°♄29'06	1.35080 AU
retrograde	2067 May 31 05:17:14	25°♄52'13					
desc. node	2067 May 31 18:17:49	25°♄51'33		superior conj	2068 Apr 03 04:16:39	14°♄15'04	-1°01'16
evening set	2067 Jun 02 23:15:12	25°♄35'26		minimum elong	2068 Apr 03 07:22:06	14°♄30'53	1°00'43
min. Earth dist.	2067 Jun 11 17:11:31	21°♄51'15	0.55053 AU	asc. node	2068 Apr 09 09:23:11	27°♄07'32	
inferior conj	2067 Jun 12 08:22:16	21°♄29'58	-3°11'17	evening rise	2068 Apr 10 21:45:07	0°♄15'38	
minimum elong	2067 Jun 12 00:35:58	21°♄40'52	3°08'59		2068 Apr 10 18:43:02	0°♄	
morning rise	2067 Jun 21 03:31:25	17°♄37'14			2068 Apr 29 00:35:28	0°♄	
min. declination.	2067 Jun 23 09:00:52	17°♄18'02	18N20'58"	evening max el	2068 Apr 29 13:52:35	0°♄33'00	20°38'01
direct	2067 Jun 24 04:48:41	17°♄16'29		max. declination.	2068 May 05 05:38:22	4°♄40'09	23N30'24"
morning max el	2067 Jul 06 01:45:15	22°♄49'38	21°31'04	retrograde	2068 May 10 18:24:18	5°♄59'19	
	2067 Jul 12 07:08:24	0°♄		evening set	2068 May 12 20:34:14	5°♄48'42	
asc. node	2067 Jul 20 11:35:58	13°♄20'54		desc. node	2068 May 17 15:19:44	4°♄14'24	
max. declination.	2067 Jul 20 19:36:17	13°♄58'24	22N45'47"	inferior conj	2068 May 22 02:17:06	1°♄50'53	-1°17'52
morning set	2067 Jul 26 11:20:50	25°♄14'09		minimum elong	2068 May 21 22:36:01	1°♄56'11	1°16'32
	2067 Jul 28 18:01:43	0°♄		min. Earth dist.	2068 May 23 05:38:25	1°♄11'33	0.55006 AU
					2068 May 25 09:08:35	30°♄	
superior conj	2067 Aug 02 19:02:34	10°♄35'45	1°41'44	morning rise	2068 May 30 23:55:04	27°♄41'53	
minimum elong	2067 Aug 02 17:39:21	10°♄28'32	1°41'39	direct	2068 Jun 03 14:39:09	27°♄13'46	
max. Earth dist.	2067 Aug 06 18:05:56	18°♄42'44	1.35065 AU	min. declination.	2068 Jun 06 01:24:40	27°♄27'09	15N40'41"
evening rise	2067 Aug 11 02:51:41	27°♄16'16			2068 Jun 12 06:15:07	0°♄	
	2067 Aug 12 13:30:14	0°♄		morning max el	2068 Jun 16 21:01:40	3°♄38'01	23°09'54
desc. node	2067 Aug 27 17:38:42	25°♄37'06			2068 Jul 04 22:19:44	0°♄	
	2067 Aug 30 16:53:37	0°♄		asc. node	2068 Jul 06 08:39:58	2°♄48'27	
evening max el	2067 Sep 15 10:06:26	19°♄17'44	26°44'36	morning set	2068 Jul 09 22:29:19	10°♄08'24	
retrograde	2067 Sep 28 10:34:52	26°♄32'23		max. declination.	2068 Jul 10 00:21:40	10°♄18'14	23N40'45"
min. declination.	2067 Sep 28 14:09:53	26°♄32'19	13S43'48"				
evening set	2067 Oct 05 02:10:45	23°♄50'27		superior conj	2068 Jul 17 00:24:57	25°♄17'39	1°31'43
min. Earth dist.	2067 Oct 09 02:36:14	19°♄45'24	0.65870 AU	minimum elong	2068 Jul 16 22:15:08	25°♄06'04	1°31'29
inferior conj	2067 Oct 10 21:22:07	17°♄39'16	-1°52'03		2068 Jul 19 05:33:04	0°♄	

max. Earth dist.	2068 Jul 19 15:12:58	0°♄50'52	1.33741 AU	min. declination.	2069 May 19 11:34:29	7°♄37'29	10N58' 2"
evening rise	2068 Jul 24 16:01:43	11°♄08'58		morning max el	2069 May 29 11:40:15	14°♄05'51	24°51'23
	2068 Aug 03 21:16:35	0°♄			2069 Jun 10 23:27:15	0°♄	
desc. node	2068 Aug 13 14:40:38	15°♄17'04		asc. node	2069 Jun 23 05:42:00	22°♄34'36	
	2068 Aug 25 06:06:57	0°♄		morning set	2069 Jun 24 10:19:46	25°♄04'35	
evening max el	2068 Aug 27 21:57:29	2°♄41'24	27°18'50		2069 Jun 26 17:37:13	0°♄	
retrograde	2068 Sep 10 09:13:21	9°♄59'58		max. declination.	2069 Jun 30 07:59:13	7°♄49'55	24N20'24"
min. declination.	2068 Sep 11 12:19:22	9°♄55'59	7S41' 5"				
evening set	2068 Sep 17 10:36:15	7°♄16'54		superior conj	2069 Jul 01 09:48:26	10°♄10'46	1°16'44
min. Earth dist.	2068 Sep 21 04:41:07	3°♄47'32	0.64520 AU	minimum elong	2069 Jul 01 07:24:00	9°♄57'39	1°16'22
inferior conj	2068 Sep 23 12:05:51	1°♄18'14	-2°50'08	max. Earth dist.	2069 Jul 02 20:25:34	13°♄18'52	1.32827 AU
minimum elong	2068 Sep 23 16:34:34	1°♄06'09	2°48'33	evening rise	2069 Jul 08 14:45:45	25°♄29'41	
	2068 Sep 24 17:30:35	30°♄			2069 Jul 10 20:25:43	0°♄	
morning rise	2068 Sep 29 23:33:19	25°♄50'51			2069 Jul 28 08:59:51	0°♄	
asc. node	2068 Oct 02 07:50:40	25°♄15'58		desc. node	2069 Jul 31 11:42:43	4°♄19'07	
direct	2068 Oct 02 14:15:44	25°♄15'35		evening max el	2069 Aug 10 07:35:01	15°♄39'01	27°24'20
max. declination.	2068 Oct 05 16:20:00	26°♄04'33	2N22' 6"	retrograde	2069 Aug 24 01:43:00	22°♄56'19	
morning max el	2068 Oct 09 02:21:21	28°♄42'37	17°59'14	min. declination.	2069 Aug 26 07:53:54	22°♄41'36	1S 3'23"
	2068 Oct 10 07:20:39	0°♄		evening set	2069 Aug 31 06:40:32	20°♄23'55	
morning set	2068 Oct 26 12:40:43	25°♄09'42		min. Earth dist.	2069 Sep 03 20:57:49	17°♄24'51	0.62829 AU
	2068 Oct 29 09:39:19	0°♄		inferior conj	2069 Sep 06 16:39:41	14°♄41'20	-3°44'26
				minimum elong	2069 Sep 06 22:00:15	14°♄28'24	3°42'58
superior conj	2068 Nov 08 23:40:42	17°♄24'04	0°04'02	morning rise	2069 Sep 13 14:45:24	9°♄31'57	
minimum elong	2068 Nov 09 00:09:24	17°♄26'00	0°03'58	direct	2069 Sep 16 01:39:53	9°♄04'11	
behind sun begin	2068 Nov 08 14:01:30	16°♄45'12		asc. node	2069 Sep 19 04:53:52	9°♄50'37	
behind sun end	2068 Nov 09 10:17:19	18°♄06'44		max. declination.	2069 Sep 20 03:13:47	10°♄21'56	7N54'24"
desc. node	2068 Nov 09 13:57:05	18°♄21'26		morning max el	2069 Sep 22 18:25:18	12°♄30'31	17°51'35
max. Earth dist.	2068 Nov 13 15:26:33	24°♄49'56	1.44692 AU		2069 Oct 04 14:46:11	0°♄	
	2068 Nov 16 22:10:09	0°♄		morning set	2069 Oct 08 23:38:27	7°♄42'31	
evening rise	2068 Nov 25 06:39:21	12°♄58'41					
	2068 Dec 06 10:40:34	0°♄		superior conj	2069 Oct 20 12:02:47	27°♄41'36	0°46'16
min. declination.	2068 Dec 07 16:41:00	1°♄52'13	25S42'26"	minimum elong	2069 Oct 20 16:08:58	27°♄58'50	0°45'43
greatest brilliancy	2068 Dec 09 14:00:59	4°♄40'12	-0.6m		2069 Oct 21 21:06:29	0°♄	
evening max el	2068 Dec 21 20:55:31	20°♄52'44	20°07'04	desc. node	2069 Oct 27 11:00:24	9°♄08'56	
asc. node	2068 Dec 29 07:05:34	25°♄24'16		max. Earth dist.	2069 Oct 27 06:58:34	8°♄52'40	1.43560 AU
retrograde	2068 Dec 29 14:31:34	25°♄24'48		evening rise	2069 Nov 04 15:11:03	22°♄03'56	
evening set	2069 Jan 02 07:32:55	24°♄04'34			2069 Nov 09 19:22:47	0°♄	
inferior conj	2069 Jan 07 17:47:54	18°♄00'47	2°48'33	min. declination.	2069 Nov 29 14:07:34	28°♄17'55	25S52' 8"
minimum elong	2069 Jan 07 15:10:31	18°♄09'37	2°47'51		2069 Nov 30 22:53:37	0°♄	
min. Earth dist.	2069 Jan 08 13:49:11	16°♄53'28	0.67028 AU	evening max el	2069 Dec 04 19:57:58	4°♄23'08	21°14'43
max. declination.	2069 Jan 09 06:29:18	15°♄58'09	19S24'15"	retrograde	2069 Dec 13 12:06:39	9°♄32'06	
morning rise	2069 Jan 12 22:36:07	11°♄47'23		asc. node	2069 Dec 16 04:07:37	8°♄52'37	
direct	2069 Jan 18 18:24:42	9°♄11'42		evening set	2069 Dec 17 15:02:46	7°♄56'36	
morning max el	2069 Jan 30 11:41:19	16°♄11'21	24°59'38	inferior conj	2069 Dec 22 23:40:28	1°♄42'34	2°10'05
min. declination.	2069 Feb 02 07:17:01	19°♄12'07	21S33'36"	minimum elong	2069 Dec 22 21:13:57	1°♄51'00	2°09'15
desc. node	2069 Feb 05 13:08:39	23°♄01'36		min. Earth dist.	2069 Dec 23 07:30:38	1°♄15'31	0.67617 AU
	2069 Feb 10 22:16:20	0°♄			2069 Dec 24 05:35:12	30°♄	
	2069 Mar 02 13:15:53	0°♄		morning rise	2069 Dec 28 03:13:31	25°♄29'37	
morning set	2069 Mar 07 15:44:02	8°♄51'55		max. declination.	2069 Dec 30 18:50:24	23°♄45'20	20S 9'47"
max. Earth dist.	2069 Mar 11 07:29:41	15°♄30'40	1.36887 AU	direct	2070 Jan 02 08:13:24	23°♄13'35	
				morning max el	2070 Jan 12 20:18:46	29°♄28'57	23°31'53
superior conj	2069 Mar 17 16:03:54	27°♄38'58	-1°27'24		2070 Jan 13 08:27:58	0°♄	
minimum elong	2069 Mar 17 20:11:20	27°♄59'14	1°26'50	desc. node	2070 Jan 23 10:11:18	12°♄13'05	
	2069 Mar 18 20:39:55	0°♄		min. declination.	2070 Jan 24 03:21:40	13°♄11'03	22S52'17"
evening rise	2069 Mar 26 00:52:28	14°♄24'31			2070 Feb 04 18:15:37	0°♄	
asc. node	2069 Mar 27 06:25:21	16°♄52'11		morning set	2070 Feb 17 01:43:36	19°♄47'45	
	2069 Apr 03 06:13:16	0°♄		max. Earth dist.	2070 Feb 21 02:05:46	26°♄42'57	1.38987 AU
evening max el	2069 Apr 12 03:41:10	12°♄09'36	19°30'34		2070 Feb 22 22:39:46	0°♄	
max. declination.	2069 Apr 19 20:23:09	16°♄34'45	19N35'15"				
retrograde	2069 Apr 21 14:55:14	16°♄44'05		superior conj	2070 Feb 28 14:17:55	10°♄19'43	-1°49'08
evening set	2069 Apr 23 16:03:52	16°♄32'11		minimum elong	2070 Feb 28 18:25:56	10°♄39'04	1°48'46
inferior conj	2069 May 02 06:07:33	12°♄26'59	0°38'54	evening rise	2070 Mar 09 20:12:28	28°♄05'11	
minimum elong	2069 May 02 07:45:35	12°♄24'21	0°38'19		2070 Mar 10 19:55:27	0°♄	
desc. node	2069 May 04 12:21:19	10°♄59'57		asc. node	2070 Mar 14 03:27:12	6°♄18'52	
min. Earth dist.	2069 May 04 20:08:16	10°♄47'45	0.55904 AU	evening max el	2070 Mar 26 03:53:03	24°♄24'17	18°42'56
morning rise	2069 May 10 20:41:05	7°♄46'00		retrograde	2070 Apr 03 04:11:03	28°♄18'46	
direct	2069 May 15 09:42:28	6°♄59'21		max. declination.	2070 Apr 03 11:28:27	28°♄18'27	13N55'34"

evening set	2070 Apr 05 10:48:59	28°♄01'02		minimum elong	2071 Feb 10 20:20:50	22°♁15'15	2°02'41
inferior conj	2070 Apr 13 05:15:41	23°♄38'35	2°10'57		2071 Feb 15 03:33:55	0°♁	
minimum elong	2070 Apr 13 09:12:51	23°♄31'11	2°09'50	evening rise	2071 Feb 21 04:26:17	11°♁09'41	
min. Earth dist.	2070 Apr 16 13:22:24	21°♄10'01	0.57554 AU	asc. node	2071 Mar 01 00:28:03	25°♁21'04	
morning rise	2070 Apr 21 04:28:37	18°♄24'12			2071 Mar 03 21:13:28	0°♄	
desc. node	2070 Apr 21 09:21:39	18°♄18'46		evening max el	2071 Mar 09 11:20:23	7°♄08'30	18°15'17
direct	2070 Apr 26 18:58:18	17°♄06'37		retrograde	2071 Mar 16 12:19:13	10°♄39'41	
min. declination.	2070 May 01 15:23:35	18°♄03'28	4N57'59"	max. declination.	2071 Mar 18 00:58:50	10°♄29'57	7N15'48"
morning max el	2070 May 11 03:29:53	24°♄39'48	26°20'24	evening set	2071 Mar 19 00:59:46	10°♄13'23	
	2070 May 16 01:04:39	0°♄		inferior conj	2071 Mar 26 01:33:12	5°♄29'57	3°09'18
	2070 Jun 03 23:35:42	0°♄		minimum elong	2071 Mar 26 04:49:41	5°♄22'47	3°08'41
morning set	2070 Jun 08 21:18:09	9°♄57'33		min. Earth dist.	2071 Mar 29 11:13:53	2°♄33'03	0.59626 AU
asc. node	2070 Jun 10 02:44:07	12°♄33'38			2071 Apr 01 23:32:06	30°♁	
				morning rise	2071 Apr 02 06:17:39	29°♁49'58	
superior conj	2070 Jun 15 21:14:38	25°♄08'30	0°57'37	desc. node	2071 Apr 08 06:23:29	27°♁56'24	
minimum elong	2070 Jun 15 19:04:42	24°♄56'35	0°57'13	direct	2071 Apr 08 16:52:59	27°♁55'52	
max. Earth dist.	2070 Jun 16 07:02:21	26°♄02'21	1.32310 AU	min. declination.	2071 Apr 14 00:10:53	29°♁08'01	1S30'23"
	2070 Jun 18 02:23:07	0°♄			2071 Apr 15 17:38:55	0°♄	
max. declination.	2070 Jun 20 18:37:34	5°♄47'39	24N51'23"	morning max el	2071 Apr 23 00:42:00	5°♄43'29	27°22'37
evening rise	2070 Jun 22 20:04:30	10°♄09'15			2071 May 11 02:09:13	0°♄	
	2070 Jul 03 03:17:16	0°♄		morning set	2071 May 24 05:39:18	24°♄40'24	
desc. node	2070 Jul 18 08:42:45	22°♄27'41			2071 May 26 18:07:16	0°♄	
evening max el	2070 Jul 23 12:37:56	27°♄59'02	26°57'05	asc. node	2071 May 27 23:45:46	2°♄39'43	
	2070 Jul 25 19:02:31	0°♄					
retrograde	2070 Aug 06 10:48:44	5°♄12'35		superior conj	2071 May 31 08:54:24	10°♄03'30	0°35'04
min. declination.	2070 Aug 10 06:34:04	4°♄33'45	5N42'51"	minimum elong	2071 May 31 07:24:44	9°♄55'16	0°34'46
evening set	2070 Aug 13 10:06:06	3°♄04'17		max. Earth dist.	2071 May 30 19:19:21	8°♄48'48	1.32177 AU
min. Earth dist.	2070 Aug 17 02:51:12	0°♄24'21	0.60877 AU	evening rise	2071 Jun 07 05:28:14	24°♄58'16	
	2070 Aug 17 14:39:35	30°♁			2071 Jun 09 15:27:05	0°♄	
inferior conj	2070 Aug 20 07:48:02	27°♄40'52	-4°29'24	max. declination.	2071 Jun 11 11:26:34	3°♄41'30	25N16'29"
minimum elong	2070 Aug 20 12:36:06	27°♄30'36	4°28'33		2071 Jun 27 04:59:36	0°♄	
morning rise	2070 Aug 27 16:53:44	22°♄52'18		desc. node	2071 Jul 05 05:42:08	9°♄19'44	
direct	2070 Aug 30 02:50:11	22°♄29'11		evening max el	2071 Jul 05 10:43:47	9°♄31'47	25°57'34
max. declination.	2070 Sep 04 17:25:31	24°♄39'38	12N57'38"	retrograde	2071 Jul 19 10:44:08	16°♄41'26	
asc. node	2070 Sep 06 01:55:59	25°♄49'14		evening set	2071 Jul 25 16:05:43	15°♄08'23	
morning max el	2070 Sep 06 07:45:15	26°♄03'09	18°02'44	min. declination.	2071 Jul 25 21:57:33	15°♄01'09	12N 1' 5"
	2070 Sep 09 17:17:05	0°♄		min. Earth dist.	2071 Jul 29 23:12:53	12°♄31'07	0.58814 AU
morning set	2070 Sep 22 03:45:59	21°♄04'35		inferior conj	2071 Aug 02 05:52:31	10°♄06'10	-4°56'04
	2070 Sep 26 23:33:33	0°♄		minimum elong	2071 Aug 02 07:47:41	10°♄02'36	4°55'58
				morning rise	2071 Aug 10 01:41:52	5°♄40'30	
superior conj	2070 Oct 02 03:04:29	9°♄17'16	1°17'05	direct	2071 Aug 12 13:25:33	5°♄19'59	
minimum elong	2070 Oct 02 07:25:17	9°♄36'30	1°16'36	morning max el	2071 Aug 20 15:10:15	9°♄11'19	18°34'04
max. Earth dist.	2070 Oct 09 17:47:58	22°♄23'04	1.41900 AU	max. declination.	2071 Aug 20 20:29:14	9°♄24'07	17N 8'21"
desc. node	2070 Oct 14 08:02:45	29°♄55'07		asc. node	2071 Aug 23 22:58:47	12°♄56'26	
	2070 Oct 14 09:15:30	0°♄			2071 Sep 03 05:09:14	0°♄	
evening rise	2070 Oct 15 10:50:57	1°♄42'43		morning set	2071 Sep 05 20:02:17	5°♄01'41	
	2070 Nov 03 06:51:01	0°♄					
evening max el	2070 Nov 17 13:46:26	17°♄55'30	22°31'17	superior conj	2071 Sep 14 16:18:34	21°♄58'33	1°36'02
min. declination.	2070 Nov 20 03:45:32	20°♄19'38	25S34'52"	minimum elong	2071 Sep 14 19:08:14	22°♄11'45	1°35'50
retrograde	2070 Nov 27 07:43:50	23°♄43'00			2071 Sep 19 01:24:40	0°♄	
evening set	2070 Dec 01 22:30:23	21°♄50'53		max. Earth dist.	2071 Sep 21 23:50:11	5°♄11'46	1.39915 AU
asc. node	2070 Dec 03 01:08:58	20°♄49'46		evening rise	2071 Sep 26 04:28:11	12°♄22'18	
inferior conj	2070 Dec 07 07:03:53	15°♄30'07	1°24'52	desc. node	2071 Oct 01 05:03:42	20°♄36'18	
minimum elong	2070 Dec 07 05:14:42	15°♄36'25	1°24'08		2071 Oct 07 05:42:49	0°♄	
min. Earth dist.	2070 Dec 07 03:46:09	15°♄41'32	0.67843 AU		2071 Oct 29 16:24:59	0°♄	
morning rise	2070 Dec 12 11:50:26	9°♄19'41		evening max el	2071 Oct 31 03:40:55	1°♄30'51	23°51'19
direct	2070 Dec 17 01:58:44	7°♄26'18		min. declination.	2071 Nov 07 13:17:41	7°♄06'14	24S10' 0"
max. declination.	2070 Dec 16 19:36:40	7°♄26'40	18S42'41"	retrograde	2071 Nov 11 00:08:47	7°♄54'16	
morning max el	2070 Dec 26 08:05:55	12°♄53'01	22°04'37	evening set	2071 Nov 16 04:20:29	5°♄45'00	
	2071 Jan 08 22:20:08	0°♄		asc. node	2071 Nov 19 22:10:54	1°♄35'49	
desc. node	2071 Jan 10 07:14:27	1°♄56'51		min. Earth dist.	2071 Nov 21 00:15:44	0°♄08'38	0.67722 AU
min. declination.	2071 Jan 15 18:51:56	9°♄57'25	23S42'46"		2071 Nov 21 02:48:32	30°♁	
morning set	2071 Jan 28 06:55:50	29°♄19'45		inferior conj	2071 Nov 21 14:11:00	29°♄21'20	0°34'12
	2071 Jan 28 16:54:19	0°♄		minimum elong	2071 Nov 21 13:22:32	29°♄24'05	0°33'51
max. Earth dist.	2071 Feb 02 23:43:34	8°♄41'46	1.41128 AU	morning rise	2071 Nov 26 22:23:32	23°♄15'33	
				direct	2071 Nov 30 22:36:20	21°♄44'45	
superior conj	2071 Feb 10 18:09:20	22°♄05'32	-2°02'46	max. declination.	2071 Dec 01 18:02:03	21°♄48'12	15S41'57"

morning max el	2071 Dec 09 02:30:41	26°  27'44"	20°45'18"	evening set	2072 Oct 30 06:51:54	19°  36'01"	
	2071 Dec 12 07:18:16	0° 		min. Earth dist.	2072 Nov 03 18:38:00	14°  34'04"	0.67268 AU
desc. node	2071 Dec 28 04:17:06	22°  03'55"		inferior conj	2072 Nov 04 19:16:32	13°  13'32"	-0°20'40"
	2072 Jan 02 10:22:35	0° 		minimum elong	2072 Nov 04 19:47:42	13°  11'50"	0°20'26"
morning set	2072 Jan 07 10:35:47	7°  43'19"		asc. node	2072 Nov 05 19:14:54	11°  55'53"	
min. declination.	2072 Jan 07 13:10:18	7°  53'20"	24S19'11"	morning rise	2072 Nov 10 08:55:07	7°  14'43"	
max. Earth dist.	2072 Jan 16 05:00:37	21°  38'25"	1.43005 AU	direct	2072 Nov 13 20:51:17	6°  04'09"	
	2072 Jan 21 07:06:31	0° 		max. declination.	2072 Nov 15 08:38:17	6°  16'13"	11S29'25"
superior conj	2072 Jan 22 22:26:34	2°  44'37"	-2°03'30"	morning max el	2072 Nov 21 04:48:46	10°  13'15"	19°38'30"
minimum elong	2072 Jan 22 20:07:32	2°  34'53"	2°03'27"	greatest brilliancy	2072 Dec 04 10:57:59	27°  15'04"	-0.7m
evening rise	2072 Feb 03 21:37:09	23°  30'09"			2072 Dec 05 21:28:02	0° 	
	2072 Feb 07 13:36:50	0° 		desc. node	2072 Dec 14 01:18:29	12°  27'08"	
asc. node	2072 Feb 15 21:29:49	13°  51'25"		morning set	2072 Dec 16 07:21:30	15°  27'54"	
evening max el	2072 Feb 20 22:50:49	20°  13'55"	18°07'18"		2072 Dec 25 08:06:41	0° 	
retrograde	2072 Feb 27 12:06:24	23°  37'35"		max. Earth dist.	2072 Dec 28 17:16:43	5°  20'26"	1.44372 AU
evening set	2072 Mar 01 06:02:27	23°  01'23"		min. declination.	2072 Dec 29 07:36:02	6°  17'18"	24S47'59"
max. declination.	2072 Feb 29 21:12:17	23°  10'18"	0N16'32"	superior conj	2073 Jan 02 00:14:55	12°  11'36"	-1°46'37"
inferior conj	2072 Mar 07 16:27:23	17°  57'04"	3°37'33"	minimum elong	2073 Jan 01 16:40:10	11°  34'08"	1°46'03"
minimum elong	2072 Mar 07 17:49:18	17°  53'38"	3°37'26"		2073 Jan 12 20:33:22	0° 	
min. Earth dist.	2072 Mar 10 16:46:23	14°  56'42"	0.61767 AU	evening rise	2073 Jan 15 19:17:55	4°  57'18"	
morning rise	2072 Mar 14 04:09:08	12°  00'45"			2073 Jan 31 10:47:53	0° 	
direct	2072 Mar 21 01:10:06	9°  34'35"		asc. node	2073 Feb 01 18:32:42	1°  39'24"	
desc. node	2072 Mar 25 03:25:29	10°  21'29"		evening max el	2073 Feb 03 11:25:09	3°  32'13"	18°18'27"
min. declination.	2072 Mar 26 19:27:39	11°  05'05"	7S43'56"	retrograde	2073 Feb 09 22:42:08	7°  01'38"	
morning max el	2072 Apr 04 04:22:04	17°  29'28"	27°49'05"	evening set	2073 Feb 12 21:48:36	6°  14'35"	
	2072 Apr 14 14:26:40	0° 		max. declination.	2073 Feb 13 03:26:40	6°  06'59"	6S30'21"
	2072 May 02 18:53:28	0° 		inferior conj	2073 Feb 18 21:49:16	0°  51'00"	3°42'48"
morning set	2072 May 07 09:28:39	9°  07'06"		minimum elong	2073 Feb 18 21:19:31	0°  52'24"	3°42'48"
max. Earth dist.	2072 May 13 05:24:45	21°  25'20"	1.32436 AU		2073 Feb 19 15:51:39	30° 	
asc. node	2072 May 13 20:48:22	22°  48'48"		min. Earth dist.	2073 Feb 21 07:46:48	28°  08'31"	0.63713 AU
superior conj	2072 May 14 19:09:11	24°  50'25"	0°09'50"	morning rise	2073 Feb 24 20:02:59	24°  45'25"	
minimum elong	2072 May 14 18:41:42	24°  47'55"	0°09'44"	direct	2073 Mar 03 18:00:54	21°  59'05"	
behind sun begin	2072 May 14 14:37:12	24°  25'43"		min. declination.	2073 Mar 10 05:59:21	24°  00'56"	13S10'38"
behind sun end	2072 May 14 22:46:12	25°  10'09"		desc. node	2073 Mar 12 00:28:03	25°  09'37"	
	2072 May 17 03:48:43	0° 		morning max el	2073 Mar 17 12:33:23	29°  54'09"	27°38'22"
evening rise	2072 May 21 16:58:08	9°  50'06"			2073 Mar 17 14:53:18	0° 	
	2072 Jun 01 02:06:00	0° 			2073 Apr 08 13:38:25	0° 	
max. declination.	2072 Jun 01 13:51:55	0°  50'23"	25N34'52"	morning set	2073 Apr 21 06:21:57	23°  09'52"	
evening max el	2072 Jun 16 02:08:38	20°  20'57"	24°32'59"		2073 Apr 24 15:16:08	0° 	
desc. node	2072 Jun 21 02:42:03	24°  25'02"		max. Earth dist.	2073 Apr 26 09:11:59	3°  38'32"	1.33106 AU
retrograde	2072 Jun 29 23:53:00	27°  23'00"		superior conj	2073 Apr 29 02:11:25	9°  23'34"	-0°17'20"
evening set	2072 Jul 04 22:22:04	26°  28'12"		minimum elong	2073 Apr 29 03:02:42	9°  28'09"	0°17'09"
min. Earth dist.	2072 Jul 10 13:10:38	23°  35'36"	0.56905 AU	asc. node	2073 Apr 30 17:51:51	12°  28'56'45"	
min. declination.	2072 Jul 11 22:52:44	22°  42'01"	16N50'39"	evening rise	2073 May 06 04:41:32	24°  38'23"	
inferior conj	2072 Jul 13 08:02:56	21°  48'22"	-4°51'20"		2073 May 08 18:34:13	0° 	
minimum elong	2072 Jul 13 04:48:19	21°  53'36"	4°51'03"	max. declination.	2073 May 23 21:44:41	25°  12'51"	25N37'16"
morning rise	2072 Jul 21 13:57:08	17°  43'59"			2073 May 27 20:04:16	0° 	
direct	2072 Jul 24 05:17:45	17°  24'51"		evening max el	2073 May 28 16:03:18	0°  49'44"	22°57'13"
morning max el	2072 Aug 02 13:42:57	21°  46'00"	19°26'41"	desc. node	2073 Jun 07 23:42:07	7°  08'06"	
max. declination.	2072 Aug 06 11:31:33	26°  08'20"	20N 8'33"	retrograde	2073 Jun 11 00:52:55	7°  29'20"	
	2072 Aug 09 05:08:24	0° 		evening set	2073 Jun 14 11:19:14	7°  03'25"	
asc. node	2072 Aug 09 20:02:20	0°  57'35"		min. Earth dist.	2073 Jun 22 01:00:31	3°  41'28"	0.55494 AU
morning set	2072 Aug 19 20:45:48	19°  23'39"		inferior conj	2073 Jun 23 14:58:04	2°  46'34"	-4°01'41"
	2072 Aug 25 03:33:19	0° 		minimum elong	2073 Jun 23 07:23:50	2°  57'33"	3°59'59"
superior conj	2072 Aug 27 22:11:47	5°  29'46"	1°44'40"		2073 Jun 28 21:50:22	30° 	
minimum elong	2072 Aug 27 23:07:23	5°  21'49"	1°44'39"	min. declination.	2073 Jul 01 02:08:37	29°  12'08"	18N43' 9"
max. Earth dist.	2072 Sep 03 03:40:13	17°  21'05"	1.37862 AU	morning rise	2073 Jul 02 05:55:40	28°  15'40'5"	
evening rise	2072 Sep 06 21:57:26	24°  08'34"		direct	2073 Jul 05 02:17:53	28°  13'45'7"	
	2072 Sep 10 06:59:55	0° 			2073 Jul 10 21:40:44	0° 	
desc. node	2072 Sep 17 02:03:49	11°  08'15"		morning max el	2073 Jul 16 01:15:04	3°  39'34"	20°40'23"
	2072 Sep 29 22:42:54	0° 		max. declination.	2073 Jul 26 07:01:13	17°  12'45"	22N 1'23"
evening max el	2072 Oct 12 15:42:39	15°  09'09"	25°08'09"	asc. node	2073 Jul 27 17:06:33	19°  40'19"	
min. declination.	2072 Oct 22 21:53:40	21°  15'21"	21S13' 7"		2073 Aug 02 03:19:45	0° 	
retrograde	2072 Oct 24 12:27:15	22°  01'14"		morning set	2073 Aug 04 03:13:16	4°  03'15"	

superior conj	2073 Aug 11 16:00:55	19°♎37'17	1°44'51	min. declination.	2074 Jun 16 08:58:10	8°♊54'36	17N30'43"
minimum elong	2073 Aug 11 15:19:44	19°♎33'46	1°44'49	morning max el	2074 Jun 28 01:34:42	14°♊49'08	22°12'00
max. Earth dist.	2073 Aug 16 10:57:55	29°♎11'44	1.35985 AU		2074 Jul 09 15:39:31	0°♊	
	2073 Aug 16 20:53:14	0°♎		asc. node	2074 Jul 14 14:09:59	8°♊53'49	
evening rise	2073 Aug 20 12:06:17	6°♎54'23		max. declination.	2074 Jul 16 06:46:45	12°♊12'45	23N11'12"
	2073 Sep 03 00:54:07	0°♎		morning set	2074 Jul 19 13:09:31	18°♊53'27	
desc. node	2073 Sep 03 23:05:14	1°♎26'00			2074 Jul 24 18:59:44	0°♎	
evening max el	2073 Sep 25 03:42:35	28°♎48'45	26°14'23				
	2073 Sep 26 10:02:06	0°♎		superior conj	2074 Jul 26 17:58:39	4°♎08'54	1°38'10
retrograde	2073 Oct 07 20:00:32	5°♎58'51		minimum elong	2074 Jul 26 16:11:36	3°♎59'30	1°38'01
min. declination.	2073 Oct 07 09:45:33	5°♎58'12	16S49'50"	max. Earth dist.	2074 Jul 30 02:34:29	11°♎08'39	1.34447 AU
evening set	2073 Oct 14 04:14:27	3°♎21'19		evening rise	2074 Aug 03 18:07:29	20°♎26'12	
	2073 Oct 17 10:56:59	30°♎			2074 Aug 08 20:27:02	0°♎	
min. Earth dist.	2073 Oct 18 08:37:17	28°♎55'09	0.66476 AU	desc. node	2074 Aug 21 20:06:34	21°♎22'02	
inferior conj	2073 Oct 19 20:31:26	27°♎04'29	-1°18'16		2074 Aug 27 23:27:57	0°♎	
minimum elong	2073 Oct 19 22:35:35	26°♎58'06	1°17'24	evening max el	2074 Sep 07 16:03:50	12°♎21'42	27°02'26
asc. node	2073 Oct 23 16:17:56	22°♎48'04		retrograde	2074 Sep 20 22:01:45	19°♎39'06	
morning rise	2073 Oct 25 17:24:36	21°♎15'15		min. declination.	2074 Sep 21 11:22:23	19°♎38'05	11S17'23"
direct	2073 Oct 28 19:12:10	20°♎21'22		evening set	2074 Sep 27 18:08:50	16°♎55'45	
max. declination.	2073 Oct 30 20:34:48	20°♎44'30	6S26'39"	min. Earth dist.	2074 Oct 01 15:46:26	13°♎05'54	0.65334 AU
morning max el	2073 Nov 04 14:06:57	24°♎07'01	18°46'51	inferior conj	2074 Oct 03 15:46:29	10°♎49'11	-2°16'53
	2073 Nov 09 11:24:34	0°♎		minimum elong	2074 Oct 03 19:26:11	10°♎38'44	2°15'29
morning set	2073 Nov 25 21:54:03	25°♎07'13		morning rise	2074 Oct 09 21:30:09	5°♎12'37	
	2073 Nov 29 00:07:00	0°♎		asc. node	2074 Oct 10 13:21:15	4°♎55'22	
desc. node	2073 Nov 30 22:22:15	3°♎02'12		direct	2074 Oct 12 15:33:19	4°♎31'33	
max. Earth dist.	2073 Dec 11 10:12:31	19°♎31'16	1.45079 AU	max. declination.	2074 Oct 15 07:27:59	5°♎09'16	0S55' 3"
				morning max el	2074 Oct 19 04:10:32	8°♎03'00	18°11'39
superior conj	2073 Dec 12 05:42:17	20°♎47'51	-1°11'04		2074 Nov 03 05:12:19	0°♎	
minimum elong	2073 Dec 11 21:23:49	20°♎15'13	1°10'06	morning set	2074 Nov 06 17:25:04	5°♎46'03	
	2073 Dec 18 01:47:56	0°♎		desc. node	2074 Nov 17 19:25:36	23°♎44'42	
min. declination.	2073 Dec 21 02:32:02	4°♎48'28	25S12' 4"				
evening rise	2073 Dec 27 18:04:52	15°♎26'45		superior conj	2074 Nov 21 08:52:45	29°♎24'24	-0°23'46
	2074 Jan 05 19:27:35	0°♎		minimum elong	2074 Nov 21 05:49:46	29°♎12'20	0°23'21
evening max el	2074 Jan 17 22:31:33	16°♎57'58	18°47'47		2074 Nov 21 17:52:32	0°♎	
asc. node	2074 Jan 19 15:35:25	18°♎32'58		max. Earth dist.	2074 Nov 24 05:15:01	3°♎54'12	1.45047 AU
retrograde	2074 Jan 24 16:03:01	20°♎44'17		evening rise	2074 Dec 07 18:20:00	25°♎03'19	
evening set	2074 Jan 27 21:02:27	19°♎45'24			2074 Dec 10 22:42:12	0°♎	
max. declination.	2074 Jan 29 00:24:45	18°♎57'08	12S36'22"	min. declination.	2074 Dec 12 23:14:06	3°♎08'12	25S32'33"
inferior conj	2074 Feb 02 13:47:26	14°♎04'25	3°31'10	greatest brilliancy	2074 Dec 20 02:02:45	14°♎03'54	-0.7m
minimum elong	2074 Feb 02 11:58:55	14°♎09'59	3°30'57		2074 Dec 31 19:34:38	0°♎	
min. Earth dist.	2074 Feb 04 08:14:59	11°♎54'08	0.65321 AU	evening max el	2075 Jan 01 06:02:08	0°♎27'25	19°33'52
morning rise	2074 Feb 08 02:30:53	7°♎54'02		asc. node	2075 Jan 06 12:36:42	4°♎17'55	
direct	2074 Feb 14 18:00:48	5°♎02'05		retrograde	2075 Jan 08 12:53:24	4°♎40'07	
min. declination.	2074 Feb 22 17:34:03	8°♎13'20	17S27'49"	evening set	2075 Jan 12 01:07:14	3°♎28'01	
desc. node	2074 Feb 26 21:31:31	11°♎45'25		max. declination.	2075 Jan 15 08:45:22	0°♎17'24	17S29'52"
morning max el	2074 Feb 27 22:18:22	12°♎46'18	26°54'48		2075 Jan 15 14:34:36	30°♎	
	2074 Mar 13 19:12:20	0°♎		inferior conj	2075 Jan 17 13:08:40	27°♎31'57	3°07'00
	2074 Apr 01 04:08:19	0°♎		minimum elong	2075 Jan 17 10:39:09	27°♎40'06	3°06'27
morning set	2074 Apr 04 17:08:01	6°♎38'43		min. Earth dist.	2075 Jan 18 16:55:49	26°♎01'00	0.66526 AU
max. Earth dist.	2074 Apr 09 02:54:22	15°♎18'43	1.34226 AU	morning rise	2075 Jan 22 19:57:13	21°♎19'18	
				direct	2075 Jan 28 23:53:41	18°♎34'51	
superior conj	2074 Apr 13 03:55:24	23°♎36'23	-0°45'20	min. declination.	2075 Feb 09 06:35:06	24°♎52'48	20S24'43"
minimum elong	2074 Apr 13 06:13:14	23°♎48'21	0°44'53	morning max el	2075 Feb 10 07:30:20	25°♎54'09	25°46'25
	2074 Apr 16 04:57:46	0°♎		desc. node	2075 Feb 13 18:35:55	29°♎38'34	
asc. node	2074 Apr 17 14:54:44	2°♎59'20			2075 Feb 14 02:03:07	0°♎	
evening rise	2074 Apr 20 14:45:30	9°♎16'50			2075 Mar 07 08:17:46	0°♎	
	2074 May 01 14:42:44	0°♎		morning set	2075 Mar 18 13:31:35	19°♎21'01	
evening max el	2074 May 10 12:08:10	11°♎30'03	21°25'04	max. Earth dist.	2075 Mar 22 09:02:16	26°♎29'40	1.35803 AU
max. declination.	2074 May 13 13:35:29	14°♎06'37	24N47'42"		2075 Mar 24 04:38:00	0°♎	
retrograde	2074 May 22 15:24:37	17°♎26'14					
evening set	2074 May 25 00:10:40	17°♎13'27		superior conj	2075 Mar 27 21:46:21	7°♎21'27	-1°12'42
desc. node	2074 May 25 20:44:03	17°♎02'48		minimum elong	2075 Mar 28 01:22:38	7°♎39'35	1°12'07
inferior conj	2074 Jun 03 09:51:12	13°♎13'34	-2°25'53	asc. node	2075 Apr 04 11:57:01	22°♎52'21	
minimum elong	2074 Jun 03 03:16:37	13°♎22'47	2°23'42	evening rise	2075 Apr 04 21:07:02	23°♎39'08	
min. Earth dist.	2074 Jun 03 13:08:17	13°♎08'58	0.54906 AU		2075 Apr 08 00:58:50	0°♎	
morning rise	2074 Jun 12 06:55:47	9°♎15'57		evening max el	2075 Apr 22 19:16:06	22°♎44'23	20°06'58
direct	2074 Jun 15 12:49:41	8°♎53'00		max. declination.	2075 Apr 29 14:13:39	27°♎11'03	22N 6'50"

retrograde	2075 May 03 05:40:24	27°♄47'16		superior conj	2076 Mar 10 04:19:03	20°♄28'35	-1°37'21
evening set	2075 May 05 06:17:26	27°♄36'42		minimum elong	2076 Mar 10 08:37:15	20°♄49'18	1°36'50
desc. node	2075 May 12 17:45:11	24°♄31'41			2076 Mar 15 01:07:54	0°♄	
inferior conj	2075 May 14 06:23:42	23°♄37'40	-0°26'44	evening rise	2076 Mar 18 21:17:24	7°♄37'19	
minimum elong	2075 May 14 05:09:22	23°♄39'31	0°26'15	asc. node	2076 Mar 21 08:57:52	12°♄30'16	
min. Earth dist.	2075 May 16 02:17:51	22°♄32'15	0.55284 AU		2076 Mar 31 14:39:30	0°♄	
morning rise	2075 May 23 02:16:40	19°♄16'29		evening max el	2076 Apr 04 13:44:43	4°♄37'37	19°07'43
direct	2075 May 27 01:56:18	18°♄41'51		max. declination.	2076 Apr 12 14:57:30	8°♄51'33	17N22'49"
min. declination.	2075 May 30 08:38:38	19°♄06'03	13N54'19"	retrograde	2076 Apr 13 08:44:50	8°♄53'19	
morning max el	2075 Jun 09 17:59:25	25°♄25'27	23°53'32	evening set	2076 Apr 15 12:00:19	8°♄39'17	
	2075 Jun 14 00:46:56	0°♄		inferior conj	2076 Apr 23 17:43:13	4°♄27'05	1°21'58
asc. node	2075 Jul 01 11:13:38	28°♄30'15		minimum elong	2076 Apr 23 20:47:14	4°♄21'49	1°20'57
	2075 Jul 02 04:47:23	0°♄		min. Earth dist.	2076 Apr 26 17:25:16	2°♄25'14	0.56538 AU
morning set	2075 Jul 04 00:47:02	3°♄49'07		desc. node	2076 Apr 28 14:47:15	1°♄14'23	
max. declination.	2075 Jul 06 13:05:52	9°♄10'44	23N58'39"		2076 Apr 30 23:27:12	30°♄	
				morning rise	2076 May 02 02:33:33	29°♄31'36	
superior conj	2075 Jul 11 01:18:00	18°♄56'12	1°25'54	direct	2076 May 07 02:29:19	28°♄33'22	
minimum elong	2075 Jul 10 22:58:12	18°♄43'37	1°25'37	min. declination.	2076 May 11 14:08:28	29°♄20'18	8N33'19"
max. Earth dist.	2075 Jul 13 03:34:34	23°♄26'07	1.33308 AU		2076 May 13 05:29:17	0°♄	
	2075 Jul 16 06:28:51	0°♄		morning max el	2076 May 21 08:26:09	5°♄52'09	25°31'45
evening rise	2075 Jul 18 11:47:10	4°♄32'07			2076 Jun 07 21:24:55	0°♄	
	2075 Aug 01 13:09:18	0°♄		asc. node	2076 Jun 17 08:15:35	18°♄22'28	
desc. node	2075 Aug 08 17:06:19	10°♄47'05		morning set	2076 Jun 17 12:27:04	18°♄44'34	
evening max el	2075 Aug 21 03:28:03	25°♄36'18	27°25'03		2076 Jun 22 17:29:11	0°♄	
	2075 Aug 26 13:23:26	0°♄					
retrograde	2075 Sep 03 17:53:17	2°♄53'51		superior conj	2076 Jun 24 11:45:28	3°♄51'40	1°09'03
min. declination.	2075 Sep 05 07:44:47	2°♄46'21	4S56'48"	minimum elong	2076 Jun 24 09:23:52	3°♄38'44	1°08'41
evening set	2075 Sep 10 21:41:53	0°♄14'04		max. Earth dist.	2076 Jun 25 11:39:54	6°♄02'26	1.32563 AU
	2075 Sep 11 05:08:26	30°♄		max. declination.	2076 Jun 25 21:45:03	6°♄57'27	24N34'14"
min. Earth dist.	2075 Sep 14 13:51:43	26°♄58'22	0.63841 AU	evening rise	2076 Jul 01 13:37:19	19°♄01'33	
inferior conj	2075 Sep 17 02:31:03	24°♄22'01	-3°13'53		2076 Jul 07 02:15:23	0°♄	
minimum elong	2075 Sep 17 07:28:29	24°♄09'12	3°12'18	desc. node	2076 Jul 25 14:06:14	29°♄28'26	
morning rise	2075 Sep 23 18:22:31	19°♄01'37			2076 Jul 26 00:02:21	0°♄	
direct	2075 Sep 26 07:14:01	18°♄29'47		evening max el	2076 Aug 02 11:23:55	8°♄18'54	27°16'48
asc. node	2075 Sep 27 10:24:38	18°♄36'14		retrograde	2076 Aug 16 07:04:32	15°♄34'16	
max. declination.	2075 Sep 29 18:02:10	19°♄29'04	4N43'34"	min. declination.	2076 Aug 19 03:28:53	15°♄11'38	1N48'14"
morning max el	2075 Oct 02 20:18:05	21°♄55'24	17°53'45	evening set	2076 Aug 23 11:13:23	13°♄10'18	
	2075 Oct 09 01:32:43	0°♄		min. Earth dist.	2076 Aug 27 01:28:20	10°♄21'24	0.62025 AU
morning set	2075 Oct 19 15:43:15	17°♄42'41		inferior conj	2076 Aug 30 01:45:59	7°♄35'56	-4°05'06
	2075 Oct 26 20:38:25	0°♄		minimum elong	2076 Aug 30 07:06:46	7°♄23'39	4°03'51
				morning rise	2076 Sep 06 04:32:47	2°♄35'14	
superior conj	2075 Nov 01 05:56:25	8°♄56'14	0°23'11	direct	2076 Sep 08 14:48:56	2°♄09'40	
minimum elong	2075 Nov 01 08:25:25	9°♄06'24	0°22'49	max. declination.	2076 Sep 13 05:46:33	3°♄45'23	10N 6'34"
desc. node	2075 Nov 04 16:26:43	14°♄31'06		asc. node	2076 Sep 13 07:27:42	3°♄48'17	
max. Earth dist.	2075 Nov 06 23:26:47	18°♄11'27	1.44289 AU	morning max el	2076 Sep 15 11:34:57	5°♄37'44	17°54'02
	2075 Nov 14 11:43:55	0°♄		morning set	2076 Oct 01 10:39:27	0°♄38'25	
evening rise	2075 Nov 17 04:12:09	4°♄09'06			2076 Oct 01 02:09:38	0°♄	
	2075 Dec 04 10:43:00	0°♄					
min. declination.	2075 Dec 04 22:13:45	0°♄41'01	25S48' 8"	superior conj	2076 Oct 12 05:57:40	19°♄48'08	1°00'50
evening max el	2075 Dec 15 08:22:20	13°♄57'39	20°34'30	minimum elong	2076 Oct 12 10:29:46	20°♄07'37	1°00'17
retrograde	2075 Dec 23 10:47:14	18°♄44'26			2076 Oct 18 07:22:43	0°♄	
asc. node	2075 Dec 24 09:38:31	18°♄39'22		max. Earth dist.	2076 Oct 19 13:30:45	2°♄03'41	1.42902 AU
evening set	2075 Dec 27 07:54:06	17°♄17'43		desc. node	2076 Oct 21 13:27:30	5°♄18'26	
inferior conj	2076 Jan 01 17:15:11	11°♄09'12	2°33'10	evening rise	2076 Oct 26 15:31:57	13°♄23'45	
minimum elong	2076 Jan 01 14:38:47	11°♄18'05	2°32'23		2076 Nov 06 13:11:25	0°♄	
min. Earth dist.	2076 Jan 02 07:51:43	10°♄19'24	0.67328 AU	min. declination.	2076 Nov 25 13:07:39	25°♄42'24	25S50'26"
morning rise	2076 Jan 06 21:13:31	4°♄56'03		evening max el	2076 Nov 27 05:00:24	27°♄28'41	21°46'27
max. declination.	2076 Jan 06 20:31:24	4°♄57'38	20S 4'55"		2076 Nov 29 22:29:27	0°♄	
direct	2076 Jan 12 10:52:38	2°♄28'16		retrograde	2076 Dec 06 07:47:49	2°♄54'17	
morning max el	2076 Jan 23 15:55:20	9°♄09'25	24°22'54	asc. node	2076 Dec 10 06:40:16	1°♄28'45	
min. declination.	2076 Jan 29 20:29:27	16°♄10'15	22S11'25"	evening set	2076 Dec 10 15:36:46	1°♄11'37	
desc. node	2076 Jan 31 15:38:37	18°♄26'08			2076 Dec 11 22:53:16	30°♄	
	2076 Feb 09 02:25:01	0°♄		inferior conj	2076 Dec 16 00:00:01	24°♄54'04	1°51'41
	2076 Feb 28 00:00:33	0°♄		minimum elong	2076 Dec 15 21:46:24	25°♄01'47	1°50'51
morning set	2076 Feb 28 13:51:20	1°♄00'14		min. Earth dist.	2076 Dec 16 02:54:23	24°♄43'59	0.67751 AU
max. Earth dist.	2076 Mar 03 06:20:24	7°♄33'17	1.37765 AU	morning rise	2076 Dec 21 03:47:38	18°♄42'02	
				max. declination.	2076 Dec 24 18:27:02	16°♄44'24	19S46'27"

direct	2076 Dec 26 02:17:56	16°♄35'45		asc. node	2077 Nov 27 03:43:21	12°♄51'38	
morning max el	2077 Jan 05 01:38:07	22°♄30'26	22°54'20	inferior conj	2077 Nov 30 07:29:37	8°♄43'27	1°03'58
	2077 Jan 11 14:07:47	0°♄		minimum elong	2077 Nov 30 06:03:32	8°♄48'24	1°03'23
desc. node	2077 Jan 17 12:41:08	7°♄52'54		min. Earth dist.	2077 Nov 29 23:37:50	9°♄10'33	0.67824 AU
min. declination.	2077 Jan 20 04:29:03	11°♄36'50	23S15'48"	morning rise	2077 Dec 05 13:27:03	2°♄34'33	
	2077 Feb 01 10:54:25	0°♄		direct	2077 Dec 09 21:24:11	0°♄51'03	
morning set	2077 Feb 08 12:03:21	11°♄21'42		max. declination.	2077 Dec 10 03:53:54	0°♄51'26	17S37'14"
max. Earth dist.	2077 Feb 13 01:42:36	19°♄04'06	1.39913 AU	morning max el	2077 Dec 18 15:57:36	5°♄58'49	21°29'35
	2077 Feb 19 06:30:45	0°♄		desc. node	2078 Jan 04 09:43:18	27°♄47'22	
					2078 Jan 05 21:54:07	0°♄	
superior conj	2077 Feb 20 19:21:45	2°♄47'36	-1°56'10	min. declination.	2078 Jan 11 22:00:39	9°♄00'11	23S59'19"
minimum elong	2077 Feb 20 22:58:21	3°♄04'08	1°55'56	morning set	2078 Jan 19 05:11:49	20°♄20'36	
evening rise	2077 Mar 02 12:23:12	21°♄04'13			2078 Jan 25 04:59:55	0°♄	
	2077 Mar 07 06:17:32	0°♄		max. Earth dist.	2078 Jan 26 01:47:48	1°♄25'32	1.41968 AU
asc. node	2077 Mar 08 05:58:30	1°♄47'43					
evening max el	2077 Mar 18 17:20:53	17°♄05'46	18°28'37	superior conj	2078 Feb 02 13:46:12	14°♄06'14	-2°04'57
retrograde	2077 Mar 26 06:28:53	20°♄48'42		minimum elong	2078 Feb 02 14:20:56	14°♄08'44	2°04'58
max. declination.	2077 Mar 27 03:53:22	20°♄45'41	11N12'26"		2078 Feb 11 12:01:34	0°♄	
evening set	2077 Mar 28 15:34:45	20°♄27'41		evening rise	2078 Feb 13 14:44:27	3°♄51'08	
inferior conj	2077 Apr 05 02:01:41	15°♄56'19	2°39'47	asc. node	2078 Feb 23 03:00:29	20°♄37'52	
minimum elong	2077 Apr 05 05:57:23	15°♄48'26	2°38'48	evening max el	2078 Mar 02 02:56:55	0°♄00'37	18°09'33
min. Earth dist.	2077 Apr 08 12:27:02	13°♄12'37	0.58408 AU		2078 Mar 02 02:42:02	0°♄	
morning rise	2077 Apr 12 17:24:39	10°♄29'25		retrograde	2078 Mar 08 21:44:27	3°♄26'49	
desc. node	2077 Apr 15 11:49:27	9°♄23'38		max. declination.	2078 Mar 10 19:38:12	3°♄10'33	4N20' 8"
direct	2077 Apr 18 17:12:41	8°♄56'33		evening set	2078 Mar 11 12:38:11	2°♄56'34	
min. declination.	2077 Apr 23 18:48:07	10°♄00'07	2N15'39"		2078 Mar 16 03:07:51	30°♄	
morning max el	2077 May 03 02:28:54	16°♄37'26	26°50'49	inferior conj	2078 Mar 18 06:45:00	28°♄04'28	3°24'29
	2077 May 14 01:49:39	0°♄		minimum elong	2078 Mar 18 09:16:05	27°♄58'37	3°24'08
	2077 May 31 05:12:25	0°♄		min. Earth dist.	2078 Mar 21 13:34:23	25°♄02'46	0.60544 AU
morning set	2077 Jun 01 22:31:49	3°♄34'10		morning rise	2078 Mar 25 03:51:53	22°♄16'21	
asc. node	2077 Jun 04 05:16:46	8°♄25'07		direct	2078 Mar 31 20:11:04	20°♄07'09	
				desc. node	2078 Apr 02 08:51:55	20°♄13'37	
superior conj	2077 Jun 08 23:30:05	18°♄49'00	0°48'25	min. declination.	2078 Apr 06 07:28:30	21°♄26'07	4S11' 1"
minimum elong	2077 Jun 08 21:34:22	18°♄38'23	0°48'03	morning max el	2078 Apr 15 02:35:02	27°♄59'32	27°38'23
max. Earth dist.	2077 Jun 08 23:24:12	18°♄48'28	1.32205 AU		2078 Apr 17 01:14:58	0°♄	
	2077 Jun 14 02:25:58	0°♄			2078 May 07 18:41:15	0°♄	
evening rise	2077 Jun 15 20:55:39	3°♄45'48		morning set	2078 May 17 05:16:20	18°♄11'58	
max. declination.	2077 Jun 16 10:34:16	4°♄57'18	25N 2'42"	asc. node	2078 May 22 02:18:53	28°♄33'39	
	2077 Jun 29 21:51:54	0°♄			2078 May 22 18:11:14	0°♄	
desc. node	2077 Jul 12 11:06:24	17°♄07'39		max. Earth dist.	2078 May 23 11:08:03	1°♄32'32	1.32235 AU
evening max el	2077 Jul 15 13:23:23	20°♄18'17	26°35'22				
retrograde	2077 Jul 29 12:44:39	27°♄31'07		superior conj	2078 May 24 10:50:43	3°♄42'25	0°24'40
min. declination.	2077 Aug 03 06:37:17	26°♄33'28	8N28'14"	minimum elong	2078 May 24 09:45:01	3°♄36'24	0°24'26
evening set	2077 Aug 05 06:05:45	25°♄36'26		evening rise	2078 May 31 07:30:15	18°♄37'43	
min. Earth dist.	2077 Aug 09 02:53:05	23°♄00'06	0.59993 AU		2078 Jun 05 21:37:12	0°♄	
inferior conj	2077 Aug 12 10:01:31	20°♄21'43	-4°43'43	max. declination.	2078 Jun 07 07:01:01	2°♄39'24	25N25'25"
minimum elong	2077 Aug 12 13:56:12	20°♄13'51	4°43'12		2078 Jun 25 19:30:55	0°♄	
morning rise	2077 Aug 19 23:48:40	15°♄43'15		evening max el	2078 Jun 27 08:26:29	1°♄32'07	25°23'59
direct	2077 Aug 22 10:08:36	15°♄21'32		desc. node	2078 Jun 29 08:06:59	3°♄19'26	
max. declination.	2077 Aug 28 23:30:31	18°♄08'42	14N50'33"	retrograde	2078 Jul 11 08:47:00	8°♄40'00	
morning max el	2077 Aug 29 22:52:11	19°♄01'29	18°13'37	evening set	2078 Jul 17 02:03:23	7°♄23'45	
asc. node	2077 Aug 31 04:30:47	20°♄18'03		min. declination.	2078 Jul 19 14:51:04	6°♄06'04	14N18'38"
	2077 Sep 07 00:08:22	0°♄		min. Earth dist.	2078 Jul 21 20:26:56	4°♄41'40	0.57960 AU
morning set	2077 Sep 14 20:27:07	14°♄16'32		inferior conj	2078 Jul 24 23:57:12	2°♄30'33	-4°58'51
	2077 Sep 23 06:07:34	0°♄		minimum elong	2078 Jul 24 23:52:08	2°♄30'42	4°58'50
					2078 Jul 28 20:34:04	30°♄	
superior conj	2077 Sep 24 07:08:35	1°♄53'51	1°26'30	morning rise	2078 Aug 02 00:09:18	28°♄14'21	
minimum elong	2077 Sep 24 10:57:38	2°♄11'09	1°26'09	direct	2078 Aug 04 13:00:33	27°♄54'42	
max. Earth dist.	2077 Oct 01 21:43:59	15°♄15'17	1.41070 AU		2078 Aug 10 17:08:32	0°♄	
evening rise	2077 Oct 06 19:41:20	23°♄25'53		morning max el	2078 Aug 13 02:57:25	1°♄57'33	18°53'47
desc. node	2077 Oct 08 10:27:47	26°♄02'51		max. declination.	2078 Aug 14 14:01:48	3°♄26'54	18N32'56"
	2077 Oct 10 21:59:44	0°♄		asc. node	2078 Aug 18 01:34:18	7°♄50'31	
	2077 Oct 31 12:53:41	0°♄		morning set	2078 Aug 29 16:45:28	28°♄25'39	
evening max el	2077 Nov 09 20:44:58	11°♄01'25	23°05'18		2078 Aug 30 11:57:04	0°♄	
min. declination.	2077 Nov 14 20:13:46	15°♄13'59	25S10' 7"				
retrograde	2077 Nov 20 02:20:42	17°♄05'38		superior conj	2078 Sep 07 04:09:32	14°♄58'35	1°40'51
evening set	2077 Nov 24 22:34:45	15°♄06'13		minimum elong	2078 Sep 07 06:10:29	15°♄08'12	1°40'44

max. Earth dist.	2078 Sep 14 01:54:42	27° $\overline{\text{M}}$ 44'27	1.39035 AU	minimum elong	2079 Jul 05 12:08:02	14° $\overline{\text{S}}$ 02'39	4°35'30
	2078 Sep 15 08:28:52	0° $\overline{\text{A}}$		min. declination.	2079 Jul 08 05:37:52	12° $\overline{\text{S}}$ 25'01	18N 3'23"
evening rise	2078 Sep 17 23:45:31	4° $\overline{\text{A}}$ 34'58		morning rise	2079 Jul 14 03:29:10	9° $\overline{\text{S}}$ 56'09	
desc. node	2078 Sep 25 07:28:11	16° $\overline{\text{A}}$ 40'49		direct	2079 Jul 16 20:52:18	9° $\overline{\text{S}}$ 37'07	
	2078 Oct 04 00:52:47	0° $\overline{\text{M}}$		morning max el	2079 Jul 26 20:54:56	14° $\overline{\text{S}}$ 15'25	19°55'26
evening max el	2078 Oct 23 09:39:41	24° $\overline{\text{M}}$ 38'17	24°24'54	max. declination.	2079 Aug 02 03:27:56	21° $\overline{\text{S}}$ 48'11	21N 2'50"
	2078 Oct 30 05:42:12	0° $\overline{\text{A}}$		asc. node	2079 Aug 04 22:37:42	26° $\overline{\text{S}}$ 10'55	
min. declination.	2078 Nov 01 04:04:10	0° $\overline{\text{A}}$ 49'36	23S 7'26"		2079 Aug 07 03:33:44	0° $\overline{\text{S}}$	
retrograde	2078 Nov 03 17:07:49	1° $\overline{\text{A}}$ 15'14		morning set	2079 Aug 13 20:15:23	12° $\overline{\text{S}}$ 56'19	
	2078 Nov 07 18:32:11	30° $\overline{\text{R}}$ $\overline{\text{M}}$					
evening set	2078 Nov 09 03:11:58	28° $\overline{\text{M}}$ 59'12		superior conj	2079 Aug 21 15:38:14	28° $\overline{\text{S}}$ 46'41	1°45'40
min. Earth dist.	2078 Nov 13 19:37:47	23° $\overline{\text{M}}$ 37'01	0.67567 AU	minimum elong	2079 Aug 21 15:49:33	28° $\overline{\text{S}}$ 47'37	1°45'39
asc. node	2078 Nov 14 00:47:21	23° $\overline{\text{M}}$ 19'45			2079 Aug 22 06:21:51	0° $\overline{\text{M}}$	
inferior conj	2078 Nov 14 13:58:53	22° $\overline{\text{M}}$ 35'29	0°11'21	max. Earth dist.	2079 Aug 27 06:46:00	9° $\overline{\text{M}}$ 43'50	1.37040 AU
minimum elong	2078 Nov 14 13:42:12	22° $\overline{\text{M}}$ 36'25	0°11'14	evening rise	2079 Aug 31 02:38:09	16° $\overline{\text{M}}$ 48'18	
transit middle	2078 Nov 14 13:42:12	22° $\overline{\text{M}}$ 36'25	0°11'14		2079 Sep 07 18:00:09	0° $\overline{\text{A}}$	
transit begin	2078 Nov 14 11:43:32	22° $\overline{\text{M}}$ 43'04		desc. node	2079 Sep 12 04:29:47	7° $\overline{\text{A}}$ 07'40	
transit end	2078 Nov 14 15:40:51	22° $\overline{\text{M}}$ 29'48			2079 Sep 28 10:45:10	0° $\overline{\text{M}}$	
morning rise	2078 Nov 20 00:14:19	16° $\overline{\text{M}}$ 32'04		evening max el	2079 Oct 05 21:46:00	8° $\overline{\text{M}}$ 18'36	25°38'08
direct	2078 Nov 23 19:00:19	15° $\overline{\text{M}}$ 10'07		min. declination.	2079 Oct 17 01:46:37	15° $\overline{\text{M}}$ 14'41	19S32'34"
max. declination.	2078 Nov 24 22:00:38	15° $\overline{\text{M}}$ 16'54	14S 3'45"	retrograde	2079 Oct 18 03:20:38	15° $\overline{\text{M}}$ 18'54	
morning max el	2078 Dec 01 13:34:41	19° $\overline{\text{M}}$ 37'37	20°15'04	evening set	2079 Oct 24 03:45:58	12° $\overline{\text{M}}$ 48'05	
	2078 Dec 09 23:29:09	0° $\overline{\text{A}}$		min. Earth dist.	2079 Oct 28 12:26:00	8° $\overline{\text{M}}$ 01'10	0.66975 AU
desc. node	2078 Dec 22 06:45:03	18° $\overline{\text{A}}$ 01'51		inferior conj	2079 Oct 29 17:41:34	6° $\overline{\text{M}}$ 27'43	-0°44'52
morning set	2078 Dec 29 02:38:45	28° $\overline{\text{A}}$ 29'14		minimum elong	2079 Oct 29 18:51:03	6° $\overline{\text{M}}$ 24'01	0°44'21
	2078 Dec 30 02:04:53	0° $\overline{\text{S}}$		asc. node	2079 Oct 31 21:51:21	3° $\overline{\text{M}}$ 46'52	
min. declination.	2079 Jan 03 16:28:16	7° $\overline{\text{S}}$ 11'20	24S32' 5"	morning rise	2079 Nov 04 10:11:33	0° $\overline{\text{M}}$ 32'33	
max. Earth dist.	2079 Jan 08 09:48:31	14° $\overline{\text{S}}$ 41'35	1.43657 AU		2079 Nov 05 10:16:38	30° $\overline{\text{R}}$ $\overline{\text{A}}$	
				direct	2079 Nov 07 17:38:57	29° $\overline{\text{A}}$ 29'21	
superior conj	2079 Jan 14 06:48:29	24° $\overline{\text{S}}$ 14'21	-1°58'43	max. declination.	2079 Nov 09 11:16:16	29° $\overline{\text{A}}$ 45'48	9S27'34"
minimum elong	2079 Jan 14 02:05:11	23° $\overline{\text{S}}$ 54'55	1°58'29		2079 Nov 10 02:51:33	0° $\overline{\text{M}}$	
	2079 Jan 17 18:02:21	0° $\approx$		morning max el	2079 Nov 14 19:00:06	3° $\overline{\text{M}}$ 26'55	19°14'34
evening rise	2079 Jan 27 00:04:50	15° $\approx$ 49'17			2079 Dec 03 16:10:44	0° $\overline{\text{A}}$	
	2079 Feb 04 05:57:01	0° $\overline{\text{H}}$		morning set	2079 Dec 08 04:31:36	7° $\overline{\text{A}}$ 00'22	
asc. node	2079 Feb 10 00:03:52	8° $\overline{\text{H}}$ 51'46		desc. node	2079 Dec 09 03:46:59	8° $\overline{\text{A}}$ 30'49	
evening max el	2079 Feb 13 15:20:04	13° $\overline{\text{H}}$ 12'53	18°09'52	max. Earth dist.	2079 Dec 22 00:51:58	28° $\overline{\text{A}}$ 39'57	1.44760 AU
retrograde	2079 Feb 20 02:31:14	16° $\overline{\text{H}}$ 37'08			2079 Dec 22 21:08:11	0° $\overline{\text{S}}$	
evening set	2079 Feb 22 22:42:59	15° $\overline{\text{H}}$ 56'21					
max. declination.	2079 Feb 22 19:38:22	15° $\overline{\text{H}}$ 59'54	2S37'31"	superior conj	2079 Dec 24 22:28:46	3° $\overline{\text{S}}$ 15'27	-1°33'38
inferior conj	2079 Mar 01 04:17:13	10° $\overline{\text{H}}$ 43'52	3°42'09	minimum elong	2079 Dec 24 13:44:09	2° $\overline{\text{S}}$ 40'45	1°32'47
minimum elong	2079 Mar 01 04:48:56	10° $\overline{\text{H}}$ 42'28	3°42'08	min. declination.	2079 Dec 26 10:58:01	5° $\overline{\text{S}}$ 40'35	24S58'42"
min. Earth dist.	2079 Mar 03 22:55:22	7° $\overline{\text{H}}$ 48'51	0.62629 AU	evening rise	2080 Jan 08 12:17:25	26° $\overline{\text{S}}$ 52'19	
morning rise	2079 Mar 07 09:47:57	4° $\overline{\text{H}}$ 43'13			2080 Jan 10 09:41:33	0° $\approx$	
direct	2079 Mar 14 08:30:09	2° $\overline{\text{H}}$ 06'34		asc. node	2080 Jan 27 21:08:07	26° $\approx$ 18'40	
desc. node	2079 Mar 20 05:55:38	3° $\overline{\text{H}}$ 43'54		evening max el	2080 Jan 28 03:37:36	26° $\approx$ 35'24	18°28'50
min. declination.	2079 Mar 20 08:43:38	3° $\overline{\text{H}}$ 47'35	10S 8'11"		2080 Feb 02 07:16:31	0° $\overline{\text{H}}$	
morning max el	2079 Mar 28 08:13:06	10° $\overline{\text{H}}$ 02'14	27°48'55	retrograde	2080 Feb 03 16:28:03	0° $\overline{\text{H}}$ 10'32	
	2079 Apr 12 22:04:50	0° $\overline{\text{Y}}$			2080 Feb 05 01:25:22	30° $\overline{\text{R}}$ $\approx$	
	2079 Apr 30 00:35:36	0° $\overline{\text{S}}$		evening set	2080 Feb 06 18:03:40	29° $\approx$ 18'23	
morning set	2079 May 01 06:28:35	2° $\overline{\text{S}}$ 29'48		max. declination.	2080 Feb 07 07:07:10	28° $\approx$ 58'37	9S10'33"
max. Earth dist.	2079 May 06 18:56:28	14° $\overline{\text{S}}$ 00'37	1.32669 AU	inferior conj	2080 Feb 12 14:35:55	23° $\approx$ 47'00	3°39'38
				minimum elong	2080 Feb 12 13:28:16	23° $\approx$ 50'18	3°39'34
superior conj	2079 May 08 19:59:08	18° $\overline{\text{S}}$ 24'52	-0°01'29	min. Earth dist.	2080 Feb 14 17:51:16	21° $\approx$ 17'15	0.64442 AU
minimum elong	2079 May 08 20:03:27	18° $\overline{\text{S}}$ 25'15	0°01'29	morning rise	2080 Feb 18 08:20:24	17° $\approx$ 39'13	
behind sun begin	2079 May 08 14:52:44	17° $\overline{\text{S}}$ 57'12		direct	2080 Feb 25 04:19:50	14° $\approx$ 49'02	
behind sun end	2079 May 09 01:14:11	18° $\overline{\text{S}}$ 53'19		min. declination.	2080 Mar 03 04:06:12	17° $\approx$ 12'42	15S 7'45"
asc. node	2079 May 08 23:20:54	18° $\overline{\text{S}}$ 43'06		desc. node	2080 Mar 06 02:57:59	19° $\approx$ 22'19	
	2079 May 14 04:08:12	0° $\overline{\text{II}}$		morning max el	2080 Mar 09 17:21:18	22° $\approx$ 40'07	27°23'34
evening rise	2079 May 15 19:20:28	3° $\overline{\text{II}}$ 29'24			2080 Mar 16 05:02:52	0° $\overline{\text{H}}$	
max. declination.	2079 May 29 13:28:41	29° $\overline{\text{II}}$ 05'28	25N39'14"		2080 Apr 05 03:47:48	0° $\overline{\text{Y}}$	
	2079 May 30 03:47:00	0° $\overline{\text{S}}$		morning set	2080 Apr 13 23:29:39	16° $\overline{\text{Y}}$ 19'34	
evening max el	2079 Jun 08 22:36:07	12° $\overline{\text{S}}$ 10'13	23°52'45	max. Earth dist.	2080 Apr 18 19:03:30	26° $\overline{\text{Y}}$ 01'47	1.33532 AU
desc. node	2079 Jun 16 05:09:29	17° $\overline{\text{S}}$ 29'21			2080 Apr 20 16:45:30	0° $\overline{\text{S}}$	
retrograde	2079 Jun 22 16:28:47	19° $\overline{\text{S}}$ 04'18					
evening set	2079 Jun 26 23:24:40	18° $\overline{\text{S}}$ 23'47		superior conj	2080 Apr 22 01:02:47	2° $\overline{\text{S}}$ 50'25	-0°29'09
min. Earth dist.	2079 Jul 03 09:13:53	15° $\overline{\text{S}}$ 20'12	0.56230 AU	minimum elong	2080 Apr 22 02:30:31	2° $\overline{\text{S}}$ 58'10	0°28'51
inferior conj	2079 Jul 05 17:37:19	13° $\overline{\text{S}}$ 54'14	-4°36'18	asc. node	2080 Apr 24 20:24:23	8° $\overline{\text{S}}$ 49'34	

evening rise	2080 Apr 29 06:37:01	18°♄14'48			2081 Mar 10 20:11:12	0°♄	
	2080 May 05 02:34:51	0°♄		morning set	2081 Mar 28 04:42:44	29°♄30'02	
max. declination.	2080 May 19 13:45:34	21°♄38'35	25N26'41"		2081 Mar 28 11:03:56	0°♄	
evening max el	2080 May 20 14:13:15	22°♄40'05	22°16'43	max. Earth dist.	2081 Apr 01 08:03:22	7°♄28'56	1.34842 AU
desc. node	2080 Jun 02 02:11:05	29°♄02'48					
retrograde	2080 Jun 02 12:00:36	29°♄03'11		superior conj	2081 Apr 05 23:42:51	16°♄51'52	-0°57'08
evening set	2080 Jun 05 09:50:58	28°♄44'27		minimum elong	2081 Apr 06 02:36:20	17°♄06'44	0°56'35
min. Earth dist.	2080 Jun 13 20:34:00	25°♄06'29	0.55142 AU	asc. node	2081 Apr 11 17:27:51	28°♄48'37	
inferior conj	2080 Jun 14 18:01:05	24°♄36'15	-3°25'52		2081 Apr 12 07:09:22	0°♄	
minimum elong	2080 Jun 14 10:05:24	24°♄47'26	3°23'41	evening rise	2081 Apr 13 15:20:20	2°♄46'59	
morning rise	2080 Jun 23 12:13:24	20°♄44'21			2081 Apr 29 09:11:58	0°♄	
min. declination.	2080 Jun 24 23:12:21	20°♄29'34	18N32'55"	evening max el	2081 May 02 14:29:27	3°♄32'30	20°49'43
direct	2080 Jun 26 12:07:12	20°♄24'09		max. declination.	2081 May 07 16:43:20	7°♄24'17	23N56'22"
morning max el	2080 Jul 08 03:24:21	25°♄49'31	21°17'27	retrograde	2081 May 14 01:19:56	9°♄06'59	
	2080 Jul 12 00:49:43	0°♄		evening set	2081 May 16 04:32:52	8°♄56'06	
max. declination.	2080 Jul 21 14:52:54	14°♄45'50	22N34'27"	desc. node	2081 May 19 23:11:05	7°♄48'51	
asc. node	2080 Jul 21 19:39:47	15°♄07'48		inferior conj	2081 May 25 11:44:26	4°♄58'14	-1°36'11
morning set	2080 Jul 28 04:25:19	27°♄41'20		minimum elong	2081 May 25 07:12:27	5°♄04'41	1°34'33
	2080 Jul 29 07:06:59	0°♄		min. Earth dist.	2081 May 26 08:58:59	4°♄27'57	0.54944 AU
				morning rise	2081 Jun 03 09:29:29	0°♄52'44	
superior conj	2080 Aug 04 13:17:27	13°♄05'38	1°42'45	direct	2081 Jun 06 21:31:03	0°♄26'19	
minimum elong	2080 Aug 04 12:04:20	12°♄59'20	1°42'42	min. declination.	2081 Jun 08 23:43:09	0°♄36'06	16N15'40"
max. Earth dist.	2080 Aug 08 17:31:07	21°♄35'55	1.35290 AU	morning max el	2081 Jun 19 23:54:59	6°♄43'39	22°54'43
evening rise	2080 Aug 13 00:03:31	29°♄54'57			2081 Jul 06 08:07:27	0°♄	
	2080 Aug 13 01:07:26	0°♄		asc. node	2081 Jul 08 16:42:07	4°♄32'15	
desc. node	2080 Aug 29 01:31:41	27°♄17'20		max. declination.	2081 Jul 11 18:41:40	10°♄47'38	23N33' 0"
	2080 Aug 30 20:56:21	0°♄		morning set	2081 Jul 12 15:17:48	12°♄35'11	
evening max el	2080 Sep 17 09:56:34	21°♄56'24	26°37'26				
retrograde	2080 Sep 30 08:24:23	29°♄10'13		superior conj	2081 Jul 19 17:51:58	27°♄45'44	1°33'35
min. declination.	2080 Sep 30 08:24:23	29°♄10'13	14S37'49"	minimum elong	2081 Jul 19 15:47:07	27°♄34'39	1°33'23
evening set	2080 Oct 06 22:14:24	26°♄29'05			2081 Jul 20 19:07:37	0°♄	
min. Earth dist.	2080 Oct 10 23:37:49	22°♄18'43	0.66038 AU	max. Earth dist.	2081 Jul 22 13:08:23	3°♄41'05	1.33909 AU
inferior conj	2080 Oct 12 16:38:12	20°♄16'20	-1°43'09	evening rise	2081 Jul 27 11:30:00	13°♄43'07	
minimum elong	2080 Oct 12 19:23:35	20°♄08'06	1°42'02		2081 Aug 05 06:04:02	0°♄	
asc. node	2080 Oct 17 18:54:28	15°♄06'55		desc. node	2081 Aug 15 22:32:10	17°♄02'12	
morning rise	2080 Oct 18 17:07:43	14°♄32'20			2081 Aug 25 19:56:55	0°♄	
direct	2080 Oct 21 15:23:09	13°♄44'20		evening max el	2081 Aug 30 21:53:26	5°♄22'59	27°15'25
max. declination.	2080 Oct 23 22:39:16	14°♄13'05	4S10' 8"	retrograde	2081 Sep 13 08:01:57	12°♄41'47	
morning max el	2080 Oct 28 06:40:42	17°♄22'29	18°29'50	min. declination.	2081 Sep 14 07:24:28	12°♄38'47	8S42'44"
	2080 Nov 06 17:05:09	0°♄		evening set	2081 Sep 20 08:10:46	9°♄58'13	
morning set	2080 Nov 17 06:59:13	16°♄48'29		min. Earth dist.	2081 Sep 24 03:05:15	6°♄23'40	0.64742 AU
desc. node	2080 Nov 25 00:48:42	29°♄09'26		inferior conj	2081 Sep 26 08:36:01	3°♄57'16	-2°41'27
	2080 Nov 25 13:34:41	0°♄		minimum elong	2081 Sep 26 12:52:39	3°♄45'33	2°39'55
					2081 Sep 30 08:38:54	30°♄	
superior conj	2080 Dec 03 00:05:38	11°♄43'29	-0°51'43	morning rise	2081 Oct 02 18:32:52	28°♄27'28	
minimum elong	2080 Dec 02 17:32:38	11°♄17'45	0°50'53	asc. node	2081 Oct 04 15:55:08	27°♄53'52	
max. Earth dist.	2080 Dec 03 19:31:56	12°♄59'48	1.45154 AU	direct	2081 Oct 05 09:59:46	27°♄50'52	
	2080 Dec 14 15:13:38	0°♄		max. declination.	2081 Oct 08 09:14:05	28°♄36'36	1N26'49"
min. declination.	2080 Dec 17 06:31:31	4°♄09'45	25S21'15"		2081 Oct 10 11:24:52	0°♄	
evening rise	2080 Dec 19 01:04:03	6°♄57'54		morning max el	2081 Oct 11 22:04:16	1°♄18'52	18°01'53
greatest brilliancy	2080 Dec 28 12:30:52	21°♄55'41	-0.8m	morning set	2081 Oct 29 14:34:44	28°♄02'54	
	2081 Jan 02 19:28:43	0°♄			2081 Oct 30 18:32:56	0°♄	
evening max el	2081 Jan 10 13:19:35	10°♄02'23	19°05'26	desc. node	2081 Nov 11 21:51:50	19°♄54'29	
asc. node	2081 Jan 13 18:11:21	12°♄44'39					
retrograde	2081 Jan 17 11:43:45	13°♄59'07		superior conj	2081 Nov 12 09:01:48	20°♄39'14	-0°03'09
evening set	2081 Jan 20 19:36:45	12°♄54'43		minimum elong	2081 Nov 12 08:39:01	20°♄37'42	0°03'07
max. declination.	2081 Jan 22 14:09:18	11°♄30'09	14S50'46"	behind sun begin	2081 Nov 11 22:14:44	19°♄56'01	
inferior conj	2081 Jan 26 10:01:56	7°♄06'43	3°22'16	behind sun end	2081 Nov 12 19:03:19	21°♄19'20	
minimum elong	2081 Jan 26 07:51:34	7°♄13'36	3°21'54	max. Earth dist.	2081 Nov 16 14:16:49	27°♄21'44	1.44808 AU
min. Earth dist.	2081 Jan 27 21:58:41	5°♄12'53	0.65877 AU		2081 Nov 18 06:26:21	0°♄	
morning rise	2081 Jan 31 19:48:54	0°♄55'00		evening rise	2081 Nov 28 17:42:06	16°♄17'38	
	2081 Feb 01 22:53:18	30°♄			2081 Dec 07 16:17:40	0°♄	
direct	2081 Feb 07 06:41:49	28°♄05'11		min. declination.	2081 Dec 09 04:18:58	2°♄16'23	25S40' 2"
	2081 Feb 13 05:24:08	0°♄		greatest brilliancy	2081 Dec 12 17:00:02	7°♄33'16	-0.6m
min. declination.	2081 Feb 16 08:06:23	2°♄12'25	18S51'43"	evening max el	2081 Dec 24 18:39:41	23°♄31'51	19°58'00
morning max el	2081 Feb 20 03:15:38	5°♄40'45	26°28'20	asc. node	2081 Dec 31 15:11:57	27°♄55'35	
desc. node	2081 Feb 21 00:00:18	6°♄33'57		retrograde	2082 Jan 01 09:21:25	27°♄58'46	

evening set	2000 Jan 05 01:02:35	26°☿40'46				2000 Nov 11 02:24:26	0°☿	
inferior conj	2000 Jan 10 11:41:42	20°☿38'55	2°53'42		min. declination.	2000 Dec 01 02:49:34	29°☿05'48	25S51'52"
minimum elong	2000 Jan 10 09:05:31	20°☿47'37	2°53'01			2000 Dec 01 19:23:22	0°☿	
max. declination.	2000 Jan 10 15:56:52	20°☿24'41	18S58'56"		evening max el	2000 Dec 07 18:36:21	7°☿02'45	21°04'01
min. Earth dist.	2000 Jan 11 09:43:02	19°☿25'19	0.66913 AU		retrograde	2000 Dec 16 07:03:03	12°☿05'44	
morning rise	2000 Jan 15 16:55:08	14°☿25'34			asc. node	2000 Dec 18 12:12:11	11°☿38'27	
direct	2000 Jan 21 14:54:04	11°☿47'15			evening set	2000 Dec 20 08:25:02	10°☿32'36	
morning max el	2000 Feb 02 12:06:27	18°☿52'39	25°12'04		inferior conj	2000 Dec 25 17:11:30	4°☿20'00	2°16'25
min. declination.	2000 Feb 04 03:22:39	20°☿35'18	21S16'28"		minimum elong	2000 Dec 25 14:41:21	4°☿28'36	2°15'34
desc. node	2000 Feb 07 21:03:45	24°☿52'27			min. Earth dist.	2000 Dec 26 02:45:56	3°☿47'02	0.67555 AU
	2000 Feb 11 22:28:43	0°☿				2000 Dec 29 00:52:45	30°☿	
	2000 Mar 03 22:08:07	0°☿			morning rise	2000 Dec 30 20:45:55	28°☿06'58	
morning set	2000 Mar 10 17:18:27	11°☿47'28			max. declination.	2000 Jan 01 23:48:24	26°☿36'29	20S13'27"
max. Earth dist.	2000 Mar 14 09:28:07	18°☿31'13	1.36592 AU		direct	2000 Jan 05 04:03:52	25°☿47'40	
	2000 Mar 20 08:48:43	0°☿				2000 Jan 13 13:40:29	0°☿	
					morning max el	2000 Jan 15 20:32:00	2°☿09'50	23°45'09
superior conj	2000 Mar 20 13:09:18	0°☿21'27	-1°23'41		min. declination.	2000 Jan 25 16:47:16	13°☿54'07	22S41'50"
minimum elong	2000 Mar 20 17:10:14	0°☿41'18	1°23'05		desc. node	2000 Jan 25 18:07:31	13°☿58'33	
evening rise	2000 Mar 28 19:20:37	16°☿59'23				2000 Feb 06 00:15:06	0°☿	
asc. node	2000 Mar 29 14:29:46	18°☿35'39			morning set	2000 Feb 20 07:14:26	22°☿55'31	
	2000 Apr 04 12:50:16	0°☿			max. Earth dist.	2000 Feb 24 04:31:06	29°☿40'32	1.38667 AU
evening max el	2000 Apr 15 02:35:19	15°☿03'21	19°39'26			2000 Feb 24 08:55:07	0°☿	
max. declination.	2000 Apr 22 14:48:41	19°☿30'46	20N21'9"					
retrograde	2000 Apr 24 19:42:20	19°☿44'51			superior conj	2000 Mar 03 13:43:48	13°☿10'01	-1°46'16
evening set	2000 Apr 26 20:25:23	19°☿33'28			minimum elong	2000 Mar 03 17:57:58	13°☿30'00	1°45'52
inferior conj	2000 May 05 13:17:31	15°☿30'26	0°22'20			2000 Mar 12 06:46:55	0°☿	
minimum elong	2000 May 05 14:15:42	15°☿28'55	0°21'59		evening rise	2000 Mar 12 16:03:24	0°☿45'15	
desc. node	2000 May 06 20:13:09	14°☿41'39			asc. node	2000 Mar 16 11:31:31	8°☿05'47	
min. Earth dist.	2000 May 07 23:12:13	13°☿59'38	0.55711 AU		evening max el	2000 Mar 29 01:26:59	27°☿13'08	18°48'43
morning rise	2000 May 14 05:30:12	10°☿54'43				2000 Apr 01 17:42:23	0°☿	
direct	2000 May 18 14:54:31	10°☿11'33			retrograde	2000 Apr 06 06:08:22	1°☿12'25	
min. declination.	2000 May 22 12:31:02	10°☿46'15	11N49'49"		max. declination.	2000 Apr 06 07:36:18	1°☿12'25	14N55'19"
morning max el	2000 Jun 01 14:47:14	17°☿12'53	24°36'41		evening set	2000 Apr 08 11:54:53	0°☿55'43	
	2000 Jun 12 01:03:09	0°☿				2000 Apr 11 00:51:20	30°☿	
asc. node	2000 Jun 25 13:43:58	24°☿16'10			inferior conj	2000 Apr 16 09:15:12	26°☿36'05	1°59'10
morning set	2000 Jun 27 03:07:08	27°☿31'33			minimum elong	2000 Apr 16 13:05:02	26°☿29'04	1°58'01
	2000 Jun 28 07:03:41	0°☿			min. Earth dist.	2000 Apr 19 15:45:18	24°☿13'53	0.57276 AU
max. declination.	2000 Jul 02 01:58:14	8°☿11'57	24N14'35"		desc. node	2000 Apr 23 17:15:14	21°☿47'03	
					morning rise	2000 Apr 24 11:07:30	21°☿26'31	
superior conj	2000 Jul 04 02:46:23	12°☿37'49	1°19'17		direct	2000 Apr 29 22:04:17	20°☿14'11	
minimum elong	2000 Jul 04 00:22:13	12°☿24'45	1°18'57		min. declination.	2000 May 04 16:28:29	21°☿08'35	5N58'55"
max. Earth dist.	2000 Jul 05 17:17:12	16°☿06'31	1.32937 AU		morning max el	2000 May 14 06:01:41	27°☿43'52	26°08'34
evening rise	2000 Jul 11 09:00:06	28°☿00'37				2000 May 16 11:11:47	0°☿	
	2000 Jul 12 08:35:20	0°☿				2000 Jun 05 10:02:00	0°☿	
	2000 Jul 29 11:19:38	0°☿			morning set	2000 Jun 11 14:19:01	12°☿25'11	
desc. node	2000 Aug 02 19:32:09	6°☿10'59			asc. node	2000 Jun 12 10:47:28	14°☿13'38	
evening max el	2000 Aug 13 08:05:23	18°☿25'51	27°25'33					
retrograde	2000 Aug 27 01:27:13	25°☿43'18			superior conj	2000 Jun 18 14:00:19	27°☿35'00	1°00'45
min. declination.	2000 Aug 29 03:04:09	25°☿30'49	2S 8'45"		minimum elong	2000 Jun 18 11:46:26	27°☿22'44	1°00'21
evening set	2000 Sep 03 06:18:29	23°☿08'38			max. Earth dist.	2000 Jun 19 03:23:00	28°☿48'30	1.32364 AU
min. Earth dist.	2000 Sep 06 20:58:16	20°☿05'26	0.63103 AU			2000 Jun 19 16:24:58	0°☿	
inferior conj	2000 Sep 09 14:51:30	17°☿23'25	-3°36'39		max. declination.	2000 Jun 22 12:00:21	6°☿07'50	24N46'46"
minimum elong	2000 Sep 09 20:08:23	17°☿10'24	3°35'08		evening rise	2000 Jun 25 13:29:35	12°☿37'46	
morning rise	2000 Sep 16 11:19:04	12°☿10'57				2000 Jul 04 11:45:14	0°☿	
direct	2000 Sep 18 22:36:21	11°☿42'15			desc. node	2000 Jul 20 16:31:27	24°☿28'41	
asc. node	2000 Sep 21 12:56:33	12°☿14'42				2000 Jul 25 16:29:56	0°☿	
max. declination.	2000 Sep 22 19:51:37	12°☿54'25	7N 1'41"		evening max el	2000 Jul 26 14:02:29	0°☿52'14	27°03'16
morning max el	2000 Sep 25 14:13:17	15°☿08'15	17°51'30		retrograde	2000 Aug 09 11:32:49	8°☿05'54	
	2000 Oct 05 22:36:33	0°☿			min. declination.	2000 Aug 13 00:35:06	7°☿32'10	4N38'36"
morning set	2000 Oct 11 22:26:50	10°☿27'03			evening set	2000 Aug 16 12:29:33	5°☿53'14	
	2000 Oct 23 06:35:47	0°☿			min. Earth dist.	2000 Aug 20 04:20:05	3°☿11'26	0.61183 AU
					inferior conj	2000 Aug 23 08:13:07	0°☿26'59	-4°23'35
superior conj	2000 Oct 23 17:15:54	0°☿44'35	0°40'33		minimum elong	2000 Aug 23 13:14:08	0°☿16'04	4°22'36
minimum elong	2000 Oct 23 21:03:51	1°☿00'26	0°40'02			2000 Aug 23 20:39:19	30°☿	
desc. node	2000 Oct 29 18:53:09	10°☿41'53			morning rise	2000 Aug 30 15:40:38	25°☿35'05	
max. Earth dist.	2000 Oct 30 06:27:40	11°☿28'30	1.43770 AU		direct	2000 Sep 02 01:39:04	25°☿11'22	
evening rise	2000 Nov 08 01:59:00	25°☿21'45			max. declination.	2000 Sep 07 09:04:58	27°☿10'52	12N11'19"

asc. node	2003 Sep 08 09:59:50	28°01'28		min. Earth dist.	2004 Aug 01 01:49:57	15°02'11	0.59116 AU
morning max el	2003 Sep 09 04:07:18	28°04'32	17°59'48	inferior conj	2004 Aug 04 09:03:39	12°05'58	-4°53'49
	2003 Sep 10 09:23:21	0°00		minimum elong	2004 Aug 04 11:35:30	12°05'31	4°53'36
morning set	2003 Sep 25 00:20:51	23°00'42		morning rise	2004 Aug 12 03:21:00	8°02'29	
	2003 Sep 28 10:19:10	0°00		direct	2004 Aug 14 14:41:25	8°02'08	
				morning max el	2004 Aug 22 12:41:16	11°02'56	18°28'08
superior conj	2003 Oct 05 04:29:02	12°09'05	1°13'12	max. declination	2004 Aug 22 08:54:15	11°02'47	16N31'57"
minimum elong	2003 Oct 05 08:56:47	12°05'28	1°12'42	asc. node	2004 Aug 25 07:03:57	14°02'59	
max. Earth dist.	2003 Oct 12 18:17:05	25°05'29	1.42175 AU		2004 Sep 03 15:00:39	0°00	
	2003 Oct 15 18:02:58	0°00		morning set	2004 Sep 07 15:08:27	7°00'35	
desc. node	2003 Oct 16 15:52:36	1°02'28					
evening rise	2003 Oct 18 19:02:18	4°02'53		superior conj	2004 Sep 16 14:52:13	24°00'42	1°33'52
	2003 Nov 04 10:29:53	0°00		minimum elong	2004 Sep 16 17:58:35	24°00'56	1°33'37
evening max el	2003 Nov 20 13:07:42	20°00'34	22°19'34		2004 Sep 19 12:15:58	0°00	
min. declination	2003 Nov 22 01:50:57	22°00'32	25S41'29"	max. Earth dist.	2004 Sep 24 01:03:23	8°02'01	1.40219 AU
retrograde	2003 Nov 30 03:02:15	26°00'16		evening rise	2004 Sep 28 09:08:59	15°02'23	
evening set	2003 Dec 04 16:00:07	24°00'26		desc. node	2004 Oct 02 12:52:59	22°02'10	
asc. node	2003 Dec 05 09:14:58	23°00'48			2004 Oct 07 12:27:00	0°00	
inferior conj	2003 Dec 10 00:28:20	18°00'53	1°32'06		2004 Oct 29 07:09:10	0°00	
minimum elong	2003 Dec 09 22:32:04	18°00'13	1°31'21	evening max el	2004 Nov 02 03:27:28	4°00'09	23°39'22
min. Earth dist.	2003 Dec 09 22:45:10	18°00'12	0.67830 AU	min. declination	2004 Nov 08 23:15:41	9°00'27	24S30'16"
morning rise	2003 Dec 15 04:55:33	11°00'56		retrograde	2004 Nov 12 20:04:37	10°00'28	
direct	2003 Dec 19 21:13:56	9°00'59		evening set	2004 Nov 17 22:12:27	8°00'21	
max. declination	2003 Dec 19 09:13:40	10°00'00	19S 4' 1"	asc. node	2004 Nov 21 06:18:05	4°00'43	
morning max el	2003 Dec 29 07:39:19	15°00'33	22°17'20	inferior conj	2004 Nov 23 07:46:07	1°00'57	0°42'11
	2004 Jan 10 00:48:07	0°00		minimum elong	2004 Nov 23 06:47:00	2°00'01	0°41'46
desc. node	2004 Jan 12 15:09:05	3°00'38		min. Earth dist.	2004 Nov 22 19:22:29	2°00'40	0.67761 AU
min. declination	2004 Jan 17 06:59:03	10°00'22	23S35'37"		2004 Nov 24 18:52:41	30°00'00	
	2004 Jan 30 01:07:29	0°00		morning rise	2004 Nov 28 15:20:10	25°00'51	
morning set	2004 Jan 31 16:52:09	2°00'40		direct	2004 Dec 02 17:29:08	24°00'17	
max. Earth dist.	2004 Feb 06 01:34:32	11°00'32	1.40820 AU	max. declination	2004 Dec 03 09:56:08	24°00'19	16S17' 1"
				morning max el	2004 Dec 11 01:01:39	29°00'06	20°56'24
superior conj	2004 Feb 13 20:40:05	25°00'05	-2°01'24		2004 Dec 11 21:42:14	0°00	
minimum elong	2004 Feb 13 23:19:11	25°00'16	2°01'19	desc. node	2004 Dec 29 12:10:52	23°00'42	
	2004 Feb 16 14:00:20	0°00			2005 Jan 02 16:55:43	0°00	
evening rise	2004 Feb 24 02:12:35	13°00'56		min. declination	2005 Jan 08 01:15:58	8°00'11	24S13'45"
asc. node	2004 Mar 02 08:34:17	27°00'12		morning set	2005 Jan 09 23:32:50	11°00'11	
	2004 Mar 04 00:13:11	0°00		max. Earth dist.	2005 Jan 18 05:34:16	24°00'20	1.42751 AU
evening max el	2004 Mar 11 08:01:30	9°00'53	18°18'05		2005 Jan 21 16:11:10	0°00	
retrograde	2004 Mar 18 11:46:15	13°00'27		superior conj	2005 Jan 25 04:43:07	5°00'54	-2°04'26
max. declination	2004 Mar 19 20:43:34	13°00'19	8N22'21"	minimum elong	2005 Jan 25 03:13:11	5°00'48	2°04'25
evening set	2004 Mar 20 23:32:46	13°00'02		evening rise	2005 Feb 05 21:54:47	26°00'23	
inferior conj	2004 Mar 28 02:32:20	8°00'22	3°02'29		2005 Feb 07 22:18:43	0°00	
minimum elong	2004 Mar 28 06:02:06	8°00'14	3°01'47	asc. node	2005 Feb 17 05:37:30	15°00'48	
min. Earth dist.	2004 Mar 31 12:44:04	5°00'27	0.59308 AU	evening max el	2005 Feb 22 19:06:51	22°00'56	18°07'15
morning rise	2004 Apr 04 10:01:41	2°00'45		retrograde	2005 Mar 01 09:32:30	26°00'20	
desc. node	2004 Apr 09 14:18:32	1°00'00		max. declination	2005 Mar 03 15:43:13	25°00'56	1N23'51"
direct	2004 Apr 10 18:03:25	0°00'56		evening set	2005 Mar 04 02:39:47	25°00'45	
min. declination	2004 Apr 16 00:01:27	2°00'06	0S27'46"	inferior conj	2005 Mar 10 14:56:49	20°00'44	3°34'50
morning max el	2004 Apr 25 02:23:46	8°00'42	27°15'29	minimum elong	2005 Mar 10 16:37:11	20°00'40	3°34'41
	2004 May 11 07:20:47	0°00		min. Earth dist.	2005 Mar 13 17:07:49	17°00'43	0.61454 AU
morning set	2004 May 25 23:11:53	27°00'10		morning rise	2005 Mar 17 04:57:53	14°00'49	
	2004 May 27 07:30:25	0°00		direct	2005 Mar 24 00:59:38	12°00'27	
asc. node	2004 May 29 07:50:42	4°00'19		desc. node	2005 Mar 27 11:22:05	13°00'00	
max. Earth dist.	2004 Jun 01 15:42:56	11°00'35	1.32174 AU	min. declination	2005 Mar 29 17:13:56	13°00'54	6S46' 1"
				morning max el	2005 Apr 07 05:17:07	20°00'22	27°47'31
superior conj	2004 Jun 02 01:45:57	12°00'30	0°38'42		2005 Apr 15 12:01:25	0°00	
minimum elong	2004 Jun 02 00:08:41	12°00'21	0°38'21		2005 May 04 05:27:01	0°00	
evening rise	2004 Jun 08 22:26:51	27°00'25		morning set	2005 May 10 03:51:55	11°00'39	
	2004 Jun 10 03:51:51	0°00		max. Earth dist.	2005 May 16 02:22:51	24°00'14	1.32370 AU
max. declination	2004 Jun 12 03:38:22	4°00'04	25N12'49"	asc. node	2005 May 16 04:54:06	24°00'28	
	2004 Jun 27 02:57:45	0°00					
desc. node	2004 Jul 06 13:32:52	11°00'34		superior conj	2005 May 17 12:21:49	27°00'19	0°13'49
evening max el	2004 Jul 07 13:10:57	12°00'31	26°08'15	minimum elong	2005 May 17 11:43:43	27°00'16	0°13'40
retrograde	2004 Jul 21 12:58:55	19°00'42		behind sun begin	2005 May 17 09:07:22	27°00'01	
evening set	2004 Jul 27 21:59:14	18°00'03		behind sun end	2005 May 17 14:20:04	27°00'30	
min. declination	2004 Jul 27 11:37:40	18°00'16	11N 4'44"				

	2085 May 18 17:42:20	0°♂	morning rise	2086 Feb 27 18:23:48	27°≈30'35
evening rise	2085 May 24 09:46:38	12°♂17'46	direct	2086 Mar 06 16:49:50	24°≈46'09
	2085 Jun 02 09:10:57	0°♂	min. declination.	2086 Mar 13 00:51:16	26°≈41'11 12S21'51"
max. declination.	2085 Jun 03 04:25:56	1°♂25'23 25N32'37"	desc. node	2086 Mar 14 08:24:48	27°≈29'44
evening max el	2085 Jun 19 05:20:15	23°♂26'34 24°46'40		2086 Mar 17 15:14:57	0°♂
desc. node	2085 Jun 23 10:35:30	26°♂58'04	morning max el	2086 Mar 20 12:54:49	2°♂42'03 27°42'14
	2085 Jun 29 12:36:51	0°♂		2086 Apr 09 20:16:41	0°♂
retrograde	2085 Jul 03 04:06:40	0°♂30'49	morning set	2086 Apr 24 01:59:55	25°♂47'24
	2085 Jul 06 20:00:04	30°♂♂		2086 Apr 26 03:57:52	0°♂
evening set	2085 Jul 08 07:43:29	29°♂30'37	max. Earth dist.	2086 Apr 29 07:17:43	6°♂31'59 1.32974 AU
min. Earth dist.	2085 Jul 13 16:35:22	26°♂41'13 0.57160 AU			
min. declination.	2085 Jul 13 23:11:26	26°♂30'42 16N14'17"	superior conj	2086 May 01 20:02:13	11°♂55'47 -0°13'07
inferior conj	2085 Jul 16 14:17:14	24°♂47'12 -4°54'44	minimum elong	2086 May 01 20:40:46	11°♂59'15 0°12'59
minimum elong	2085 Jul 16 11:53:15	24°♂51'09 4°54'35	behind sun begin	2086 May 01 17:35:09	11°♂42'37
morning rise	2085 Jul 24 18:44:54	20°♂40'06	behind sun end	2086 May 01 23:46:23	12°♂15'52
direct	2085 Jul 27 09:21:03	20°♂20'55	asc. node	2086 May 03 01:57:31	14°♂36'57
morning max el	2085 Aug 05 12:47:30	24°♂36'56 19°17'33	evening rise	2086 May 08 21:36:03	27°♂07'33
max. declination.	2085 Aug 08 16:12:06	28°♂02'08 19N44' 2"		2086 May 10 06:28:47	0°♂
	2085 Aug 10 04:20:55	0°♂	max. declination.	2086 May 25 12:52:09	26°♂41'35 25N39' 5"
asc. node	2085 Aug 12 04:07:39	2°♂53'33		2086 May 28 03:20:15	0°♂
morning set	2085 Aug 22 14:50:10	21°♂54'51	evening max el	2086 May 31 19:07:18	3°♂57'23 23°11'39
	2085 Aug 26 15:54:12	0°♂	desc. node	2086 Jun 10 07:37:58	10°♂05'31
			retrograde	2086 Jun 14 06:56:39	10°♂41'27
superior conj	2085 Aug 30 18:39:08	8°♂07'25 1°43'56	evening set	2086 Jun 17 22:24:00	10°♂12'20
minimum elong	2085 Aug 30 19:51:37	8°♂13'18 1°43'54	min. Earth dist.	2086 Jun 25 04:44:41	6°♂55'27 0.55661 AU
max. Earth dist.	2085 Sep 06 04:43:11	20°♂15'00 1.38158 AU	inferior conj	2086 Jun 26 23:54:00	5°♂52'16 -4°12'27
evening rise	2085 Sep 09 23:15:52	27°♂00'19	minimum elong	2086 Jun 26 16:43:18	6°♂02'49 4°10'58
	2085 Sep 11 16:38:12	0°♂	min. declination.	2086 Jul 03 06:32:10	2°♂40'00 18N38'54"
desc. node	2085 Sep 19 09:54:14	12°♂44'31	morning rise	2086 Jul 05 13:34:12	1°♂58'42
	2085 Oct 01 00:39:40	0°♂	direct	2086 Jul 08 09:01:27	1°♂39'42
evening max el	2085 Oct 15 15:36:21	17°♂47'45 24°57'10	morning max el	2086 Jul 19 02:02:58	6°♂37'20 20°28'04
min. declination.	2085 Oct 25 13:07:27	24°♂23'31 21S48'26"	max. declination.	2086 Jul 28 04:05:57	18°♂20'40 21N46'19"
retrograde	2085 Oct 27 09:08:35	24°♂36'31	asc. node	2086 Jul 30 01:10:36	21°♂30'51
evening set	2085 Nov 02 01:21:09	22°♂13'34		2086 Aug 03 14:28:40	0°♂
min. Earth dist.	2085 Nov 06 14:17:00	17°♂06'18 0.67357 AU	morning set	2086 Aug 06 20:34:30	6°♂32'16
inferior conj	2085 Nov 07 13:17:46	15°♂50'31 -0°12'07			
minimum elong	2085 Nov 07 13:35:55	15°♂49'31 0°11'59	superior conj	2086 Aug 14 10:55:37	22°♂10'10 1°45'16
transit middle	2085 Nov 07 13:35:55	15°♂49'31 0°11'59	minimum elong	2086 Aug 14 10:27:22	22°♂07'46 1°45'16
transit begin	2085 Nov 07 11:44:02	15°♂55'40		2086 Aug 18 09:04:35	0°♂
transit end	2085 Nov 07 15:27:49	15°♂43'23	max. Earth dist.	2086 Aug 19 11:02:26	2°♂06'21 1.36250 AU
asc. node	2085 Nov 08 03:21:29	15°♂04'23	evening rise	2086 Aug 23 10:37:55	9°♂38'08
morning rise	2085 Nov 13 02:00:20	9°♂50'23		2086 Sep 04 07:47:09	0°♂
direct	2085 Nov 16 15:36:48	8°♂37'02	desc. node	2086 Sep 06 06:56:11	3°♂04'49
max. declination.	2085 Nov 18 01:16:27	8°♂47'41 12S14'45"		2086 Sep 26 16:36:34	0°♂
morning max el	2085 Nov 24 02:10:16	12°♂50'43 19°47'29	evening max el	2086 Sep 28 03:36:54	1°♂27'37 26°05'30
	2085 Dec 07 02:15:52	0°♂	min. declination.	2086 Oct 10 03:14:12	8°♂34'19 17S37'56"
greatest brilliancy	2085 Dec 07 13:04:02	0°♂40'03 -0.7m	retrograde	2086 Oct 10 17:24:26	8°♂35'35
desc. node	2085 Dec 16 09:11:52	14°♂03'13	evening set	2086 Oct 16 23:36:00	5°♂59'45
morning set	2085 Dec 19 19:37:47	19°♂20'32	min. Earth dist.	2086 Oct 21 05:06:33	1°♂28'03 0.66617 AU
	2085 Dec 26 16:03:23	0°♂		2086 Oct 22 09:23:57	30°♂♂
min. declination.	2085 Dec 30 19:45:31	6°♂32'55 24S43'30"	inferior conj	2086 Oct 22 15:13:58	29°♂41'47 -1°09'24
max. Earth dist.	2085 Dec 31 16:34:00	7°♂55'31 1.44208 AU	minimum elong	2086 Oct 22 17:03:33	29°♂36'06 1°08'38
			asc. node	2086 Oct 26 00:23:39	25°♂48'08
superior conj	2086 Jan 05 10:23:49	15°♂31'48 -1°50'26	morning rise	2086 Oct 28 10:55:21	23°♂50'49
minimum elong	2086 Jan 05 03:27:51	15°♂03'47 1°49'57	direct	2086 Oct 31 14:06:17	22°♂54'38
	2086 Jan 14 05:17:08	0°♂	max. declination.	2086 Nov 02 13:27:58	23°♂15'56 7S18'38"
evening rise	2086 Jan 18 22:43:37	7°≈59'05	morning max el	2086 Nov 07 10:28:10	26°♂43'06 18°53'27
	2086 Feb 01 08:30:08	0°♂		2086 Nov 10 08:03:42	0°♂
asc. node	2086 Feb 04 02:40:02	3°♂43'38	morning set	2086 Nov 29 06:21:47	28°♂20'07
evening max el	2086 Feb 06 07:42:52	6°♂13'19 18°15'41		2086 Nov 30 07:47:07	0°♂
retrograde	2086 Feb 12 18:45:25	9°♂41'04	desc. node	2086 Dec 03 06:12:49	4°♂36'35
evening set	2086 Feb 15 17:03:37	8°♂55'44	max. Earth dist.	2086 Dec 14 08:58:01	22°♂03'11 1.45021 AU
max. declination.	2086 Feb 15 20:13:21	8°♂51'38 5S26'57"			
inferior conj	2086 Feb 21 18:24:48	3°♂35'04 3°43'12	superior conj	2086 Dec 15 17:59:24	24°♂13'05 -1°17'28
minimum elong	2086 Feb 21 18:10:16	3°♂35'45 3°43'11	minimum elong	2086 Dec 15 09:19:53	23°♂39'00 1°16'30
min. Earth dist.	2086 Feb 24 06:43:37	0°♂48'41 0.63439 AU		2086 Dec 19 09:54:45	0°♂
	2086 Feb 25 01:16:54	30°♂≈	min. declination.	2086 Dec 22 14:35:16	5°♂04'15 25S 8'15"

evening rise	2086 Dec 31 00:59:00	18°☾36'59		superior conj	2087 Nov 24 20:03:44	2°☿45'38	-0°31'11
	2087 Jan 07 01:33:44	0°☿		minimum elong	2087 Nov 24 16:02:07	2°☿29'44	0°30'38
evening max el	2087 Jan 20 19:11:30	19°☿38'25	18°42'21	max. Earth dist.	2087 Nov 27 04:15:54	6°☿26'59	1.45100 AU
asc. node	2087 Jan 21 23:41:50	20°☿46'21		evening rise	2087 Dec 11 04:21:08	28°☿20'59	
retrograde	2087 Jan 27 11:13:20	23°☿21'22			2087 Dec 12 05:43:53	0°☿	
evening set	2087 Jan 30 15:19:38	22°☿24'14		min. declination.	2087 Dec 14 10:56:50	3°☿27'34	25S29'19"
max. declination.	2087 Jan 31 14:12:09	21°☿45'33	11S41'30"	greatest brilliancy	2087 Dec 22 19:25:25	16°☿23'19	-0.7m
inferior conj	2087 Feb 05 08:58:45	16°☿45'47	3°33'50		2088 Jan 01 09:31:29	0°☿	
minimum elong	2087 Feb 05 07:19:42	16°☿50'48	3°33'38	evening max el	2088 Jan 04 03:21:07	3°☿07'36	19°25'58
min. Earth dist.	2087 Feb 07 05:44:05	14°☿30'07	0.65106 AU	asc. node	2088 Jan 08 20:42:53	6°☿42'46	
morning rise	2087 Feb 10 22:53:36	10°☿36'04		retrograde	2088 Jan 11 07:43:31	7°☿15'45	
direct	2087 Feb 17 15:47:26	7°☿44'04		evening set	2088 Jan 14 18:49:03	6°☿05'37	
min. declination.	2087 Feb 25 07:18:30	10°☿39'01	16S51'16"	max. declination.	2088 Jan 17 13:20:07	3°☿34'42	16S49'23"
desc. node	2087 Mar 01 05:26:30	13°☿51'19		inferior conj	2088 Jan 20 07:23:24	0°☿11'30	3°11'20
morning max el	2087 Mar 02 22:26:44	15°☿30'22	27°03'08	minimum elong	2088 Jan 20 04:57:43	0°☿19'24	3°10'49
	2087 Mar 14 20:35:04	0°☿			2088 Jan 20 10:55:32	30°☿	
	2087 Apr 02 14:17:55	0°☿		min. Earth dist.	2088 Jan 21 13:12:54	28°☿34'39	0.66374 AU
morning set	2087 Apr 07 14:27:15	9°☿21'46		morning rise	2088 Jan 25 14:52:45	23°☿59'08	
max. Earth dist.	2087 Apr 12 02:39:16	18°☿17'16	1.34028 AU	direct	2088 Jan 31 20:44:21	21°☿13'03	
				min. declination.	2088 Feb 11 09:58:46	26°☿46'16	20S 1' 0"
superior conj	2087 Apr 15 22:39:51	26°☿12'01	-0°41'05	morning max el	2088 Feb 13 07:50:06	28°☿36'42	25°57'50
minimum elong	2087 Apr 16 00:44:35	26°☿22'54	0°40'39		2088 Feb 14 15:57:18	0°☿	
	2087 Apr 17 17:59:33	0°☿		desc. node	2088 Feb 16 02:27:30	1°☿34'19	
asc. node	2087 Apr 19 22:59:11	4°☿40'41			2088 Mar 07 15:20:07	0°☿	
evening rise	2087 Apr 23 07:59:31	11°☿47'49		morning set	2088 Mar 20 13:14:15	22°☿11'24	
	2087 May 02 18:40:36	0°☿		max. Earth dist.	2088 Mar 24 10:32:41	29°☿31'49	1.35544 AU
evening max el	2087 May 13 13:52:17	14°☿34'12	21°38'07		2088 Mar 24 16:23:01	0°☿	
max. declination.	2087 May 15 16:20:31	16°☿26'45	25N 2' 2"	superior conj	2088 Mar 29 17:48:58	10°☿01'15	-1°08'42
retrograde	2087 May 25 22:18:30	20°☿37'38		minimum elong	2088 Mar 29 21:14:57	10°☿18'37	1°08'06
evening set	2087 May 28 09:57:25	20°☿23'42		asc. node	2088 Apr 05 20:01:18	24°☿35'17	
desc. node	2087 May 28 04:40:19	20°☿26'01		evening rise	2088 Apr 06 15:00:13	26°☿12'37	
inferior conj	2087 Jun 06 19:47:45	16°☿22'16	-2°42'45		2088 Apr 08 11:48:53	0°☿	
minimum elong	2087 Jun 06 12:40:18	16°☿32'14	2°40'28	evening max el	2088 Apr 24 19:05:54	25°☿41'52	20°17'26
min. Earth dist.	2087 Jun 06 16:34:53	16°☿26'46	0.54938 AU	max. declination.	2088 May 01 05:26:18	0°☿04'09	22N41'30"
morning rise	2087 Jun 15 16:18:16	12°☿26'46			2088 May 01 01:12:24	0°☿	
direct	2087 Jun 18 20:30:48	12°☿04'40		retrograde	2088 May 05 11:48:30	0°☿52'52	
min. declination.	2087 Jun 19 03:40:43	12°☿04'52	17N52'59"	evening set	2088 May 07 12:44:09	0°☿42'23	
morning max el	2087 Jul 01 03:49:40	17°☿52'46	21°57'22		2088 May 10 04:10:21	30°☿	
	2087 Jul 10 19:22:03	0°☿		desc. node	2088 May 14 01:40:37	28°☿11'49	
asc. node	2087 Jul 16 22:12:46	10°☿40'11		inferior conj	2088 May 16 15:00:58	26°☿44'02	-0°44'48
max. declination.	2087 Jul 18 01:25:43	12°☿51'25	23N 1'22"	minimum elong	2088 May 16 12:54:49	26°☿47'08	0°44'01
morning set	2087 Jul 22 06:04:51	21°☿21'07		min. Earth dist.	2088 May 18 05:19:12	25°☿47'47	0.55167 AU
	2087 Jul 26 08:26:29	0°☿		morning rise	2088 May 25 11:43:01	22°☿27'31	
superior conj	2087 Jul 29 11:49:21	6°☿38'30	1°39'33	direct	2088 May 29 08:08:02	21°☿55'24	
minimum elong	2087 Jul 29 10:10:14	6°☿29'49	1°39'26	min. declination.	2088 Jun 01 08:32:08	22°☿15'48	14N37' 7"
max. Earth dist.	2087 Aug 02 01:22:58	14°☿01'39	1.34657 AU	morning max el	2088 Jun 11 21:01:23	28°☿32'16	23°38'19
evening rise	2087 Aug 06 14:31:49	23°☿03'36			2088 Jun 13 08:35:49	0°☿	
	2087 Aug 10 06:56:50	0°☿		asc. node	2088 Jul 02 19:15:11	0°☿13'24	
desc. node	2087 Aug 24 03:58:28	23°☿04'39			2088 Jul 02 16:35:33	0°☿	
	2087 Aug 28 23:44:08	0°☿		morning set	2088 Jul 05 17:32:33	6°☿16'13	
evening max el	2087 Sep 10 16:02:19	15°☿02'23	26°56'43	max. declination.	2088 Jul 07 07:18:19	9°☿36'34	23N51'31"
retrograde	2087 Sep 23 20:09:50	22°☿18'48		superior conj	2088 Jul 12 18:28:29	21°☿23'47	1°28'05
min. declination.	2087 Sep 24 06:00:20	22°☿18'15	12S15' 7"	minimum elong	2088 Jul 12 16:11:36	21°☿11'30	1°27'49
evening set	2087 Sep 30 14:45:19	19°☿35'51		max. Earth dist.	2088 Jul 15 01:06:06	26°☿15'57	1.33451 AU
min. Earth dist.	2087 Oct 04 13:22:42	15°☿40'40	0.65528 AU		2088 Jul 16 19:39:14	0°☿	
inferior conj	2087 Oct 06 11:30:09	13°☿27'38	-2°08'01	evening rise	2088 Jul 20 06:39:15	7°☿04'52	
minimum elong	2087 Oct 06 14:56:04	13°☿17'44	2°06'41		2088 Aug 01 19:49:16	0°☿	
morning rise	2087 Oct 12 15:49:03	7°☿48'58		desc. node	2088 Aug 10 00:59:39	12°☿35'13	
asc. node	2087 Oct 12 21:24:50	7°☿42'10		evening max el	2088 Aug 23 03:36:23	28°☿20'12	27°23'31
direct	2087 Oct 15 10:54:42	7°☿06'09			2088 Aug 24 23:33:59	0°☿	
max. declination.	2087 Oct 18 00:25:15	7°☿41'20	1S50'14"	retrograde	2088 Sep 05 16:54:00	5°☿38'02	
morning max el	2087 Oct 21 23:57:30	10°☿39'02	18°15'49	min. declination.	2088 Sep 07 02:52:38	5°☿31'56	6S 0'39"
	2087 Nov 04 12:27:02	0°☿		evening set	2088 Sep 12 20:03:14	2°☿56'51	
morning set	2087 Nov 09 21:30:50	8°☿46'17			2088 Sep 16 03:33:05	30°☿	
desc. node	2087 Nov 20 03:14:55	25°☿18'15		min. Earth dist.	2088 Sep 16 12:50:06	29°☿36'32	0.64085 AU
	2087 Nov 23 02:09:57	0°☿					

inferior conj	2088 Sep 18 23:40:09	27° $\mathbb{M}$ 02'29	-3°05'32	desc. node	2089 Jul 27 21:59:52	1° $\mathbb{M}$ 24'34	
minimum elong	2088 Sep 19 04:28:32	26° $\mathbb{M}$ 49'53	3°03'56	evening max el	2089 Aug 05 12:09:31	11° $\mathbb{M}$ 08'28	27°20'08
morning rise	2088 Sep 25 13:58:17	21° $\mathbb{M}$ 39'37		retrograde	2089 Aug 19 07:20:17	18° $\mathbb{M}$ 24'39	
direct	2088 Sep 28 03:26:53	21° $\mathbb{M}$ 06'37		min. declination.	2089 Aug 21 22:23:34	18° $\mathbb{M}$ 05'10	0N42'15"
asc. node	2088 Sep 28 18:27:06	21° $\mathbb{M}$ 08'37		evening set	2089 Aug 26 12:00:07	15° $\mathbb{M}$ 57'19	
max. declination.	2088 Oct 01 10:56:57	22° $\mathbb{M}$ 01'55	3N48'42"	min. Earth dist.	2089 Aug 30 02:03:47	13° $\mathbb{M}$ 05'13	0.62312 AU
morning max el	2088 Oct 04 15:58:55	24° $\mathbb{M}$ 32'27	17°55'19	inferior conj	2089 Sep 02 00:52:24	10° $\mathbb{M}$ 20'02	-3°58'01
	2088 Oct 09 02:48:14	0° $\mathbb{L}$		minimum elong	2089 Sep 02 06:15:32	10° $\mathbb{M}$ 07'26	3°56'40
morning set	2088 Oct 21 16:10:41	20° $\mathbb{L}$ 32'21		morning rise	2089 Sep 09 02:01:46	5° $\mathbb{M}$ 16'18	
	2088 Oct 27 05:50:15	0° $\mathbb{M}$		direct	2089 Sep 11 12:27:11	4° $\mathbb{M}$ 50'02	
				asc. node	2089 Sep 15 15:30:23	6° $\mathbb{M}$ 07'46	
superior conj	2088 Nov 03 13:34:00	12° $\mathbb{M}$ 06'49	0°16'31	max. declination.	2089 Sep 15 22:14:35	6° $\mathbb{M}$ 18'36	9N15'38"
minimum elong	2088 Nov 03 15:24:35	12° $\mathbb{M}$ 14'18	0°16'16	morning max el	2089 Sep 18 07:39:36	8° $\mathbb{M}$ 17'17	17°52'47
desc. node	2088 Nov 06 00:16:08	16° $\mathbb{M}$ 04'14			2089 Oct 02 11:46:21	0° $\mathbb{L}$	
max. Earth dist.	2088 Nov 08 22:48:02	20° $\mathbb{M}$ 46'16	1.44444 AU	morning set	2089 Oct 04 08:27:44	3° $\mathbb{L}$ 20'28	
	2088 Nov 14 19:31:41	0° $\mathbb{M}$					
evening rise	2088 Nov 19 15:29:53	7° $\mathbb{M}$ 29'03		superior conj	2089 Oct 15 09:33:05	22° $\mathbb{L}$ 46'52	0°55'53
	2088 Dec 04 13:58:00	0° $\mathbb{L}$		minimum elong	2089 Oct 15 13:59:37	23° $\mathbb{L}$ 05'48	0°55'19
min. declination.	2088 Dec 05 09:48:48	1° $\mathbb{L}$ 12'12	25S46' 0"		2089 Oct 19 16:37:10	0° $\mathbb{M}$	
evening max el	2088 Dec 17 06:27:50	16° $\mathbb{L}$ 37'28	20°24'35	max. Earth dist.	2089 Oct 22 13:22:27	4° $\mathbb{M}$ 42'16	1.43143 AU
retrograde	2088 Dec 25 05:40:00	21° $\mathbb{L}$ 19'00		desc. node	2089 Oct 23 21:18:14	6° $\mathbb{M}$ 51'39	
asc. node	2088 Dec 25 17:44:21	21° $\mathbb{L}$ 17'36		evening rise	2089 Oct 30 01:29:13	16° $\mathbb{M}$ 39'23	
evening set	2088 Dec 29 01:19:33	19° $\mathbb{L}$ 54'32			2089 Nov 07 19:08:15	0° $\mathbb{M}$	
inferior conj	2089 Jan 03 10:56:34	13° $\mathbb{L}$ 47'33	2°38'48	min. declination.	2089 Nov 27 04:15:48	26° $\mathbb{L}$ 50'31	25S51'45"
minimum elong	2089 Jan 03 08:19:23	13° $\mathbb{L}$ 56'27	2°38'03	evening max el	2089 Nov 30 03:54:59	0° $\mathbb{L}$ 08'02	21°35'11
min. Earth dist.	2089 Jan 04 03:23:43	12° $\mathbb{L}$ 51'43	0.67234 AU		2089 Nov 30 00:46:38	0° $\mathbb{L}$	
max. declination.	2089 Jan 07 09:23:14	8° $\mathbb{L}$ 48'17	19S53'44"	retrograde	2089 Dec 09 02:58:21	5° $\mathbb{L}$ 27'57	
morning rise	2089 Jan 08 15:09:12	7° $\mathbb{L}$ 34'17		asc. node	2089 Dec 12 14:46:03	4° $\mathbb{L}$ 20'50	
direct	2089 Jan 14 06:57:13	5° $\mathbb{L}$ 03'41		evening set	2089 Dec 13 09:01:29	3° $\mathbb{L}$ 47'49	
morning max el	2089 Jan 25 16:23:51	11° $\mathbb{L}$ 51'28	24°35'53		2089 Dec 16 21:37:46	30° $\mathbb{R}$	
min. declination.	2089 Jan 30 12:47:39	17° $\mathbb{L}$ 14'08	21S57'15"	inferior conj	2089 Dec 18 17:27:42	27° $\mathbb{L}$ 31'20	1°58'23
desc. node	2089 Feb 01 23:29:31	20° $\mathbb{L}$ 14'42		minimum elong	2089 Dec 18 15:08:47	27° $\mathbb{L}$ 39'21	1°57'34
	2089 Feb 09 05:42:47	0° $\mathbb{M}$		min. Earth dist.	2089 Dec 18 22:03:39	27° $\mathbb{L}$ 15'24	0.67711 AU
	2089 Feb 28 09:29:53	0° $\mathbb{M}$		morning rise	2089 Dec 23 21:06:34	21° $\mathbb{L}$ 18'54	
morning set	2089 Mar 02 16:58:40	4° $\mathbb{M}$ 01'06		max. declination.	2089 Dec 27 04:40:11	19° $\mathbb{L}$ 23'36	19S57'45"
max. Earth dist.	2089 Mar 06 08:49:10	10° $\mathbb{M}$ 34'23	1.37456 AU	direct	2089 Dec 28 21:51:33	19° $\mathbb{L}$ 09'11	
				morning max el	2090 Jan 08 01:42:59	25° $\mathbb{L}$ 11'19	23°07'24
superior conj	2089 Mar 13 02:17:03	23° $\mathbb{M}$ 14'23	-1°33'55		2090 Jan 12 09:57:57	0° $\mathbb{L}$	
minimum elong	2089 Mar 13 06:33:08	23° $\mathbb{M}$ 35'06	1°33'23	desc. node	2090 Jan 19 20:32:28	9° $\mathbb{L}$ 36'09	
	2089 Mar 16 12:57:31	0° $\mathbb{Y}$		min. declination.	2090 Jan 21 17:07:29	12° $\mathbb{L}$ 10'32	23S 6'37"
evening rise	2089 Mar 21 16:16:57	10° $\mathbb{Y}$ 14'45			2090 Feb 02 18:04:39	0° $\mathbb{M}$	
asc. node	2089 Mar 23 17:03:42	14° $\mathbb{Y}$ 15'46		morning set	2090 Feb 11 19:26:58	14° $\mathbb{M}$ 34'59	
	2089 Apr 01 12:52:16	0° $\mathbb{L}$		max. Earth dist.	2090 Feb 16 03:46:55	21° $\mathbb{M}$ 58'10	1.39590 AU
evening max el	2089 Apr 07 11:56:14	7° $\mathbb{L}$ 29'04	19°15'15		2090 Feb 20 17:01:42	0° $\mathbb{M}$	
max. declination.	2089 Apr 15 10:32:29	11° $\mathbb{L}$ 47'33	18N15'14"				
retrograde	2089 Apr 16 12:30:03	11° $\mathbb{L}$ 51'12		superior conj	2090 Feb 23 19:58:33	5° $\mathbb{M}$ 41'19	-1°53'51
evening set	2089 Apr 18 14:52:17	11° $\mathbb{L}$ 38'01		minimum elong	2090 Feb 23 23:49:18	5° $\mathbb{M}$ 59'04	1°53'35
inferior conj	2089 Apr 26 23:32:44	7° $\mathbb{L}$ 28'23	1°07'15	evening rise	2090 Mar 05 08:58:28	23° $\mathbb{M}$ 46'44	
minimum elong	2089 Apr 27 02:10:44	7° $\mathbb{L}$ 23'58	1°06'22		2090 Mar 08 15:23:07	0° $\mathbb{Y}$	
min. Earth dist.	2089 Apr 29 20:12:34	5° $\mathbb{L}$ 34'10	0.56298 AU	asc. node	2090 Mar 10 14:06:21	3° $\mathbb{Y}$ 36'43	
desc. node	2089 Apr 30 22:41:00	4° $\mathbb{L}$ 52'34		evening max el	2090 Mar 21 14:27:43	19° $\mathbb{Y}$ 52'51	18°33'12
morning rise	2089 May 05 10:31:30	2° $\mathbb{L}$ 37'56		retrograde	2090 Mar 29 07:16:59	23° $\mathbb{Y}$ 39'29	
direct	2089 May 10 06:32:14	1° $\mathbb{L}$ 44'03		max. declination.	2090 Mar 29 23:59:11	23° $\mathbb{Y}$ 37'42	12N15'38"
min. declination.	2089 May 14 15:16:57	2° $\mathbb{L}$ 28'10	9N29'47"	evening set	2090 Mar 31 15:28:58	23° $\mathbb{Y}$ 19'42	
morning max el	2089 May 24 11:27:47	8° $\mathbb{L}$ 58'56	25°18'05	inferior conj	2090 Apr 08 04:41:39	18° $\mathbb{Y}$ 51'35	2°30'09
	2089 Jun 09 03:58:56	0° $\mathbb{I}$		minimum elong	2090 Apr 08 08:40:58	18° $\mathbb{Y}$ 43'46	2°29'07
asc. node	2089 Jun 19 16:18:56	20° $\mathbb{I}$ 03'51		min. Earth dist.	2090 Apr 11 14:40:00	16° $\mathbb{Y}$ 12'37	0.58101 AU
morning set	2089 Jun 20 05:20:06	21° $\mathbb{I}$ 12'20		morning rise	2090 Apr 15 22:50:24	13° $\mathbb{Y}$ 28'46	
	2089 Jun 24 07:21:57	0° $\mathbb{L}$		desc. node	2090 Apr 17 19:44:13	12° $\mathbb{Y}$ 42'08	
				direct	2090 Apr 21 19:31:13	12° $\mathbb{Y}$ 01'17	
superior conj	2089 Jun 27 04:37:36	6° $\mathbb{L}$ 18'53	1°11'54	min. declination.	2090 Apr 26 19:28:03	13° $\mathbb{Y}$ 02'38	3N18' 2"
minimum elong	2089 Jun 27 02:14:24	6° $\mathbb{L}$ 05'50	1°11'31	morning max el	2090 May 06 04:43:54	19° $\mathbb{Y}$ 39'58	26°40'50
max. declination.	2089 Jun 27 15:31:49	7° $\mathbb{L}$ 18'28	24N28'35"		2090 May 14 23:48:33	0° $\mathbb{L}$	
max. Earth dist.	2089 Jun 28 08:16:12	8° $\mathbb{L}$ 49'46	1.32642 AU		2090 Jun 01 17:11:45	0° $\mathbb{I}$	
evening rise	2089 Jul 04 07:28:35	21° $\mathbb{L}$ 31'37		morning set	2090 Jun 04 15:46:19	6° $\mathbb{I}$ 03'06	
	2089 Jul 08 13:11:36	0° $\mathbb{I}$		asc. node	2090 Jun 06 13:22:02	10° $\mathbb{I}$ 05'16	
	2089 Jul 26 20:21:04	0° $\mathbb{M}$					

superior conj	2090 Jun 11 16:18:53	21° Π 16'23	0°51'48	direct	2091 Apr 03 20:57:50	23° X 04'54	
minimum elong	2090 Jun 11 14:17:34	21° Π 05'15	0°51'24	desc. node	2091 Apr 04 16:46:32	23° X 06'47	
max. Earth dist.	2090 Jun 11 19:38:36	21° Π 34'43	1.32229 AU	min. declination.	2091 Apr 09 06:39:44	24° X 21'15	3S 9'40"
	2090 Jun 15 15:59:45	0° S			2091 Apr 17 04:41:21	0° Y	
max. declination.	2090 Jun 18 03:29:58	5° S 18'15	24N58' 3"	morning max el	2091 Apr 18 03:49:05	0° Y 55'41	27°33'34
evening rise	2090 Jun 18 14:08:15	6° S 14'14			2091 May 09 02:45:52	0° X	
	2090 Jul 01 03:17:21	0° Q		morning set	2091 May 19 23:07:41	20° X 42'42	
desc. node	2090 Jul 14 19:01:51	19° Q 14'44		asc. node	2091 May 24 10:26:23	0° Π 13'06	
evening max el	2090 Jul 18 15:15:08	23° Q 15'04	26°43'41		2091 May 24 08:01:29	0° Π	
	2090 Jul 29 05:29:38	0° W		max. Earth dist.	2091 May 26 07:37:42	4° Π 19'37	1.32210 AU
retrograde	2090 Aug 01 14:19:48	0° W 28'21					
	2090 Aug 04 21:38:45	30° R Q		superior conj	2091 May 27 03:50:18	6° Π 10'30	0°28'28
min. declination.	2090 Aug 05 23:32:44	29° Q 38'27	7N25'45"	minimum elong	2091 May 27 02:35:33	6° Π 03'39	0°28'12
evening set	2090 Aug 08 10:00:36	28° Q 28'39		evening rise	2091 Jun 03 00:22:57	21° Π 05'16	
min. Earth dist.	2090 Aug 12 05:05:25	25° Q 51'25	0.60302 AU		2091 Jun 07 08:28:44	0° S	
inferior conj	2090 Aug 15 11:38:28	23° Q 10'48	-4°39'08	max. declination.	2091 Jun 08 22:32:55	3° S 05'51	25N21'48"
minimum elong	2090 Aug 15 15:55:41	23° Q 01'59	4°38'30		2091 Jun 26 04:15:35	0° Q	
morning rise	2090 Aug 22 23:48:43	18° Q 28'46		evening max el	2091 Jun 30 11:23:26	4° Q 35'10	25°36'12
direct	2090 Aug 25 09:53:34	18° Q 06'39		desc. node	2091 Jul 01 16:02:50	5° Q 41'13	
max. declination.	2090 Aug 31 14:18:51	20° Q 38'47	14N 7'32"	retrograde	2091 Jul 14 11:44:56	11° Q 43'38	
morning max el	2090 Sep 01 19:46:01	21° Q 44'22	18°09'22	evening set	2091 Jul 20 09:28:50	10° Q 21'35	
asc. node	2090 Sep 02 12:34:44	22° Q 26'40		min. declination.	2091 Jul 22 00:27:29	9° Q 32'33	13N28'30"
	2090 Sep 08 04:39:02	0° W		min. Earth dist.	2091 Jul 24 23:36:10	7° Q 41'29	0.58254 AU
morning set	2090 Sep 17 16:23:37	16° W 53'10		inferior conj	2091 Jul 28 04:27:01	5° Q 25'14	-4°58'39
	2090 Sep 24 17:08:06	0° S		minimum elong	2091 Jul 28 05:07:21	5° Q 24'02	4°58'39
				morning rise	2091 Aug 05 03:07:47	1° Q 05'41	
superior conj	2090 Sep 27 07:18:19	4° S 42'19	1°23'22	direct	2091 Aug 07 15:33:49	0° Q 45'45	
minimum elong	2090 Sep 27 11:20:11	5° S 00'26	1°22'57	morning max el	2091 Aug 16 01:02:48	4° Q 44'25	18°46'26
max. Earth dist.	2090 Oct 04 22:22:53	18° S 00'04	1.41367 AU	max. declination.	2091 Aug 16 23:47:02	5° Q 41'03	18N 1'14"
evening rise	2090 Oct 10 02:30:16	26° S 32'50		asc. node	2091 Aug 20 09:38:38	9° Q 50'05	
desc. node	2090 Oct 10 18:19:46	27° S 36'43			2091 Aug 31 23:27:23	0° W	
	2090 Oct 12 06:07:29	0° M		morning set	2091 Sep 01 11:22:06	0° W 58'06	
	2090 Nov 01 13:28:12	0° X					
evening max el	2090 Nov 12 20:22:14	13° X 40'36	22°53'24	superior conj	2091 Sep 10 01:42:18	17° W 38'52	1°39'19
min. declination.	2090 Nov 17 00:04:05	17° X 18'45	25S21'26"	minimum elong	2091 Sep 10 04:00:49	17° W 49'47	1°39'11
retrograde	2090 Nov 22 21:57:59	19° X 39'09			2091 Sep 16 19:05:14	0° S	
evening set	2090 Nov 27 16:13:50	17° X 42'21		max. Earth dist.	2091 Sep 17 03:08:43	0° S 35'45	1.39344 AU
asc. node	2090 Nov 29 11:49:07	15° X 55'29		evening rise	2091 Sep 21 02:56:24	7° S 31'31	
inferior conj	2090 Dec 03 00:59:29	11° X 20'13	1°11'34	desc. node	2091 Sep 27 15:21:30	18° S 15'50	
minimum elong	2090 Dec 02 23:24:51	11° X 25'40	1°10'55		2091 Oct 05 06:17:05	0° M	
min. Earth dist.	2090 Dec 02 18:44:26	11° X 41'48	0.67835 AU	evening max el	2091 Oct 26 09:38:13	27° M 16'58	24°13'17
morning rise	2090 Dec 08 06:28:41	5° X 10'40			2091 Oct 29 09:03:21	0° X	
direct	2090 Dec 12 16:34:38	3° X 23'40		min. declination.	2091 Nov 03 16:45:48	3° X 16'43	23S33'43"
max. declination.	2090 Dec 12 18:53:17	3° X 23'43	18S 4'19"	retrograde	2091 Nov 06 13:17:34	3° X 49'12	
morning max el	2090 Dec 21 15:05:33	8° X 38'02	21°41'39	evening set	2091 Nov 11 21:18:05	1° X 35'34	
desc. node	2091 Jan 06 17:34:59	29° X 26'57			2091 Nov 13 12:31:42	30° R M	
	2091 Jan 07 02:41:37	0° S		asc. node	2091 Nov 16 08:52:52	26° M 28'44	
min. declination.	2091 Jan 13 10:07:41	9° S 21'35	23S52'36"	min. Earth dist.	2091 Nov 16 14:57:15	26° M 08'23	0.67629 AU
morning set	2091 Jan 22 16:45:09	23° S 45'18		inferior conj	2091 Nov 17 07:44:33	25° M 11'49	0°19'37
	2091 Jan 26 13:43:43	0° S		minimum elong	2091 Nov 17 07:15:58	25° M 13'26	0°19'25
max. Earth dist.	2091 Jan 29 02:51:05	4° S 11'20	1.41678 AU	morning rise	2091 Nov 22 17:14:01	19° M 07'31	
				direct	2091 Nov 26 13:54:06	17° M 42'28	
superior conj	2091 Feb 05 17:51:23	17° S 10'05	-2°04'29	max. declination.	2091 Nov 27 14:23:28	17° M 48'00	14S43'10"
minimum elong	2091 Feb 05 19:04:00	17° S 15'23	2°04'28	morning max el	2091 Dec 04 11:31:41	22° M 15'02	20°25'20
	2091 Feb 12 22:08:55	0° X			2091 Dec 10 23:39:51	0° X	
evening rise	2091 Feb 16 13:31:04	6° X 40'18		desc. node	2091 Dec 24 14:36:11	19° X 38'37	
asc. node	2091 Feb 25 11:09:24	22° X 31'35			2091 Dec 31 09:31:57	0° S	
	2091 Mar 02 12:50:11	0° Y		morning set	2092 Jan 01 15:49:44	1° S 57'06	
evening max el	2091 Mar 04 23:28:36	2° Y 44'46	18°11'10	min. declination.	2092 Jan 05 04:40:04	7° S 27'47	24S26'44"
retrograde	2091 Mar 11 20:10:53	6° Y 12'18		max. Earth dist.	2092 Jan 11 09:56:50	17° S 20'38	1.43440 AU
max. declination.	2091 Mar 13 14:54:18	5° Y 58'31	5N27'39"				
evening set	2091 Mar 14 10:19:20	5° Y 43'27		superior conj	2092 Jan 17 14:50:03	27° S 28'39	-2°00'50
inferior conj	2091 Mar 21 06:37:18	0° Y 54'26	3°19'33	minimum elong	2092 Jan 17 10:58:00	27° S 12'37	2°00'40
minimum elong	2091 Mar 21 09:25:32	0° Y 48'03	3°19'07		2092 Jan 19 03:11:17	0° S	
	2091 Mar 22 06:25:01	30° R X		evening rise	2092 Jan 30 01:38:56	18° S 45'56	
min. Earth dist.	2091 Mar 24 14:39:41	27° X 53'54	0.60225 AU		2092 Feb 05 12:32:15	0° X	
morning rise	2091 Mar 28 06:22:47	25° X 09'01		asc. node	2092 Feb 12 08:12:25	10° X 51'15	

evening max el	2002 Feb 16 11:36:31	15° K 54'28	18°08'36	max. Earth dist.	2002 Dec 24 00:14:56	1° Z 14'12	1.44642 AU
retrograde	2002 Feb 22 23:16:31	19° K 18'13					
evening set	2002 Feb 25 18:43:02	18° K 38'59		superior conj	2002 Dec 27 09:51:53	6° Z 38'06	-1°38'38
max. declination.	2002 Feb 25 13:28:15	18° K 44'46	1S31'17"	minimum elong	2002 Dec 27 01:24:11	6° Z 04'23	1°37'54
inferior conj	2002 Mar 03 01:53:34	13° K 29'18	3°40'54	min. declination.	2002 Dec 26 23:12:54	5° Z 55'40	24S54'12"
minimum elong	2002 Mar 03 02:43:02	13° K 27'09	3°40'51	evening rise	2003 Jan 10 17:13:23	29° Z 57'20	
min. Earth dist.	2002 Mar 05 22:34:51	10° K 32'04	0.62333 AU		2003 Jan 10 17:51:58	0° $\approx$	
morning rise	2002 Mar 09 09:30:25	7° K 30'06		asc. node	2003 Jan 29 05:13:24	28° $\approx$ 25'30	
direct	2002 Mar 16 07:47:49	4° K 56'55		evening max el	2003 Jan 29 23:59:36	29° $\approx$ 15'07	18°24'48
desc. node	2002 Mar 21 13:48:20	6° K 14'09			2003 Jan 30 18:32:50	0° K	
min. declination.	2002 Mar 22 05:34:05	6° K 33'37	9S13'35"	retrograde	2003 Feb 05 12:07:07	2° K 48'02	
morning max el	2002 Mar 30 08:45:40	12° K 52'14	27°49'44	evening set	2003 Feb 08 12:49:36	1° K 57'42	
	2002 Apr 13 00:49:30	0° Y		max. declination.	2003 Feb 08 22:54:51	1° K 43'01	8S10'18"
	2002 Apr 30 12:25:13	0° X			2003 Feb 11 03:39:23	30° R $\approx$	
morning set	2002 May 03 01:17:59	5° X 03'40		inferior conj	2003 Feb 14 10:30:16	26° $\approx$ 28'54	3°41'04
max. Earth dist.	2002 May 08 16:22:37	16° X 51'06	1.32583 AU	minimum elong	2003 Feb 14 09:35:29	26° $\approx$ 31'33	3°41'01
asc. node	2002 May 10 07:28:51	20° X 22'21		min. Earth dist.	2003 Feb 16 16:05:11	23° $\approx$ 54'28	0.64194 AU
				morning rise	2003 Feb 20 05:44:11	20° $\approx$ 21'45	
superior conj	2002 May 10 13:24:36	20° X 54'32	0°02'37	direct	2003 Feb 27 02:31:22	17° $\approx$ 32'43	
minimum elong	2002 May 10 13:16:52	20° X 53'50	0°02'36	min. declination.	2003 Mar 05 21:14:18	19° $\approx$ 47'03	14S23'55"
behind sun begin	2002 May 10 08:10:08	20° X 26'05		desc. node	2003 Mar 08 10:50:27	21° $\approx$ 34'57	
behind sun end	2002 May 10 18:23:36	21° X 21'37		morning max el	2003 Mar 12 17:38:40	25° $\approx$ 25'26	27°29'26
	2002 May 14 17:38:17	0° II			2003 Mar 16 22:34:32	0° K	
evening rise	2002 May 17 12:08:54	5° II 57'07			2003 Apr 06 12:26:44	0° Y	
max. declination.	2002 May 30 03:38:46	29° II 50'46	25N37'45"	morning set	2003 Apr 16 19:42:44	18° Y 57'53	
	2002 May 30 05:57:00	0° Z		max. Earth dist.	2003 Apr 21 17:51:16	28° Y 56'31	1.33372 AU
evening max el	2002 Jun 11 01:53:02	15° Z 16'33	24°06'55		2003 Apr 22 06:02:04	0° X	
desc. node	2002 Jun 17 13:04:01	20° Z 11'34					
retrograde	2002 Jun 24 21:16:04	22° Z 13'34		superior conj	2003 Apr 24 19:09:35	5° X 22'30	-0°24'55
evening set	2002 Jun 29 09:44:49	21° Z 28'18		minimum elong	2003 Apr 24 20:24:10	5° X 29'07	0°24'39
min. Earth dist.	2002 Jul 05 12:38:09	18° Z 28'51	0.56456 AU	asc. node	2003 Apr 27 04:31:16	10° X 29'14	
inferior conj	2002 Jul 08 01:02:34	16° Z 55'16	-4°42'43	evening rise	2003 May 01 23:34:42	20° X 43'23	
minimum elong	2002 Jul 07 20:18:40	17° Z 02'38	4°42'08		2003 May 06 12:35:49	0° II	
min. declination.	2002 Jul 09 01:56:08	16° Z 16'48	17N40' 0"	max. declination.	2003 May 21 08:09:42	23° II 16'56	25N32' 0"
morning rise	2002 Jul 16 09:32:00	12° Z 55'12		evening max el	2003 May 23 16:40:09	25° II 45'05	22°30'43
direct	2002 Jul 19 02:14:11	12° Z 36'07			2003 May 29 03:20:39	0° Z	
morning max el	2002 Jul 28 20:36:47	17° Z 08'05	19°44'59	desc. node	2003 Jun 04 10:05:16	2° Z 10'18	
max. declination.	2002 Aug 03 04:24:46	23° Z 21'56	20N42'36"	retrograde	2003 Jun 05 18:39:15	2° Z 14'31	
asc. node	2002 Aug 06 06:41:38	28° Z 03'47		evening set	2003 Jun 08 20:39:41	1° Z 53'33	
	2002 Aug 07 10:32:41	0° Q			2003 Jun 13 22:41:54	30° R II	
morning set	2002 Aug 15 13:57:29	15° Q 25'57		min. Earth dist.	2003 Jun 17 00:01:31	28° II 21'30	0.55247 AU
	2002 Aug 22 18:57:25	0° P		inferior conj	2003 Jun 18 03:31:33	27° II 42'28	-3°39'38
				minimum elong	2003 Jun 17 19:35:10	27° II 53'45	3°37'33
superior conj	2002 Aug 23 11:19:34	1° P 21'23	1°45'27	morning rise	2003 Jun 26 20:39:59	23° II 50'50	
minimum elong	2002 Aug 23 11:46:18	1° P 23'36	1°45'28	min. declination.	2003 Jun 27 10:55:38	23° II 43'46	18N40'16"
max. Earth dist.	2002 Aug 29 07:41:16	12° P 38'53	1.37321 AU	direct	2003 Jun 29 19:12:42	23° II 31'07	
evening rise	2002 Sep 02 02:37:46	19° P 35'40		morning max el	2003 Jul 11 04:50:53	28° II 48'57	21°04'09
	2002 Sep 08 02:49:45	0° A			2003 Jul 12 10:02:35	0° Z	
desc. node	2002 Sep 13 12:22:56	8° A 44'45		max. declination.	2003 Jul 23 10:33:49	15° Z 38'27	22N21'51"
	2002 Sep 28 08:38:31	0° M		asc. node	2003 Jul 24 03:43:54	16° Z 55'42	
evening max el	2002 Oct 07 21:42:05	10° M 56'52	25°27'52	morning set	2003 Jul 30 21:31:29	0° Q 08'58	
min. declination.	2002 Oct 18 18:07:09	17° M 48'28	20S13'20"		2003 Jul 30 19:47:03	0° Q	
retrograde	2002 Oct 20 00:12:40	17° M 54'21					
evening set	2002 Oct 25 22:34:17	15° M 25'24		superior conj	2003 Aug 07 07:39:57	15° Q 36'21	1°43'37
min. Earth dist.	2002 Oct 30 08:18:24	10° M 33'16	0.67087 AU	minimum elong	2003 Aug 07 06:37:37	15° Q 30'59	1°43'35
inferior conj	2002 Oct 31 11:57:04	9° M 04'14	-0°36'07	max. Earth dist.	2003 Aug 11 17:03:55	24° Q 29'32	1.35524 AU
minimum elong	2002 Oct 31 12:52:46	9° M 01'15	0°35'43		2003 Aug 14 12:55:07	0° P	
asc. node	2002 Nov 02 05:54:07	6° M 52'21		evening rise	2003 Aug 15 21:32:23	2° P 34'54	
morning rise	2002 Nov 06 03:25:18	3° M 07'48		desc. node	2003 Aug 31 09:24:22	28° P 57'19	
direct	2002 Nov 09 12:25:35	2° M 02'05			2003 Sep 01 01:54:04	0° A	
max. declination.	2002 Nov 11 04:02:36	2° M 16'56	10S15'51"	evening max el	2003 Sep 20 09:42:16	24° A 34'56	26°29'46
morning max el	2002 Nov 16 15:54:25	6° M 03'19	19°22'37		2003 Sep 27 07:43:20	0° M	
greatest brilliancy	2002 Nov 30 08:47:14	24° M 32'17	-0.7m	retrograde	2003 Oct 03 06:08:04	1° M 47'50	
	2002 Dec 03 22:44:10	0° X		min. declination.	2003 Oct 03 02:32:43	1° M 47'46	15S30'12"
desc. node	2002 Dec 10 11:38:13	10° X 05'40			2003 Oct 08 14:33:35	30° R A	
morning set	2002 Dec 10 15:26:53	10° X 20'27		evening set	2003 Oct 09 18:04:36	29° A 07'44	
	2002 Dec 23 05:27:19	0° Z		min. Earth dist.	2003 Oct 13 20:28:20	24° A 51'54	0.66198 AU

inferior conj	2093 Oct 15 11:42:12	22°♄53'27	-1°34'15	desc. node	2094 Aug 18 06:25:50	18°♎46'18	
minimum elong	2093 Oct 15 14:13:03	22°♄45'51	1°33'13		2094 Aug 26 13:57:29	0°♄	
asc. node	2093 Oct 20 02:55:12	18°♄01'15		evening max el	2094 Sep 02 21:48:11	8°♄03'44	27°11'23
morning rise	2093 Oct 21 10:54:43	17°♄07'38		retrograde	2094 Sep 16 06:40:01	15°♄22'17	
direct	2093 Oct 24 10:20:03	16°♄17'42		min. declination.	2094 Sep 17 02:26:52	15°♄20'06	9S43' 6"
max. declination.	2093 Oct 26 15:30:56	16°♄44'24	5S 3'40"	evening set	2094 Sep 23 05:27:04	12°♄38'36	
morning max el	2093 Oct 31 02:47:03	19°♄58'16	18°35'27	min. Earth dist.	2094 Sep 27 01:16:18	8°♄58'42	0.64958 AU
	2093 Nov 07 20:52:03	0°♄		inferior conj	2094 Sep 29 04:51:21	6°♄35'29	-2°32'47
morning set	2093 Nov 20 13:34:07	19°♄55'20		minimum elong	2094 Sep 29 08:55:28	6°♄24'11	2°31'16
	2093 Nov 26 21:46:53	0°♄		morning rise	2094 Oct 05 13:18:34	1°♄03'15	
desc. node	2093 Nov 27 08:39:41	0°♄43'02		asc. node	2094 Oct 06 23:57:25	0°♄33'21	
				direct	2094 Oct 08 05:35:05	0°♄25'12	
superior conj	2093 Dec 06 12:22:40	15°♄07'47	-0°58'48	max. declination.	2094 Oct 11 02:07:11	1°♄07'56	0N31'44"
minimum elong	2093 Dec 06 05:05:03	14°♄39'09	0°57'54	morning max el	2094 Oct 14 17:44:33	3°♄54'19	18°04'53
max. Earth dist.	2093 Dec 06 18:16:06	15°♄30'54	1.45144 AU		2094 Nov 01 03:14:07	0°♄	
	2093 Dec 15 23:15:57	0°♄		morning set	2094 Nov 01 16:56:33	0°♄57'03	
min. declination.	2093 Dec 18 18:35:37	4°♄26'16	25S17'28"	desc. node	2094 Nov 14 05:41:33	21°♄26'57	
evening rise	2093 Dec 22 09:27:16	10°♄11'06					
greatest brilliancy	2093 Dec 31 00:46:37	23°♄54'56	-0.8m	superior conj	2094 Nov 15 18:50:08	23°♄55'20	-0°10'25
	2094 Jan 03 22:43:38	0°♄		minimum elong	2094 Nov 15 17:32:09	23°♄50'09	0°10'14
evening max el	2094 Jan 13 10:10:23	12°♄41'33	18°58'54	behind sun begin	2094 Nov 15 09:08:17	23°♄16'40	
asc. node	2094 Jan 16 02:13:29	15°♄01'29		behind sun end	2094 Nov 16 01:56:00	24°♄23'37	
retrograde	2094 Jan 20 06:46:04	16°♄34'34		max. Earth dist.	2094 Nov 19 12:59:30	29°♄52'40	1.44906 AU
evening set	2094 Jan 23 13:35:28	15°♄32'08			2094 Nov 19 14:51:01	0°♄	
max. declination.	2094 Jan 25 01:40:49	14°♄23'34	14S 1' 5"	evening rise	2094 Dec 02 04:27:43	19°♄35'40	
inferior conj	2094 Jan 29 04:46:28	9°♄46'31	3°25'41		2094 Dec 08 22:32:23	0°♄	
minimum elong	2094 Jan 29 02:43:07	9°♄52'59	3°25'21	min. declination.	2094 Dec 10 16:06:18	2°♄39'01	25S37' 6"
min. Earth dist.	2094 Jan 30 18:59:34	7°♄46'45	0.65690 AU	greatest brilliancy	2094 Dec 15 15:52:08	10°♄11'28	-0.7m
morning rise	2094 Feb 03 15:30:26	3°♄35'09		evening max el	2094 Dec 27 16:17:33	26°♄10'59	19°49'18
direct	2094 Feb 10 04:03:35	0°♄44'18			2095 Jan 01 16:52:43	0°♄	
min. declination.	2094 Feb 18 18:19:48	4°♄27'06	18S20'43"	asc. node	2095 Jan 02 23:14:31	0°♄24'45	
morning max el	2094 Feb 23 03:33:40	8°♄23'25	26°38'06	retrograde	2095 Jan 04 04:09:27	0°♄32'46	
desc. node	2094 Feb 23 07:52:14	8°♄34'18			2095 Jan 06 13:05:41	30°♄	
	2094 Mar 12 00:54:27	0°♄		evening set	2095 Jan 07 18:35:01	29°♄16'56	
	2094 Mar 29 22:14:19	0°♄		max. declination.	2095 Jan 12 07:58:31	24°♄28'34	18S28'21"
morning set	2094 Mar 31 02:58:13	2°♄15'11		inferior conj	2095 Jan 13 05:40:42	23°♄17'05	2°58'36
max. Earth dist.	2094 Apr 04 08:30:45	10°♄28'07	1.34611 AU	minimum elong	2095 Jan 13 03:06:06	23°♄25'38	2°57'59
				min. Earth dist.	2095 Jan 14 05:42:19	21°♄57'19	0.66786 AU
superior conj	2094 Apr 08 18:56:30	19°♄28'04	-0°52'56	morning rise	2095 Jan 18 11:23:07	17°♄03'55	
minimum elong	2094 Apr 08 21:37:34	19°♄41'57	0°52'26	direct	2095 Jan 24 11:31:41	14°♄23'12	
	2094 Apr 13 19:54:15	0°♄		morning max el	2095 Feb 05 12:28:52	21°♄33'52	25°24'19
asc. node	2094 Apr 14 01:33:13	0°♄29'37		min. declination.	2095 Feb 06 01:32:34	22°♄07'15	20S57'28"
evening rise	2094 Apr 16 08:48:00	5°♄17'54		desc. node	2095 Feb 10 04:54:18	26°♄44'02	
	2094 Apr 30 02:39:04	0°♄			2095 Feb 12 20:52:21	0°♄	
evening max el	2094 May 05 15:23:41	6°♄33'05	21°01'50		2095 Mar 05 06:39:13	0°♄	
max. declination.	2094 May 10 01:41:04	10°♄03'20	24N18'39"	morning set	2095 Mar 13 18:22:46	14°♄41'21	
retrograde	2094 May 17 08:18:07	12°♄15'29		max. Earth dist.	2095 Mar 17 11:22:01	21°♄32'08	1.36308 AU
evening set	2094 May 19 13:01:51	12°♄04'09			2095 Mar 21 20:56:54	0°♄	
desc. node	2094 May 22 07:06:05	11°♄20'27					
inferior conj	2094 May 28 21:24:44	8°♄05'55	-1°54'15	superior conj	2095 Mar 23 09:58:51	3°♄03'00	-1°19'50
minimum elong	2094 May 28 16:04:40	8°♄13'28	1°52'23	minimum elong	2095 Mar 23 13:52:04	3°♄22'21	1°19'15
min. Earth dist.	2094 May 29 12:25:15	7°♄44'41	0.54908 AU	evening rise	2095 Mar 31 13:40:05	19°♄33'33	
morning rise	2094 Jun 06 19:04:21	4°♄03'32		asc. node	2095 Mar 31 22:35:10	20°♄18'37	
direct	2094 Jun 10 04:43:09	3°♄38'32			2095 Apr 05 21:11:00	0°♄	
min. declination.	2094 Jun 11 21:15:24	3°♄44'56	16N46'57"	evening max el	2095 Apr 18 01:46:14	17°♄57'47	19°48'41
morning max el	2094 Jun 23 02:40:58	9°♄48'42	22°39'33	max. declination.	2095 Apr 25 08:28:46	22°♄26'05	21N 3'53"
	2094 Jul 07 16:46:31	0°♄		retrograde	2095 Apr 28 00:50:17	22°♄46'29	
asc. node	2094 Jul 11 00:46:53	6°♄16'33		evening set	2095 Apr 30 01:19:37	22°♄35'29	
max. declination.	2094 Jul 13 13:11:48	11°♄19'24	23N24'26"	inferior conj	2095 May 08 20:53:10	18°♄34'13	0°05'15
morning set	2094 Jul 15 08:06:06	15°♄01'45		minimum elong	2095 May 08 21:07:09	18°♄33'52	0°05'10
				transit middle	2095 May 08 21:07:09	18°♄33'52	0°05'10
superior conj	2094 Jul 22 11:23:12	0°♄13'49	1°35'20	transit begin	2095 May 08 17:22:26	18°♄39'38	
minimum elong	2094 Jul 22 09:24:07	0°♄03'16	1°35'09	transit end	2095 May 09 00:51:52	18°♄28'05	
	2094 Jul 22 08:47:10	0°♄		desc. node	2095 May 09 04:07:50	18°♄23'02	
max. Earth dist.	2094 Jul 25 11:12:47	6°♄31'24	1.34091 AU	min. Earth dist.	2095 May 11 02:15:03	17°♄12'09	0.55542 AU
evening rise	2094 Jul 30 07:10:30	16°♄17'43		morning rise	2095 May 17 14:33:43	14°♄03'44	
	2094 Aug 06 15:30:15	0°♄		direct	2095 May 21 20:30:04	13°♄23'46	

min. declination.	2005 May 25 13:26:12	13°♄54'56	12N39' 2"	max. declination.	2006 Apr 08 03:41:26	4°♄06'19	15N53' 1"
morning max el	2005 Jun 04 17:53:25	20°♄19'17	24°21'45	evening set	2006 Apr 10 13:27:52	3°♄50'43	
	2005 Jun 12 23:51:56	0°♄			2006 Apr 17 22:54:15	30°♄♂	
asc. node	2005 Jun 27 21:50:45	25°♄57'44		inferior conj	2006 Apr 18 13:42:55	29°♄33'43	1°46'35
morning set	2005 Jun 29 19:54:12	29°♄57'44		minimum elong	2006 Apr 18 17:20:48	29°♄27'14	1°45'27
	2005 Jun 29 20:19:57	0°♄		min. Earth dist.	2006 Apr 21 18:12:03	27°♄18'19	0.57010 AU
max. declination.	2005 Jul 03 20:12:22	8°♄34'55	24N 8'13"	desc. node	2006 Apr 25 01:09:05	25°♄17'25	
				morning rise	2006 Apr 26 18:08:06	24°♄29'02	
superior conj	2005 Jul 06 19:47:56	15°♄04'12	1°21'46	direct	2006 May 02 01:20:44	23°♄21'49	
minimum elong	2005 Jul 06 17:24:44	14°♄51'15	1°21'26	min. declination.	2006 May 06 17:44:14	24°♄13'48	6N58'36"
max. Earth dist.	2005 Jul 08 14:19:13	18°♄53'56	1.33059 AU		2006 May 15 12:27:11	0°♄	
	2005 Jul 13 21:15:43	0°♄		morning max el	2006 May 16 08:40:58	0°♄47'43	25°56'14
evening rise	2005 Jul 14 03:24:57	0°♄31'17			2006 Jun 05 19:49:38	0°♄	
	2005 Jul 30 15:23:17	0°♄		morning set	2006 Jun 13 07:19:12	14°♄51'54	
desc. node	2005 Aug 05 03:27:25	8°♄01'08		asc. node	2006 Jun 13 18:55:17	15°♄53'16	
evening max el	2005 Aug 16 08:32:20	21°♄11'11	27°26'02				
retrograde	2005 Aug 30 00:57:42	28°♄28'37		superior conj	2006 Jun 20 06:48:59	0°♄00'40	1°03'49
min. declination.	2005 Aug 31 22:13:57	28°♄18'06	3S13'31"	minimum elong	2006 Jun 20 04:31:47	29°♄48'07	1°03'24
evening set	2005 Sep 06 05:35:11	25°♄51'59			2006 Jun 20 06:41:42	0°♄	
min. Earth dist.	2005 Sep 09 20:44:18	22°♄44'29	0.63368 AU	max. Earth dist.	2006 Jun 20 23:48:50	1°♄33'56	1.32425 AU
inferior conj	2005 Sep 12 12:46:47	20°♄04'18	-3°28'45	max. declination.	2006 Jun 23 05:43:43	6°♄28'05	24N41'47"
minimum elong	2005 Sep 12 17:58:19	19°♄51'16	3°27'11	evening rise	2006 Jun 27 07:03:00	15°♄05'44	
morning rise	2005 Sep 19 07:37:10	14°♄48'53			2006 Jul 04 21:15:38	0°♄	
direct	2005 Sep 21 19:22:39	14°♄19'10		desc. node	2006 Jul 22 00:28:49	26°♄27'38	
asc. node	2005 Sep 23 21:01:15	14°♄40'06			2006 Jul 25 00:41:35	0°♄	
max. declination.	2005 Sep 25 12:40:01	15°♄26'21	6N 8'29"	evening max el	2006 Jul 28 15:17:44	3°♄43'34	27°08'43
morning max el	2005 Sep 28 09:57:40	17°♄44'54	17°51'52	retrograde	2006 Aug 11 12:07:08	10°♄57'37	
	2005 Oct 07 05:29:25	0°♄		min. declination.	2006 Aug 14 18:49:51	10°♄28'22	3N33'53"
morning set	2005 Oct 14 21:39:14	13°♄12'09		evening set	2006 Aug 18 14:27:17	8°♄40'48	
	2005 Oct 24 16:13:55	0°♄		min. Earth dist.	2006 Aug 22 05:33:24	5°♄56'51	0.61480 AU
				inferior conj	2006 Aug 25 08:17:38	3°♄11'45	-4°17'22
superior conj	2005 Oct 26 23:04:35	3°♄48'43	0°34'35	minimum elong	2006 Aug 25 13:28:27	3°♄00'16	4°16'17
minimum elong	2005 Oct 27 02:29:07	4°♄02'50	0°34'07		2006 Aug 29 06:44:07	30°♄♂	
desc. node	2005 Nov 01 02:44:30	12°♄14'07		morning rise	2006 Sep 01 14:07:22	28°♄16'42	
max. Earth dist.	2005 Nov 02 05:58:16	14°♄03'36	1.43965 AU	direct	2006 Sep 04 00:11:04	27°♄52'21	
evening rise	2005 Nov 11 12:58:48	28°♄39'39		max. declination.	2006 Sep 09 01:09:36	29°♄42'16	11N23'45"
	2005 Nov 12 09:50:57	0°♄			2006 Sep 09 10:39:03	0°♄	
min. declination.	2005 Dec 02 15:13:09	29°♄47'55	25S50'46"	asc. node	2006 Sep 09 18:06:24	0°♄14'49	
	2005 Dec 02 18:46:24	0°♄		morning max el	2006 Sep 11 00:26:07	1°♄22'48	17°57'23
evening max el	2005 Dec 10 17:08:12	9°♄41'55	20°53'30	morning set	2006 Sep 26 21:17:47	26°♄21'32	
retrograde	2005 Dec 19 01:57:03	14°♄39'04			2006 Sep 28 21:02:00	0°♄	
asc. node	2005 Dec 20 20:16:46	14°♄21'44					
evening set	2005 Dec 23 01:48:47	13°♄08'11		superior conj	2006 Oct 07 06:30:01	15°♄02'11	1°09'02
inferior conj	2005 Dec 28 10:45:12	6°♄57'02	2°22'31	minimum elong	2006 Oct 07 11:02:03	15°♄21'56	1°08'30
minimum elong	2005 Dec 28 08:12:23	7°♄05'46	2°21'43	max. Earth dist.	2006 Oct 14 18:48:01	27°♄46'56	1.42439 AU
min. Earth dist.	2005 Dec 28 22:04:15	6°♄18'12	0.67485 AU		2006 Oct 16 03:12:02	0°♄	
morning rise	2006 Jan 02 14:24:48	0°♄43'59		desc. node	2006 Oct 17 23:46:06	3°♄01'01	
	2006 Jan 03 11:51:38	30°♄♂		evening rise	2006 Oct 21 03:47:25	8°♄04'39	
max. declination.	2006 Jan 04 01:48:30	29°♄35'44	20S13'25"		2006 Nov 04 15:05:38	0°♄	
direct	2006 Jan 07 23:58:27	28°♄21'34		evening max el	2006 Nov 22 12:23:57	23°♄13'37	22°07'52
	2006 Jan 13 00:37:03	0°♄		min. declination.	2006 Nov 22 21:51:28	23°♄37'20	25S46'14"
morning max el	2006 Jan 18 20:49:18	4°♄50'22	23°58'24	retrograde	2006 Dec 01 22:20:08	28°♄49'38	
min. declination.	2006 Jan 27 06:58:18	14°♄42'01	22S30'25"	evening set	2006 Dec 06 09:29:40	27°♄02'20	
desc. node	2006 Jan 28 01:56:33	15°♄43'51		asc. node	2006 Dec 06 17:20:29	26°♄45'49	
	2006 Feb 07 05:46:16	0°♄		inferior conj	2006 Dec 11 17:54:41	20°♄43'07	1°39'12
morning set	2006 Feb 23 12:05:56	26°♄00'44		minimum elong	2006 Dec 11 15:51:32	20°♄50'13	1°38'26
	2006 Feb 25 19:09:14	0°♄		min. Earth dist.	2006 Dec 11 17:46:12	20°♄43'36	0.67812 AU
max. Earth dist.	2006 Feb 27 07:04:33	2°♄38'54	1.38352 AU	morning rise	2006 Dec 16 22:05:18	14°♄31'56	
				max. declination.	2006 Dec 20 22:09:58	12°♄34'48	19S22'37"
superior conj	2006 Mar 05 12:46:32	15°♄58'31	-1°43'14	direct	2006 Dec 21 16:33:08	12°♄31'53	
minimum elong	2006 Mar 05 17:04:18	16°♄18'56	1°42'48	morning max el	2006 Dec 31 07:24:21	18°♄12'42	22°30'08
	2006 Mar 12 18:09:15	0°♄			2007 Jan 10 02:15:54	0°♄	
evening rise	2006 Mar 14 11:41:44	3°♄24'02		desc. node	2007 Jan 13 22:58:32	5°♄18'54	
asc. node	2006 Mar 17 19:38:22	9°♄51'42		min. declination.	2007 Jan 17 19:20:58	10°♄49'37	23S27'58"
	2006 Mar 30 22:24:56	0°♄			2007 Jan 30 09:17:01	0°♄	
evening max el	2006 Mar 30 23:10:46	0°♄01'51	18°54'54	morning set	2007 Feb 03 02:09:42	5°♄58'14	
retrograde	2006 Apr 08 08:33:36	4°♄06'27		max. Earth dist.	2007 Feb 08 03:34:15	14°♄23'38	1.40508 AU

superior conj	2007 Feb 15 22:39:00	28° 28 02'06	-1°59'48	morning max el	2007 Dec 13 23:43:08	1° 27 44'38	21°07'41
minimum elong	2007 Feb 16 01:41:53	28° 28 15'51	1°59'38	desc. node	2007 Dec 31 20:01:17	25° 25 19'37	
	2007 Feb 17 00:42:13	0° 0			2008 Jan 03 23:20:51	0° 0	
evening rise	2007 Feb 25 23:39:17	16° 16 40'43		min. declination.	2008 Jan 09 13:27:41	8° 8 29'56	24S 8' 5"
asc. node	2007 Mar 04 16:40:48	29° 29 02'26		morning set	2008 Jan 13 12:11:02	14° 14 38'05	
	2007 Mar 05 05:52:47	0° 0		max. Earth dist.	2008 Jan 21 06:06:01	27° 27 02'07	1.42487 AU
evening max el	2007 Mar 14 04:46:55	12° 12 38'11	18°21'20		2008 Jan 23 01:23:47	0° 0	
retrograde	2007 Mar 21 11:35:10	16° 16 14'55					
max. declination.	2007 Mar 22 16:40:48	16° 16 09'05	9N27'51"	superior conj	2008 Jan 28 10:21:22	9° 9 01'33	-2°04'59
evening set	2007 Mar 23 22:25:09	15° 15 51'25		minimum elong	2008 Jan 28 09:38:28	8° 8 58'30	2°04'58
inferior conj	2007 Mar 31 03:56:22	11° 11 14'04	2°55'00	evening rise	2008 Feb 08 21:44:23	29° 29 15'04	
minimum elong	2007 Mar 31 07:37:28	11° 11 06'21	2°54'12		2008 Feb 09 07:43:16	0° 0	
min. Earth dist.	2007 Apr 03 14:26:42	8° 8 23'05	0.58988 AU	asc. node	2008 Feb 19 13:42:21	17° 17 43'28	
morning rise	2007 Apr 07 14:10:42	5° 5 40'26		evening max el	2008 Feb 25 15:23:09	25° 25 38'10	18°07'41
desc. node	2007 Apr 11 22:10:25	4° 4 06'55		retrograde	2008 Mar 04 07:10:39	29° 29 02'45	
direct	2007 Apr 13 19:27:17	3° 3 57'26		evening set	2008 Mar 06 23:29:43	28° 28 29'44	
min. declination.	2007 Apr 18 23:58:51	5° 5 05'09	0N34'40"	max. declination.	2008 Mar 06 10:28:16	28° 28 41'36	2N30'58"
morning max el	2007 Apr 28 04:17:34	11° 11 41'42	27°07'39	inferior conj	2008 Mar 13 13:44:15	23° 23 31'44	3°31'38
	2007 May 12 11:17:32	0° 0		minimum elong	2008 Mar 13 15:42:58	23° 23 26'57	3°31'24
morning set	2007 May 28 16:41:22	29° 29 38'51		min. Earth dist.	2008 Mar 16 17:41:26	20° 20 29'36	0.61135 AU
	2007 May 28 20:43:53	0° 0		morning rise	2008 Mar 20 06:09:42	17° 17 39'08	
asc. node	2007 May 31 15:59:26	5° 5 58'03		direct	2008 Mar 27 01:03:12	15° 15 21'20	
				desc. node	2008 Mar 29 19:11:50	15° 15 42'33	
superior conj	2007 Jun 04 18:37:43	14° 14 57'19	0°42'14	min. declination.	2008 Apr 01 15:15:55	16° 16 45'12	5S47'23"
minimum elong	2007 Jun 04 16:53:19	14° 14 47'44	0°41'54	morning max el	2008 Apr 10 06:15:39	23° 23 15'39	27°45'09
max. Earth dist.	2007 Jun 04 12:06:11	14° 14 21'23	1.32173 AU		2008 Apr 16 06:47:16	0° 0	
evening rise	2007 Jun 11 15:29:39	29° 29 52'50			2008 May 05 15:32:57	0° 0	
	2007 Jun 11 16:51:12	0° 0		morning set	2008 May 12 22:05:26	14° 14 11'39	
max. declination.	2007 Jun 13 20:12:16	4° 4 26'52	25N 8'52"	asc. node	2008 May 18 13:00:46	26° 26 07'03	
	2007 Jun 28 03:48:18	0° 0		max. Earth dist.	2008 May 18 23:09:46	27° 27 02'15	1.32311 AU
desc. node	2007 Jul 08 21:29:38	13° 13 04'54					
evening max el	2007 Jul 10 15:28:21	15° 15 30'29	26°18'22	superior conj	2008 May 20 05:29:28	29° 29 47'49	0°17'45
retrograde	2007 Jul 24 15:06:36	22° 22 04'52		minimum elong	2008 May 20 04:41:00	29° 29 43'24	0°17'35
min. declination.	2007 Jul 30 02:17:22	21° 21 02'734	10N 6'27"		2008 May 20 07:42:57	0° 0	
evening set	2007 Jul 31 03:22:49	20° 20 05'708		evening rise	2008 May 27 02:33:51	14° 14 44'48	
min. Earth dist.	2007 Aug 04 04:20:45	18° 18 02'120	0.59420 AU		2008 Jun 03 17:40:38	0° 0	
inferior conj	2007 Aug 07 11:50:13	15° 15 04'8'33	-4°50'49	max. declination.	2008 Jun 04 19:18:08	1° 1 57'15	25N30' 3"
minimum elong	2007 Aug 07 14:55:19	15° 15 04'2'35	4°50'32	evening max el	2008 Jun 22 08:29:38	26° 26 03'117	25°00'07
morning rise	2007 Aug 15 04:34:56	11° 11 01'6'23		desc. node	2008 Jun 25 18:30:45	29° 29 02'733	
direct	2007 Aug 17 15:31:05	10° 10 05'5'18			2008 Jun 26 12:11:41	0° 0	
max. declination.	2007 Aug 24 22:16:03	14° 14 01'2'44	15N53'46"	retrograde	2008 Jul 06 08:03:05	3° 3 03'715	
morning max el	2007 Aug 25 10:03:59	14° 14 04'0'06	18°22'39	evening set	2008 Jul 11 16:39:42	2° 2 03'130	
asc. node	2007 Aug 27 15:10:36	17° 17 03'26		min. declination.	2008 Jul 16 02:10:33	0° 0 13'22	15N33'36"
	2007 Sep 05 00:05:46	0° 0			2008 Jul 16 10:39:27	30° 30 R0	
morning set	2007 Sep 10 10:27:26	10° 10 07'10'00		min. Earth dist.	2008 Jul 16 20:02:09	29° 29 04'4'58	0.57430 AU
				inferior conj	2008 Jul 19 20:06:58	27° 27 04'4'33	-4°57'09
superior conj	2007 Sep 19 13:50:25	27° 27 07'26'42	1°31'26	minimum elong	2008 Jul 19 18:33:44	27° 27 04'7'10	4°57'05
minimum elong	2007 Sep 19 17:12:49	27° 27 07'42'12	1°31'09	morning rise	2008 Jul 27 23:06:30	23° 23 03'4'27	
	2007 Sep 20 23:22:32	0° 0		direct	2008 Jul 30 13:01:17	23° 23 05'15'10	
max. Earth dist.	2007 Sep 27 02:08:43	10° 10 04'9'02	1.40521 AU	morning max el	2008 Aug 08 11:36:51	27° 27 02'6'21	19°08'48
evening rise	2007 Oct 01 14:23:00	18° 18 02'5'10			2008 Aug 10 21:39:55	0° 0	
desc. node	2007 Oct 04 20:49:01	23° 23 04'4'34		max. declination.	2008 Aug 10 22:24:25	0° 0 02'11	19N17'44"
	2007 Oct 08 19:52:04	0° 0		asc. node	2008 Aug 14 12:13:14	4° 4 04'9'58	
	2007 Oct 30 02:08:48	0° 0		morning set	2008 Aug 25 08:59:50	24° 24 02'5'33	
evening max el	2007 Nov 05 03:09:20	6° 6 47'46	23°27'25		2008 Aug 28 04:15:13	0° 0	
min. declination.	2007 Nov 11 07:56:13	11° 11 45'33	24S47'51"				
retrograde	2007 Nov 15 15:57:04	13° 13 47'01'36		superior conj	2008 Sep 02 15:20:08	10° 10 04'4'54	1°43'00
evening set	2007 Nov 20 16:00:40	10° 10 47'57'18		minimum elong	2008 Sep 02 16:49:44	10° 10 05'2'07	1°42'56
asc. node	2007 Nov 23 14:22:29	7° 7 47'50'09		max. Earth dist.	2008 Sep 09 05:41:44	23° 23 07'0'05	1.38462 AU
min. Earth dist.	2007 Nov 25 14:28:49	5° 5 47'10'57	0.67794 AU	evening rise	2008 Sep 13 01:00:48	29° 29 05'2'31	
inferior conj	2007 Nov 26 01:19:12	4° 4 47'33'49	0°50'05		2008 Sep 13 02:44:08	0° 0	
minimum elong	2007 Nov 26 00:10:09	4° 4 47'37'45	0°49'36	desc. node	2008 Sep 21 17:51:24	14° 14 02'0'06	
	2007 Nov 29 16:01:44	30° 30 R0			2008 Oct 02 04:00:43	0° 0	
morning rise	2007 Dec 01 08:17:24	28° 28 02'6'22		evening max el	2008 Oct 18 15:30:14	20° 20 02'5'30	24°46'01
direct	2007 Dec 05 12:24:24	26° 26 04'9'11		min. declination.	2008 Oct 28 03:49:16	26° 26 05'3'41	22S21'11"
max. declination.	2007 Dec 06 01:35:05	26° 26 05'0'45	16S49'49"	retrograde	2008 Oct 30 05:41:28	27° 27 05'10'31	
	2007 Dec 12 03:03:11	0° 0		evening set	2008 Nov 04 19:42:21	24° 24 04'9'59	

min. Earth dist.	2008 Nov 09 09:50:45	19° $\mathbb{M}$ 37'23	0.67437 AU	superior conj	2009 Aug 17 05:57:22	24° $\mathbb{Q}$ 42'24	1°45'33
inferior conj	2008 Nov 10 07:13:10	18° $\mathbb{M}$ 26'31	-0°03'36	minimum elong	2009 Aug 17 05:42:40	24° $\mathbb{Q}$ 41'10	1°45'33
minimum elong	2008 Nov 10 07:18:20	18° $\mathbb{M}$ 26'14	0°03'35		2009 Aug 19 21:35:22	0° $\mathbb{M}$	
transit middle	2008 Nov 10 07:18:20	18° $\mathbb{M}$ 26'14	0°03'35	max. Earth dist.	2009 Aug 22 11:16:55	5° $\mathbb{M}$ 00'08	1.36520 AU
transit begin	2008 Nov 10 04:37:07	18° $\mathbb{M}$ 35'09		evening rise	2009 Aug 26 09:27:13	12° $\mathbb{M}$ 21'45	
transit end	2008 Nov 10 09:59:33	18° $\mathbb{M}$ 17'19			2009 Sep 05 15:28:09	0° $\mathbb{Q}$	
asc. node	2008 Nov 10 11:23:54	18° $\mathbb{M}$ 12'39		desc. node	2009 Sep 08 14:52:16	4° $\mathbb{Q}$ 42'42	
morning rise	2008 Nov 15 19:01:13	12° $\mathbb{M}$ 25'08			2009 Sep 27 05:58:28	0° $\mathbb{M}$	
direct	2008 Nov 19 10:22:45	11° $\mathbb{M}$ 08'52		evening max el	2009 Oct 01 03:33:19	4° $\mathbb{M}$ 05'43	25°56'18
max. declination.	2008 Nov 20 17:46:47	11° $\mathbb{M}$ 18'06	12S58'20"	min. declination.	2009 Oct 12 20:28:07	11° $\mathbb{M}$ 09'05	18S24' 0"
morning max el	2008 Nov 26 23:37:35	15° $\mathbb{M}$ 27'15	19°56'48	retrograde	2009 Oct 13 14:37:00	11° $\mathbb{M}$ 11'09	
	2008 Dec 08 06:20:00	0° $\mathbb{M}$		evening set	2009 Oct 19 18:46:28	8° $\mathbb{M}$ 37'05	
desc. node	2008 Dec 18 17:04:40	15° $\mathbb{M}$ 38'52		min. Earth dist.	2009 Oct 24 01:24:06	3° $\mathbb{M}$ 59'59	0.66747 AU
morning set	2008 Dec 23 08:14:55	22° $\mathbb{M}$ 45'53		inferior conj	2009 Oct 25 09:47:09	2° $\mathbb{M}$ 18'11	-1°00'35
	2008 Dec 28 00:02:51	0° $\mathbb{Q}$		minimum elong	2009 Oct 25 11:22:16	2° $\mathbb{M}$ 13'12	0°59'55
min. declination.	2009 Jan 01 07:54:44	6° $\mathbb{Q}$ 48'07	24S38'59"		2009 Oct 27 06:59:00	30° $\mathbb{R}$ $\mathbb{Q}$	
max. Earth dist.	2009 Jan 03 15:57:25	10° $\mathbb{Q}$ 30'37	1.44028 AU	asc. node	2009 Oct 28 08:26:14	28° $\mathbb{Q}$ 49'13	
				morning rise	2009 Oct 31 04:18:35	26° $\mathbb{Q}$ 25'37	
superior conj	2009 Jan 08 20:02:38	18° $\mathbb{Q}$ 50'10	-1°53'48	direct	2009 Nov 03 08:57:07	25° $\mathbb{Q}$ 27'01	
minimum elong	2009 Jan 08 13:50:36	18° $\mathbb{Q}$ 24'57	1°53'26	max. declination.	2009 Nov 05 06:16:59	25° $\mathbb{Q}$ 46'33	8S 9'28"
	2009 Jan 15 14:17:27	0° $\mathbb{M}$		morning max el	2009 Nov 10 06:54:25	29° $\mathbb{Q}$ 18'29	19°00'31
evening rise	2009 Jan 22 01:39:18	10° $\mathbb{M}$ 59'00			2009 Nov 10 22:55:01	0° $\mathbb{M}$	
	2009 Feb 02 10:07:44	0° $\mathbb{M}$			2009 Dec 01 15:20:10	0° $\mathbb{M}$	
asc. node	2009 Feb 06 10:42:41	5° $\mathbb{M}$ 45'54		morning set	2009 Dec 02 15:32:02	1° $\mathbb{M}$ 34'40	
evening max el	2009 Feb 09 03:58:33	8° $\mathbb{M}$ 54'04	18°13'16	desc. node	2009 Dec 05 14:07:23	6° $\mathbb{M}$ 10'55	
retrograde	2009 Feb 15 14:53:50	12° $\mathbb{M}$ 20'19		max. Earth dist.	2009 Dec 17 08:00:24	24° $\mathbb{M}$ 35'46	1.44946 AU
evening set	2009 Feb 18 12:26:28	11° $\mathbb{M}$ 36'37					
max. declination.	2009 Feb 18 13:13:44	11° $\mathbb{M}$ 35'38	4S22'56"	superior conj	2009 Dec 19 06:11:16	27° $\mathbb{M}$ 37'44	-1°23'33
inferior conj	2009 Feb 24 15:12:25	6° $\mathbb{M}$ 18'51	3°43'09	minimum elong	2009 Dec 18 21:20:23	27° $\mathbb{M}$ 02'49	1°22'37
minimum elong	2009 Feb 24 15:13:52	6° $\mathbb{M}$ 18'48	3°43'09		2009 Dec 20 18:12:02	0° $\mathbb{Q}$	
min. Earth dist.	2009 Feb 27 05:49:34	3° $\mathbb{M}$ 29'00	0.63163 AU	min. declination.	2009 Dec 24 02:39:08	5° $\mathbb{Q}$ 19'16	25S 4'29"
morning rise	2009 Mar 02 17:03:07	0° $\mathbb{M}$ 15'37		evening rise	2100 Jan 03 07:25:18	21° $\mathbb{Q}$ 45'45	
	2009 Mar 03 01:16:54	30° $\mathbb{R}$ $\mathbb{M}$			2100 Jan 08 08:29:21	0° $\mathbb{M}$	
direct	2009 Mar 09 15:46:28	27° $\mathbb{M}$ 33'32		evening max el	2100 Jan 23 15:47:46	22° $\mathbb{M}$ 18'31	18°37'14
min. declination.	2009 Mar 15 20:22:53	29° $\mathbb{M}$ 22'38	11S31'40"	asc. node	2100 Jan 24 07:43:33	22° $\mathbb{M}$ 57'36	
desc. node	2009 Mar 16 16:13:06	29° $\mathbb{M}$ 51'34		retrograde	2100 Jan 30 06:29:15	25° $\mathbb{M}$ 58'21	
	2009 Mar 16 21:39:07	0° $\mathbb{M}$		evening set	2100 Feb 02 09:43:26	25° $\mathbb{M}$ 02'58	
morning max el	2009 Mar 23 13:12:01	5° $\mathbb{M}$ 29'39	27°45'16	max. declination.	2100 Feb 03 04:32:06	24° $\mathbb{M}$ 32'25	10S45'12"
	2009 Apr 11 02:10:05	0° $\mathbb{Y}$		inferior conj	2100 Feb 08 04:19:47	19° $\mathbb{M}$ 26'59	3°36'08
morning set	2009 Apr 26 21:19:19	28° $\mathbb{Y}$ 23'04		minimum elong	2100 Feb 08 02:51:13	19° $\mathbb{M}$ 31'25	3°35'59
	2009 Apr 27 16:33:34	0° $\mathbb{R}$		min. Earth dist.	2100 Feb 10 03:21:28	17° $\mathbb{M}$ 06'13	0.64885 AU
max. Earth dist.	2009 May 02 05:05:52	9° $\mathbb{R}$ 23'39	1.32858 AU	morning rise	2100 Feb 13 19:31:22	13° $\mathbb{M}$ 17'55	
				direct	2100 Feb 20 13:39:01	10° $\mathbb{M}$ 26'18	
superior conj	2009 May 04 13:41:35	14° $\mathbb{R}$ 26'32	-0°08'56	min. declination.	2100 Feb 27 22:20:19	13° $\mathbb{M}$ 07'39	16S12'53"
minimum elong	2009 May 04 14:07:31	14° $\mathbb{R}$ 28'51	0°08'50	desc. node	2100 Mar 03 13:15:35	15° $\mathbb{M}$ 58'23	
behind sun begin	2009 May 04 09:43:00	14° $\mathbb{R}$ 05'06		morning max el	2100 Mar 05 22:35:36	18° $\mathbb{M}$ 14'22	27°10'50
behind sun end	2009 May 04 18:32:02	14° $\mathbb{R}$ 52'38			2100 Mar 15 20:31:49	0° $\mathbb{M}$	
asc. node	2009 May 05 10:02:16	16° $\mathbb{R}$ 16'20					
evening rise	2009 May 11 14:24:10	29° $\mathbb{R}$ 35'27					
	2009 May 11 19:02:49	0° $\mathbb{II}$					
max. declination.	2009 May 27 03:27:01	27° $\mathbb{II}$ 43'56	25N39'56"				
	2009 May 28 18:58:28	0° $\mathbb{Q}$					
evening max el	2009 Jun 03 22:14:14	7° $\mathbb{Q}$ 04'00	23°26'03				
desc. node	2009 Jun 12 15:29:15	12° $\mathbb{Q}$ 57'49					
retrograde	2009 Jun 17 12:33:59	13° $\mathbb{Q}$ 51'53					
evening set	2009 Jun 21 09:17:09	13° $\mathbb{Q}$ 19'10					
min. Earth dist.	2009 Jun 28 08:20:59	10° $\mathbb{Q}$ 07'08	0.55847 AU				
inferior conj	2009 Jun 30 08:24:17	8° $\mathbb{Q}$ 55'51	-4°22'04				
minimum elong	2009 Jun 30 01:43:53	9° $\mathbb{Q}$ 05'48	4°20'49				
min. declination.	2009 Jul 05 07:46:56	6° $\mathbb{Q}$ 13'52	18N29'48"				
morning rise	2009 Jul 08 20:44:53	5° $\mathbb{Q}$ 00'54					
direct	2009 Jul 11 15:27:23	4° $\mathbb{Q}$ 41'54					
morning max el	2009 Jul 22 02:28:35	9° $\mathbb{Q}$ 32'41	20°16'13				
max. declination.	2009 Jul 30 02:01:45	19° $\mathbb{Q}$ 34'32	21N30'15"				
asc. node	2009 Aug 01 09:16:01	23° $\mathbb{Q}$ 21'23					
	2009 Aug 05 00:57:17	0° $\mathbb{Q}$					
morning set	2009 Aug 09 13:56:45	9° $\mathbb{Q}$ 00'22					